Northern Nut Growers Association Report of the Proceedings at the 41st Annual Meeting

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Northern Nut Growers Association

Incorporated

Affiliated with the American Horticultural Society

41st ANNUAL REPORT

Annual Meeting at

PLEASANT VALLEY, NEW YORK

August 28, 29 and 30, 1950

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SECOND DAY'S SESSION

Officers of the Association

1951

~President~—William M. Rohrbacher, M.D., 811 E. College, Iowa City, Iowa

~Vice-President~—Dr. L. H. MacDaniels, Cornell University, Ithaca, New York

~Treasurer~—Sterling A. Smith, 630 W. South St., Vermilion, Ohio

~Secretary~—J. C. McDaniel, Dept. of Horticulture, U. of I., Urbana, Illinois

~Additional Directors~—Mildred Jones Langdoc (Ill.) and H. F. Stoke (Va.)

~Nominating Committee~—Dr. H. L. Crane, (Chairman) Plant Industry Station, Beltsville, Maryland; Spencer B. Chase, Norris, Tenn.; Raymond E. Silvis, Massillon, Ohio

EXECUTIVE APPOINTMENTS, 1950-51

~Program~—Dr. A. S. Colby, Chm. (Ill.); J. C. McDaniel (Ill.); Prof. Geo. L. Slate (N. Y.); Royal Oakes (Ill.); Prof. W. D. Armstrong (Princeton, Ky.); Dr. H. L. Crane (Md.); D. C. Snyder (Ia.); W. W. Magill (Ky.); Prof. F. L. O'Rourke (Mich.); Ira M. Kyhl (Ia.); H. Gleason Mattoon (Pa.)

~Publications~—Editorial Section: Dr. Lewis E. Theiss, Chm. (Pa.); Dr. W. C. Deming (Conn.); Dr. J. Russell Smith (Pa.); Prof. George L. Slate (N. Y.); H. F. Stoke (Va.); John Davidson (O.); Dr. L. H. MacDaniels (Dept. of Floriculture and Ornamental Horticulture, Cornell University, Ithaca, N. Y.)

Printing Section—John Davidson, Chm. (O.); J. C. McDaniel (Ill.); Prof. George L. Slate (N. Y.); Carl F. Prell (Ind.)

~Place of Meeting~—J. F. Wilkinson, Chm. (Ind.); R. P. Allaman (Pa.); John A. Gerstenmaier (O.)

~Varieties and Contests~—Spencer B. Chase, Chm. (Tenn.); G. J. Korn, (Mich.); J. F. Wilkinson (Ind.); A. G. Hirschi (Okla.); L. Walter Sherman (Mich.); Sylvester Shessler (O.); Dr. L. H. MacDaniels (N. Y.); Fayette Etter (Pa.); Gilbert L. Smith (N. Y.)

Standards and Judging Section of this Committee—Spencer B. Chase, Chm. (Tenn.); Dr. L. H. MacDaniels (N. Y.); Dr. J. Russell Smith (Pa.)

~Survey and Research~—H. F. Stoke, Chm. (Va.); and the State and Foreign Vice-presidents.

~Membership~—D. C. Snyder, Chm. (Ia.); Stephen Bernath (N. Y.); Sterling A. Smith (O.); Raymond E. Silvis (O.); Carroll D. Bush (Wash.)

~Exhibits~—J. F. Wilkinson, Chm. (Ind.); R. P. Allaman (Pa.); Fayette Etter (Pa.); A. G. Hirschi (Okla.); G. J. Korn (Mich.); H. F. Stoke

(Va.); G. H. Corsan (Ont.); Edwin W. Lemke (Mich.); Carl Weschcke (Minn.)

~Necrology~ Mrs. Herbert Negus, Chm. (Md.); Mrs. C. A. Reed (D. C.); Mrs.
G. A. Zimmerman (Pa.)

~Auditing~ Raymond E. Silvis (O.); Carl F. Walker (O.)

~Finance~ Sterling A. Smith, Chm. (O.); Carl Weschcke (Minn.)

~Legal Adviser~ Sargent Wellman (Mass.)

~Official Journal~ American Fruit Grower, Willoughby, Ohio

State and Foreign Vice Presidents

Alabama Edward L. Hiles, Loxley
Alberta, Canada A. L. Young, Brooks
Belgium R. Vanderwaeren, Bierbeekstraat, 310,
Korbeek-Lo
British Columbia, Canada J. U. Gellatly, Box 19, Westbank
California Thos. R. Haig, M.D., 3021 Highland Ave.,
Carlsbad
Connecticut A. M. Huntington, Stanerigg Farms, Bethel
Delaware Lewis Wilkins, Route 1, Newark
Denmark Count F. M. Knuth, Knuthenborg, Bandholm
District of Columbia Edwin L. Ford, 3634 Austin St., S.E.,
Washington, 20
Florida C. A. Avant, 960 N.W. 10th Avenue, Miami

Georgia William J. Wilson, North Anderson Ave.,
Fort Valley
Hong Kong P. W. Wang, 6 Des Voeux Rd., Central
Idaho Lynn Dryden, Peck
Illinois Royal Oakes, Bluffs (Scott County)
Indiana Ford Wallick, Route 4, Peru
Iowa Ira M. Kyhl, Box 236, Sabula
Kansas Dr. Clyde Gray, 1045 Central Avenue, Horton
Louisiana Dr. Harald E. Hammar, 608 Court House,
Shreveport
Maryland Blaine McCollum, White Hall
Massachusetts S. Lathrop Davenport, 24 Creeper Hill Rd.,
North Grafton
Michigan Gilbert Becker, Climax
Minnesota R. E. Hodgson, Southeastern Exp. Station,
Waseca
Mississippi James R. Meyer, Delta Branch Exper. Station,
Stoneville
Missouri Ralph Richterkessing, Route 1, Saint Charles
Nebraska Harvey W. Hess, Box 209, Hebron
New Hampshire Matthew Lahti, Locust Lane Farm, Wolfeboro
New Jersey Mrs. Alan R. Buckwalter, Route 1, Flemington
New Mexico Rev. Titus Gehring, P. O. Box 177, Lumberton
New York George Salzer, 169 Garford Road, Rochester 9
North Carolina Dr. R. T. Dunstan, Greensboro College,
Greensboro
North Dakota Homer L. Bradley, Long Lake Refuge, Moffit
Ohio A. A. Bungart, Avon
Oklahoma A. G. Hirschi, 414 N. Robinson,
Oklahoma City

Ontario, Canada George H. Corsan, Echo Valley, Toronto 18
Oregon Harry L. Pearcy, Route 2, Box 190, Salem
Pennsylvania R. P. Allaman, Route 86, Harrisburg
Prince Edward Island, Canada Robert Snazelle, Forest Nursery, Rt. 5, Charlottetown
Rhode Island Philip Allen, 178 Dorance St., Providence
South Carolina John T. Bregger, P. O. Box 1018, Clemson
South Dakota Herman Richter, Madison
Tennessee W. Jobe Robinson, Route 7, Jackson
Texas Kaufman Florida, Box 154, Rotan
Utah Harlan D. Petterson, 2076 Jefferson Avenue, Ogden
Vermont Joseph N. Collins, Route 3, Putney
Virginia H. R. Gibbs, Linden
Washington Carroll D. Bush, Grapeview
West Virginia Wilbert M. Frye, Pleasant Dale
Wisconsin C. F. Ladwig, 2221 St. Laurence, Beloit

Attendance at the 1950 Meeting

Pleasant Valley, New York

Dr. J. Alfred Adams, New York State Agricultural Experiment Station, Route 33, Poughkeepsie, New York
Mr. R. P. Allaman, 8032 16th St., Harrisburg, Pennsylvania
Mrs. R. P. Allaman, 8032 16th St., Harrisburg, Pennsylvania
Mr. R. D. Anthony, State College, Pennsylvania
Mrs. Lillian V. Armstrong, 40 Earl Street, Toronto, Canada

(Now Mrs. George Hebden Corsan)
Mr. Richard Barcus, Massillon, Ohio
Mr. Alfred L. Barlow, 13079 Flanders Ave., Detroit 5, Michigan
Mrs. Irene M. Barlow, 13079 Flanders Avenue, Detroit 5, Michigan
Miss Betty Barlow, 13079 Flanders Ave., Detroit 5, Michigan
Mr. Leon Barlow, 13079 Flanders Ave., Detroit 5, Michigan
Mrs. Alice M. Bernath, Pleasant Valley, New York
Mr. Stephen Bernath, R. D. 3, Poughkeepsie, New York
Mr. Charles B. Berst, Erie, Pennsylvania
Mr. Harold Blake, Saddle River, New Jersey
Mr. Harold Blake, Jr., Saddle River, New Jersey
Mrs. Katherine Blake, Saddle River, New Jersey
Mr. George Brand, R. D. 45, Lincoln, Nebr. (Now in California)
Mr. William G. Brooks, Monroe, New York
Mrs. Alan R. Buckwalter, Flemington, New Jersey
Mr. Redmond M. Burr, 320 S. 5th Avenue, Ann Arbor, Michigan
Mrs. R. M. Burr, 320 S. 5th Avenue, Ann Arbor, Michigan
Mr. David H. Caldwell, 217 W. Hickory Street, Canastota, New York
(New York State College of Forestry)
Mr. Spencer B. Chase, Norris, Tennessee
Mr. William S. Clarke, Jr., Box 167, State College, Pennsylvania
Dr. Arthur S. Colby, University of Illinois, Urbana, Illinois
Mrs. Arthur S. Colby, Urbana, Illinois
Mr. George Hebden Corsan, Echo Valley, Toronto 18, Ontario
Mr. George E. Craig, Dundas, Ohio
Dr. H. L. Crane, Plant Industry Station, Beltsville, Maryland
Mrs. H. L. Crane, Hyattsville, Maryland
Mr. L. H. Dowell, 529 North Avenue, N.E., Massillon, Ohio
Mr. Aaron L. Ebling, R. D. 2, Reading, Pennsylvania
Mr. Ralph W. Emerson, Highland Park, Michigan

Mr. Edwin L. Ford, Washington, D. C.
Mr. Wilbert M. Frye, Pleasant Dale, West Virginia
Mr. Charles Gerstenmaier, Massillon, Ohio
Mr. John A. Gerstenmaier, Massillon, Ohio
Mrs. J. A. Gerstenmaier, Massillon, Ohio
Mrs. Bessie J. Gibbs, Linden, Virginia
Mr. H. R. Gibbs, Linden, Virginia
Mr. Ralph Gibson, Williamsport, Pennsylvania
Mr. S. H. Graham, Bostwick Road, Ithaca, New York
Mrs. S. H. Graham, Bostwick Road, Ithaca, New York
Mr. Henry Gressel, R. D. 2, Mohawk, New York
Mrs. Nora Gressel, R. D. 2, Mohawk, New York
Mr. Earl C. Haines, Shanks, West Virginia
Mr. Walter Hasbrouck, New Paltz, New York
Mrs. Walter Hasbrouck, New Paltz, New York
Mr. Andrew Kerr, Barnstable, Massachusetts
Mr. Frank M. Kintzel, Cincinnati, Ohio
Mr. Ira M. Kyhl, Sabula, Iowa
Miss Bertha Landis, 425 Marion Avenue, Mansfield, Ohio
Mr. James D. Lawrence, R. D. 3, Middletown, New York
Mr. Frederick L. Lehr, Hamden, Connecticut
Mr. James Lowerre, R. D. 3, Middletown, New York
Dr. L. H. MacDaniels, Ithaca, New York
Prof. J. C. McDaniel, 104 Horticultural Field Laboratory,
University of Illinois, Urbana, Illinois
Mr. J. W. McKay, Plant Industry Station, Beltsville, Maryland
Mr. Elwood Miller, Hazleton, Pennsylvania
Mrs. Elwood Miller, Hazleton, Pennsylvania
Mr. Louis Miller, Cassopolis, Michigan
Dr. James K. Mossman, Ramapo, New York

Mrs. Herbert Negus, Mount Ranier, Maryland
Mr. Royal Oakes, Bluffs, Illinois
Mrs. Royal Oakes, Bluffs, Illinois
Mr. F. L. O'Rourke, Hidden Lake Gardens, Michigan State College,
Tipton, Michigan
Mr. John H. Page, Dundas, Ohio
Mr. Philip P. Parkinson, 567 Broadway, Newark, New Jersey
Mrs. Philip P. Parkinson, 567 Broadway, Newark, New Jersey
Mr. Christ Pataky, Jr., Mansfield, Ohio
Mrs. Christ Pataky, Mansfield, Ohio
Mr. Gordon Porter, Windsor, Ontario
Mrs. Penelope Porter, Windsor, Ontario
Mrs. C. A. Reed, 7309 Piney Branch Road, Washington 12, D. C.
Mr. John Rick, 438 Penn Street, Reading, Pennsylvania
Dr. William M. Rohrbacher, Iowa City, Iowa
Mrs. Elizabeth I. Rohrbacher, Iowa City, Iowa
Mr. George Salzer, Rochester, New York
Mrs. George Salzer, Rochester, New York
Mr. Rodman Salzer, Rochester, New York
Mr. L. Walter Sherman, Harrisburg, Pennsylvania
Mrs. L. W. Sherman, Harrisburg, Pennsylvania
(The Shermans now in Michigan)
Mr. Raymond E. Silvis and Family, Massillon, Ohio
Mr. George L. Slate, Geneva, New York
Mr. Douglas A. Smith, Vermilion, Ohio
Mr. Gilbert L. Smith, Millerton, New York
Mr. Jay L. Smith, Chester, New York
Mr. Sterling A. Smith, 630 W. South Street, Vermilion, Ohio
Mr. Harwood Steiger, Red Hook, New York
Mrs. Sophie H. Steiger, Red Hook, New York

Mr. H. F. Stoke, 1436 Watts Avenue, Roanoke, Virginia
Mrs. H. F. Stoke, 1436 Watts Avenue, Roanoke, Virginia
Mr. Alfred Szego, 77-15A 37th Avenue, Jackson Heights, New York, N. Y.
Prof. T. J. Talbert, Columbia, Missouri
Dr. Lewis E. Theiss, Lewisburg, Pennsylvania
Dr. Frank A. Washick, Philadelphia 11, Pennsylvania
Mr. Harry R. Weber, Cincinnati, Ohio
Mr. Sargent H. Wellman, Topsfield, Massachusetts
Mrs. Laura L. Whiteford, Pleasant Valley, Duchess County, New York
Mr. J. F. Wilkinson, Rockport, Indiana,
Mr. William J. Wilson, Fort Valley, Georgia
Mrs. William J. Wilson, Fort Valley, Georgia
Mrs. G. A. Zimmerman, Route 1, Linglestown, Pennsylvania

Complete membership list is in back of this volume.

CONSTITUTION

of the

NORTHERN NUT GROWERS ASSOCIATION, INCORPORATED

(As adopted September 13, 1948)

NAME

~Article I.~ This Society shall be known as the Northern Nut Growers Association, Incorporated. It is strictly a non-profit organization.

PURPOSES

~Article II.~ The purposes of this Association shall be to promote interest in the nut bearing plants; scientific research in their breeding and culture; standardization of varietal names; the dissemination of information concerning the above and such other purposes as may advance the culture of nut bearing plants, particularly in the North Temperate Zone.

MEMBERS

~Article III.~ Membership in this Association shall be open to all persons interested in supporting the purposes of the Association. Classes of members are as follows: Annual members, Contributing members, Life members, Honorary members, and Perpetual members. Applications for membership in the Association shall be presented to the secretary or the treasurer in writing, accompanied by the required dues.

OFFICERS

~Article IV.~ The elected officers of this Association shall consist of a President, Vice-president, a Secretary and a Treasurer or a combined Secretary-treasurer as the Association may designate.

BOARD OF DIRECTORS

~Article V.~ The Board of Directors shall consist of six members of the Association who shall be the officers of the Association and the two preceding elected presidents. If the offices of Secretary and Treasurer are combined, the three past presidents shall serve on the Board of Directors.

There shall be a State Vice-president for each state, dependency, or country represented in the membership of the Association, who shall be appointed by the President.

AMENDMENTS TO THE CONSTITUTION

~Article VI.~ This constitution may be amended by a two-thirds vote of the members present at any annual meeting, notice of such amendment having been read at the previous annual meeting, or copy of the proposed amendments having been mailed by the Secretary, or by any member to each member thirty days before the date of the annual meeting.

BY-LAWS

(Revised and adopted at Norris, Tennessee, September 13, 1948)

SECTION I.—MEMBERSHIP

Classes of membership are defined as follows:

~Article 1. Annual members.~ Persons who are interested in the purposes of the Association who pay annual dues of Three Dollars ($3.00).

~Article 2. Contributing members.~ Persons who are interested in the purposes of the Association who pay annual dues of Ten Dollars ($10.00) or more.

~Article 3. Life members.~ Persons who are interested in the purposes of the Association who contribute Seventy Five Dollars ($75.00) to its support and who shall, after such contribution, pay no annual dues.

~Article 4. Honorary members.~ Those whom the Association has elected as honorary members in recognition of their achievements in the special fields of the Association and who shall pay no dues.

~Article 5. Perpetual members.~ "Perpetual" membership is eligible to any one who leaves at least five hundred dollars to the Association and such membership on payment of said sum to the Association shall entitle the name of the deceased to be forever enrolled in the list of members as "Perpetual" with the words "In Memoriam" added thereto. Funds received therefor shall be invested by the Treasurer in interest bearing securities legal for trust funds in the District of Columbia. Only the interest shall be expended by the Association. When such funds are in the treasury the Treasurer shall be bonded. Provided: that in the event the Association become defunct or dissolves, then, in that event, the Treasurer shall turn over any funds held in his hands for this purpose for such uses, individuals or companies that the donor may designate at the time he makes the bequest of the donation.

SECTION II.—DUTIES OF OFFICERS

~Article 1.~ The President shall preside at all meetings of the Association and Board of Directors, and may call meetings of the Board of Directors when he believes it to be to the best interests of the Association. He shall appoint the State Vice-presidents; the standing committees, except the Nominating Committee, and such special committees as the Association may authorize.

~Article 2.~ Vice-president. In the absence of the President, the Vice-president shall perform the duties of the President.

~Article 3.~ Secretary. The Secretary shall be the active executive officer of the Association. He shall conduct the correspondence relating to the Association's interests, assist in obtaining memberships and otherwise actively forward the interests of the Association, and report to the Annual Meeting and from time to time to meetings of the Board of Directors as they may request.

~Article 4.~ Treasurer. The Treasurer shall receive and record memberships, receive and account for all moneys of the Association and shall pay all bills approved by the President or the Secretary. He shall give such security as the Board of Directors may require or may legally be required, shall invest life memberships or other funds as the Board of Directors may direct, subject to legal restrictions and in accordance with the law, and shall submit a verified account of receipts and disbursements to the Annual meeting and such current accounts as the Board of Directors may from time to time require. Before the final business session of the Annual Meeting of the Association, the accounts of the Treasurer shall be submitted for examination to the Auditing Committee appointed by the President at the opening session of the Annual Meeting.

~Article 5.~ The Board of Directors shall manage the affairs of the association between meetings. Four members, including at least two elected officers, shall be considered a quorum.

SECTION III.—ELECTIONS

~Article 1.~ The Officers shall be elected at the Annual Meeting and hold office for one year beginning immediately following the close of the Annual Meeting.

~Article 2.~ The Nominating Committee shall present a slate of officers on the first day of the Annual Meeting and the election shall take place at the closing session. Nominations for any office may be presented from the floor at the time the slate is presented or immediately preceding the election.

~Article 3.~ For the purpose of nominating officers for the year 1949 and thereafter, a committee of five members shall be elected annually at the preceding Annual Meeting.

~Article 4.~ A quorum at a regularly called Annual Meeting shall be fifteen (15) members and must include at least two of the elected officers.

~Article 5.~ All classes of members whose dues are paid shall be eligible to vote and hold office.

SECTION IV.—FINANCIAL MATTERS

~Article 1.~ The fiscal year of the Association shall extend from October 1st through the following September 30th. All annual memberships shall begin October 1st.

~Article 2.~ The names of all members whose dues have not been paid by
January 1st shall be dropped from the rolls of the Society. Notices of
non-payment of dues shall be mailed to delinquent members on or about
December 1st.

~Article 3.~ The Annual Report shall be sent to only those members who have paid their dues for the current year. Members whose dues have not been paid by January 1st shall be considered delinquent. They will not be entitled to receive the publication or other benefits of the Association until dues are paid.

SECTION V.—MEETINGS

~Article 1.~ The place and time of the Annual Meeting shall be selected by the membership in session or, in the event of no selection being made at this time, the Board of Directors shall choose the place and time for the holding of the annual convention. Such other meetings as may seem desirable may be called by the President and Board of Directors.

SECTION VI.—PUBLICATIONS

~Article 1.~ The Association shall publish a report each fiscal year and such other publications as may be authorized by the Association.

~Article 2.~ The publishing of the report shall be the responsibility of the Committee on Publications.

SECTION VII.—AWARDS

~Article 1.~ The Association may provide suitable awards for outstanding contributions to the cultivation of nut bearing plants and suitable recognition for meritorious exhibits as may be appropriate.

SECTION VIII.—STANDING COMMITTEES

As soon as practical after the Annual Meeting of the Association, the President shall appoint the following standing committees:

1. Membership 2. Auditing 3. Publications 4. Survey 5. Program 6. Research 7. Exhibit 8. Varieties and Contests

SECTION IX.—REGIONAL GROUPS AND AFFILIATED SOCIETIES.

~Article 1.~ The Association shall encourage the formation of regional groups of its members, who may elect their own officers and organize their own local field days and other programs. They may publish their proceedings and selected papers in the yearbooks of the parent society subject to review of the Association's Committee on Publications.

~Article 2.~ Any independent regional association of nut growers may affiliate with the Northern Nut Growers Association provided one-fourth of its members are also members of the Northern Nut Growers Association. Such affiliated societies shall pay an annual affiliation fee of $3.00 to the Northern Nut Growers Association. Papers presented at the meetings of the regional society may be published in the proceedings of the parent society subject to review of the Association's Committee on Publications.

SECTION X—AMENDMENTS TO BY-LAWS

~Article 1.~ These by-laws may be amended at any Annual Meeting by a two-thirds vote of the members present provided such amendments shall have been submitted to the membership in writing at least thirty-days prior to that meeting.

REPORT OF THE PROCEEDINGS at the Forty-First Annual Meeting of the
Northern Nut Growers Association, Inc.

Held at PLEASANT VALLEY, DUTCHESS COUNTY, NEW YORK on AUGUST 28, 29 and 30, 1950

TOGETHER WITH OTHER PAPERS ON NUT CULTURE

MONDAY MORNING SESSION

The meeting was called to order by the Vice-President, Dr. L. H. MacDaniels, in the absence of the President.

DR. MacDANIELS: I have here the official gavel of The Northern Nut Growers Association, which was sent to me by Mildred Jones Langdoc, who unfortunately is not able to come to this meeting. She, of course, is our president. She expected to come until fairly recently but on her doctor's orders changed her plans and wrote to me a very short time ago asking me if I would preside at this meeting.

Does anyone present know the history of this gavel?

MR. GEORGE SLATE: It was presented to the Association by Mr. Littlepage, and was made from Indiana pecan wood.

DR. MacDANIELS: But anyway here it is, and we declare the Association in session.

This morning the meeting is quite brief. We will start the meeting with the report from the Secretary, Mr. McDaniel.

Secretary's Report

J. C. McDaniel

MR. J. C. McDANIEL: My report before the meeting will be very brief. It may be extended a little later for the publication.

The last count for this Association's membership made last week shows the Association has 575 paid members, plus 20 subscribers and one foreign exchange membership, totalling 596. There have been a few more members come in since then, so I might say we have in round figures about 600 members to date in 1950, a few less than last year.

I probably owe the members an explanation on the delay in the printing of the Fortieth Annual Report. That was finally taken up by the printing company and should be printed by now. It was ready to put on the press—in fact, some of it was on the press when I left Nashville two weeks ago, and we have every reason to believe that it will be ready for mailing in about another week. The Treasurer said he heard me say that six months ago. That's six months nearer to being the truth now.

I requested that the printer send up two copies, whether they are bound or not, so they may be in to show you later during the meeting.

I believe that's about all I will say at this time, Mr. President.

DR. MacDANIELS: This matter of the report not being here I know is the cause of considerable dissatisfaction, and it arises out of our attempt to get the report printed cheaply. We have had the same trouble before. The Corse Press did this at one time and did it cheaply, because they would work it in with the other business. The last time they did it, and other business was so heavy that it was delayed.

The printers who do it at Nashville also did the Legislative printing and other things cut in, so that it was not carried on. Now, I think that we have some ideas in mind for printers for the next issue, so that if we get the papers in on time, the report will be coming out fairly promptly.

Is the Treasurer ready with his report? Mr. Sterling Smith.

Treasurer's Report

Sept. 1, 1949 to Aug. 25, 1950

RECEIPTS:

Annual Membership Dues $1,689.55

(Contributing Members: Arp Nursery Co. and
Mr. Hjalmar W. Johnson
$10.00 each)
Life Membership (Herschel L. Boll) 75.00

Contributions
Mr. A. M. Huntington 50.00
Mr. Geo. L. Slate 2.00

Sale of Reports 186.00
Interest on U. S. Bonds 31.25
Worcester County (Mass.) Hort. Society 25.00
Advertisement 5.00
Miscellaneous 18.00

Total Income $2,081.80

DISBURSEMENTS:

U. S. Bond "G" $ 500.00
American Fruit Grower Subscriptions 224.00
Supplies, Stationery, etc. for Secretary 96.75
Secretary's 50c per Member 275.00
Secretary's Expense 88.00
Treasurer's Expense 66.52
Reporting Beltsville Meeting 60.00
Mr. Reed's Memorial 10.00
Bank Service Charge 3.33
Miscellaneous 21.00

Total Disbursements $1,344.60

Cash on deposit at Erie County United Bank $2,292.97
Petty Cash on Hand 12.70
Disbursements 1,344.60

Total $3,650.27

On Hand Sept. 1, 1949 $1,568.47
Receipts Sept. 1. 1949, to Aug. 25, 1950 2,081.80

Total $3,650.27

U. S. Bonds in Safety Deposit Box $3,000.00

DR. MacDANIELS: Thank you, Mr. Smith. I think it is usual to accept the report and then refer it, I believe, to an auditing committee.

A MEMBER: I so move.

DR. MacDANIELS: It is moved that the report be accepted and turned over to the auditing committee.

MR. SZEGO: Second.

DR. MacDANIELS: Seconded. Any remarks? (No response.)

(A vote was taken on the motion, and it was carried unanimously.)

DR. MacDANIELS: I'd like to appoint Mr. Royal Oakes and Mr. Weber as Auditing Committee, and I think they report at the final business session, which comes at the banquet.

I will say that matter of $25.00 I didn't know anything about, except now I recall the circumstances. At the convention I took over what was left of the exhibits—nobody wanted them—and took them back to Ithaca, thinking I would send them to the Massachusetts Horticultural Society. I didn't have time to do that, but I did send them to Worcester (Mass.) Horticulture Society, and apparently I was out of the country and they sent the award to the Treasurer, and that accounts for the $25.00. It's the first I have heard of it, but anyway, we have it.

The treasurer's report indicates we have some little surplus in the treasury, but after our report is paid for, that will be reduced to the amount of about $800.00. That is the net surplus at the present time, and if we face the facts of the matter, it means that we are not living within our income, that is, with printing costs going up. The reports used to cost $600.00, instead of $1400.00, and what not.

The reason we have kept going has been the use of life memberships and the extra contribution of Mr. Archer Huntington.

The matter of deficit financing seems to be good for the Government, but I don't think it is any good for the society. I think, however, we can adjust our affairs so as to get along. It is proposed we make a change in the by-laws which will set up another type of membership. That is, at the present time we have an annual membership of $3.00 and a contributing membership of $10.00 and life membership for $75.00. Taking the pattern from some other societies, it at least was discussed that we put up a membership of $5.00, which was a sustaining membership, and anybody who felt that he could do that easily could do so, not receiving any additional benefits, except, perhaps, a star in front of his name,—just considering it a contribution to the society.

What we had in mind is that we know that there are some of the membership that find the $3.00 is plenty high enough. There are others to whom probably it means another dinner, or something of that kind, and it doesn't make so much difference. And what we propose to do is to make it easy for those who can to give that additional support.

That amendment will be proposed at the last business meeting in some form, and it will have to go over until the next meeting, according to our constitution, which provides for the amendment of the by-laws.

Mr. Secretary, do we have a report of the editor?

MR. J. C. McDANIEL: Yes, I have that here, a short report from Dr. Lewis
E. Theiss, who will be at the meeting in the morning.

Report of Publications and Publicity

DR. LEWIS E. THEISS, Chairman

The annual Report, which should be issued very soon, will speak for itself. Delay more than usual was occasioned by an effort to make the publication fully complete. To that end, printing was held up so that, for one thing, we could include Dr. J. Russell Smith's remarkable summary or survey of nut experimentation in the U. S. and Canada.

We cannot overemphasize the great services of our secretary, Mr. McDaniel, in the preparation of this work. He collected the material, forwarded it to me for editing, did much editing himself, secured the printing contract, and in general oversaw the production of the volume.

To edit the manuscripts for a book of this size is in itself quite a chore. Proof reading is a great burden. In the preparation of this Report, we have had the hearty cooperation and help of Mrs. Herbert Negus (Md.); Professor George Slate (New York); Dr. A. S. Colby (Ill.); Mr. Spencer Chase (Tenn.); and Mr. Alfred Barlow (Mich.). We are indebted to all of these members for their fine support. We hope that this present issue will be a worthy successor to the many fine ones that have preceded it.

LEWIS E. THEISS, Chairman Publications Committee Read at meeting 8/28/50.

MR. J. C. McDANIEL: I might say, by the way, it will be 8 pages larger than last year's, totalling 232 pages.

DR. MacDANIELS: Thank you, Mr. Secretary.

The question is going to arise as to the size of our report. That is, the reports up to the last two have been something less than 200 pages, I believe. This one is running over considerably, and the question comes up as to whether or not we should economize by reducing the size of the report. It was the general opinion of the Directors in discussing the matter that perhaps somewhat closer editing should be done, but we realize that for many members of the Association the report is the one tangible thing that they get out of the whole picture and that the reports should be kept, certainly, at a good length and high grade.

I think those are all of the officers' reports. Are there reports of the committees? Program Committee, Mr. Slate, do you have a brief report?

MR. GEORGE L. SLATE: The report of the Program Committee has been published, and the programs are on this table in the rear of the room.

DR. MacDANIELS: Brief and to the point. In other words, Mr. Slate has written around to the persons who are going to be on the program, sort of cranking them up. This society is in a situation where its members don't just flock to the call of requests for papers, and they have to be solicited. Well, Mr. Slate has done a very good job of soliciting papers, and the report speaks for itself in the program which has been prepared.

Reports of any special committees? Do we have a committee on contests?—of the Carpathian walnut contest?

MR. McDANIEL: I believe that will be taken up in the afternoon program.

DR. MacDANIELS: The matter of old business. Do we have any old business,
Mr. Secretary?

MR. McDANIEL: I don't know of any that's carried over now.

Discussion on Time and Place of Meeting

DR. MacDANIELS: Coming to new business. There is always the time and the place of the next convention. I think that that is usually in the hands of a committee, but in the open meeting the matter is discussed, and we are open for any suggestions.

I have heard that Dr. Colby of Illinois is going to have a suggestion that we come to Illinois.

MR. McDANIEL: That's my understanding, and he should be here a little later.

DR. MacDANIELS: Anybody else have any suggestions?

I think, with regard to our time and place of meeting, we have in mind alternating between the East, and the Middle West. The center of membership appears to be about Central Ohio, is that right? And I don't think we have gone any farther west than Center Point, Iowa.

MR. WEBER: That was back in 1930.

DR. MacDANIELS: That probably is about as far West as we are going to get, unless we get a lot of members out farther.

Now, suggestions that have been made have been that next year the meeting would be in Illinois—at the University of Illinois—and the year following somewhere in the East, possibly Pennsylvania, although we haven't been invited to Pennsylvania. I don't know whether we can get one or not. And the next year west again, possibly Michigan, and beyond that we haven't thought. But I think there is a real advantage in having time blocked out in advance for at least two years so that people can make their plans as to where they will go. That is, I think often in planning vacations and what not, it goes that far ahead.

MR. JAY SMITH: Mr. Chairman, the last week in August seems to be better than the first week in September, from the point of view of the school openings in early September.

MR. WELLMAN: I think we should wait a little while and see what kind of attendance we get at this meeting this time of the year.

MR. RICK: If we could arrange it, we'd like to appeal to the membership to have a meeting in Lancaster County. I think Mr. Hostetter has quite a number of things that could be shown and perhaps some others in the neighborhood that might make it quite interesting.

DR. MacDANIELS: We can refer that to the committee.

MR. ALLAMAN: Mr. President, I think that is a very fine suggestion. One of our nut growers in Pennsylvania lives in Lancaster County, and he has told me he has 29,000 nut trees, including filberts, and is still planting.

DR. MacDANIELS: That sounds almost like the Government debt, only not quite.

We will let that matter go until the committee reports when Dr. Colby arrives.

Is there any other business which we ought to transact at this time? If not, I think the next item is the president's address, which has just arrived. Mrs. Bernath just brought it in. It just came in under the wire, I guess.

DR. CRANE: Mr. Stoke has just come in.

DR. MacDANIELS: We will have the report of the nominating committee, Mr.
Stoke.

Report of Nominating Committee

MR. STOKE: We bore in mind when we were making nominations for the presidency that we will probably hold our next meeting in the West, so we have nominated Dr. William Rohrbacher of Iowa for president, and Dr. MacDaniels, our perennial vice-president be nominated again and hope that we get him across next year as president. He has served a pretty good apprenticeship. Our secretary, J. C. McDaniel, has been nominated for re-election and Sterling Smith as treasurer. The last two ex-presidents will be on the Board of Directors. Those, with the other officers named, constitute our entire Board of Directors.

DR. MacDANIELS: Thank you, Mr. Stoke.

You have heard the report of the nominating committee.

DR. CRANE: Move that they be accepted.

MR. ALLAMAN: Second.

DR. MacDANIELS: Are there remarks? (No response.) If not, we will take a vote.

(Whereupon, a vote was taken on the motion, and it was carried unanimously.)

DR. MacDANIELS: The election comes at the time of the banquet, and nominations may be made from the floor at the time of election.

Dr. Colby, I believe, came in. Do you want to say something about Illinois as a meeting place for next year. Dr. Colby of the University of Illinois.

DR. COLBY: I don't know whether there was any malice aforethought in that committee nomination! Before I left Urbana a few weeks ago, Dean H. P. Rusk of our College of Agriculture asked me to invite you people to come to Urbana, Illinois for your meeting next year. So that, Mr. President, is an official invitation. We hope that you can all come. I see some of our Illinois friends here, and we are all working together to provide an interesting meeting at that time.

Now, as to the date, that will have to be settled a little later.

DR. MacDANIELS: Thanks very much, Dr. Colby. That makes it official.

MR. WEBER: Mr. President, I move we accept the invitation.

MR. JAY SMITH: I second.

DR. MacDANIELS: Moved and seconded we go to Illinois, the time to be arranged by the committee. Any remarks? (No response.)

(Whereupon, a vote was taken on the motion, and it was carried unanimously.)

DR. MacDANIELS: That fixes that, and the time will depend somewhat on the availability of dormitories. If the meeting is held the last week in August, the dormitories would be available, would they not?

Mr. Weber: Get away from the Labor Day problem, too.

DR. MacDANIELS: Any other business? Has anyone else come in in the meantime who has a report?

If not, we will go ahead with the next item, which is the President's Address, and I will ask Mr. Weber of Cincinnati to read this. I am much pleased to do this because of Mr. Weber's friendship for the president.

President's Address

MILDRED JONES LANGDOC, Erie, Illinois

I have been a member of this organization for a good many years, and I have always had a deep interest in its success. Our members are in a position to encourage the planting of good varieties of nut trees which may some day be appreciated even more for food and other uses as our population increases than we as a nation appreciate them today. Tree crops are a means of conserving our soils, both from the point of erosion and moisture holding content. I like the opportunity we have to be far-sighted in encouraging the planting of nut trees which will play a large part in the future well-being of our country.

Our N.N.G.A., as it is today, has been built on the unselfish efforts of a number of far-sighted men who had an ideal and a will to see that ideal accomplished. I think I was fortunate to know a number of the early founders of the organization either through their visits to my home where my father and they would talk their favorite subject of nut varieties known, just found, or the ideal variety they hoped they'd locate—perhaps in the next nut contest. In lighter mood—usually around the dinner table—they would sometimes reminisce about this joke or that which some member played on another. Altogether our early founders and officers were really great men, bringing experiences from various walks of life. Today we have a still wider variety of occupations listed among our membership, and an even greater opportunity to make acquaintances and friends. I hope every member will make full use of his leisure time here at this convention to make new acquaintances and to renew old ones. Knowing the members as I do, I know you will treasure these acquaintances during your entire lifetime.

The Association can serve its members in a number of ways, but I would place special emphasis on our reports carrying from year to year a progressive report on varieties. In other words, I think our survey reports are one important part of our means of learning about the performance of varieties in various sections of the country where they are being tried. I would urge every member to make a definite effort to co-operate with the survey committee in sending the information they require, because these men making the survey are busy men, too, just like the rest of us, and they have to make a real effort to find time to tabulate the information they receive, and they want to receive more, so they are willing to do their part to tabulate the information which will help us as an organization to be more definite about encouraging or discouraging the planting of a certain variety.

There is a question in my mind whether the very best nut so far as cracking quality is concerned will be the best variety for the average home planter. I think we should consider whether the variety will bear good crops consistently, and if it doesn't bear well—why? Perhaps it is a matter of soil condition which can be corrected, a matter of a variety being planted in a climate where it cannot bear well, and perhaps elevation above sea level is another factor. We may even find with the hickories and walnuts that certain varieties will perform better with certain other varieties as pollinators.

When we think of these things there is much to be done in the evaluation of varieties, although there has been a start in the right direction.

It seems to me that nut contests at regular intervals help to stimulate interest in better varieties of nuts and we do gain a certain amount of free advertising through newspapers and magazines. The results of the contest should state, in my opinion, the comparison of the varieties selected as the best of the contest with the ratings of varieties already named and now in propagation. This would mean using the same score card always. Remembering that the very best rated cracking nut is not always the best bearing variety, it would help to accompany this variety report with data as to the location of the tree—soil it is growing in—soil type—good drainage or a damp location—rainfall during the year—days between frost—whether the tree has had good care or not—whether it's a heavy bearer—and any other information which may have a bearing upon the health and vigor of the tree. If notes can be taken on the blooming and bearing habit of other trees of the same species close by which may influence this particular variety through cross-pollination, then we would have a good record immediately on each variety.

I realize in stating the above that we must rely on the human mind which colors and evaluates everything our senses perceive, so it's up to us as individuals to try constantly to train ourselves to evaluate a variety as it really is. I feel that much of the success of our organization in the gathering of nut tree varieties has been due to an honest effort towards reporting only facts and we will do well to enlist the aid of our college trained scientific minds to help us individuals in asking ourselves the necessary questions about our nut tree varieties.

According to the phrase "Life begins at 40," we are now just beginning to live as an organization. Let us then examine every means to set our course towards the definite goal of evaluating the worth of all the named varieties of northern grown nut trees, let us report our findings without prejudice, let us continue to make our annual reports so necessary as a clearing house for the year's progress in nut culture, so valuable, that anyone interested in nut culture can't afford not belonging to and being an active part of our group. I would especially like to see other active state groups as the Ohio group all bringing together their yearly information in one book form—our Annual Report. The Ohio group deserves special recognition on the wisdom of their officers to work towards a unified northern nut growers group, each helping the other where they can.

I want to express my appreciation to our Secretary, Mr. McDaniel, for his work this year which can be doubly appreciated by those who know the excellent job he has performed in spite of many adversities. I hope he will continue as Secretary.

Our Treasurer, Mr. Smith, has been right on the job, and we can all be of special help to him by sending or giving to him here and now our dues for the coming year. We would not waste any time by paying our dues promptly, but we would save a tremendous amount of time for him. We can in this way make his association and work for us most pleasant and in that way show him how much we appreciate his help. I express the hope that Mr. Smith will be our Treasurer for a long time.

I want to thank the Board of Directors and all of the committees who have labored so faithfully during the year. Our convention program for this year is evidence that our Program Committee has spent much time in thought, correspondence and work and we all appreciate and give them our hearty thanks.

Since I cannot be with you this year, Dr. MacDaniels has consented to occupy the Chair and the 41st annual meeting will now go forward under his able direction. I am with you in thought.

Sincerely,
MILDRED JONES LANGDOC

* * * * *

MR. WEBER: By the way, since I am on the floor and I am on my feet, I will pass this attendance record. Will you all please sign your names and addresses. It doesn't bind you to anything.

MR. CORSAN: You might tell the audience—there are some strangers here—who the president is whose address you just read.

MR. WEBER: I read her name, the former Mildred Jones, whose father was the late J. F. Jones who was one of the pioneers in the propagating of nut trees, and was formerly living in Lancaster County, Pennsylvania, three miles south of Lancaster on U. S. 222. His daughter continued his work after his death, has since married and is now living out at Erie, Illinois, which is west of Chicago near the Mississippi River. Her name now is Langdoc.

DR. MacDANIELS: Our president brought out two points in which I most heartily concur. One is our search for new varieties and the evaluation of varieties, and the other, the more extensive rating of the varieties we already have. There will be this round-table this evening on evaluation of varieties, of which Dr. Crane will be the chairman.

Association Sends Greetings to Dr. Deming

DR. McKAY: I'd like to bring up this matter—I'd like to make this in the form of a motion, that in view of the long and active service of Dr. W. C.

Deming to this organization, I think it would be appropriate for this organization to send him greetings. I would like to make that in the form of a motion.

MR. BERNATH: I second it.

DR. MacDANIELS: Moved and seconded to send Dr. Deming greetings from the meeting. We had hoped that he would be here. He may come yet, unless somebody knows definitely to the contrary. George Slate saw him a while ago and said he hopes to get here.[1]

[1] Dr. Deming was present at the lunch stop on the Wassaic State School grounds during the third day's tour.—Ed.

MR. WEBER: I have just been informed that Dr. Deming will be 89 years old on September first.

DR. MacDANIELS: That's something.

How old is Mr. Corsan?

MR. WEBER: The question arises: How old is Mr. Corsan? The gentleman is here, and he will speak for himself.

Talk by the Oldest Member

MR. CORSAN: I don't know how old I am. I know I was born near Rockport, New York, and my father brought me across the river to Hamilton, Ontario, when I was seven, and according to my aunts and uncles and people who told me, they say I was born June 11, 1857. So here I am kicking around, but I am not blowing how long I will live. I don't know, but I will try my best.

I have joined the Vegetarian Society many years ago, and I am still hanging onto that idea, and I hope that we have a vegetarian banquet some of these times, because nearly all vegetarian associations are very deeply interested in the Northern Nut Growers Association. That's what they all told me at the convention at Lake Geneva last August a year ago. And I just came back from visiting Rodale. I thought I'd see Rodale. He looks a good deal like this gentleman here (indicating Mr. Bernath), our friend here, about the size and appearance of him. But he is of the greatest ancestry in the world. He is Jewish, and he doesn't know exactly how to eat, because he has jowls and dewlaps and he is too fat, but he is a very fine man; beautiful, clear, honest eyes, he has, and I hope to have him consider the planting of nut trees on his place. He has a disgraceful looking place in comparison to mine.

This year my place is just loaded down with nuts, except filberts. Last year I had so many filberts that I have half a ton left over yet. And I want to see people beautify the country. I started off one day with a thought that came to my head. I heard that there were a half a million widows and orphans buried in the Hudson Hill Cemetery. And I thought: Why, those dead people can be working; they can be doing something. Let them feed the roots of the Japanese heartnut. And as a try, I sent them 1100 seeds just as a start. And the Japanese heartnut, a stranger to this country, isn't anywhere near any other nut, and it grows true to form, and a lot of the trees are much hardier up on Lake Ontario. It does not grow well on the north of the lake, but south of the lake it grows enormous crops every year, and the nuts come out whole. But there is a better shaped nut without that kind of groove in the center, and it's the father or the mother—father, probably—of the finest heartnuts in the world, and there is nothing that beats a heartnut for eating. Every time I sell heartnuts to eat I have ruined myself, because they won't eat any other nut. So that shows just exactly what the general public thinks of it. Even Italians. There I have a half a ton of filberts. I bring the heartnuts down to Florida, the Fairchild and my hybrid trees and butternuts and Japanese heartnuts, and I have a package of almonds and another package of brazil nuts, and I let them taste those. They are woody in comparison to our heartnuts and hybrids. They are not anything, they are just like so much wood in comparison.

Now, I have received from John W. Fowler, Secretary to Albert Williams of the Department of Corrections on 100 Center Street. New York, a beautiful letter accepting those nuts, and I had my housekeeper—I was down in Florida—send them to them early in February, and they are planted. And the breezes going up and down the Hudson are going to wave the two-foot-long leaves of the most beautiful deciduous trees in the world, the Japanese heartnut, healthiest, hardiest nut in the world, and these dead people will be feeding them. Just think! five thousand children without a name or number. Now, they have erected a monument just recently, but the real monuments are the living trees. I am going to send them a lot more, because I want to see them working. I might come back and eat some of these nuts myself.

* * * * *

DR. MacDANIELS: Thank you, Mr. Corsan.

(Applause.)

DR. MacDANIELS: Mr. Corsan is certainly well on his way to being a hundred, and I think if eating nuts and other vegetables will do that, more of us ought to pay attention.

I think we voted on that motion. I think it was unanimous that we send this greeting to Dr. Deming in his eighty-ninth year.

(The following telegram was sent to Dr. Deming:

"AT THIS FORTY-FIRST ANNUAL MEETING OF THE NORTHERN NUT GROWERS ASSOCIATION IN CONVENTION ASSEMBLED AT PLEASANT VALLEY, NEW YORK, THE MEMBERS SEND YOU THEIR LOVE AND ALSO EXTEND THEIR BEST WISHES FOR YOUR CONTINUANCE OF GOOD HEALTH.")

Any other business?

MR. McDANIEL: There is one elective committee that probably will need to be acted on, which is always done at the meeting before, and that's the nominations committee for next year. That's elective.

DR. MacDANIELS: The Resolutions Committee. Mr. Allaman, will you take chairmanship for that? And Mr. Porter of Windsor, will you help Mr. Allaman on the Resolutions Committee?

MR. PORTER: Do I act now, in this meeting?

DR. MacDANIELS: Yes, during the time you are here work out with Mr. Allaman the resolutions that pertain to this particular meeting.

Anything else? If not, this first session is adjourned. Meet again promptly this afternoon at one o'clock,

(Whereupon, at 10:40 o'clock, a.m., the meeting was recessed, to reconvene at 1:00 o'clock, p.m. of the same day.)

MONDAY AFTERNOON SESSION

DR. MacDANIELS: I will call the meeting to order, the afternoon session. This afternoon we have the session given over mostly to the Carpathian walnut. The first paper, by Spencer Chase of Norris, Tennessee.

MR. CHASE: First, with the president's permission, I thought perhaps a short report of the 1949 contest would be in order. As you probably recall, we conducted a Persian walnut contest last year for Northern Nut Growers members only. In this contest we had 31 entries submitted.

The 1949 Persian Walnut Contest with Notes from Persian Walnut Growers

SPENCER B. CHASE, Contest Chairman Tennessee Valley Authority Norris,
Tennessee

The Persian Walnut Contest of 1949 attracted 31 entries from Association members. The following sent nut samples: E. W. Lemke (Michigan) (4), Ray McKinster (Ohio) (1), S. Shessler (Ohio) (2), F. S. Hill (N. Y.) (3), R. C. Lorenz (Ohio) (1), Benton and Smith Nut Tree Nursery (N. Y.) (16), A. S. Colby (Illinois) (2), E. M. Shelton (Ohio) (1), and N. W. Fateley (Indiana) (1). The Contest Committee appreciates their interest in this informal contest.

It was not practical for all of the judges to convene at one place to evaluate the samples. Therefore, the following system was used: One nut from each sample was sent to H. F. Stoke (Va.), Gilbert Becker (Michigan), G. J. Korn (Michigan), and J. C. McDaniel. These four judges were asked to select the best five of the 31 entries. The Chairman then made the final selections based on their findings. Therefore, the samples were actually subjected to five evaluations. The results indicate that this method was very satisfactory.

First place went to the sample submitted by Ray McKinster, Columbus, Ohio., It is significant that four of the five judges selected this sample as the best entry. Mr. McKinster reports that his tree is a Carpathian obtained as seed from the Wisconsin Horticultural Society in 1939. The 11 year old tree has a circumference of 26 inches at the base and has withstood 17 degrees below zero without injury. It began bearing in 1944 and yielded approximately one-half bushel in 1949. The yield is an estimate since squirrels play havoc with the crop. The nuts weighed 12.9 grams with 6.8 grams of kernel. Four judges considered this an outstanding Carpathian.

Second place went to a sample submitted by Sylvester Shessler, Genoa, Ohio. Three judges selected this sample for second place, one placed it first and the other selected it for third place. Again it was significant that the judges were in close agreement. The parent tree is growing in Clay Center, Ohio, and is estimated to be 50 years old. It began bearing in 1920. It yielded an estimated two bushels in 1947, three pecks in 1948, and one bushel in 1949. It has withstood 15 degrees below zero without damage. The source of this seedling is unknown. The nut weighed 8.8. grams with 5.2 grams of kernel. The nut is round with a smooth shell and has a very attractive kernel. This selection has been named ~Hansen~.

Third place, after some disagreement, also went to Mr. Shessler for his entry now named ~Jacobs~. This sample received one vote for second place and one for third place. Two judges agreed on another sample for third place but in a comparative test involving more nuts the Jacobs sample was selected. The nut weighed 12.8 grams with 6.0 grams of kernel. The parent Jacobs tree is located in Elmore, Ohio, and is estimated to be 70 years old.

Bearing since 1915, it yielded an estimated 300 pounds in 1947, 100 pounds in 1948, and 200 pounds in 1949. The tree has withstood 15 degrees below zero. The seed which produced this tree came from Germany.

Fourth and fifth places were awarded to samples S-66 and S-XD submitted by Benton and Smith Nut Tree Nurseries, Millerton, N. Y. Three judges selected these two entries for fourth and fifth places while the other two judges selected other entries. S-66 weighed 13.3 grams with 6.2 grams of kernel. S-XD weighed 12.6 grams with 7.1 grams of kernel. Both selections were raised from Carpathian walnuts obtained from the Wisconsin Horticultural Society in 1935. The nuts entered in the contest came from 9-year old grafted trees located at the Wassaic State School, Wassaic, N. Y. They began bearing a few nuts at six years of age. Both have withstood 34 degrees below zero.

In addition to the five prize winners other entries are worthy of mention. Four additional Benton, and Smith selections (S-61, S-25, S-9, S-32), selection Illinois 10 from Dr. Colby, and a sample from Mr. Lorenz were all considered in the first five by at least one judge. The Carpathian sample from N. W. Fateley was outstanding for size of nut and kernel. Unfortunately, the kernels were shriveled. Since this sample arrived late all of the judges did not have an opportunity to evaluate it. Mr. Lemke also entered a very large Persian walnut. It was considered for third place by two judges but was discarded in the final judging because of shriveled kernels. Both of these large selections should be tested further.

It must be borne in mind that in this, as in all similar contests, only nut characteristics of one year's crop could be evaluated. Whether these selections are adapted to our varying conditions will have to be determined. In other words, this contest should be considered as a preliminary exploration and not as a final selection of suitable varieties.

Following is a summary table containing data on the prize winners:

Results of Persian Walnut Contest

Rank	Entry	Name and Address	Nut Weight	Kernel Weight	Kernel Percentage
1	No. 1	Ray McKinster, 1632 S. 4th St., Columbus 7, Ohio	12.9	6.8	52.7
2	Hansen	S. M. Shessler, RFD, Genoa, Ohio	8.8	5.2	59.6
3	Jacobs	S. M. Shessler, RFD, Genoa, Ohio	12.8	6.0	46.8
4	S-66	Benton & Smith Nut Tree Nursery, Rt. 2, Millerton, New York			
5	S-XD	Benton & Smith Nut Tree Nursery, Rt. 2, Millerton, New York	15.6	7.1	45.8

To obtain information on the culture of hardy Persian walnut a questionnaire was sent to members known to have experience with ~Juglans regia~. The following information, based on the reports of thirteen growers, should prove valuable to those interested in testing Persian walnut.

The members contacted are testing 35 named varieties in addition to many seedlings. Of the varieties, Broadview appears to be represented in more plantings than any other variety. Gilbert Becker (Michigan) has most of the named Crath selections in addition to seedlings. H. F. Stoke (Virginia) has a large assortment of Crath and other Persian varieties.

Fayette Etter (Pennsylvania) reports that he has approximately 150 Persian walnut trees while Royal Oakes (Illinois), Sylvester Shessler, and Gilbert Becker each report 60 trees. Many others have from 25 to 40 grafts or trees while Ray McKinster has only one seedling Carpathian which took top honors in the contest. Most of these members have been testing Persian varieties for more than 13 years. Mr. Stoke has some trees 20 years old.

~Yields~—Most trees reported on began bearing at five to eight years. Topworked trees start bearing several years sooner. It is generally agreed that Persian varieties bear annually. Many trees are bearing only small nut crops. Lack of pollination is given as a reason for these low yields. In addition, winter injury and spring frosts can seriously reduce nut crops. Apparently, none of the trees have borne more than a bushel of nuts at 12 years of age. Accurate records of nut crops were generally lacking. Since this is a very important factor in the selection of varieties, growers should keep accurate yield records for each variety. Where pests are a factor in reducing final yield, a crop estimate should be made early in the season.

~Varieties~—Mr. Stoke considers Bedford, Broadview and Lancaster best under his conditions. Mr. Becker's choice is McDermid but he thinks Crath No. 1 a potential commercial variety. Mr. Oakes likes Crath No. 1 and Ill. No. 3. Mr. Etter lists Burtner and Alleman as his best varieties. Mr. Fateley especially favors one tree because of nut and bearing qualities. Other growers have not as yet evaluated their varieties.

~Hardiness~—Only several growers in the colder regions felt that lack of winter hardiness was a serious limiting factor with their varieties. Those with winter temperatures ranging from 10 to 23 degrees below zero report little damage. Spring frosts are serious to many, especially in the southern states.

~Pests~—Several insects causing damage to Persian walnut were reported. The butternut curculio was most frequently mentioned. Others included leaf hoppers, tent caterpillars, and husk maggots. Few effective control measures have been developed. Squirrels are an ever present threat to nut crops in some localities, as are blackbirds.

~Cultural Practices~—Most growers apply varying amounts of fertilizer or manure to their trees in some form or other. Few mulch their trees. All do some pruning, mainly of a corrective nature.

~Pollination~—Most growers agree that usually, but not always, pistillate flowers are produced several years before the occurrence of catkins. Generally, Persian varieties do not adequately pollinate themselves but exceptions are reported. The problem is one of variable dichogamy. Some varieties shed pollen before pistillate flowers are receptive; others shed pollen when pistillate flowers are no longer receptive. This unfortunate situation probably explains the low yields experienced by some growers. Mr. Stoke lists the flowering dates of 13 varieties in the 1942 NNGA Annual Report which clearly illustrates dichogamy in Persian walnut.

Some varieties are considered sufficiently self-pollinating to produce at least light crops. However, this may be influenced by weather conditions. During an unusually warm spring catkins develop more rapidly than terminal growth containing the pistillate flowers. Mr. Stoke reports that ~Bedford~ produces both flowers simultaneously and that ~Caesar~ is practically self-pollinating. Mr. Etter finds ~Burtner~ fully self-pollinating and ~Alleman~ partially. Mr. McKinster's tree is apparently self-pollinating.

To overcome dichogamy it is necessary to have varieties which pollinate one another. Again Mr. Stoke's list referred to above is useful in selecting varieties for cross-pollination. Mr. Becker finds that ~Crath No. 1~ and ~Carpathian D~ pollinate each other under his conditions.

More information on the pollination of Persian varieties is definitely needed. Members are urged to record the flowering date of their varieties. Such information will be very helpful in variety selection.

~Handling the Nut Crop~—The nuts are harvested and dried promptly. Methods of drying vary. Some have drying screens in which the nuts are placed several layers deep. Some dry the nuts in the sun; others prefer a shady place. Following drying, the nuts are stored in a cool place.

At least one grower has enough walnuts to sell locally; others feel that local markets would take all they could produce. Many of the growers sell the nuts for seed purposes. Of course, all have a supply for home use.

~Future Prospects~—Growers see good prospects for Persian walnut in most of their respective regions if improved varieties are developed. Many growers are planning to increase the size of their plantings with promising varieties. Others would like more trees but lack the necessary space.

The 1949 contest uncovered several very promising selections. The 1950 National Contest should produce many more.

(Applause.)

DR. MacDANIELS: I believe, Mr. Chase, your second paper has to do with the 1950 Carpathian walnut contest, which is just a matter of explanation, I take it, as to what is going to happen.

Plans for the 1950 Carpathian Walnut Contest

SPENCER CHASE, Norris, Tenn.

MR. CHASE: The 1950 contest plans have not been fully formulated. Our main problem will be one of advertising. Our good secretary has agreed to help out on that. Mr. Sherman and Dr. Anthony have agreed to help out in their region. I was successful in getting Mr. Neal of the ~Southern Agriculturist~ to promise to give us a little Southern publicity on contest.

MR. McDANIEL: I wrote him; also wrote Mr. Niven of the ~Progressive Farmer~ at Memphis and Chet Randolph with the ~Prairie Farmer~ at Chicago.

MR. CHASE: As I say, we plan on handling it the same as we did the 1949 contest. It will be simply the submission of entries. We may want to consider the method of judging a little further.

The problem of prize money needs to be resolved, how much the Association is going to offer—feels that they could stand to offer—for first, second, or how many prizes we are going to have. That's about all that we have to report now concerning the contest. But we do need, before we can proceed too far, some commitment on prize money. Last year we did not offer prizes simply because it was for the membership, and there has been some question whether prizes are necessary. Of course, it wasn't necessary from the Association standpoint, but it probably will stimulate some others not in the Association to submit samples from their trees.

Do any of the contest committee or members have any suggestions? We'd be very happy to have them.

DR. MacDANIELS: Will this include all Persian walnuts?

MR. CHASE: That was another problem that came up the last time, and we talked about it as being a Carpathian contest, and we decided, who can tell a Carpathian from another Persian, and we decided to make it a Persian walnut contest.

DR. MacDANIELS: No Persian walnut will be refused?

MR. CHASE: Yes, sir.

DR. MacDANIELS: Should they be sent to you?

MR. CHASE: Yes.

DR. MacDANIELS: Mr. Spencer Chase at Norris.

MR. CHASE: Then, shall we exclude the Northwestern states?

MR. McDANIEL: Last year we limited it to those trees which stood at least zero temperature. That would eliminate most of California, at least.

DR. MacDANIELS: That makes sense.

MR. SHERMAN: How many nuts are expected?

MR. CHASE: Last year we asked and received fifteen. We'd like to have twenty-five. That gives us a better opportunity for the tasting department. We have a lot of tasters. We don't have many crackers, but a lot of tasters.

MR. McDANIEL: I found that the mice in the State Capitol at Nashville weren't very particular as to variety. They took to any that were open.

DR. MacDANIELS: Are we men, or are we mice?

MR. CHASE: In case you didn't notice, downstairs we have all the entries in the contest with the exception of some which human mice got from me, two samples, I believe. But all the rest I managed to save. And I, of course, have not seen too many Persian walnuts, being down there where the spring frost gets them. I was very favorably impressed by the appearance of all these samples. We simply picked five, as I said, and pointed out that this should be considered a preliminary finding and not definite, but all those samples were fine. Some were, of course, more bitter to the taste than others. That's where we lost a lot of nuts, trying to find out the least bitter. But many were an improvement on the commercial varieties, as far as I was concerned.

I think if we all get active on hunting out these Persians the way we have blacks, we can make very good progress.

MR. McDANIEL: Even on appearance I think some of them beat what you see in the stores.

MR. CHASE: Yes, on appearance. Of course, some of them were handed back and forth and competing against each other, that's what happened.

DR. McKAY: I'd like to ask how much importance you ascribe to tree characteristics and not the nut itself.

MR. CHASE: I asked for that information and tabulated it, and it didn't mean much. We found we couldn't do it. So then we came back to the nut first.

Carpathian Scions for Testing~

There is one other point I might mention. Last year you may recall that I reported on our planting of Carpathian seedlings at Norris, some 500 of them, which were frosted every single year. We have babied them along now for almost ten years, and I don't see any prospects of getting any nuts on them.

Now, among those 500 there must be one good one, and I will be very happy to collect scion wood of all those trees and send it to members who are willing to top-work them and see what they will do. So if any of you folks are interested in some of these varieties—not varieties yet, but seedlings—I'd like to see them fruit, and I am sure we never will at Norris.

DR. MacDANIELS: Where did you get the seed?

MR. CHASE: From the Wisconsin State Horticultural Society.

DR. MacDANIELS: In other words, it's just as good seed as any other.

MR. FRYE: You are in a frost pocket.

MR. CHASE: The whole place is a frost pocket. They are up on the hill—the frosty spot.

A MEMBER: When were they planted?

MR. CHASE: In the spring of 1939.

MR. CORSAN: Let me understand that. You say there are 500 trees that did nothing at all?

MR. CHASE: We have approximately 500 of the Crath seedlings, and each year they are frosted.

MR. CORSAN: Let me explain that. I have had the same trouble. Mr. Crath, not knowing the nature of my place, put some of the best nuts in wet places, in frost pockets, but he had two rows of one kind of nut that grew very rapidly the first year, but they are not any bigger now, and that was many years ago, back in 1935 they were planted. And there were about 80 varieties he got from Russia, he being able to speak four Russian dialects, his father being the Burbank of Russia and the gardener to the Czar, he had a lot of information, and he knew just what he was doing. But he was too hopeful and got some varieties from the foothills, some up a little higher, some up half way, some up towards the snow line, and they are tremendously hardy.

Now, I have given these nut trees away to people south of Lake Ontario. You see, I am north of Lake Ontario, and those are around St. Catherines. There trees will grow and succeed. I have been told there is no check by frost on them. I have given a lots of those away. But with me they are absolutely worthless north of the Lake, and there is a vast difference in them.

Now, I thought, looking at a great, big nut, the Rumanian giant, thought sure a nut that big would be bitter. I thought sure that it wouldn't be hardy, but at any rate, I planted a few, and I have a nearly perfect reproduction of those nuts, and one is very hardy and very productive, and the other is not quite so hardy. It's a huge nut and not so productive. However, size has nothing to do with it. I noticed a certain type and shape of nut was sometimes quite tender, and then again the same shape of nut but different variety was quite hardy.

I sold a lot of trees in varying sizes, keeping the small and the runts and those that were injured by the tractor and other trees for myself, but I have enough varieties every year to come down and see some wonderful results.

For instance, I slashed one up badly to dwarf it, and it had a little, wee nut that big (indicating). When I cracked that nut, the shell was crammed full of meat, and it was exceedingly sweet, and it tasted like a hickory nut. So I cut my own throat, as it were.

* * * * *

DR. MacDANIELS: Mr. Chase's problem right now is to get these trees out somewhere where they can be tested further, and he has asked any of you if you want scions to get in touch with him.

MR. CORSAN: I say, send them south.

DR. MacDANIELS: The farther south you go the worse they are.

MR. H. F. STOKE: May I also say a word? Also send them north. Sometimes the winter sun will start the growth activity, and then wind comes along and kills it. The original Crath that was started in Toronto, I had it killed back to five-year-old wood thick as my wrist one winter, when the sun moved it to activity. It was hardy in Toronto, but it wasn't hardy in Roanoke, Virginia.

DR. MacDANIELS: Let's have a showing of hands of those who have that trouble, starting in the spring and freezing back. (Showing of hands.) About five or six.

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The next paper will be, "The Persian Walnut in Pennsylvania and Ohio," Mr. L. Walter Sherman.

MR. SHERMAN: Ladies and gentlemen, Mr. Chairman: First I'd like to tell you who I am. Some of you have been to my place and know who I am, but last fall Pennsylvania started something new—a little bit different. They put on a survey of the nut trees of Pennsylvania. Two of us were selected for the job, and I would like to introduce Dr. Anthony—stand up so they can see. He and I were the two that were selected to put on the tree crop survey of that State of Pennsylvania.

Pennsylvania is a big state, and there is lots to see. They not only made it a survey of the nut trees, but any trees that are potential food for wildlife. Well, that made it the acorns and the honeylocust and, well, what have you, How big a job they hung on two fellows! Well, we have done the best we can, and we want to bring you this afternoon just a little of those results.

The Persian Walnut in Pennsylvania and Ohio

L. WALTER SHERMAN, Harrisburg, Pennsylvania Pennsylvania Department of
Agriculture Tree Crop Survey, Harrisburg, Pennsylvania

As members of the Northern Nut Growers Association, most of you are familiar with the early history of the Persian walnut, its introduction into the United States by the early settlers, and how it finally found a home in California. You also know of the more recent introduction into this country of nuts and other material from the Carpathian Mountains by the Rev. Mr. Crath, who was assisted by members of your organization. (1)

These recent Crath introductions are supposed to be much hardier than the former ones, and probably able to establish themselves in northern United States and southern Canada.

When the Pennsylvania legislature authorized a survey of the nut trees of the state, very few people realize the foothold that the Persian walnut already had in Pennsylvania.

Early in this survey, we visited Fayette Etter, who is Pennsylvania's Luther Burbank with nut trees. He is well informed concerning the Persian walnut in his section, and he surprised us by his estimate of several thousand trees in his county of Franklin. The adjoining counties of Adams, York, and Lancaster, along the southern border of the state, have fully as many trees of this species, so it is a very conservative estimate that there are ten thousand of these trees in Pennsylvania. These are located, for the most part, in the southeastern corner of the state below one thousand feet elevation.

Local grown Persian walnuts were found on sale last fall in the farm markets of York, Lancaster, and Harrisburg and at many grocery stores. Wherever we found such local nuts on sale, we asked where and by whom they were grown. Many of them came from Halifax and Linglestown, in Dauphin County; from Lampeter, Lancaster County; and from Seven Valleys, York County.

Farther investigation revealed the facts that in all but one of the centers of production, the trees were seedling trees and that there were from four to 23 trees planted relatively close together. In one instance, a lone tree produced the nuts being sold, and in another case the nuts were from several grafted trees.

The lone tree, which produced three bushels in 1949, was of interest. Investigation revealed that the nearest Persian walnut tree was at least a city block distant. Was this lone tree self pollinating or receiving pollen from a tree this far away? We still are not sure of the answer.

Jacob Houser, of Lampeter, was selling Pomeroy seedling nuts and nuts from three Rush Persian walnuts grafted on black walnut stock. They were growing close enough for cross-pollination.

Driving through the counties of southeastern Pennsylvania, we found many thousand seedling Persian walnut trees as shade trees about the farm homes. Investigations revealed that most of these trees never produced any nuts. Repeatedly we are told that, "my tree never has any nuts, but a certain tree on an adjoining farm always produces," or "I have two trees, one of which bears quite regularly but the other never has borne." They are the same age and both seem to be growing equally well. Some produce only a few handfulls of nuts when they should be producing five to ten bushels, judging by their size.

You as nut growers know the answer, but the general public does not. Even some of you have made the mistake of planting one tree by itself and expecting it to produce. This seldom happens. Mixed plantings of several varieties or several seedlings planted close together is the safe rule to plant by.

I know of one planting of ten grafted trees of one variety of Persian walnuts, now twenty years old, that has never produced any nuts even though they are planted so that cross-pollination would be expected. In 1950 only a few catkins developed. These produced pollen early and were on the ground before the pistilate bloom opened and was receptive. I never saw a nicer pistillate bloom on any Persian walnuts than these trees had, yet not a single nut set. They are in the center of a fifty-five acre black walnut orchard, and when the pistillate bloom was at its peak, the black walnuts surrounding were shedding pollen. Do not try to tell me that native black walnuts will satisfactorily pollinate the Persian walnut. After this demonstration, I know different. Were all the Persian walnut trees of Pennsylvania properly pollinated, the crop of nuts, in my estimation, would be increased a hundredfold over what it is normally. Lack of pollination is probably the greatest factor causing non-production in our Persian walnuts. It is far more important that the fertility factor which is so important in production of the common black walnut. (2)

Fayette Etter and Milo Paden both feel that the Broadview variety is self-pollinating, but even this variety may prove to be benefited by cross pollination.

The Persian walnut has developed in Pennsylvania and Ohio in a rather interesting pattern. Trees planted fifty to a hundred and fifty years ago managed to live and produce nuts. From these trees, seedlings were grown and planted by neighbors and friends. These trees and their seedlings in turn have now grown to producing age. Some few that produce good crops of nuts you hear about, but the vast majority are just non-producing shade trees. Until you look for them you little realize how numerous they are.

At Linglestown, Dauphin County, however, we find a striking exception to this. Here all the trees are productive. The question there is not why don't my trees produce, but is quite spirited as to who harvests the largest crop and best nuts.

About seventy-five years ago Alfred Kleopfer planted some Persian walnuts of unknown origin, but probably from Germany. He grew three trees which were planted, one beside the village blacksmith shop, one across the street, and the third at a neighbor's. One tree lived for only a short time. The blacksmith shop has been replaced by a modern dwelling but the walnut tree was saved and has grown to be a tree 6' 6" in circumference and probably 60 feet high. The one across the street is of nearly equal size but the top has been damaged by storm and the tree is not as tall.

These two trees were able to cross-pollinate and one tree was especially productive. Miles Bolton recognized its value and began growing seedling trees and distributing them to his neighbors. Some of them were quite skeptical and even refused to take them as a gift and plant them. However, he got the village pretty well planted to Persian walnut trees, so that today there are 145 nice trees within the village, and two small orchards on farms nearby.

Standing in the village square, one can see at least six Persian walnut trees higher than the house tops. Pollination is not a problem, and all trees are good producers. Young trees are in demand for planting, and seedling trees, coming up in the flower beds, compost piles, fence corners, and other places where squirrels have hidden nuts, are carefully transplanted to permanent locations.

The story of the development of the Persian walnut at Linglestown, with minor variations of course, can be repeated many times in southeastern Pennsylvania. In Linglestown, the development has been concentrated within a village, whereas in most places it has been spread over a farming community, with less opportunity for cross-pollination. The result has been a very high percentage of barren trees. However, Persian walnut seedling trees have taken over and are making good in this milder climate area of Pennsylvania.

About the same can be said of northern Ohio, though the development is probably 50 years behind that in Pennsylvania. The climate there apparently is not so well suited to the Persian walnut, and fewer trees have been able to thrive. A few, however, are growing nicely and their seedlings are rapidly spreading. The Jacobs tree at Elmore, Ohio, produced 300 pounds of nuts in 1947, at 30 years of age, and many nuts from this tree are being planted. The Ohio Nut Growers are propagating vegetatively from the outstanding trees and rapid development is taking place. Named varieties are thus being developed from superior trees, and future success will be based on these named varieties rather than on seedlings.

During the last few years, some of the seedlings developed from the Crath Carpathian importations are coming into bearing in parts of

Pennsylvania and Ohio, and wherever I have seen them they look very promising indeed. The Crath Carpathians are doing well at Mt. Jackson, Lawrence County, Pennsylvania, along with Broadview, for Riley Paden and Howard Butler. A. W. Robinson, of Pittsburgh, has five trees of Crath seedlings, two of which are in bearing. All these trees seem to be perfectly hardy. The nuts of course vary, but all are good.

Riley Paden, at Mt. Jackson, is grafting Broadview on black walnut stock, and for him this variety is doing well. He has about forty trees of it from two to fifteen years of age. His prize fifteen-year-old tree produced one bushel of nuts in 1949. A sample of these nuts is on the table for your inspection. Paden says he can grow Broadview anywhere peaches will do well. Fayette Etter at Lemasters, Franklin County, considers Broadview too bitter flavored for him. He thinks Burtner, which is a local seedling, superior for his section to all other varieties that he has tested.

With an estimated ten thousand Persian walnut seedlings growing in Pennsylvania, the Pennsylvania nut growers are faced with a big task to sort out the best and get them tested in different sections of the state. We should find the best half dozen varieties for each section.

The Persian walnut is established in Pennsylvania and in northern Ohio. There are not just a few scattered trees having a hard time to survive but there are many thousands of them, growing vigorously, some producing big crops of fine nuts, others not producing any. They are ready now for the intelligent development you can give to them. Nature has gone about as far as she will without your assistance. The job now is up to you nut growers.

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Discussion

MR. FRYE: How about butternuts for pollenization?

MR. SHERMAN: I don't know. I have one hybrid, and that's a sample downstairs that I think is an English walnut crossed with a butternut. The nut looks like a butternut; the tree looks like an English walnut, but it has the butternut bark. They will occasionally pollinate, I think, but don't depend on them.

MR. CORSAN: I'll tell you how you can tell. That butternut-English walnut cross is the most powerful tree I ever came across, especially for good wood. I got a tremendous one.

MR. STOKE: I produced, I think, 22 seedling trees from the Lancaster Persian walnut. About five per cent are hybrids. There was one strong-growing black × Persian hybrid that I am sure of. There are three or four very dwarfish trees that undoubtedly were crossed with the heartnut. They were all dwarf. I haven't been able to get one to bear. I have had one grafted five or six years on a black walnut, but that was the heartnut and not the butternut.

MR. SHERMAN: That study of the hybrid is another story and really doesn't belong in this discussion at all.

MR. CORSAN: Here is a point on that. When they are only that high (indicating)—if they are only babies, I can tell them. You know, occasionally. Look at the leaflets on the compound leaf, and if there are over seven, they are hybrids, and if they are extra vigorous growing, they are hybrids, because they occasionally pollenize.

MR. SHERMAN: Those are all characteristics of the hybrids, but here is what I want to bring out now, and Dr. Anthony is going to stress it on his chestnuts a little bit later: You people have a wealth of material to select from. Nature has gone about so far, and I am just a believer enough in what the Bible says, that God made the heavens and the earth and put man here to tend and keep it, and made him master of everything above the earth and every creeping thing on the earth and everything beneath the earth, and it is up to you fellows to direct intelligently this mass of material you have to direct. You have got nuts growing where they are hardy, you have got big nuts, you have got little nuts, you have got everything under the sun you can think of. What more do you want for a nice job ahead? It's up to you fellows to do. It's going to be not a one-year job, not a two-year job, not a five-year job; you will be at this, and your children and your grandchildren.

MR. CORSAN: Make you live long.

MR. SHERMAN: Maybe you will live long enough, but it's a century's job, and not the job for one man's lifetime.

(Loud applause.)

DR. MacDANIELS: Thank you, Mr. Sherman. Any questions?

MR. CHASE: Yes, sir. I want to ask Mr. Sherman, should I be thinking about receiving 10,000 entries in this contest?

MR. SHERMAN: No, because there aren't 10,000 trees producing. Out of that 10,000 maybe there are a thousand of them producing. The nine thousand others are nothing but shade trees, and never produce any nuts. You don't hear of them, but if you travel through York, Lancaster, and Adams Counties down there and look for Persian walnuts, you will find them on—I was going to say 50 per cent of the farm homes. You can see them along the road everywhere.

My wife travels with me a good deal of the time. She will say, "Why don't you stop and look at that Persian walnut? There are some over there. Why don't you stop there?"

A MEMBER: Don't they bloom a month later than most of the others?

MR. CORSAN: Did you find a good French variety?

MR. SHERMAN: But those French varieties—I can't take you to a good French variety in Southeastern Pennsylvania that has been producing the nuts. They produce the nuts, but folks won't even pick them up.

A MEMBER: They are good for pollen.

MR. SHERMAN: If you want a good pollenizer go to Fayette Etter and get his Burtner. It's a very late pollen producer. This year I took some buds from his Burtner and put them in the top of those ten trees in that 55-acre black walnut orchard to see if I can't do something. Maybe it won't stick—maybe I hadn't better tell you.

MR. CORSAN: Mr. Chairman, there is one point raised by the last speaker that's not understood; that the young black walnut trees, when they first blossom, they come out with a mass of male blossoms. Then the English walnut, when it comes out, it sometimes comes out with a mass of pistillate flowers which people might not know are the female flowers. They make the nuts, but there is not even one catkin. I have seen that time and again.

Those trees in Russia would be dependent upon larger trees to pollinate them. But here you have young trees, and you have to wait till they get a certain growth, and then they produce their catkins.

DR. MacDANIELS: Thank you, Mr. Corsan.

The next paper, by Mr. J. F. Wilkinson of Rockport, Indiana, "Observations and Experiences with the Persian Walnut in Southern Indiana." Mr. Wilkinson.

(Paper not available for this Report.)

DR. MacDANIELS: We have a choice of doing several different things. There are several other papers we have here, the authors of which are not present. Then the other possibility would be to go on and have some papers that require the use of the lantern, as long as we have this all fixed up.

Perhaps the thing to do is to have Dr. Anthony's paper on chestnuts, using the lantern, and then have these other papers on the Persian walnut summarized after that. Does that seem to be a reasonable thing to do?

(Chorus of yeses.)

DR. MacDANIELS: We will go ahead on that basis, then. Dr. Anthony has the talk on chestnuts.

(This talk, withdrawn for revision, may appear in next Report.)

MR. CORSAN: Dr. Anthony, I knew Captain Sober very well, and he showed me quite a group—a double handful—of Korean sweet chestnuts. They were a little thicker than the native Pennsylvania chestnut, they are rounder and a little larger, but they weren't as large as some of the Chinese or nearly as large as the Japanese. What about those nuts, because, you see, the blight killed all his Paragon chestnuts—you know, the cross between the European and the American chestnuts—killed them all off completely, as it did with me.

DR. ANTHONY: In our detective work we were instructed to follow down that plantation. Mrs. Sober is still alive, living in Lewisburg. The planting has practically disappeared. I am going over there next week. It is still with the man who wrote "Chestnut Culture in Pennsylvania." MR. CORSAN: It broke his heart.

DR. ANTHONY: We are going over there next week, but I think that whole planting has disappeared. When these things change hands, another man comes in who is not interested, and things disappear very rapidly.

(Continue with paper.)

MR. CORSAN: I want to tell you how to keep the deer out of the chestnut orchard. Plant filberts five feet apart all around the place, and after while just put one single electrified wire five feet from the ground, and the deer won't get in through that.

DR. ANTHONY: Glad to hear that, because deer is one of our problems.

(Continue with paper.)

DR. ANTHONY: There is a tree beside the blacksmith shop, and the old man used to go there early in the morning as a boy to get chestnuts. Today he has taken down the old blacksmith shop and built a home, but he preserved that tree in Linglestown. It practically covers his house, six feet six inches in trunk circumference, 60 feet high and a spread of 60 feet. It isn't too long before we will have chestnuts that big to eat alongside the old blacksmith shop.

DR. MacDANIELS. It is about three o'clock. We will take a five-minute recess.

(Whereupon, a short recess was taken.)

DR. MacDANIELS: For the first paper after the recess, we will call on Sargent Wellman to speak to us about the Persian walnuts in England. Mr. Wellman.

Notes on Persian Walnuts in England

SARGENT WELLMAN, Topsfield, Massachusetts

MR. WELLMAN: Members of the Association: I was fortunate enough to be in England last summer, and I agreed that I would say a few words about nut growing there. What I am really going to do is largely to read you a few things from some articles that I found there.

I was very much impressed with the little interest that there is in nut growing in England, and I was very much surprised at it. Of course, you all know that the walnut grows there. The chestnut grows there. There are some fine, marvelous trees in Kew Gardens, of course, that I saw, and if you read the English poets, you will remember how they talk about chestnut blossoms on chestnut trees, but curiously enough, there is now very little interest.

MR. McDANIEL. When they speak of the blossom, they speak of the horsechestnut, do they not?

MR. WELLMAN: Not always, but there are pink flowered horsechestnuts in France, particularly, whole avenues of pink ones. The cob nut, as they call the filbert, is very common there, grown in hedges. One year when I was in England previously I brought home a few in my pocket, and I have a seedling which grew from one of those, which is comparable to the filberts I have, but apparently there is no interest in that, so far as I can see—I mean, any investigation and any experimentation and encouragement of its planting. But there is about the walnut. That's the one nut tree in which they are interested.

I picked up two reports, both of them made by Elizabeth M. Glenn, who is the woman connected with the East Malling Station down in Kent and is the one person who is doing more with walnut work than anybody else, as far as I could find out. Unfortunately, the day I was there she was on vacation, so I couldn't see her, but they were very kind to me and took me around and showed me everything.

As you know, the East Malling Station is the place where they have done all that work with apple root stocks. This one is a reprint from the annual report for the East Malling Station for 1946. And then "The Men of the Trees," which is a forestry society there which some of you may have heard of, have reprinted in the Autumn, 1949, number another article by Elizabeth Glenn on "The Selection and Propagation of Walnuts." And I think if I make a few comments and read a few things from these, you will be interested.

She says, "The earliest record of a walnut tree in England is 1562, but remains of walnut shells have been found in Roman villas, and it is probable that the Romans planted some nuts and raised trees in this country."

She says, "There is a large tree of it"—black walnut—"at Kew, near the entrance to the Rock Garden." Of course there are some rootstocks, and they are all specimen trees, but they are not used for nuts. She says somewhere here, "In this country the nuts are of little value, although in America they are used for confectionery purposes."

The East Malling Station is really a fruit research station, as I said, and they are the ones who are primarily interested in walnut crops and not timber production. "However, there is no reason why a tree shouldn't produce both good crops and good timber."

"The French, have been grafting walnuts for well over 100 years, and the famous Grenoble nuts all come from grafted trees of named varieties." She emphasizes the fact that almost all of the English walnuts are grown on seedling trees and are very much inferior to those that come from the Continent and from this country. And of course that was the purpose of their work, to encourage the use of grafted trees.

I was interested in this sentence: "The late Mr. Howard Spence began the survey and collection of good varieties growing in this country and abroad, and collaborated with East Malling in the trial of selected varieties." He was always interested in our society and was an honorary member of it for a good many years prior to his death.

I was interested in the fact that the problems that they have over there in the way of climate and some other things are very similar to our problems.

She speaks a good deal about the matter of climate. I will come to that as I go along.

"Work on walnuts, started at East Malling in 1925, soon showed that the budding or grafting of walnuts out of doors was far too chancy in this climate to be relied upon as a means of raising young trees," so that all their grafting is done in the greenhouse, and they don't try to do anything outdoors.

"Outdoor grafting can be done successfully only where the mean temperature from May to September is above 65° F." Then she gives a description of the greenhouse grafting, bringing in the seedlings and potting them in November, in the fall, and then starting along in February in grafting, and then taking them out and planting them in the spring. I won't go into that; there is nothing particularly interesting I think, for us about that.

Patch budding she also describes…. She says it's a much cheaper method than grafting under glass but at the moment the results are far less reliable.

"The walnut will tolerate a wide range of soils so long as the drainage is good and the soil is not too acid. Lime should be applied before planting, unless there is plenty present in the soil.

"The site should not be in a valley or frost hole, because, although the dormant tree is quite hardy and can stand severe frost, the young growths and catkins are very easily killed by spring frosts." They are talking about the same problem we have. In fact, in spite of the fact that the weather is warmer than in Boston and New England, they don't have the severe winters, but they do have this late frost.

Manuring. They recommend mulching with farmyard manure or compost put on the soil and worked in and no artificial nitrogen because that again gives too much late growth, and you have trouble with killing back.

She goes over the problems that we have been talking about this afternoon, about the time of leafing out in the spring and what the difference in the varieties is and the effects of that on the winter killing.

Now, I am not going to read much more. I will just read over the names of the varieties which may interest you. This first article, the 1946 one, lists Franquette, Mayette, Meylanaise, Chaberte, Excelsior of Taynton, Northdown, Clawnut, and Secrett. The latter article, which was published last year, says that in 1929, with the help of Dr. Taylor, the Royal Horticultural Society held a walnut competition. "Over 700 entries were received and were subjected to severe tests. Most of the nuts were far below the required standards, but five Were selected for propagation and further tests. The owners of the trees from which these nuts came supplied scion wood to raise grafted trees for trial at East Malling." The best ones came from a tree which they called "Champion of Ixworth." The second one was called "Excelsior of Taynton," which was in the list I read previously. Another variety is called "Lady Irene." I am not going into the description of these varieties here, because if any of you are interested, you can get hold of these publications and get it. She lists the Stutton seedling and then the Northdown Clawnut.

Also in this article she mentions the French varieties, of course, which were mentioned before.

Well, I thought it might just interest you that in another part of the world they are doing the same sort of thing we are, and they are having the same sort of problems and working on it. (Applause.)

DR. MacDANIELS: Several of these papers which were scheduled will be either summarized or read. One of them will be read now by Mr. Silvis of Ohio. The paper is by Carl Weschcke.

Prospects for Persian Walnuts in the Vicinity of St. Paul, Minnesota

CARL WESCHCKE

Although I was asked to prepare a paper on the Carpathian walnut, I feel that my other experiences with Persian or so-called English walnut (the botanical name of which is *Juglans regia*) are also of some value to those who might be tempted to try this species of walnut in cold climates.

When I first started my experiments with nut bearing trees, I included the English walnut among the possibilities for our section. Mr. J. F. Jones of Lancaster, Pennsylvania, gave me much information and a great deal of help in trying out what he considered hardy strains. There was a walnut tree in Boston, known as the Boston walnut, of which he sent scions, and which I grafted on butternut. This was about the year 1920, and was included in my grafting experiments together with black walnut, heartnut, hickories, and hybrids between hickory and pecan. Later on, he sent me scionwood from other known hardy varieties which I placed on butternut, and many of these made tremendous growths but were winterkilled the very first winter. None of the English walnut with which I continued experiments lived over the first winter until I received scionwood from Prof. James Neilson of Canada, who sent the Broadview. These Broadview scions were grafted on butternut and black walnut, and a few of the scions survived for possibly three seasons, even producing staminate and pistillate blossoms and small nuts which grew only to about the size of a quarter and then dropped off.

Clarence A. Reed arranged to have some small seedling Chinese strain of *Juglans regia* sent from Chico, California; these were planted in favorable places and survived a few winters. I also planted seeds of the Chinese strains which gave me no better results than the seedlings.

Then I bought walnuts from A. C. Pomeroy, of Lockport, New York. These were even more tender than other varieties with which I had experimented, although they were very much publicized by Mr. Pomeroy in the Nut Grower during that era as being extra hardy, because they were growing near the south shore of Lake Erie.

I next went to Mr. Jones, who was then selling quite a quantity of Wiltz Mayette and Franquette strains of English walnut grafted on black walnut. These proved to be among the most tender varieties I have ever tested here. Then he sent me scions of the Hall and Holden varieties, which he felt were considerably more winter hardy, but here they failed to survive even one winter.

We have not neglected the Rush English walnut either, which was tested in a similar manner without any good practical results.

This now brings us to the convention at Geneva, New York, in 1936 when the Rev. Mr. Crath and George H. Corsan presented a new strain of English walnuts, known as the Carpathian strain, originating in the Carpathian mountains in Poland. This so impressed me that after talking it over with my father we decided to finance a trip into the same region that Mr. Crath had been in, to locate new and better varieties for a real test. The story of the Rev. Mr. Crath's and my adventure along these lines, during the winter of 1936-37, has been printed in the records of the Northern Nut Growers Association, and I will bring out only the high spots that seem to be important 14 years later.

In the shipments of hardy material collected were some 4,000 scions of possibly a dozen different good strains of what Mr. Crath considered hardiest and best. In addition to that, there were around 500 trees ranging in size from small whips of one foot long to some that were over eight feet; also there were some 400 pounds of nuts to be planted to produce seedlings.

These nuts had been gathered from superior hardy trees with the expectation that the seedlings would produce nearly true imitations of their parents in the quality of their fruit and hardiness. These seedling nuts produced somewhat over 12,000 seedling trees, which were planted in about six large strips of land so as to give room for cultivation. The 500 trees received from Poland were planted in favorable locations and many of them are still alive. The scionwood was put on native butternut and black walnut. Some of it was grafted to young nursery stock, but most of it was put on large mature trees, being top worked. Grafting was started in April and continued into the early part of June. The later grafts were much more successful than the earlier ones, although some of the April grafts grew and flourished. Many of these grafts bore flowers and had little nutlets but none of them ripened nuts. After about three seasons some of the grafts that continued to live produced a few nuts.

Three varieties were practically mature, and then the native insect pests caught up with them. Also, there was a black rot or wilt which I am fairly sure was walnut bacteriosis disease, although specimens sent out to competent authorities did not corroborate this diagnosis. What turned out to be the butternut curculio attacked all grafted and seedling trees with such vigor that there was no way to combat it. I sprayed some of the grafted specimens and kept it up for several years, trying to hold on to them, but it became too much for me and my equipment; I doubt now whether any amount of poison would have saved the trees because the butternut curculio is difficult to kill with poison. One of the varieties, known as the

Kremenetz, grafted on black walnut, was sent to Harry Weber. It thrives and bears nice crops at his country estate in Cleves, Ohio, near Cincinnati. He has sent me scions of this variety, and this spring I grafted them back on black walnut, as the butternut curculio is not nearly as bad as it was when there was so much English walnut foliage for them to feed on (this foliage is their choice over all other foliage). These insect pests also wiped out several heartnut varieties which came from J. U. Gellatly, of Westbank, B. C, Canada; for next to English walnuts the curculio loves heartnut foliage and its new branch growth.

We have about 60 to 70 acres of woods which contain a large percentage of butternut, therefore it is next to impossible to wipe out their native food. I doubt very much whether this would have benefited the situation at all, as the curculio would have then centered all its activities on the English walnut foliage and perhaps have attacked hickories, pecans, and black walnuts, on which they sometimes try their appetites. Hybrids between butternut and black walnut are viciously attacked by this curculio. Hybrids between English walnut and other species of walnut which I have here also become a prey to curculio. So there is no trick species which would be immune to their attack.

The English walnut usually vegetates too early in the spring to escape some of our late frosts. Because this new growth generally contains the flowers, the fruiting of such trees would be very unreliable and only occasional. We even have trouble with black walnut and butternut in this respect. The hickory is much better, and the pecan is even later in respect to vegetation. I mention this because even though everything had gone well it is doubtful whether reliable crops of English walnuts would ever have been produced from the so-called hardy Carpathian series.

A year or so following the experiment with the Carpathian walnut, I imported about 100 pounds of seeds from Austria. They came in two different lots: one of them was more expensive than the other seed, and it proved to be much the hardier. The larger lot of smaller seeds was not as hardy. Although we have several hundred trees of this better seed lot which remain alive, they are no better off in any respect than the Carpathian seedlings. In fact, I could not see much difference between the behavior of these seedlings and the behavior of the Carpathian walnut strain.

While in California in 1939 I picked up about five pounds of seeds from a hardy tree growing in the Sierra Nevadas in Sonora, also some native black walnuts. These survived a few years but finally were winter-killed entirely, root and all. The Carpathians are never killed out entirely but continue to grow from the root systems, even though they are frozen back to the ground; but the insect and the fungus have destroyed many thousands of the original group of trees so that there are today perhaps between 1000 and 2000 living trees, which sprout up each spring and kill back each fall with clock-like regularity. Among these; However, are a few outstanding varieties which extend some hope that there may be among these survivors one or more trees which resist the butternut curculio and have become acclimated, to such an extent that they do not entirely kill back but only a little of their new growth is killed. These specimens usually are the ones that make a shorter growth during the summer, in fact have more of a tendency to be a genuine dwarf type of tree. Three such seeding trees were known to have sprouted from exceptionally large and very thin-shelled walnuts, which I believe the Rev. Mr. Crath calls the giant type.

I will now summarize and express my own private opinion regarding the future possibilities of introducing the English walnut into such an extreme northern latitude as we are in. First, experiments started thirty years ago, which period gives a reasonable period of time that any man should feel is necessary to devote to giving a species a try-out. Secondly, we have used material from every reasonably known source. Third, persons in charge had a reasonable amount of skill and success with other varieties to have insured success if the material had been responsive. My opinion, for what it may be worth, is that the species is out of its range in this northern latitude, more particularly because it is too tender to fight its own battles as to insect life which attacks it, particularly the butternut curculio. Grasshoppers, leaf eating insects, and worms of different sorts, also attack it more than they do other nut tree foliage. The possibilities of a break in the strong cycle of insect life is a hopeful prospect which we are helping by breeding tens of thousands of toads and frogs. This might allow some, of the more vigorous specimens to acquire sufficient size to overcome this weakness. In my opinion, the climate itself is not the main governing factor which would kill out all hope of raising English walnuts here; but certainly, coupled with the disastrous attack of insect life and susceptibility to blight, these three foes are almost insurmountable. And then in view of the early vegetating habit of these species, there is the possibility that even though you had a hardy tree, immune to insects, you would never get much fruit.

Discussion

DR. MacDANIELS: Remember, the climate up around St. Paul is a bit rugged, and I think that work of that kind is certainly of value to give us an idea of the limits at which we can grow these trees, but I don't think that we have by any means explored the whole field.

In the Morris collection at Ithaca there is a little Persian walnut about the size of the end of this finger (indicating), a very small nut, that was given to Dr. Morris by a consul from the interior of Asia up in the Himalaya Mountains in Tibet, from of an elevation of about 10,000 feet. That little walnut had a hard shell, harder than some of our shellbark hickory nuts, and a bound kernel that I would say was much less promising than many of the nuts which we discard.

Somewhere, it seems to me, in this vast range of material we ought to be able to find some variety or clone of these species that would be adapted to practically every part of the United States. There at Ithaca we have the difficulties with the Persian walnut mainly of winter cold. That is the absolute low temperature that wipes out the trees, now that I have seen them come and go in my place there and in the vicinity. The old Pomeroy strain is killed at about 20 below zero Fahrenheit. It stayed there in fairly good condition up in the Lockport region until the extreme cold of 1933-34. Once the temperatures went down to nearly 30 below zero, except for a small region around the Niagara peninsula, where it hit only 12. Those trees are still there in that little circumscribed area around Niagara, and we saw a picture of one of them in Mr. Sherman's collection. But the Pomeroy trees, I have learned—I haven't seen them myself—were practically wiped out, as were the others, in what was thought to be the protected area along Lake Erie.

I remember the trees on the Whitecroft farm along Keuka Lake. Some of you saw those when the Nut Growers Association met at Geneva. They are on a bench close to Keuka Lake, which up to 1933-34 had not been frozen over for many years. They had grown, produced good crops, were in excellent condition, but that year the temperature went down to about 30 below zero and stayed there for a number of days. The lake froze over, and the trees were severely damaged. A California redwood which was there—had been there for 80 years—was killed outright, and so it goes.

Now, just for these Carpathian strains it seems to me that we have pretty well—perhaps you might say—licked this question of winter cold; that is, at least down to perhaps 30, 35 below zero Fahrenheit, but we certainly haven't licked the problem of early vegetation. That is, it starts out with warm days in the spring, the shoots get about this long (indicating), you get temperature going down to, say, 26, 27, 28, and your shoots are all killed back and you have lost your year's crop. So that's the problem which in the selection of varieties for this northern country, we have got to keep in mind, as I think that's one thing to look for among your Carpathian trees. It's one which will mature its foliage in the fall fairly early and which does not start out too quickly in the spring.

Now, we know there are some that don't start out in the spring, like these Chinese types, but what we want is a combination of short-season, late-starting, winter-hardy walnuts, and I think we can find them if we keep at it.

I didn't start out to talk so long, but I thought that was perhaps a sort of a summary of some of these things which we are looking for.

DR. CRANE: I'd just like to make a few comments. There is one thing that you have got to be very careful about, I think, in watching for these late-blooming Persian walnut trees that start in to grow, in Oregon, particularly, although the same thing is true in some areas of California where we are growing large quantities of Persian walnuts. You know that a deficiency of boron will cause trees to go into a condition which the growers out there now call "sleepers." They will stay dormant for quite a long period of time in the spring before they start growth. That's due to a severe boron deficiency.

Now, we have a lot of boron deficiency here in the East, and in areas in which we have trouble with growing vegetables, like cauliflower that has a hollow stem, or beets or turnips that split and crack, or where we have so-called drouth spot or internal corking in apples, you can be sure that you can't grow a Persian walnut, because the boron requirement alone is many, many times that of an apple or of most vegetables.

In Oregon on the same soils where we are growing apples, we put on a half a pound of borax per tree to control boron deficiency on apples. On walnuts we have to use anywhere from five to ten or twelve pounds for a tree of the same size. We have to have a boron content in walnuts very, very much higher than that of apples. We have got to be careful about that.

So if you do find late-sleeping walnut trees, or walnut trees that are late in starting to grow, you will probably find that is a result of boron deficiency.

MR. CORSAN: Mr. Chairman, I visited the Pomeroy Nursery in 1934. I had, in my own planting, about a score of trees and they were a most amazing sight. The big trees were all seriously damaged by that 1933-34 winter, as were all Ben Davis apple orchards. So what amazed both of us was the fact that Pomeroy's young trees weren't dead.[2] Of the Pomeroy, all the big trees were dead. I ordered some more from him, and I planted them, but the trees froze down to the ground. Just as a very few varieties of the Crath Carpathians did. They froze twigs and they froze buds and sometimes they froze the trunk. Only a couple of Carpathian varieties froze down to the ground, but every one of the Pomeroy did. I was quite sorry, because I had a Chinese English walnut from North China that was extremely hardy and lived through that winter almost undamaged. The nut, though, had a bitter tang, and Pomeroy's nuts were quite sweet and delicious, but I haven't a Pomeroy on the place. They are all stone dead.

[2] See Mr. Gellatly's paper in this volume.—Ed

DR. MacDANIELS: Thank you, Mr. Corsan.

Mr. Harry Weber will give us a paper by Gilbert Becker on Persian and black walnuts in Michigan.

Grafted Black and Persian Walnuts in Michigan

GILBERT BECKER, Climax, Michigan

The performance of grafted Persian walnuts in southwestern Michigan has been so satisfactory that I would not hesitate to recommend them, in preference to grafted black walnuts. One of the nicest things about grafted Persian walnuts is that when they start to produce nuts, they bear *every* year—there is not an off-season, as with the black walnut. Our locality may be especially suitable to them. Our skies are cloudy, and it is cool through much of the spring, thus preventing early growth before conditions are right for the buds to develop unhampered by late spring frosts. We have had an occasional late freeze that caused the lower nuts to drop, while the higher ones remained on the tree, unharmed.

In this article I would like to answer briefly our most often asked question, as to which varieties do we think best from our experience with them? Our climate must be quite different from that found around Ithaca, New York, because we have never had winter injury in certain Persian varieties, as occurs in that area. (And we had 26 below zero in February, 1949.) An instance of this difference is in regard to the McDermid variety, which happens to be our choice. We honestly believe the Crath No. 1 variety to have great commercial possibilities, because of its heavy production of large, thin-shelled nuts, of average quality. The Broadview is another. The Carpathian "D", apparently, pollinates the Crath No. 1 well. This one, however, is small, with a very white kernel that is sweet. We have many other varieties producing, some with their first crop this year; but we are not able to recommend any of them yet.

The black walnut varieties must be rather limited, because of the brooming disease trouble; so we select from those that are quite able to resist it, or that seem immune to the trouble. The Thomas and Grundy varieties lead with us, and two other local nuts, the Adams and the Climax, rate high in our estimation. We have some nice grafts of the Homeland bearing their third crop, which we like very much, and they appear disease free. The Elmer Myers, Michigan, and other varieties are now badly affected with brooming disease.

Several years ago I reported on my observations on the brooming disease. Now, I wish to report a little more upon the subject, especially in regard to how certain varieties have withstood its ravages. I hesitate to make any estimation as to how prevalent the disease is in the wild black walnut today, for it could be quite a controversial subject, with some claiming I was very wrong. Anyway, many of our native walnuts are now affected. Outward appearances are often very deceiving; but, when one cuts the top off a seedling and attempts to graft it, he may be amazed at the broomy growth that soon appears from the stock, should his graft fail to take. Trees that appear healthy, but have made slow or poor growth are often affected. Short, twiggy, upright growth that soon becomes dead or partly so, and arises from the main framework of an apparently healthy tree, is one of the signs that disease is there.

I have claimed there are two, or possibly, three forms of brooming disease, and I am still as convinced as ever. The so-called "witches-broom," as commonly seen in the Japanese walnut, is the form most people seem to think of. The second form is the rapid-growing type, that lops, or arches downward, is gray or green in color of wood, is very brittle and easily broken in the wind, ripping off good sized limbs, and winter-injures badly. An investigation, will, however, show much dead wood comes before severe weather. This form has some broomy, upright growth, like the first, but it is never bunched. The other, or possibly, the third form, is the latent type that doesn't seem to do much harm, merely causing poorly filled nuts.

The latent form is difficult to note, and can be detected only by the many short, dead, or partly dead, upright twigs scattered along the main framework of older trees. Cutting off part of the top will cause the typical growth to arise, thus identifying itself.

Early observation showed that certain walnut varieties were almost unaffected, or could even be immune, to the brooming disease. Different limbs of a large tree were topworked to the Thomas and the Allen varieties of black walnut. The Allen "took" the disease at once, while the Thomas grew thriftily and has always produced good crops of nuts. Later, the Calhoun variety was grafted on some lower limbs, and has remained healthy. The diseased Allen grafts are still in the tree, are now 15 years old, and are more or less alive, but in very poor condition, with the signs as found in what I call the latent form. In 1938, the McDermid Persian walnut was grafted into this same tree, and its grafts produced good crops of nuts.

I wish to cite another instance of how little the Persian walnut is affected, regardless of variety. In 1938 a large black walnut near the house was grafted with Persian grafts, on stubs that had failed the previous year. The tree had the second, or rapid growing form, of brooming disease. I have pictures showing how badly the 1938 grafts took the rapid growing form of growth; while two 1937 Persian grafts showed no signs of trouble. The tree started to bear in 1941, and has made remarkable growth. It is now one of the nicest Persian walnut trees I have, bearing heavily every year. It is about 30 feet tall and 20 feet broad, with no apparent signs that it was ever affected.

I feel we should recognize the fact that eradication of brooming disease is impossible; but one should plant, or graft, those varieties known to bear good crops in spite of this trouble. The Thomas and Grundy black walnuts do very well here, as well as the two local nuts mentioned. I do not know of any Persian varieties affected. I do not have any Persian trees with the typical broomy bunch, as is so often seen in the Japanese walnut, and its hybrids. The native black walnuts, when affected, seem to fail to fill properly, are immature, and watery, black veined, and worthless at harvest time, shriveling to a dark, hard, kernel when cured.

I think this answers the oft-asked question, "Why do not my black walnuts fill as they used to?" There is a strange relation to the filling of the native black walnut and the days of 1934 and 1935, when we had the great walnut caterpillar scourge!—when the trees were stripped of all their leaves. Ever since, we have had the brooming disease to contend with. One could jump to the conclusion that improper filling and this trouble were caused by a lack of certain nutrients; but seedlings in nursery rows are often affected, even where they are given every care.

At one time this spring I thought I had found a new way of "bench-grafting" walnuts. Seven grafts, on black root, were made in December, and were planted directly in a frost-proof coldframe, as lilacs can be grafted. All seven grafts made good growth, that is, over three inches, by early May, but failed later. There is only one alive today, I do not think this an impossible method, but there must be a better way of handling to give success, such as attention to shading and careful watering. One may find more on this subject in "Propagation of Trees, Shrubs, and Conifers," by Wilfrid G. Sheat.

In our greenhouse work we have used several nutrient preparations, with poor to good results. There is one that has proved quite remarkable, and may be of use to the nut grower. Our concern has been to promote greener, healthier leaves, and the product "Ra-Pid-Gro" is most outstanding. Our tests in regards to nut growing are very limited. A pan of Chinese chestnut seed mixed in pure sand was set under the greenhouse bench last winter.

The seed sprouted too early to be planted out, and trees have been left inside. Since the sand had no food value, Ra-Pid-Gro was applied to the leaves, allowing the drippings to go into the sand throughout the summer. Today, the little seedlings are indeed nice. Outside, a Persian walnut had yellow-toned leaves, and Ra-Pid-Gro was applied—now the leaves are green! It is amazing how quickly yellow leaves will become green. This appears to be a very useful product.

I believe we can have scions too dormant to graft! Last winter I had to make a new scion-box for storage, so copied it after the Harrington method, sinking it in the ground north of some evergreens. Scions have kept perfectly—maybe too perfectly—because they were absolutely dormant at grafting time, and have given poor success. It was rather late to save scionwood when I received an order to cut some of Mr. Hostetter's "Special Thomas" wood, so I cut a little extra for myself, and some wood from a little seedling Persian walnut that I wished to hasten by topworking. The buds were very much swollen that day, and the terminal buds were partly expanded. At grafting time I was quite surprised to find the wood I had cut late to be in exactly the same condition as it was the day I cut it. When grafted, every scion grew—all nine grafts made of the little Persian walnut were smaller than a lead pencil—and were *pithy* as well! This experience is so encouraging, I hope to have most of my wood in this advanced condition another year. Absolutely dormant wood might well be brought out of storage several days before grafting, in order to get it adjusted from winter to summer conditions.

DR. MacDANIELS: I think Dr. Crane is going to talk about the bunch disease tomorrow morning and will give us some indication about the work that has been done with that.

This matter of dormancy of scions we could probably get into an argument about, but that isn't the subject right now.

MR. CORSAN: I find that you mustn't go cutting back much. They don't like to be pruned. They are an open tree that grows a branch here, a branch there. They don't get anything like the dense branches of, say, the Turkish tree hazel. They are the very opposite, and they don't want to be pruned, and if you go pruning them, they are likely to have the witches'-broom.

MR. McDANIEL: There is another paper by Mr. Ward of Lafayette, Indiana, "The Carpathian Walnut in Indiana." The first part of it, the introduction, covers pretty much the same thing we have heard before from some of the other speakers about the Carpathian strains in this country.

The Carpathian Walnut in Indiana

W. B. WARD

Extension Horticulturist, Purdue University

West Lafayette, Indiana

The Carpathian or hardy Persian walnuts (*Juglans regia*), as grown in Indiana, are nearly all seedling trees resulting from the desire of some hobbyists to try something new. Other than a few exceptions, most of the seedling trees were planted during the period of 1934 to 1938. Credit is due to the Wisconsin Horticultural Society in offering the seedling nuts for sale and from these plantings numerous trees grew and fruited. A few test winters, with the temperature as low as minus 20 degrees F., left only those trees hardy in wood and bud. The seedling trees under observation have been fruiting for the past six to eight years, with some trees producing as much as five to six bushels of nuts per year.

The tree grows best in well drained, fertile soil and a bluegrass sod. Small amounts of nitrate fertilizer, about the same quantity used on fruit trees, have stimulated growth and no doubt have helped in the sizing up of the nuts. The tree does not do well under cultivation or mulching, as winter injury to the tree has been recorded when compared to bluegrass sod. There is also a possibility that the tree will respond to applications of liquid or soluble nitrates when mixed in spray materials. Six walnut trees were sprayed with "Nu Green" on May 9th and May 28th, 1950, using the same mixture as is recommended for apples—five pounds per 100 gallons of spray mix. These trees were observed weekly, and by late August had made more growth and gave better response than trees in comparable unsprayed rows. As the walnut trees are of different varieties, no definite comparisons may be drawn, but the trees so sprayed outgrew the unsprayed plot, although both plots had received a spring application of fertilizer of equal amount.

Set of Fruit Depends on Pollination

The best yields of fruit are found on trees that have a good pollinator close by. Oftentimes the catkins of the Persians dry up, fail to shed pollen when the pistillate flowers are receptive or fail to produce staminate flowers. It was noted early this spring that the catkins on the Persians were very few. Pollen was gathered from the butternut (*Juglans cinerea*) for pollinating the pistillate flowers that opened early. The mid-season flowers were pollenized with black walnut (*Juglans nigra*), and the later blooms were fertilized with pollen from the heartnut (*Juglans sieboldiana cordiformis*). Many of the pistillate flowers were bagged and remained receptive for a long period.

The best set of fruit on trees this year is on trees that have either the black walnut or the heartnut near by as pollinizers. The pollen from the butternut seemed to dwarf the fruit size on those trees where the pistillate flowers were bagged in the Purdue planting. We have little doubt that the Persian walnut develops a preponderance of pistillate flowers and relies on pollen from kindred species for a good set of fruit.

Nut Displays Have Educational Value

The interest in the Persian walnut in Indiana has developed to the extent that several commercial fruit growers have set out small acreages. Most of the trees are seedlings from trees previously fruited, although several growers have budded or grafted the better seedlings on the native black walnut. The public has become enthused through the various displays at local and state fairs and through the state nut show now being held annually. The exhibits have brought out some very desirable seedlings, each listed under the owner's name. Some of the seedling nuts have averaged about two inches in diameter, and 12 year old trees have produced as much as 50 pounds of cured nuts.

The largest Persian walnut tree found in Indiana is at Lafayette, it being 12 inches in diameter and possibly 40 feet high. This tree has been fruiting for the past 15 years. There are probably five or six bushels of nuts on this large tree at the present time. This tree was placed as a yard tree for its ornamental value and for the fruit.

Numerous persons have inquired about the Persian walnut as a specimen tree in their landscaping program and the demand far exceeds the supply. As many of the elms, oaks, and some chestnuts are going out from disease troubles, the Persians may be used as a replacement. The food value of the walnut compares very favorably with that of other native nuts, according to Dr. A. S. Colby, of the University of Illinois.

	% Water	% Protein	% Fat	% Carbo-hydrate	% Ash	No. Calories per Pound
Persian walnut	2.8	16.7	64.4	14.8	1.3	3305
Black walnut	2.5	27.6	56.3	11.7	1.9	3105
Hickory nut	3.7	15.4	67.4	11.4	2.1	3495
Pecan	3.0	11.0	71.2	13.3	1.5	3633

Nut Data Important in Classification

Three students enrolled in Horticulture have classified several of the seedlings. Paul Bauer, 1947-48, and Edward Burns and Gilbert Whitsel, 1949-50, have been using such information for their special project work as graduate and undergraduate students. These workers found a difference in the habits and performance of the seedling trees and two such examples follow.

Nut Data Sheet

1. Common Name: *Fateley No. 1*

2. Scientific Name: *Juglans regia*

3. Source or Owner: *Nolan Fateley* City: *Franklin* State: *Indiana*

4. Average Size: inches 1.7x1.8

5. Average Number Per Lb.: 23

6. Average Wt. Each Nut: 15.8 *gm.*

7. Shell
Texture: *Wrinkled and furrowed*
Crackability: *Very good, thin shell*
Separation: *Very good*
Average Wt. Per Nut: 7.1 *gm.*

8. Kernel
Color: *Light tan*
Quality: *Very good, bland*
Average Wt. Per Nut: 8.7 *gm.*

9. Percent Kernel: 40.5%

10. Remarks:
Exceptionally large, well formed kernel, appealing taste. Bore 50 lb. 1949. Tree set as 1 year seedling 1939. (*Carpathian strain.*)

Nut Data Sheet

1. Common Name: *Fateley No. 3*

2. Scientific Name: *Juglans regia*

3. Source or Owner: *Nolan Fateley* City: *Franklin* State: *Indiana*

4. Average Size: inches 1.3x1.54 *long*

5. Average Number Per Lb.: 34

6. Average Wt. Each Nut: 12.3 *gm.*

7. Shell
   Texture: *Smoothly wrinkled*
   Crackability: *Very good, paper thin shell*
   Separation: *Very good to best*
   Average Wt. Per Nut: 6.9 *gm.*

8. Kernel
   Color: *Light tan*
   Quality: *Good, desirable taste*
   Average Wt. Per Nut: 6.4 *gm.*

9. Percent Kernel: 54.5%

10. Remarks:

Fairly large, well filled, attractive shape and size with a thin shell. This seedling placed first at the Indiana State Fair and the State Nut Show, 1949. Tree medium in size, planted as one year seedling in 1939. This tree bore 24 pounds of cured nuts in 1949 and has been in good production for 7 years. (Carpathian strain.)

The descriptions given of the two Fateley trees are typical of some of the forty seedlings coming from various parts of Indiana, as shown in the following list.

The distribution of the Persian walnut to the public depends on the ability of the nurserymen to propagate and list the available varieties or unnamed seedlings. There is a great demand and a wonderful opportunity for the hardy Persian walnuts all over the Middle West or where apples will produce, not only for the nutritious fruits but for the ornamental value and for something different.

Indiana Counties with Carpathian Walnuts Under Observation and Test

(North to South and West to East on Map)

Northern

Porter (on Lake Michigan)
Elkhart (adjoins Michigan)
La Grange (adjoins Michigan)
Kosciusko
Whitley
Allen (adjoins Ohio)
Miami (Peru here)
Wells

Central

Tippecanoe (Lafayette here)
Carroll
Howard
Grant
Delaware
Henry
Wayne (adjoins Ohio)
Marion (Indianapolis here)
Rush
Johnson (Franklin here)

Southern

Greene (Linton here)
Monroe (Bloomington here)
Brown
Gibson (adjoins Illinois)
Pike
Posey (adjoins Illinois and Kentucky)
Vanderburg (Evansville here)
Warrick
Spencer (Rockport here)
Harrison
(Last 5 counties are on Ohio river,
opposite Kentucky.)

DR. MacDANIELS: Is Mr. I. W. Short of Taunton, Massachusetts here, or does he have his paper here?

MR. McDANIEL: I haven't received it.

There is a paper here, however, "Notes on Nut Growing in New Hampshire," by Matthew Lahti of Boston, Massachusetts. Mr. Wellman.

MR. WELLMAN: This is very short. It is just a report of bad winters in New Hampshire. Mr. Lahti I knew in Boston. His farm is in Wolfeboro, New Hampshire, about 75 or a hundred miles north of Boston.

Notes on Nut Growing in New Hampshire

MATTHEW LAHTI, Wolfeboro, New Hampshire

I will bring up to date my experience on nut growing in Wolfeboro, N. H., and supplement my reports for the years 1947 and 1948.

We had late frosts this spring, so that there is not a peach on any of my peach trees this year. This may also account for the fact that there are no black walnuts either on the Tasterite, the wood of which has withstood the winters very well, or on the Thomas. The Thomas black walnut which I reported in 1948 as having suffered no winter injury the previous winter, apparently did suffer considerable damage, which became evident later. It has borne no nuts since, and there is a lot of dead wood this year and the leaves are sickly looking. I am afraid that the tree is going to die.

The filberts, Medium Long, Red Lambert, and No. 128 Rush x Barcelona, which started to bear in 1947, have since then borne a few nuts each year, but the crop is not heavy enough to recommend them for planting in our climate. While the wood suffers no winter injury, the catkins for the most part get winter killed and, consequently, there is a very sparse crop. What is needed for northern latitudes is a filbert that will ripen in our fairly short growing season, and whose catkins are immune to winter kill. The Winkler seems to be more hardy than the others, but the nuts do not ripen. This year even the Winkler catkins were killed, although the catkins of a wild hazel growing nearby were not.

I have two Crath Persian walnuts planted in 1938 which are the survivors of perhaps a dozen seedlings. These two trees have shown no injury. One is bearing seven nuts this year for the first time, and the other one, bearing for the second year, has 80 nuts on it at the present time. Last year the squirrels got all the nuts so that I could not evaluate them, but I will take precautions to save some this year.

The Broadview Persian walnut has thirty nuts on it this year, but the wood of the Broadview definitely is not hardy in our climate.

Summing up my experience with the various nut trees as previously reported, I would say that our climate is not suited for commercial nut growing, but for home use named varieties of butternuts and hickories that crack out easily and possibly one or two of the Crath walnuts should give satisfactory results. My chief difficulty with hickories has been the poor union at the graft, resulting in slow starvation and death in a few years. I have only three left out of approximately 25 trees that I have planted.

MR. CORSAN: A professor from the University of New Hampshire wrote to me that they were very much interested in planting a nut arboretum. Does anybody know what result came of it? I sent them some hybrids of the Japanese heartnut (female blossom) crossed with our native butternut (male blossom).

DR. MacDANIELS: I guess they are somewhat interested. They have very little possibility of growing very much except the butternuts, and sometimes hybrid filberts.

MR. WELLMAN: I have a friend who is up a little farther north than that, in Woodsville, and they have been urging him to set out filberts for wildlife food there, and he has shown me some of those that he has started. It's been quite a movement up there. I don't know how wide. He has about a hundred seedlings that are used for propagation by the state.

Is the Farmer Missing Something?

JOHN DAVIDSON, Xenia, Ohio

(Read by title)

The farmer is a specialist; a producer of edible crops. Like any other specialist, his thinking tends to be channeled along the lines of his specialty, to the exclusion of other lines.

For example, the average farmer probably knows little and cares less about teleology, metaphysics, or, let us say, forestry. He is a farmer. He makes his living by raising crops. And yet, a better knowledge and practice of forestry will not only make him a better farmer wherever he is located but, in certain locations, this knowledge and practice is absolutely essential to his continued existence.

In a recent decision of the U. S. Supreme Court upholding a decision made by the Supreme Court of the State of Washington, a principle has been approved which may have a profound influence upon our future well-being. It affirmed the constitutionality of a Washington State law which requires the owners of land used for commercial logging to provide for its reforestation.

Such a law is novel indeed. What? May private owners of the earth's resources not use or destroy them as they see fit? The court, in effect, says they have no such right. In the court's own words, the "inviolate compact between the dead, the living, and the unborn requires that we leave to the unborn something more than debts and depleted natural resources. Surely, where natural resources can be utilized, *and at the same time perpetuated* for future generations, what has been called 'constitutional morality' requires that we do so."

The New York Times, in commenting upon this revolutionary but perfectly sane decision, says: "Time is truly running short; the annual cut of saw-timber, with natural losses, is 50% greater than annual growth…. If the individual forestland owner is too lazy, short-sighted, or indifferent to act, the Federal Government will have to enter the picture."

It is a complex picture. The American farm owner is, by every implication, also involved along with the forestland owner. He, too, has a duty to the unborn, but it is an opportunity as well as a duty. It is only because of what J. Russell Smith calls his insane obstinacy, that the average farmer is now operating a one-story agriculture in place of a two-story agriculture. If he were thinking and doing more about his debt to the unborn, he would also be serving himself better.

I am convinced that the farmer is the key man in forest husbandry. And the best way to interest him in tree planting is through his specialty—through *crop* production. A *two-story* agriculture! Tree crops along with other crops!

The farmers' education along this line has been very inadequate. We have been very stupid. Can we never learn to begin, as Hitler began—as the Russians are even now beginning—with the nation's children?

Perhaps we are learning a little. It is heartening to know that school and community forests are fast increasing in number, notably in New England. When fully used and well managed, they can work a revolution in the thinking of the young people who are so fortunate as to have some of their schooling out in the open. These future American leaders are learning at first hand through the ways of the woods how to make the work of their hands live far beyond the span of their lives.

Perhaps, as the result of this training early in life, a new interest among the farmers will emerge and some of our sins of omission will be remedied. As a planter of trees for the future, the American farmer, both of yesterday and today, has notoriously, thoughtlessly, and disastrously failed both his children and himself. By all standards, he should be the first-ranking tree planter of the land. As a matter of fact, it is practically impossible to interest the average farmer at all. State experiment stations and forestry departments make some effort to stimulate interest in the planting of trees by furnishing seedling stocks of forest trees at nominal prices and by issuing occasional bulletins. However well intentioned and, within their limits, well done these bulletins may be, the fact remains that in proportion to their numbers, farmers are still not notable planters of trees. Perhaps one reason for this failure is that most of the literature upon the subject seems aimed at lumbermen, and not at farmers. As to the bulletins which are aimed primarily at the farmer, examples of advice on forestry which is given in these rather too specialized and somewhat near-sighted publications are typically of the following kind: "Fence off the woodlot and never pasture it," "Use your best land for field crops; your waste land for trees." "You are interested in nuts? You can not have nuts and timber, too."

It is evident that these rules are prepared by foresters—not farmers. Is it any wonder that the inquiring farmer finds them rather frustrating?

It should be remembered that practices which are valid and helpful in the care of an already existing forest or woodlot where mature growth is periodically harvested and where young sprouts are encouraged for replenishment may be of little use in the management of an entirely new planting of certain kinds of trees where cultivation, at least for a time, is necessary. Deep-rooted trees, for example. Such rules have been of little use to me in my own planting of American black walnuts upon an Ohio farm. Indeed, to have followed them would have been disastrous.

My planting is not large. It is modest enough to be within the power of nearly any farmer. It has been treated as a farmer would treat it, without too much pampering. We now have a few more than three thousand trees planted upon forty acres. Most of them are now fifteen years old. Here are some of the things we have learned in fifteen years from our trees:

1. Trees spaced 80 feet apart in good deep soil have not made as much growth as seedling black walnut trees spaced 8 feet apart in rows 20 feet apart, also in good soil. However, these wider spaced trees are grafted pecans and Persian walnuts.

2. The seedling trees which stand in good soil have made surprisingly good growth. Some better than 8 inches diameter, breast height. One measured tree has grown 7 feet 1/2 inch this year to date—Aug. 20. (No fertilizer used, but cultivated.) Those which stand in shallow, thin soil are dwarfs, worthless. Walnuts have deep taproots. They need deep, rich earth.

3. Trees grown from planted seed make the best timber trees. Upon the other hand, if production of known quality is the primary objective, grafted trees of known varieties must be planted. The seedling *of good parentage* is an exciting gamble. It may be, and usually is a commonplace producer of nuts. Upon the other hand, it is more likely than the tree of poor parentage to win a place among the named varieties, set aside for propagation by budding or grafting upon other stocks.

4. Walnut seedlings like human beings tend to show marked inherited trends, erratic and undependable though they may be. Thus, seedlings grown from vigorous and upright trees *tend* to be vigorous and upright. Conversely, trees of poor parentage, either as timber or nut producers, will tend to reproduce the poor characteristics of their parent. This is more markedly true where the parent tree stands isolated from the pollen of other walnut trees of the same species.

5. I have found no real evidence that walnuts of our planting are toxic to other trees standing immediately beside them. To test this, we planted a few apple, peach, and plum trees in the walnut rows. They still stand literally arm in arm. This is, of course, all wrong. No tree should be so crowded. The apple trees monopolize space by excessive lateral growth. The plums send up unwanted shoots from their roots. The peach trees are passing out. Two or three of the apple trees are half dead. Others still live, but I am not very hopeful that, after the walnut trees are more mature, any of the apples will survive. The usual diseases and insects, plus shading by the walnuts seems to account for most if not all of the dead trees to date.

6. Grass growth is excellent right up to the trunks of all of the trees. It has never been necessary for us to lose the use of the land upon which the trees are planted. While the trees were young, of course, no pasturing was permitted. The land between the rows was cultivated. In these strips we raised berries and other crops. Now that the trees are tall enough to be beyond the danger of damage from livestock, we graze the pasture under and between the trees. No damage is evident from trampled earth (the walnuts are deep-rooted) and the hazard of fire is eliminated because there is now no need to mow excessive grass, weeds, or brush.

7. The most precocious seedling walnuts began to bear nuts at about 7 years of age. New bearers are coming in each year. All are still counted as adolescent trees, yet, last fall, picking up the nuts from none but trees marked for their better quality of nuts, we gathered some 40 bushels of nuts in the shell.

8. Today, we can count about 2,000 walnut trees which promise to be of good timber quality 35 years hence. At a reasonable estimate, 1,000 trees will then survive, be 50 years old, be worth $50.00 each, at present prices. Total, $50,000.00. This represents an annual increment in value of

$1,000.00 per year for the 20 acres which are closely planted to black walnuts. Can the average farmer *save* that much in his lifetime? Can even the exceptional farmer do it on 20 acres? With as little investment of money and work? If so, how?

Any farmer can do as well, or better, without losing a single immediately productive acre. Why doesn't he?

The answer is in the very nature of the farmer's business. As has already been said, he is primarily a producer of food. If trees stand in the way, he chops them down. He has always chopped them down. It has become a habit. If the farmer is to be persuaded to change his ways and turn to planting trees, instead of destroying them, I repeat, the entering wedge into his interest will be, I believe, through dual-purpose trees—trees for food crops, as well as for timber crops. Of these species, the black walnut of eastern America is probably the most outstanding one of all, at least in the mid-section of America. The butternut—"white walnut"—flourishes better in the north. The chestnut is another—a tree almost literally raised from the dead by the efforts of a few miracle workers like Dr. Arthur H. Graves of the Connecticut Experiment Station, who, with others of his kind, has been in the throes of producing a blight-resistant, tall-growing hybrid timber tree out of the bushy Chinese chestnut, a producer of the sweetest of nuts. The pecan, too, is being pushed northward. Great groves of wild pecans have firmly established themselves along the Ohio River. Their timber is fair; not wonderful. The mulberry tree is still another. The American species produces a timber which is remarkably durable under ground. Its fruit is not sufficiently appreciated. It makes an unsurpassed jam or jelly or pie when combined with a tart fruit like the cherry, grape, or currant. And who does not know the precious wood of the wild cherry? Its rosy warmth of color is the pride of the "antique" connoisseur; its fruit beloved by birds and squirrels; its juice, the secret of the cherry cordial. Even that foreigner, the

Persian "English" walnut, of Carpathian strains, is pushing north into Canada and the East Coast region. Its wood, too, under the name of "Circassian," is famous for its figured beauty[3].

[3] Some of the "Circassian walnut" is another genus, the wingnut (Pterocarya).—Ed.

One might go on and on with a list of trees and tree crops easily available, mostly native, all of which should be both figuratively and actually right down the farmer's alley.

Perhaps the education which can come through the agency of many school forests will in good time turn the attention of young and impressionable minds to the potential wealth to be found in the trees. Normally, the young, who, of all people, should be forward-looking, are least concerned with the long-term future. They are not given to making plans or building estates for their grandchildren. As a consequence, the planting of trees is traditionally taken over by the aged, or at least by the mature. This is all wrong. The young farmer who plants interesting trees is preparing for some of the most exciting and prideful moments in the years which follow. And he is also building, at low cost, and with little labor, a priceless estate.

How to Lose Money in Manufacturing Filbert Nut Butter

CARL WESCHCKE, St. Paul, Minnesota

Inasmuch as there are so many words of wisdom and advice showing the reader how to make money in different ways, I have started a new line of caption with the hope that it might serve as a warning for those who would stick their necks out, as the term applies to those people who venture beyond safe margins of restraint. Since this is a recital of facts, and since Professor George L. Slate has requested me to report on my experiences, I submit the following for what interest it may hold for the readers.

Most ventures are backed by optimism of some sort or other, coupled with some experience, capital, hopes, and ambition. The project which sparked the entrance into the manufacture of filbert butter was the success that I was having with hybridizing our best native hazels with the best known filberts, such as crossing of the wild American hazel with Barcelona, DuChilly, Italian Red, Purple Aveline, Red Aveline, White Aveline, also filbert strains from J. U. Gellatly of Westbank, B. C., Canada, and strains from J. F. Jones, hybrids, European strains of filberts from the Carpathian mountains, and any right pollen which could be obtained from known filbert parents. Today we have over 2,000 seedling hybrids of which between 500 and 600 have come into bearing. Some of these are really surprising varieties of the combination hazels and filberts, but a complete history of the hybridization work and the results really deserve a separate account to be published some time in the future. I merely mention this because the success of these plants in producing nuts leads me to contemplate the future production of these hybrid nuts, called Hazilberts,[4] on a large scale.

[4] Another coined name, by Mr. Gellatly, is "Filazel."

My problem was to engineer a scheme whereby I could interest farmers in setting out small acreages of these plants and guarantee that there would be a market when the plants produced nuts, which would be in about three years from the time they were planted. Seeing that the filbert producers in the west were struggling for a better market, since conditions were not too favorable for the filbert in its competition with the foreign nuts and the California produced Persian walnut, I decided that nuts in the shell were a little bit old-fashioned. Many of our prominent members of the NNGA have from time to time advised the marketing of nut kernels rather than nuts in their natural containers, and I thought a step in the right direction would be to manufacture a ready-to-eat product from the kernels. And what could be nicer than a butter similar to peanut butter?

So I began scouring the market for a grinding machine that would grind filberts to the consistency of a smooth peanut butter. My first machine was a Hobart peanut grinder. When buying this machine the mistake I made was to let the agent of the manufacturer demonstrate how good it was to grind Spanish peanuts; I should have had it tested on filberts as they are much tougher, even though they do carry more oil. This machine was installed, but it was a complete failure and I decided to buy more expensive machinery, and also put in a cracking plant and buy the nuts by the ton or carload, if necessary, directly from the growers on the Pacific Coast or through their organization, the Northwest Nut Growers. I located a satisfactory machine for the purpose, which required about 7 horse power to run. Since this was during the war and no motors of the right speed and power were available at the time, I set up my own generating plant, using a 25 kilowatt generator driven by a Diesel engine which generated direct current so that I could use direct current motors which I already had among my machinery supplies. Then a separating machine, which required a 10 horse power motor just to operate the fan, which is part of that equipment, was purchased. Also, a nut cracking machine was secured from a West Coast manufacturer. Along with tanks and containers and other necessary equipment, all set up in a little factory building I had available for that purpose, I commenced the manufacture of filbert butter on a commercial scale.

The product was declared by every one to be excellent. We were quite sure of this since we had taken pains to buy up any product that purported to be a nut butter, and had tested those products in many ways to assure ourselves that we had a product superior to anything that we could find on the market at that time. The Owens Illinois Glass Company designed our label and gave us the benefit of their experience with containers. Then we placed our initial order for glass containers and re-shipping cases. Every detail in handling this material was properly taken care of, to insure that if the orders came rolling in we would be able to supply the demand and have our shipments reach the consumer in first class shape.

Then we initiated an advertising campaign, coupled with sampling, and received many fine letters which encouraged us to hire a salesman who sold the product to the stores in the Twin City area so as to have proper distribution. Advertising was done also in two national magazines, so we sat back, hopefully anticipating the big orders that we were soon to receive. The reorders from the local stores came in slowly, too slowly for our set-up. We received suggestions from the store keepers and from other persons that perhaps the product was too high priced, so we made experiments in other towns where we set the price so low that there was no profit. In fact, there would be a loss of money were we to do business on that basis. Yet there was no stimulation of sales due to this reduction in price.

Many good suggestions came in; among these was the suggestion that the product lent itself nicely to an ice cream topping; by mixing it with honey or with syrup we interested our largest manufacturer of ice cream in this locality and he did a lot of experimental selling. He was very cooperative. He also sold it in his branch stores as milk shakes; everybody liked it. No complaints whatsoever except that the manager said it was too expensive to compete with a chocolate flavor on which he made much more money. Finally this whole thing fizzled out and was discontinued.

The next experiment was with candy; as a candy center it was one of the finest tasting confections that had ever been made, but the oil which would ooze through the chocolate coatings prevented the practical use of it. You see, the filbert has about 65% oil, and when it is ground into a fine, creamy butter, this oil will come out and sometimes be an inch or more in depth over the top of the butter in the glass container in which it was marketed. So we investigated several methods by which we could eliminate the oil. We could pour it off and sell the oil separately; we could emulsify the product with the addition of certain emulsifiers, so as to keep the oil mixed with the starch and protein of the filbert nut. We tried many ways; there is only one method that we haven't used and that is to combine solidified or hydrogenized peanut oil with the filbert butter in order to prevent this liquid oil from rising to to the top of the product. The reason we did not do this is quite apparent—we did not want to mix peanuts and filberts, as we considered peanut butter a cheaper and inferior product. We could not hope to compete with peanut butter with the prices already set for peanut butter recognized by the trade.

Among the products that came to our attention, however, was one which had both filbert butter and solidified peanut oil in it. When we tested this product among many of our friends, they declared it tasted too much like peanut butter. It spoiled the delicate, fine flavor of the natural filbert butter (which we were marketing without adding any sort of seasoning, and without roasting the product the way peanuts are roasted before they are ground into butter.)

Now, if any of you readers think that we have left out something important which would have insured the success had we done it that way, we would certainly like to hear from you, or we have some nice machinery that we will sell cheap in case you want to experiment with it yourself. I would be the last one to condemn the future possibility of producing a commercial nut butter, and yet it is strange that the only successful nut butter is not a nut butter at all. Peanut butter is not a nut butter because peanuts are a legume like a pea or bean. To my knowledge, we do not have any nut butters on the market today with the exception of the cashew nut butter, which recently had a distribution in our locality, but which seems now to have run its course much as our products did. We bought the cashew butter and tried to interest everybody to use it, just to see whether it was any different than our product in its popularity. In our meager tests we found that the filbert butter was slightly more popular than the cashew, since the cashew reminded people too much of peanuts again. It was also very expensive. However, there must be a way to make a satisfactory butter out of filberts or hybrid nuts, as they carry the hope of the cheapest nut product, which is fundamentally necessary to manufacture a popular food item.

The method of propagation of the Hazilbert is by layers instead of grafting—layering is a cheaper and more satisfactory method. Also, the nuts are the most satisfactory to crack as they have no inner partitions which would require intricate machinery to extract the kernel. Their keeping quality is excellent; we have tested this out over a number of years, and filbert butter properly processed will easily keep a year without turning rancid or having an unfavorable flavor. The tonnage of nuts that can be produced on an acre of land is unbelievably high. I have measured individual plants and their production, and the area that they covered, and it is safe to say that we can expect to produce a ton of nuts in the shell per acre in favorable locations on good deep soil. Even at 10c per pound for the nuts this is a good return. New methods of gathering the nuts after they fall from the involucre or husk are being discovered and improved by the western growers from time to time, so that the old expensive method of hand-picking is being eliminated. This should make the filbert even cheaper to harvest.

It is not my intention here to discourage the manufacture of filbert butter, but to point out the difficulty that I have had personally to promote the idea in a commercial way. Neither is it my intention to stimulate too much interest in the planting of the new filbert varieties which are still under test. I feel that it is necessary to test a plant for at least a five-year period before it can be singled out as a plant to propagate. We have not yet reached the point where we care to sell these plants, as much better ones might crop up among the untested plants, which number over 1000, and which have never yet had a chance to bear so as to show what they can do. At some future time I expect to write an article on filbert hybrid culture (Hazilberts) for the whole central, north, and northeastern part of the United States, and at that time I believe that tests will have progressed to such a point that recommendations can be made.

DR. MacDANIELS: There was one more paper that the Secretary has that was not scheduled, from Mr. Elton E. Papple, of Ontario. Title, "Filberts, Walnuts, and Chestnuts on the Niagara Peninsula."

Filberts, Walnuts and Chestnuts on the Niagara Peninsula

ELTON E. PAPPLE, Cainsville, Ontario

My brother and I have been interested in growing nut trees for some time, and have had some interesting experiences and some success. A few years ago, Mr. Slate sent us from Geneva some varieties of filberts which he considered quite hardy. We purchased some from Mr. Gellatly in Westbank, British Columbia, some from Mr. Troup, Jordan Station, Ontario (near Vineland); also from J. F. Jones Nursery, then in Lancaster, Pa. Mr. Slate sent us scionwood and we grafted these scions in the spring and layered them shortly afterwards. By the following spring they were rooted well enough to be planted out in the nursery row. This gave us our material to work with, and about the third year we started making crosses between different varieties. The first year we obtained quite a few crosses, and had a good number of these seeds to germinate in the spring after taking from stratified storage and planting them in the nursery row. These trees have now started to come into bearing, and they promise to be better than their parents in some instances.

We made a number of crosses since, but we have been very busy and the young trees of these crosses have just about perished through neglect. In this last lot we had a cross of the filbert on the beak or horn hazel[5], and of a cluster of three, had one to grow, which in turn was promptly eaten off by a rabbit or rodent of some description. The reason for this cross originally, was that, so far as we could see in the last fifteen years the male catkins never winter-kill; whereas filbert trees are subject to this hazard. Some of the filbert varieties have the ability to withstand changeable weather and not lose all of their catkins. Others will winter-kill in the wood as well. We have removed all our Barcelona and Du Chilly trees because they winter-killed almost one hundred percent.

[5] Corylus rostrata.—Ed.

With the experience we have had with filberts, we believe that before they could be commercialized, it would be necessary to have hardy catkins that will withstand changeable weather: not altogether resistance to extreme cold, but to temperatures that vary from warm to freezing in a few hours. A mulch does help where the warm period is for a short duration; but last winter we had a week or more of warm weather in January, with rain and then a cold snap. Even then, some of the catkins on the German varieties and others came through fairly well.

Selection of varieties for machine cracking or eating from the shell should determine varieties one should grow, but hardiness should be the key factor in selecting varieties. The following table shows some of the crosses we made. Most of these seedlings have borne a few nuts to date, but we cannot give anything definite as to whether the catkins are hardier than those of the parents.

Table of Crosses:

Female	Male
Italian Red	Medium Long
" "	Red Lambert
Medium Long	" "
Cosford	" "
"	Vollkugel
Comet	Cosford
"	Vollkugel
Craig	Red Lambert
Gellatly	Vollkugel
Carey	Red Lambert
Fertile de Coutard	" "
Barcelona	Vollkugel
Seedling (W)	Red Lambert
" (E)	Vollkugel

I would like to make a few remarks on our heartnut and Carpathian walnut trees. Most of the heartnut varieties came from B. C. and we think that Mr. Gellatly has some of the best obtainable anywhere in North America. The Bates heartnut from J. F. Jones Nursery seems to be very hardy here, and quality of nut is very good. We have found—comparing a heartnut rootstock which grows two weeks later in the fall than some of our black walnuts—that the same variety of heartnut will live one hundred percent on black walnut stock and winter-kill severely on the heartnut rootstock. We believe that the root system for the north, either heartnut or black, should be carefully selected for its growth habits before considering its use as material for rootstock in grafting or budding. I might add here that we also found that if the variety of heartnut was not hardy, it did not help any in regard to hardiness to use black walnut at the rootstock. There is a good crop of heartnuts on the trees here this year.

In grafting Carpathian walnuts on black, we found that some varieties graft or take more readily than others. Also some would give a better union. The Broadview winter-kills with us, but it is not hard to graft it almost one hundred percent. We have quite a number of the Carpathians bearing and they seem to be quite hardy, of good size and quality, and bear every year. As the catkins were killed on all but one variety, due to the unseasonable weather experienced last winter, there will be only a light crop. The hardy variety has late blooming male catkins which might account for its catkin hardiness. It is of good size and excellent flavor. Possibilities for commercial planting of these Carpathian varieties in the north appear promising in favored localities.

Our Chinese chestnut trees seem to be hardy and this year have produced a few burs for the first time. We have planted out about sixty young trees this year and they are all growing nicely. The weather has been wet and just the thing to get them started.

Our hickory trees, which we grafted, are growing well and we set some more out last year. When we started grafting hickories, we had one hundred percent failure, but kept at it until we got almost a perfect take. The hickory seems very slow in forming a union. A lot can happen to the graft before it gets started. Filberts graft as easily as apple. Our findings in grafting nut trees are that any amateur can graft apple trees, but nut trees are something different. We have a number of odds and ends besides what has been mentioned.

Being a member of the N.N.G.A. has helped us in growing nut trees, and the information in the Annual Reports should help anyone who has just become interested in growing nut trees. The information is up-to-date and fairly accurate. All one has to do is apply his findings to his own planting.

MR. CORSAN: Doctor, in that same neighborhood is a man who called on me who has a nut aboretum of 40 acres on Grand Island in the Niagara River. That's above Niagara Falls, of course. I thought he'd call again, but I didn't get his name, or at least I have lost it, and what do you think he is growing in the way of nuts? Can anybody guess:

A MEMBER: Coconuts!

A MEMBER: Peanuts!

MR. CORSAN: I am growing coconuts in Florida—but on that one 40-acre tract on Grand Island, New York—he lives in Buffalo—he is growing evergreen nuts from Swiss stone pine (*Pinus cembra*), Korean pine, Philippine pine, *Pinus Lambertiana*, *Pinus Monophylla*, *Pinus edulis* and Digger pine (*Jeffreyi*). He is growing these evergreen pine nuts, and he says he is making very good success of it.

MR. STERLING SMITH: Chas. F. Flanigen is his name. He's a member.

MR. WEBER: I'd like to ask the members, or those present, whether they have failed to sign the registry of attendance.

DR. MacDANIELS: That ends the formal program this afternoon. It's always been a criticism that things are too crowded. We have an opportunity now for about half an hour to visit, look over exhibits and then later on we will meet at six o'clock at The Stone Chimney.

(Whereupon, at 4:35 p.m., the Monday afternoon session was closed.)

MONDAY EVENING SESSION

DR. MacDANIELS: Without any question at all, I think, the most important single consideration in determining the planting of nuts is the matter of varieties, and I know that Dr. Crane has some ideas along that line which he wishes to develop, and without any further talk on my part, I will introduce Dr. Harley Crane, United States Department of Agriculture.

(Applause.)

Nut Varieties: A Round Table Discussion

H. L. CRANE, Chairman

DR. CRANE: Mr. President, members of the Northern Nut Growers' Association: I think it is, without a question of doubt, of the greatest importance that we consider this question of varieties. After all, a variety of any plant, in my opinion—which I think can be well supported—is the most important thing that anyone can consider when it comes to planting or developing a nut tree or a fruit tree or anything in the fruit line. We can cultivate and fertilize and spray and do everything that is needed to be done today in a modern fruit or nut orchard farm, but if the variety is not suited to the climate, if it is not a good variety, all our efforts that we make towards developing a good tree and bringing it into fruiting are wasted.

I know that every one of you appreciates old varieties of corn and just what has been done in our new varieties of hybrid corn, how hybrid corn has changed the variety situation. Now it's hybrid this and hybrid that, because hybrid varieties are generally superb.

Now, at this time in our nut work we are a long way yet from growing good hybrid varieties, and I feel that there has been an effort on the part of a lot of people to capitalize on the word "hybrid," because hybrid corn has been such a success; and we figured that by carrying it over into other plants, particularly the nut trees, we would get the same remarkable performance from hybrid nuts that we do from hybrid corn. But that is not the case.

We will come to that some day in the future, maybe—not in our lifetime, but we will have hybrid varieties, because, after all, our great improvements that have come in most of our plants, in corn and in wheat, and in other plants, have come through the mixing of the genes, or the characters that we have differing between species.

In our nuts, now, with the exception of hicans, we are still dealing with pure species, and most, if not quite all, of our hicans are worthless at the present time, largely because of sterility.

A good variety is the most outstanding thing that a horticulturist can get or can have, because of the fact that it does have the character in it which will make good growth. It will set a lot of nuts, it will carry them through to maturity and it fills them, and if a variety doesn't do that, it's not a good variety. Then after we get the nuts filled, cracking quality, eating quality or oil content, and all these things come next.

Now, this brings us next to the very important consideration of how are we going to get a new good variety? Well, we can do that by selecting from seedling nuts, or we can make controlled pollinations, crossing different varieties, or varieties of different species, planting the nuts or growing new trees and then selecting out of them those that have the desirable characters.

But the first thing that we have got to do after we have either selected the nut or made the hybrid and selected the nut is to evaluate the nut as to whether it does have the first character, or proper characters, that we ought to have in the nut. Does the crop ripen evenly? Whether it hulls readily or comes free of the husk is a minor consideration, provided that the nut itself has the desired characteristics. By that I mean, does it have a good, large kernel which is well filled and bright in color, or good flavor free from any objectionable characters? How about its shell, percentage of shell in relation to kernel? Those are some of the things that we have first got to consider.

That's what we can do in holding our contests to find good varieties. Those are the ones submitted by growers and others. They are in competition with nuts from other sources, and then the committee, or someone, goes over and rates them, and places them, just as has been done by Mr. Chase and others in their Carpathian walnut contest for members of the Northern Nut Growers' Association.

Now, at the present time we have no standard method for evaluating the nut. It's the opinion of the judges that do the scoring or rating which determines the placing that the nuts get. Well, now, that's one of the things that we members of the Northern Nut Growers' Association have been working on for a long while, but we still haven't arrived at any definite place.

Well, then, what's the next step that we take up? The next thing we do, some growers find out that a Persian walnut from Mr. Shessler, for example, placed second in the contest this year. They will get some scions from Mr. Shessler, or somebody else, and they will make a few grafts and grow some trees, and then they will make a study of these nuts and find out how well they do and what they are like under their conditions, and that's about as far as it goes.

Well, now, we cannot continue to do that kind of a job, as I see it. If we go back over the reports of the Northern Nut Growers' Association we will find that this matter of varieties is discussed in a very large majority of the papers that have been presented. But those that have taken part in investigations and in advising the public, like those in the Extension Services of the colleges, those teaching in the universities, those doing research, like myself, anybody who has to answer correspondence from would-be nut growers, almost always get the question, "What variety should I plant?" Then they put it up to me or Dr. McKay, or Dr. Colby, and think that you could just name right and left, and they ask, "What varieties shall we plant?" They put you right down on the spot. Here you are, you are supposed to be a real expert, know all things, and they are asking you for advice, and they will take that advice and carry it out.

Now, today it puts a fellow in an awfully hot spot, because as you read the reports of the Northern Nut Growers' Association you find that there is absolutely no unanimity of opinion. Every grower is absolutely certain in his ideas, and they are different from every other grower's.

Well, you can't recommend them all. It's really impossible. Now, this is one of the things that the Northern Nut Growers have been dealing with all of these years. This is the forty-first annual meeting. You'd have thought in 41 years we'd have come up with something, but we haven't yet. Now, I feel that it's about time that we stop and take stock of our situation.

I am not going to do the talking tonight, I am just making a few suggestions and trying to direct the thought a little bit. But one of the nuts that we have done so much with and have said so much about in our reports is the black walnut. It's very interesting to read the reports on varieties of black walnuts and how those who have grown black walnuts differ in their opinion, regardless. Well, I don't know. When I get a letter coming in from most anywhere in the country wanting to know what variety of black walnut to plant, do you know what I tell them?

MR. CALDWELL: Let them find out for themselves.

DR. CRANE: No, sir, they will never find out, not in their lifetime. I tell them to plant Thomas. Thomas, Thomas Thomas! Why?

MR. KINTZEL: Because we know more about that than any other.

DR. CRANE: That is right. I expect there are four or five times as many Thomas walnuts propagated and sold by nurserymen in the United States as all other varieties.

MR. CORSAN: It always has a bigger crop, too.

DR. CRANE: It bears, that's one thing. It may not always fill, but Thomas is a good variety. But we in the Nut Growers' Association haven't the nerve to come out and say the Thomas is a good variety. It has its faults. I know I am going to be wrong in a lot of cases by planting Thomas.

MR. CORSAN: But don't plant it outside the peach belt.

DR. CRANE: Well, the peach belt is an awful lot of territory. I know I am going to be wrong, but I know I am going to be safer with Thomas variety than I would be with some of the others.

Now, I think that it's time, and I think that the biggest thing that the Northern Nut Growers' Association can do is to give very serious thought and take action at this meeting some way looking towards the Association's giving consideration to methods and means whereby we can properly evaluate varieties that we have that are growing so that we can recommend and tell others the varieties that they should grow.

You know, here is the situation exactly. In the territory of the Northern Nut Growers we don't have a commercial industry at the present time. I doubt if there is a single family of the Northern Nut Growers who are here that depend on the sale of nuts for their living. Well, when your living depends on something, you take an awful lot of interest in it. And that has been true in the case of apples, for example. I don't know how many there are, but twenty years ago or more there have been fifteen or sixteen thousand apple varieties that have been described and have been planted and propagated, and you can name all of the commercial apple varieties grown in the United States almost on the fingers of your hands. That is, the important ones. Oh, the list has grown, would probably take in 200, but that 190 hardly make a drop in the bucket as compared to the ten big ones.

Well, the same thing is true with peaches. The Elberta peach just is completely outstanding. It's a big commercial peach. Now, in all of the Association here, almost every paper that is presented always has some commercial aspect mentioned in the paper, but we could never have any commercial industry as long as we are fooling with a lot of these varieties with nobody giving them the serious consideration that they deserve, in an effort to properly evaluate them.

This evaluation of a variety is our problem. I have given an awful lot of thought to it over the years and how to get around it, how to come up with the proper answers within the near future so that we can be of help to others and stop a lot of our amateurs, those who are attracted to the industry, from making mistakes and getting discouraged. That is the problem. And that is the thing that I want all of you to be thinking about tonight and help us with the suggestions.

Now, we could just start almost, I expect, in dogfights, if we were to conduct this round table to get to discussing the different qualities or desirability or other aspects among varieties, and each fellow would be right, because I know there wouldn't be agreement. It would make an interesting round table, but I don't know how constructive it would be. So I have tried in these preliminary remarks to get you to thinking about this problem, of evaluation.

Now, there is one other way that we could go about it. For years we have had in the Northern Nut Growers Association a group of officers that are known under the title of State Vice-Presidents, and I think if you judge by their performance in the past, the main reason that we have had these State Vice-presidents is that we were attempting to confer some honor on somebody, the honor being in having them so designated and their names published as State Vice-presidents in the proceedings. In many cases their performance hasn't warranted that honor, because, after all, a vice-president is supposed to be a working vice-president, not an ornament. The ornament is supposed to be the president, if we have any such thing. At least, that's what I have heard. I have never been president. And I have thought that if in the consideration of our State Vice-presidents we select the ones who are particularly active and very much interested in this variety problem and in the Northern Nut Growers' Association, that we might take up this variety problem and get us information by two ways.

One would be through surveys made in their states by contact with the growers, either personal contacts or by letters. Then those reports could be assembled, and we could have our variety committee over all, so the Association could attempt to evaluate. That would be one start.

Another thing would be that our State Vice-president in collaboration with the President, would appoint a state committee. Now, we have a lot of growers in some states that are vitally interested. In Pennsylvania, for example, and in Ohio and New York we have a lot of growers who are members of this or state associations that are vitally interested in this thing. You have a State Vice-president appointing a committee in collaboration with the president of the National to evaluate the variety situation as it exists in their state.

Now, we would expect them to do some honest work on this thing and come up with a report in which the different members could agree. Then we would be nearer getting unanimity of opinions. We have got to get this some way so that we can agree upon what we do with the answers to individuals better than we have been doing in the past.

There may be some error to this. Well, you see, I know that some of you must be familiar with the New Jersey Peach Testing Association. I am not sure just what the name of it is, but it's something like that.

A MEMBER: New Jersey Peach Council.

DR. CRANE: It has been a great power and a great help in regard to the selection and evaluation of peach varieties in the State of New Jersey. In New Jersey the experiment station has had a peach breeding program going for a number of years. They have done outstanding work, and they have brought out some very good varieties. Well, the station has selected the good ones and discarded the poor ones, or what they thought were the poor ones. They call in members of this Peach Growers' Council, and they have the peaches evaluated. They are passing them on to the fruit growers. "Do you think, in your opinion, that this would be a good peach for us to grow? Is it better? Does it have better flavor than other peach varieties?" They will, out of that group, select some of these new ones, maybe. Then the New Jersey Experiment station will see to it that the trees of these varieties are propagated, and they are given to the members of that Association in order that they can plant them under their conditions and grow them to fruiting and see how they do.

Well, then, this committee still continues to evaluate them, and if the members of the Association say, "Well, that's a variety we should grow," then they will grow it. If they feel it isn't as good as some they already have, they throw it away and that's the end of it. But they don't clutter up the variety situation with a lot of poor stuff. And they make profits, because always two heads are better than one, even though one is a sheep's head, as the old saying goes. Well, when you get four or five or more in a group and they agree, you can be sure that their opinion is far better than five individual opinions or judgments.

I am very anxious to see that tonight we agree in open discussion of this whole variety evaluation problem and that we start work some way, somehow, towards working out some means whereby we can properly and more effectively and more quickly evaluate our varieties than we have up to this time. Now, that's the end of my story. The talk and the rest of it is up to you folks.

Mr. Anthony and Mr. Sherman have been working over here in Pennsylvania. They have found a lot of new material known only to a few people. They are just wringing their hands over there to know how in this wide world this stuff can be evaluated, the good saved, and that which is not worthy of doing anything with, well, "just pass it up" and let it go. That's the way we make profits.

Their experience is no different from all the rest. We have nut growers with whom I have had correspondence in years past who want to propagate material that this Association should have flatly condemned years ago, because the majority of the group here knows it is worthless, but they just haven't done it. Now, it's time that we change this thing, or I will tell you frankly in a lot of ways the Nut Growers' Association has become a social institution, rather than one which we learn from and recommend practices to the new groups that are coming on to keep them from making mistakes.

Now, I have talked from the bottom of my heart tonight, and I want some of the rest of you here to express your opinions and give suggestions as to how we might do that.

MR. WEBER: Dr. Crane, I think I will start the ball rolling, and I think Ohio has taken the lead in the very thing you have been talking about. It's the Northern Ohio group. They have been very active in finding out the better nut varieties that were suitable to Ohio conditions, both the black walnuts and the hickories. They have conducted contests, both for black walnut and hickories. They practice what they preach. They have traded their information. They are up in the northern part, and I am down in the southern part, too far to be included with them, so I am not blowing my own horn; I am blowing it for the other fellows. And I think they are a worthwhile group, and if you look to the membership in this Association in Ohio, I think it has the largest membership. And you get that Northern Ohio group, they test out varieties, and a man will fight for a particular one in his group against the variety from another. And so they are not afraid to stand up and say what they think.

But having done that, we need the aid of our different state agriculture groups. You must have a place where they can go and put those trees on a testing ground so the people can go there and see them. You can go there to this Ohio experiment station and you will see this variety growing, or you go over to the other branch and see this variety growing, and then when they find the state has taken it up, it gives them confidence more than a fellow blowing his horn for one variety against another variety.

You have to get the members in their own states to form their own local organizations and carry out what you have been talking about here and find out in their particular states which are the best varieties. And then you get a starting point, and each individual state's agricultural experiment station should take it up, follow it up, if they have the funds. Where if one individual gives his mite and then his health fails or life fails, why, he has contributed his mite, and it will be perpetuated. But if it's on my place or someone else's place, the next fellow doesn't appreciate it, and if they need the wood handy, down comes that tree. It has no memories from then on, and it's not perpetuated.

So I think some of the Northern Ohio members—I think Mr. Smith is here, are there any other members? Silvis—deserve a lot of credit.

MR. McDANIEL: I would like particularly to hear if the Northern Ohio group has got together on a discard list. Have they agreed on any one variety they don't want to plant?

MR. STERLING SMITH: I am glad you brought out the black walnut. I am more familiar with it than with other species, and I have been personally thinking along your line for several years. We have in black walnuts probably over 200. I started to count them up one time. I got 196, and I know there were more than that, I don't know how many. And among those nearly 200 varieties of black walnuts I am confident there must be 150 at least that aren't worth being grown—that is, in Northern Ohio. They may be good in some other places, or they may be worthwhile for experimental purposes. But to grow them for commercial means or for home use, they are not good varieties. And I have suggested to different ones eliminating them, or trying to work out, say, maybe 25 or 50 and then from those 50 try to pick out ten. There has not much been done on it. There is a lot of difficulty in a situation like that.

DR. CRANE: That's right.

MR. STERLING SMITH: Here is one thing: What one person has varieties which correspond with what his neighbor or somebody ten miles down the road will have? We will take Grundy, for example, or Rohwer, some of those. Two or three of them might have that, but the ten or fifteen other members in the near vicinity won't have that variety. That's one of the difficulties.

And I have thought personally that there should be some sort of committee set up along the line you suggested, not necessarily on state lines, but more on zone or regional lines.

DR. CRANE: Yes, sir, that's what I mean.

MR. STERLING SMITH: Because those suitable in Northern Ohio wouldn't necessarily be suitable in Southern Ohio, and so with any of the states along that tier of states. And I think there should be some type of committee set up to judge these different varieties as far as we can, and also to enlarge their testing plan.

Mr. Shessler, I believe, has somewhere in the neighborhood of 100 under test, maybe three or four of the same tree. For myself, I don't know exactly what I do have, somewhere between 40 and 50 varieties, but there are only about 10 or 12 of them bearing. And I have of late years started working on that line, having sort of a test orchard, having one or two trees of the several varieties so I can find out what to plant.

Not too many years ago I was in the position of the amateur who wanted to know what to plant. Should I plant Stabler, Ohio, Thomas? It was just like you spoke about concerning the inquiries that you have. I have earnestly read all the reports and have earnestly looked where I could get them in time for the current year. I read so I would know what the new varieties are and what different people's opinions on them were. And I think there should be a central committee, probably like you suggested.

And another suggestion I would like to make would be that before we permit, as far as possible, any further new varieties of black walnut to be mentioned or published, that they be passed upon by several of the members, oh, maybe ten of the members, at least, to learn what their opinion is before they are mentioned. Lots of times one or two persons have a good opinion of the nut, and immediately something is published about it, and as you say, immediately a half dozen fellows write for it, as in your Persian walnut contest. And it would be better if that nut weren't allowed to be named until it has been passed upon by a qualified group of, we will say, experts. And that same condition should be carried out with the Persian walnut and the hickories and northern pecans and other groups of nuts we are interested in.

MR. CORSAN: I'd like to suggest that we get started on this matter of varieties, because we can say an awful lot and then say nothing. I have tested a great many varieties of black walnuts, and as soon as I hear people talk about the Stabler walnut, I know they know nothing about nuts at all, because the Stabler has a crop on it only about once in twenty years, and then it's a small crop. It's a very good nut to eat and crack, but it's not for crops. As this gentleman says, the Thomas. We all know the Thomas. There is one point about the Thomas, you have got to keep it within just the northern limits of the peach belt where the peach will grow. There are years that come around when the Thomas will not mature. The frost will come on.

It has a very thick outer shell, the hull, and the hull comes off the nut itself quite clean. And then we hear people talking about the Ohio. Now, what about it? Well, it's a monster nut when you look at it on the tree, but knock the thick hull off of it, the strong, sturdy hull, and there's only a little nut in it. Yet you have something that cracks well enough. The nuts I would condemn right away are the Ohio and Stabler. No doubt about it.

Now the Cresco, very, very rich! That tree will actually kill itself, just overbearing. You know a tree can kill itself. Some people kill themselves having 24 or 30 children, but that's about what that tree will do.

Then we have the nut that years ago I saw, the Snyder, and I said to Mr. Snyder, "Look, it's a sure nut." He said, "Never saw it." He looked at it, examined it, and it's a marvelous nut. I think I have the backing of our friend, Mr. Gilbert Smith. I think he'd back me in saying that that is one of the best nuts in the world, even with the Thomas.

But we don't quite want to reduce—comb down the list of varieties like the apple grower has. When you go to Boston and ask a peddler or hawker about "apples," he won't know what you are talking about. Apples?—they wonder what the word is. It is "McIntosh." They will go around the street shouting, "McIntosh, McIntosh." You won't hear the word "apple" in Boston, it's "McIntosh."

Now, let's get down to nuts, and let us know our nuts.

MR. CALDWELL: (New York State College of Forestry.) I suppose this is my first time at a meeting of this sort, and probably I should observe with a critical mind. But when you speak about a committee to pass upon varieties, immediately I start wondering exactly what you mean by a variety, and then I start wondering what your approach is in picking that so-called variety.

First of all, a "variety" that you use is not really a variety. It is just a vegetation of one particular tree that you happened upon. You decided by chance it was a tree you wanted to use and then passed it around to your friends and decided you want it.

DR. CRANE: I want to correct you, for one reason: It is truly a horticultural variety or clone that has just as much standing or identity as the botanist's or forester's "variety."

MR. CALDWELL: It is a clone, and I agree with you, but a variety seems—

DR. CRANE: You are speaking from the forester's point of view.

* * * * *

MR. CALDWELL: That's why I make this other statement.

DR. CRANE: When you have got something by controlled breeding, you don't know when you have got it. That's the whole story in a nutshell.

Now, I am going to tell you about using controlled breeding. We started almond breeding in California, where we have one of the biggest commercial nut industries in the country. We started almond breeding in 1920 with the best known almonds. In the 30 years of almond breeding we have introduced two varieties. We had a panel of 125 commercial almond growers who decided on those two varieties out of more than 20,000 known controlled crosses that were made of trees that were grown to fruiting. But it took a panel of 125 commercial growers to determine whether or not these two varieties, the Jordanolo and the Harpareil, were commercial varieties.

Those two varieties were planted. The nurserymen planted them, the grower took them over, and they couldn't grow enough trees to supply the demand. These two varieties have been introduced for commercial planting now for 14 years. Of the two, one has stood the test of time, and it stands now as probably the second most important almond variety in all the United States, has been taken to foreign countries and is being extensively propagated. One of them made the grade, the Jordanolo. The Harpareil is still in the running, but it is down with the 30 or 40 varieties that are of lesser importance.

MR. CALDWELL: Can you reproduce that result?

DR. CRANE: No.

MR. CALDWELL: Then you don't know what that is or the happenstance that got it.

DR. CRANE: Certainly, because you don't know about breeding nut trees.

MR. CALDWELL: That's what I say should be learned.

DR. CRANE: In the first place, the chromosomes are so small and there are so many, that you can't identify them, and you can't tell which genes, and they have got a heterozygous population, and the variety is self-sterile and has to be cross-pollinated, so there is only one way from a horticultural standpoint by which we can do anything, and that is through clones.

DR. MacDANIELS: I think we are getting a little bit off.

DR. CRANE: We are off, way off.

DR. MacDANIELS: How to get a new variety I don't think is what we are trying to decide this evening. As I have looked at this whole field of what we are trying to do, I think we have analogies that we can point to. I think any project of this kind in nut varieties goes through various stages. The first is finding what material there is that is available that you can use. The next is the evaluation of that material to see what's worth keeping, and setting up your standards of what you are trying to get, and then from then on out perhaps breeding that sort of thing.

Now, as far as we are concerned, it seems to me the Northern Nut Growers' Association made a pretty good stab at surveying the materials available. In other words, I think an additional nut contest is not going to turn up the perfect nut. That is, we have one contest after another, and the ones that win the first prizes as the best nuts we can find are not markedly better. There is no great difference away from the average that we have had in the others.

I think that's a valuable thing to keep going along so we don't miss a trick and let anything be lost. But the next thing is to take these things that we have selected and evaluate them, and it seems tome that's exactly where we stand at the present time.

I also think that we should not in this situation get ideas that are too big. That is, if you get something that's impossible, you are licked before you start. If you have got to wait before you do anything and make a complete study of chromosomes of any one of these nut trees, 99.44 percent of the Northern Nut Growers Association might as well quit doing it. I am not capable of doing it, and Dr. McKay is probably the only one that is capable of looking at these things from that standpoint. But we have, it seems to me, to use the machinery we have and take some definite action which will be of some value within a year or perhaps two.

I agree that this idea of putting the State Vice-presidents to work is a very good thing. I think each one could if we could find the right man—take his state and divide it into two parts, and also take in groups of growers of nut trees that are members, and all the others that we can find, and get their pooled opinions on what varieties are available, together with the record of these varieties in that particular locality.

Then I think on the basis of one of the committees we have, that is, our standards and judging subcommittee, we could set that up in such a way that they could evaluate things about which there is some doubt.

But before we do that, we have got to clear the decks and adopt judging standards, standards by which we wish to work or to evaluate different varieties. I don't know whether anyone else has done more judging than I have or not, but I know I have given this a lot of attention through the years.

We had one system of judging which was worked out some years ago and was based on previous judging systems, and they went to a point where it seemed to me and to the others who were working along with me that they just didn't have any real basis in the factual situation that warranted its continuance; that is, a system which was based on percentages of kernel and penalties for empty nuts or flavor, and other things which could not be effectively measured. And they quit with that system and started out on a new tack. And to do that we got Dr. Atwood, who is head of the Department of Plant Breeding Genetics at Cornell, to go through some extensive tests which he applied as a biometrical statistical method, to find out what is the sample which will give you specific results and then to measure the qualities that give you what you want. And I think we are nearer that than before. But I think the schedules are relatively simple and haven't been used to any great extent. They need further testing.

But it seems to me that the Association as such must decide whether we want that schedule, making it an official schedule and going ahead on that basis.

Now, a judging schedule for nuts will not tell you anything about the tree; it will just tell you the characteristics of the sample. That's the first thing you want to find out: Is the nut itself intrinsically the type of thing you want to deal with? Then whether the tree bears annually or whether it alternates, or what diseases it is subject to. Those are other matters.

So I think this is a way out, or at least I suggested the plan we could go along with of putting the vice-presidents to work and setting up a committee under the title of judging and standards and try to bring out a report at the next session. It seems to me that would be right practical.

Where we go from there in production of new varieties I think should be a subject for a round table discussion sometime. I think the gentleman in forestry has a good idea. I think we will get a long way if you have proper control of the first elements of the first varieties, and from them we can build up. But it seems to me we have to be practical about things that we can do, then go ahead and do them.

DR. CRANE: Thank you, thank you.

DR. COLBY: I would like to add one point, that we must "zone" all these varieties. In a state as long as Illinois, over 400 miles long, growing conditions are different in the south than in the north. In the north we don't find that Thomas fills out very well and that's true also at Urbana in the central section of the state. Beck and Booth and some of the smaller nuts do fill out. The zones I mentioned may well run across several states where environmental conditions are similar.

I recall a little survey I made when I was honored by being president of your association several years ago, in which I tried to list all of the work that was in progress at the different national and state experiment stations, and most of those stations were carrying on some work in nut growing. I am sure that if you check that matter now, several years later, you would find that many more are carrying on investigations of that nature. They have expanded as much as their facilities will permit. For example, just the other day I visited the station at the University of New Hampshire, and there they were growing chestnut trees from seed that had been brought in from Korea. Little trees just two years from the seed were full of burs this year. Whether they are going to fill a place in New Hampshire remains to be seen. They were not as yet attacked by blight, but, of course, the trees were small, and there were no cracks in the bark as yet.

I am sure that most of the station workers know that you at Beltsville are extremely interested in testing new nuts as they become available. In cooperation with other workers it may be found that this variety is good in ~this~ zone and that variety is good in ~that~ zone. Nurserymen might well include maps of such zones in their catalogs.

DR. ANTHONY: Now that the experiences of the Northern Ohio growers has been brought up and you have mentioned many times your own experience as the Northern Nut Growers, I think the Northern Ohio group, a closely knit group, rather closely geographically related, has worked for almost twenty years, and hasn't gotten too far, and this organization has worked for 41 years and hasn't gotten too far. So that if we want to get anywhere, we must have a more closely knit organization with a better financial backing back of it and a better sense of responsibility back of it.

DR. CRANE: That's right.

DR. ANTHONY: You have mentioned the New Jersey Peach Council. We have been talking to our own Pennsylvania nut growers just as we have been talking to you today, telling them that they had a marvelous opportunity in all of these seedlings that we have been finding around the state. I think we have got them quite stirred up. But now they are considering the possibilities of organizing along the line of New Jersey Peach Council, a nut tester's council, which will be an off-shoot and part of the Pennsylvania Nut Growers Association.

Now, why have such a thing? Why have it in Pennsylvania? Why not have it as an organization of the Northern Nut Growers. The problem of varieties actually in its final analysis is a local problem. We have one area in Pennsylvania where on one side of the river it's McIntosh and the other side of the river it's Stayman. There are meteorological differences on each side of the Susquehanna River at Scranton-Wilkes Barre where the varieties shift. In the northern area we go from the northern hardwood with the beech-birch-sugar maple, into the oaks right in the state, with a third of the state in the northern hardwoods and the rest of the state in the oaks. We have no idea that any one variety of black walnuts or English walnuts or chestnuts will fill our needs any more than we know that any one apple will fill our needs, that one grape or one cherry will fill our needs, even one peach, not even the Elberta.

So it comes down to a regional problem, and for that reason I think that the state should be the logical center for your close knit organization to test your varieties.

There is another reason. I don't believe that any group of growers facing a problem of this magnitude can get very far unless you secure continuity by tying your organizations in some way to your state experiment station. I think you have got to have your continuity by making your tie-up there.

DR. CRANE: That's right.

DR. ANTHONY: I have said a number of times in our own group that one of the great disadvantages of our amateur nut growers in Pennsylvania is that most of them are 70 years old or older. That's fine for them, but it's hard on the industry, because just the time that they should be giving us the most valuable returns, they aren't there. So to secure the continuity you want, you are going to have to tie in your experiments with the experiment station. You are going to have to make a group, you are going to have to incorporate, because you are going to face the problem of propagation. You might have one good tree, and it's of no value for you, and you have got to plant it in more than one spot to know how good it is.

If the Delicious apple or Grimes Golden had appeared in our seedling blocks, we'd have thrown them away. I know we have thrown many things out at Geneva which in other places might have survived. We took a number of those and planted them in Pennsylvania and found them worthy of naming. That means you have got to propagate in more than one place and you have got to propagate in conditions where you know you have got the demand.

And all of that means that you have got to have a tight legal organization. Valuable as the Northern Nut Growers Association is, I don't think you are going to get it out of your present organization. I think you have got to find some way to condense your stuff into some tighter organization. In Pennsylvania I think it's going to be a nut tester's council, legally organized, financially responsible, tied up to the experiment station, if we can make it just as the New Jersey council is.

The New Jersey council was a success because they had the best possible tie-up between Morris Plains, 15 or 20 miles on the other side, and a good nursery in between. That's why they made a success.

The New York State Fruit Testing Association is a success because they have had continuity. Mr. King has been manager of that association for 25 years, I think, and you have a legal organization doing its own propagation where they know the material is true to name.

Use your vice-presidents all you can, use every committee that you have but you have to have something that's tighter.

DR. CRANE: Thank you. Just one comment that I want to make. You have suggested an awful big camel to get over. Now, we are trying to start. If we could just get a little start towards the end we could grow into it.

DR. ANTHONY: We have got to start.

MR. O'ROURKE: I am one of those unfortunate ones who is supposed to know everything when an inquiry comes in to the college. I happen to have the privilege of answering the nut inquiries at Michigan State College. The first thing people want to know is, "what varieties do I plant?" The second is, "Where do I buy them?" I am very sorry to say I can answer neither one of those questions at the present time satisfactorily to myself, nor to the people of the State of Michigan, and I feel that we do need action, and we need it quick in order that we can select a certain number of varieties that we can conscientiously recommend to the grower, and also a very few varieties to recommend to the nurserymen of the state so that they will propagate them and make them available to prospective customers.

MR. SLATE: I want to support Mr. Anthony's remarks that there are too many old men testing nut tree varieties.

DR. ANTHONY: Not too many, no.

MR. SLATE: And there are too many squirrels involved. If a man gets the idea that he is going to take up the nuts, by the time he accumulates a collection of nuts, when these come into bearing the squirrels get most of the nuts, and they don't seem to be very much concerned about evaluation. Then the man dies and the collection goes to pot. There must be some continuity, and as far as I can see, that will have to come through state experiment stations.

Now, just how you are going to get the experiment stations started in testing nut tree varieties, I don't really know. Many of the projects at the experiment stations are there because they are catering to the larger industries in the state, and sometimes the projects are there because somebody in an administrative position has an idea which he wishes to see developed.

Now, I would like to comment on the remark of our forester friend here, and I think he won't take offense at what I am going to say. It seems to me that the foresters are not in a good position to criticize the horticulturists. The forester's knowledge of variety improvement for a long, long time has been based upon the problem of lots of seed from certain geographical areas, and I feel sure that foresters as a class have only very, very recently become aware of the importance of the clone as we use it in horticulture.

Now, horticulturists, that is, pomologists, nut culturists, people who deal with ornamentals, have been keenly aware of the horticultural clone for a long, long time. There have been brought improvements into our cultivated plants through the hybridization of clones that all of the horticulturists are familiar with. The blueberry work done by the Department of Agriculture is probably the most striking example of this work, because it was all carried out during the lifetime of one man.

I feel that we will not get much further in searching for wild nuts. We have had contests for hickories and black walnuts, and I doubt whether we have made any very substantial increases. I feel certain, and I know there are a number here who will back me up, that future improvements, if they are to be really substantial—that is, if they are to be substantial advances over what we already have—such improvements will have to come through breeding work.

DR. McKAY: Mr. Chairman, I have been listening to these remarks, and I have been trying to think of some comment that could be made in connection with some practical suggestions that we could arrive at tonight, a starting point, perhaps, in connection with the chairman's remarks about doing something tonight at this meeting. I'd like to say that it seems to me that the thing we could probably do right now to start things off would be to have this regional committee or this group that represents a wide area, decide on, say, five varieties based on all the evidence that can be obtained as to which five would be most likely to succeed over a wide area.

Now, the chairman has commented at length on our lack of unanimity when it comes to varieties. I think most of that problem has come out of the fact that our information is all based on little, piecemeal bits of work done here and there, and it does not refer to variety testing over a wide area. Now with all due respect to Dr. Anthony's remarks about varieties being a local situation, we still have, as mentioned by the chairman, the apple situation. The varieties in the final analysis are going to be adopted over a wide area, and if our nurserymen and all our growers could know or understand that these five varieties have been selected by opinion of people that ought to know that those five varieties stand the best chance to succeed over a wide area, then we would have something definite to tie to.

The way it is now, we in our office feel that Thomas is probably the most widely adapted variety of black walnut we have, and probably the best performing variety. We are not sure, but that's our opinion. I might mention another variety, the Stabler. I think most people would agree that that is a variety that used to be thought well of, yet is no more, and so it is out of the picture. Those two varieties we have information about, based on a wide area of territory.

Now, it seems to me, coming down to something specific, what we could do here, or as soon as we can get to it, would be to have a large committee, a committee representing opinion over a wide area, come to some conclusion about the five varieties that will be the ones to test and to grow over a wide area and give our nurserymen or our growers something to tie to in the matter of selecting varieties to grow.

DR. CRANE: Thank you, Dr. McKay. There is one other comment that I want to make. I think that if we were to take a vote tonight in here, get an expression on the variety Stabler, we'd say, "Yes, it's a curious nut, it's a curiosity. Some trees sometimes bear single-lobe nuts in varying proportions. It is a fine nut when you get it, but they don't bear enough and they don't bear regularly enough. That is the criticism of the Stabler."

Yet we have nurserymen, lots of them, that are propagating Stabler and still selling them to people.

MR. McDANIEL: I know one nursery which has recently discontinued it.
That's Armstrong, way out in California.

MR. CALDWELL: Why doesn't it produce a good nut? Can you answer that question?

DR. CRANE: It does produce a good nut ~when~ it produces.

MR. CALDWELL: If it doesn't produce all the while, why doesn't it? If you can solve that—

DR. CRANE: Why didn't you grow up to a six-foot-six guy weighing 250 pounds?

MR. CALDWELL: It would be physically impossible for me to do so with my constitution, which is what I am trying to apply to the nut trees.

MR. WILKINSON: Don't condemn it over all territories[6]. At my place, the Stabler produces nuts as regular as the Thomas, and in the nursery it outsells the Thomas two to one, if not more. I have handled nut sales for Mr. Weber's orchard, one of the largest black walnut orchards in the United States. When the people come there we will crack a Stabler walnut to make a customer out of them, and we have to get on to something else to keep them from buying all the Stablers first. And if I were planting a hundred walnut trees today, the majority of them would be Stabler. They have been bearing since 1918 when I started producing Stabler walnuts.

[6] The territory giving best reports on Stabler lies along the Mississippi and Ohio rivers from about Cincinnati to no farther south than Memphis.—J.C.McD.

DR. CRANE: That's what we are talking about tonight.

MR. CALDWELL: Yet your committee throws the thing out.

MR. CHASE: I'd like to say a few words. First off, I am in agreement with the idea of some sort of a regional testing set-up.

Now here we are getting into discussion about individual varieties, and that is not the purpose of this, as I understand, but all of you gentlemen have been propagating the various varieties simply because one has become available to you at a certain time, and you have grafted it. Our committee on varieties, of which I am a member, probably should be criticized, because we have not gathered that information from the folks who have grafted trees, and they are scattered over the region. We don't need the regional set-up, it's already set up. In other words, if we have varieties to be tested, we could have selected members in our group to graft it, if they do not already have it grafted. In a few years we can get some pretty definite information on a few varieties.

Now, in 1938, in our work we recognized the advisability of quickly doing something about the 100-and-some varieties existing in the proceedings, and finally we have culled that down to, I think, 43, which, on the basis of nut characteristics only, are very close together. Now, we started out in 1938 and established four or five test plantings containing the first ten varieties. Ten trees of ten varieties, a hundred trees in the planting. It took quite an area.

Since that time we have set out variety test plantings of 43 varieties scattered over seven states at various geographical locations within the seven states.

MR. KINTZEL: How many trees do you have in a planting now?

MR. CHASE: Twenty-five now. Twenty-five of five varieties. This work is being carried on at the state experiment stations in the Tennessee Valley. In fact, they have become more and more interested in the testing program which we have been trying to get them interested in, and we hope to have some information for our region on some of these varieties, the better varieties as we consider them.

But back to this problem. I think it is very simple to set out. I think the Varieties Committee—I believe Dr. Crane is chairman—

DR. MacDANIELS: You are chairman.

MR. CHASE: No. It has a job on its hands: first to find out what our members have. Certainly they are spread over the region we are interested in, aren't they? Well, it simply becomes a secretary's job to canvass our membership to find out which varieties we have, so that the Varieties Committee can go to work.

Let's be realistic. We are not going to influence all the experiment stations to do this work. It is not going to be practicable for them. They probably would very much like to do it, but it's not in the picture, as I see it now. Therefore, we are not going to wait, as our forester would have us wait, until we breed one. Let's get these good ones that we have got and cull them out so Dr. Crane can answer a letter without having a guilty conscience.

DR. CRANE: That's right. Folks, I want to make one comment on Mr. Chase's remarks—also Mr. Slate's remarks, about tying this work up to the experiment stations. There is one thing that, in my experience, we can't place too much dependence on. Of course, in the Department of Agriculture our main interests that we are likely to contend with are our four major nut industries in the country. That is pecans, Persian walnuts, filberts and almonds. In the case of those, we can get very little help from the experiment stations, with the possible exception of California.

MR. CORSAN: There is lots of truth in that.

DR. CRANE: They haven't got the interest in it. They haven't got the money, they haven't got the support. They depend more on the U. S.

Department of Agriculture. Well, the Department of Agriculture can't carry it. Hence, it comes back to growers. The grower organizations, even in the great state of California, with all their great wealth and abundance, go to the California experiment stations more than to any other experiment stations in the United States. But the commercial growers out there have already set up organizations for the testing of these varieties and for trial plantings. You can't come back to the experiment stations and just as has been pointed out, many of the experiment stations have only one or two or, at most, three different kinds of nuts of their own. They have got to go out just the same as we do ~with the growers~; we co-operate with them. And we have already got a lot of these experimental plantings. There is Sterling Smith with—I have forgotten how many he said—60 walnut varieties, and Mr. Shessler with a hundred, there in Ohio.

I'd like to know from Sterling Smith and Mr. Shessler which are the best five walnut varieties.

MR. KINTZEL: In that section?

DR. CRANE: In that section, that's what I want to know.

MR. CORSAN: That's what we are here for tonight. Let us talk it over.

MR. WEBER: Put the question to him, Dr. Crane, and let him tell you what he thinks to be his best five. Put him on the spot right now.

DR. CRANE: That would be just a waste of time, because that would be his opinion. It's just like what Mr. Wilkinson says, that if he were planting a hundred walnut trees they would be Stablers.

MR. WEBER: In his particular locality.

MR. CORSAN: And he may be quite right in that locality. I am not going to dispute it.

DR. CRANE: But we want to know how some other folks agree with him and study this situation over and find out why Stabler was doing its stuff right there.

MR. CALDWELL: That's what I asked you.

DR. CRANE: And how much evidence did he base his conclusion on? That's what we have got to discover.

MR. CORSAN: I base my conclusion on the experiment station that put out the Redhaven peaches. Dr. George Slate here has made a very big point, and it went to pot. Those words there are what we have got to be careful about, that our institution doesn't go to pot. I have started affairs that went with a fury, and when I let go of them, they just went to pot.

Take Michigan State College's Bird Sanctuary, the W. K. Kellogg Bird Sanctuary. What is it now? A colorless affair. It's gone to pot, and we want to see that the nut growers don't allow ~their~ institutions to go to pot.

DR. CRANE: That's right: You hit the nail on the head, there, but it's up to the nut growers to see that they don't. And how many experiment stations or their actions have been influenced by the Northern Nut Growers Association?

MR. CORSAN: I have built upon the experience of J. F. Jones and Neilson and Professor Slate and all of them. Now, here is what I did. I picked out a section of land that floods every spring, about four times the width of this room and has sometimes eight feet of water. Now, nobody is going to build houses on that and tear my nut trees down. They are there forever, and it will always be a nut haven, and nobody will be able to destroy it. Now I have got to be careful to see that it doesn't go to pot, as Professor Slate said, by selecting some brains to succeed me, to carry on. Is that right, Professor Slate?

PROFESSOR SLATE: (Nods.)

MR. SILVIS: We can't spend too much time thinking about the atomic bomb. We can't think too much about getting an organization to start this, it just takes somebody to go ahead and do it. We don't need experiment stations to develop the nut, either. The nut was here a long time before the experiment station was ever developed.

I wrote in a letter here two or three or maybe four years ago—I think it was after the Norris meeting, to every vice-president in NNGA that commercial possibilities of a nut must first be apparent before any experiment station is interested, because then money is involved, capital has been invested. Before capital can be invested must come coordination. Coordination is labor. That's grafting or flowering, or whatever you want to call it—back-breaking exercise.

I still think we have the organization here. We don't need to argue about any more organization. We have organization right here in our own State Vice-presidents. I tried to bring that out, the suggestion as to the fact that I thought maybe the State Vice-president would serve on a perpetual committee, if he lived into perpetuity, to get these zones within his state. If Illinois is 400 miles long and he has 16 zones of climate, let him get 16 plantings of the same kind of a nut in those 16 zones. The same way with Texas, the same way with Montana or Ohio.

MR. SHERMAN: I think both Mr. Stoke and Mr. Davidson thought that it might be a good idea to give somebody a job instead of an honorary position by naming a State vice-president for that sort of a job. Now, we have got to start somewhere, and that would be a good place to start: give somebody something to do, like some of these other dead people that will feed these nuts that Corsan was telling us about this afternoon.

But the commercial possibilities are always apparent. You can subsidize them, you know. If you can get enough money behind it, you can subsidize it. I think our problem still is the same as it was before: We are still trying to find out what the other guy has that's better than our own. And if we have got five nuts that are any good, I'd like to know about them myself.

DR. CRANK: That's right.

MR. SILVIS: I will make this statement in favor of the Homeland black walnut—if we are on black walnuts. I came in a little late on account of the mud here. The Homeland is growing in Massillon, and Mr. Stoke sent me the scions. All it did was produce staminate bloom. I gave some of the wood to John Gerstenmaier in Massillon. It is doing very well.

I also favor the Thomas black walnut, and I think the hickories and everything else have commercial possibilities. Just let somebody go ahead and correlate these factors. Life is very short. I have copies of these letters, four letters out of 50 or 60 that I prepared.

DR. CRANE: Mr. Jay Smith. We are going to have to limit this to not over three minutes' time.

MR. JAY SMITH: My experience is somewhat limited. I have a few seedling trees that are good, and I have a few named varieties that seem to be good. I just want to point out one reason why we should have a number of varieties. One of my choice varieties in my back yard has five nuts on it this year, and it has produced a good crop other years. And the answer seems to be that the pollen came out during a period of very rainy weather and the tree did not fertilize. Now, other trees apparently blossomed before or after, mostly after, but this one was a rather early blooming tree, and I have more nuts on other types of trees.

One of my good seedling trees has very few nuts on this year. Possibly that might be for a similar reason. So regardless of how good these varieties may be, we must have several varieties. Don't put all your eggs in one basket.

I have some good filberts that came from Geneva, and they have had trouble with wood damage due to the beetles laying eggs in the wood, and the beetles may possibly have come from nearby willows. And I have had some of the willow growing, too, because I thought it looked nice. Now I have cut down all of the willow, and there is some birch in the neighborhood, and I understand the birch harbors this same thing, some variety of Agrilus beetle,[7] and we have a lot of angles to work on in order to get rid of our drawbacks. And we have the matters of season and soil and elevation. It's quite a big problem.

[7] Agrilus anxius Gory, the bronze birch borer.

DR. CRANE: It ~is~ a big problem, but we will never settle it the way we are going. We have got to do better.

MR. STOKE: I don't know whether I have anything that is really pertinent to say. The thought I had in mind should have come sooner. That is: Why are we growing nuts? There are two angles from which we can approach that, two natural angles. Here is the angle of the amateur that wants to grow nuts to eat. After all, that's what I suppose they are for. There is the commercial grower who wants to grow them to make a profit, and I think we should approach our subject, evaluation of nuts, from either one of those two angles, or work along two different channels. I think that's very necessary.

You take the Elberta peach. If you want a peach in your back yard, you are not going to plant Elberta peaches to eat. If you want to make a commercial success, you are going to plant the Elberta, if you know anything about it. Are we commercial nut growers, or do we grow them for home consumption? Go downstairs and look at the nuts we judged last year and the eye appeal of some that didn't rate at all would sell those nuts ahead of the prize winner. But if you want to grow them to eat, those three prize winners are the best nuts down there.

And if we thrash over this field, I think we have got a definite idea of what we are after, and I think we should have had that to start with.

DR. CRANE: That's right, and there is one other point of view, too. There is a third reason for growing nut trees. That is simply for the ornamental value. That hasn't been dealt with.

MR. WELLMAN: I'd just like to ask a question. There has been some reference to apples here. I don't know very much about it, but I understand that the American Pomological Society got out a list of apples nearly a century ago, which they have kept changing and adding to and subtracting from over all of that time. Is there any analogy there that would help us in anything we can do? They made mistakes and put apples on there that they are sorry they put on and they have had to take off. People don't use those varieties in one part or another part of the country for some reason. Is there any reason why we shouldn't follow some suggestion such as that, stick our necks out and go ahead?

DR. CRANE: That is right, no reason in the world why you can't.

MR. SHERMAN: I'd like to do some commenting. You are doing here tonight what you have done at the last meeting. You have talked varieties. I thought the purpose of that was to get a committee appointed some way, some organization that will say, "Here are certain varieties that should be tested. Make arrangements to propagate those varieties and have them tested."

I made a demonstration right downstairs here; some of you witnessed it. You have got some black walnuts that you are cracking. I went out to the car and got some that would crack in four nice quarters that laid out. I tried it again. Sure, they cracked and cracked good. Where can I get some trees? There are a lot of you right here who would take them just that quick (snapping fingers), take them home and test them.

This meeting was to get an organization or discuss a means of getting an organization that will get those trees propagated and spread out for testing. Now, I think it's just as simple as A, B, C. It's a prolonged job. You have got to have an organization that's going to perpetuate itself for the next century, because if you start that organization right it will be here a hundred years from now, and you will be just as busy a hundred years from now as you are right now.

What that committee has got to be, whether it is a statewide or a nationwide, Northern Nut Growers or Pennsylvania Nut Growers or Ohio Nut Growers, is a committee of five—I will say five, you can make it 10 or 15—that will say, "Now, for Ohio here are ten varieties that we think should be tested. Get 50 trees of each of those ten propagated and spread out over Ohio and find out where they will grow." That will apply for some of Western Pennsylvania, too. It isn't just state lines, understand, but the main thing is to get that variety tested before your nurseryman is spreading it all over everywhere.

And how can you get it tested? You have got to have some trees propagated, and you have got to have some nurseryman who knows about the propagation. And I will say a lot of you nurserymen, and there are a lot of you here, take it or leave it, don't know how to propagate a decent black walnut tree. I have had them sent to me with a 6-inch sprout growing in the top of a club. I have had others two years old with a nice whip five feet high, one-year-old growth. You have got to have good trees. You have got to have a nurseryman who knows how to propagate those ten and send them out.

Now, the next meeting was to find out what sort of an organization you have got to have to get that done, not talk about a Stabler, whether this is good or that is good. That's what you have been doing for 40 years.

MR. SLATE: It takes more than a committee, it takes land, labor, tools, supervisory people.

MR. SHERMAN: I can point to 25 members that will take ten varieties that they will test—and pay for them.

MR. O'ROURKE: I would like to say, are we going to wait until we test all of those varieties? We have no information to answer all those letters that are coming in. We want something, not tomorrow, we want something today, that we can give them, information which, at least to the best of our knowledge of today is accurate. And the only way we can get that accurate information is to get a committee together in each region.

MR. SHERMAN: That won't take care of the future. That will answer our present questions to the best of our knowledge, but we want an organization that will take care of the future.

DR. CRANE: There is one other thing that I should mention. We in the Department of Agriculture have released a number of new varieties. We have got others coming on, not only your chestnuts, but filberts and others, pecans, and so on. But we haven't got any organization in any way, shape or form. We can put these out with the growers who test them, but gee whiz, we have put them out and put them out; and look what kind of information we get. We haven't got facilities or the money or anything else to follow up. We have got to have some organization some way, somehow, that could take this material and test it, at least give some idea as to how it performed.

Now, then, the question is what kind of an organization? If the Northern Nut Growers is not the one that should do it, what kind of an organization can be effective to do it?

MR. CORSAN: Now I'd just like to say one more thing tonight. That chestnut blight, I honestly believe, was a godsend to this country. I can remember way back when I'd go into a store and buy a lot of these Paragon chestnuts in New York City in the finest grocery store, and they were crammed full of weevils. Now, the chestnut blight came, and it has about annihilated the weevil, because there was no chestnut to weevil in. And I would like to have some report about the weevil.

MR. WILSON: They are in Georgia.

MR. McDANIEL: They are in Virginia and Indiana.

DR. MacDANIELS: Mr. Chairman, I suppose I should have the chair. This is a committee of the whole.

DR. CRANE: That's right.

DR. MacDANIELS: I have a right to speak,

DR. CRANE: That's right.

DR. MacDANIELS: I say we have always come down to the point, here we are, where do we go from here and what do we do next? There, in a word, "Here we are." Lots of discussion, much of it irrelevant. I will just propose, along the lines I spoke before, that what comes out of this is that We recommend to the incoming president to organize a survey and testing campaign along the lines that seem to meet with some agreement; namely, getting the state vice-presidents busy in finding out the regional evaluation of different varieties.

Supposing we try black walnuts; just one species for this year, and that he organize his state according to zones and come up with that information with regard to that state.

And the other thing would be that these findings be sent to the committee. We have a committee on surveys and one on judging and standards, and let that be compiled by them jointly or set up in some way that would seem to be effective and come up next year with this overall evaluation along those lines.

I'd make that motion.

DR. COLBY: Second the motion.

DR. MacDANIELS: Any discussions?

DR. ANTHONY: In Pennsylvania two of us have worked full time for a year, and I am not sure we'd be able to evaluate the black walnut yet.

DR. CRANE: We are not evaluating the black walnut, though.

DR. ANTHONY: You are asking one man to do that, your vice-president.

DR. CRANE: He is to appoint a committee.

DR. MacDANIELS: Any way he chooses to mark them out.

DR. ANTHONY: He is organizing a nut tester association.

DR. MacDANIELS: No, an evaluation association. As I would say, you have the Ohio Association already formed; that would be their problem to come up with an answer for their state. We have the Pennsylvania organization already organized. They will come up with some sort of evaluation: No. 1, Thomas, No. 2, whatever it is, No. 3, whatever it is. Now, in your other states we don't have an organization; do it some other way. I don't care how they do it.

DR. CRANE: There are some others in these other states, too, that are already formed.

Any other discussion?

(Whereupon, a vote on the motion was called for, and it was carried unanimously.)

MR. SILVIS: Just one thing. It was made with the express purpose that we start maybe just the black walnut. At the same time in certain areas you may as well raise a hickory or a Persian right along with the black walnut, or the filbert.

MR. McDANIEL: No objection, but this year we are surveying the black walnut named varieties only.

MR. SALZER: I am just a buck private in the rear rank, but we have been having little local meetings in New York, and they appointed me vice-president for the State of New York, the Empire State, and here Ohio has their organization, Pennsylvania has their organization. What am I going to do? I can work Western New York, but I have got to have someone to help me in Eastern New York.

DR. MacDANIELS: Take the membership list and take the men who can do it.

DR. CRANE: There are a lot of good men in Eastern New York.

Now, if there isn't anything else, I will turn the meeting back to Dr. MacDaniels.

DR. MacDANIELS: Thank you, Dr. Crane. I think these talks are good for the soul. We can let our hair down and know what we all think. And I do think it's important that we do make some progress on this particular problem. I think this is one way to do it. There may be a half dozen ways and other ways better, but at least you have to agree on something and go on from there.

Now, the meeting in the morning begins at nine o'clock, the full program.

If there is no further business, then, this session is adjourned.

(Whereupon, at 10 o'clock, p.m., the meeting was adjourned, to reconvene at 9 o'clock, a. m. the following day, August 29, 1950.)

TUESDAY MORNING SESSION

August 29, 1950

DR. MacDANIELS: I want to make the remark that this isn't church, you can sit up front if you want to.

The first paper this morning has to do with a nut tree disease that is bothering a good many of us, I think, particularly in Michigan, as you recall from Mr. Becker's paper, the Bunch Disease of Walnuts, by Dr. H. L. Crane and Dr. J. W. McKay. I don't know which one is going to give it. Dr. McKay?

The Bunch Disease of Walnuts

Discussion

(Manuscript too late for publication.)

(Drs. Crane and McKay reported that there had been little further development in knowledge regarding the walnut bunch disease since 1948, when G. F. Gravatt and Donald C. Stout of the U.S.D.A. Division of Forest

Pathology reported on it with illustrations at the N.N.G.A. meeting (see our report for 1948 pp. 63-66.) Since then the state of California has prohibited the entry of all walnut nursery trees and scions from the Rocky Mountain states or farther east.—Ed.)

DR. CRANE: I'd like to make one additional remark. You see, we call this trouble "bunch disease" rather than "brooming," to distinguish it from other diseases that are caused by known parasites. We have a disease very similar to this one affecting walnuts and pecan and hickory, and that one has been studied more carefully than has the bunch disease. It is unquestionably caused by virus, and in our pecan orchards we have a situation that exists that is a parallel to what it is in the black walnut. The variety Stuart practically never has shown any symptom of the bunch disease. Yet it performs very much like a lot of our black walnuts do. They just don't bear; they don't have the proper foliage; they don't make the proper kind of growth. So we are not sure whether they are symptomless carriers, that is, in terms of the lack of expression of virus growth and this bunchy condition on them.

Really, we feel that all people that are interested in the walnuts and that are trying to grow them should make careful observations on these trees to study just what the situation is, how it develops, and note the performance of these trees that become diseased; because we feel that it's a much more serious thing than people appreciate at the present time.

In much of Eastern Shore Maryland and of the area around Washington and Beltsville and over in Virginia, a great majority of the trees are affected by it, particularly Japanese walnuts of all types and the butternuts. I feel it is so bad on Japanese walnuts and butternuts that they shouldn't be propagated in the area.

MR. McDANIEL: I had the bunch growth developed on a new species this year in my planting in north Alabama, a 12-year-old tree of ~Juglans rupestris~. It is a growth that looks practically the same as the bunch disease on the Japanese walnut. I believe that's the first time it's been observed on that species. There are no butternuts or Japanese walnuts on the farm. There are dozens of black walnuts (seedlings and several varieties) none of which show the bunch symptoms. However, it is typically developed on some Japanese trees a few miles away.

At Whiteville, Tenn., Dr. Aubrey Richards has a suspicious looking tree among some two year old seedlings of ~Juglans major~ from Arizona seeds.

MR. CHASE: I'd like to add to that, too, Mac. In our walnut arboretum we had some ~rupestris~, and I had been suspicious of its being diseased for a number of years. I finally have decided that it had the bunch disease, and those trees down at Norris have all passed out.

MR. McDANIEL: My tree came from Norris, 10 years ago.

DR. MacDANIELS: ~Juglans rupestris~ killed by the disease.

MR. STOKE: Just because this is a little contradictory to what you have heard, I want to say that my experience has been this: I have an old nursery—well, there is a butternut in the row and also heartnut—Japs. One of those Japs has had the bunch disease for six or eight years. None of the others has been affected. It was a variety I wanted to perpetuate. I took an apparently healthy scion from that and put it on another tree, and that grafted tree also had the disease. But there has been no evidence of contagion from this Jap to the other Japanese, butternuts and black walnut in the same planting in the immediate neighborhood—in fact, they crowd each other. That's a statement of fact.

I spoke a little while ago of an old black walnut tree that had that disease for a number of years and none other in that planting had it.

MR. O'ROURKE: Is there any correlation between the age of the tree and the expression of the disease?

DR. McKAY: It's been our observation that we haven't had it in our nursery to any extent. We have seen it in the nursery of J. Russell Smith on Persian walnut. It, to my knowledge, is the only place where we have seen it on nursery trees. It may be that our nursery happened to be free of the inoculum, because it's been about a mile from the orchards.

MR. O'ROURKE: Would you by any chance think it might be seed borne?

DR. McKAY: We have no information on that virus.

MR. GILBERT SMITH: I have one statement to put in at this time. Dr. Crane questioned whether the Japanese walnut should be grown. I wonder if the Japanese walnut might not be a safeguard in the area where they don't have the disease, in that you will detect the disease the quickest on the Japanese walnut, and in that way anyone would become wise to it, rather than if it was in the black walnut. It might be so insidious that it could be well spread before persons knew they had it at all. I wonder if the Japanese walnut, through its quickness in showing the disease, might not be a safeguard to the other walnuts?

DR. MacDANIELS: That's a technique that's used with some other plants.

MR. CORSAN: I go on the principle that a tree that's well fed might not resist every disease, but it will resist a great many diseases and most of the diseases, if it's well fed. Now, the feeding of trees is very important. I noticed that in going back and forth between Florida and Toronto. I examine the pecan situation every fall and spring, and just to think of Stuarts—you know the size of Stuart pecan—coming in good, big crop of nuts that size (indicating with fingers). Can you see that? And you know that is less than half the size the Stuart should be. It's a great nut for cracking by machinery. In fact, a lot of people grow nothing but Stuart. And last year they had such a crop. Last year I pointed to a farm right near the highway. "Do you see that? For years I have been trying to get you to put that sawdust, which is nearly 40 feet high in a pile, around your pecans and see the vast difference in your pecans." You know there was no rain down there all last summer, and the pecans were half the proper size. Now, that sawdust would keep the moisture in. I am a great believer in the use of sawdust. It's a tree product itself and it has some of the constituents of what the pecan should feed on.

As Dr. Waite told us one time in Washington—you will probably remember the remark he made about the pecan trees in an orchard which were absolutely fruitless year after year. He went through that orchard, and he saw a pecan here and a pecan there that had a good, big crop right among the empty trees. He examined them and found signs driven into the trees, and some of the signs were put up with zinc covered nails. Those signs that had the steel covered nails had no nuts on, but those that had zinc in had a huge crop. It excited the growth of the female blossom.

Now, we have got an awful lot to discover, as you gentlemen say in this nut culture, way beyond the imagination of the human mind.

DR. MacDANIELS: We had better limit discussion to this particular problem. Is there more comment?

MR. McDANIEL: On that problem, I have observed the brooming in the heartnut seedlings about three years old, which were seedlings of the Fodermaier variety growing at Norris in the late 30's. Brooming developed in some of them in either the second or third year from seed.

DR. MacDANIELS: That answers their remark about the young trees.

MR. SLATE: A plant that is well fed and making very vigorous growth may be more attractive to the insect vector. Therefore, a healthy tree might take it.

MR. McDANIEL: These trees were very vigorous.

DR. MacDANIELS: How many growers of nut trees have this bunch disease on their property?

MR. KINTZEL: Black walnuts?

DR. MacDANIELS: On anything at all. (Showing of hands.) There are at least a dozen.

When Mr. Burgart up in Michigan finds out that the limiting factor practically cleans him out, there is this question of bunch disease with witches'-broom resulting from ground deficiency. I know in the Wright plantings in the vicinity of Westfield they had brooming trees of the Japanese walnut which apparently recovered after treatment with zinc. And, of course, we know on the West Coast you get witches'-broom in the Persian walnut which cannot be cured by zinc.

Is there any other discussion on this point?

(No response.)

We will go on to the next paper.

MR. CORSAN: Anybody passing through Toronto can drop in and see my
Japanese walnuts with 24 to the cluster and not a sign of bunch disease.

DR. MacDANIELS: Yes, you may not have the bunch disease near you. We hope you haven't.

The next paper is by J. A. Adams, who is from the Experiment Station here at Poughkeepsie. This experiment station is a branch of the Geneva Agricultural Experiment Station. I believe that's right, isn't it, Mr. Adams?

MR. ADAMS: That's right, and it is concerned primarily with the fruits down here in this region.

DR. MacDANIELS: His subject is "Some Observations on the Japanese Beetle on Nut Trees." Let me say Mr. Adams would like to show some slides, but it didn't seem feasible to close this window down.

The Japanese Beetle and Nut Growing

J. A. ADAMS

Associate Professor of Entomology, New York State Agricultural Experiment Station, Geneva and Poughkeepsie, New York

It is a pleasure to attend this meeting of the Northern Nut Growers. Association and to take part in your program. I shall discuss the

Japanese beetle as it seems to affect nut culture, and outline our methods of control.

The Japanese beetle evidently came into this country in the soil about some roots of plants imported to a nursery near Philadelphia nearly 40 years ago. Since 1916, its distribution, habits, and control have been closely studied by the federal Japanese Beetle Laboratory at Moorestown, New Jersey. The insect has become generally distributed in the coastal area, as far north as Massachusetts, as far south as Virginia, and as far west as West Virginia. Beyond these limits, it has established local colonies in New Hampshire, Vermont, Western New York, Ohio, Michigan, and North Carolina. In most of the states affected there is an investigator who, like myself, carries on local studies, more or less in cooperation with the federal laboratory. In New York we now have, in addition to the generally infested areas on Long Island and in the Hudson Valley, about 50 isolated infestations in the central and western parts of the state.

Might I have a showing of hands by those who have Japanese beetle already? (Showing of hands.) There is quite a sprinkling of you who have them. Many of you do not have them yet, but, since the insect is spreading every year, you can expect them some day, especially if you live in the Northeast. It is expected that this pest will not thrive in the drier central States, but it might become established in the Pacific States some day, unless prevented.

You can see these beetles anywhere in and around Poughkeepsie. From Poughkeepsie I have watched them spread in the past few years to Pleasant Valley and eastward. This morning as I parked my vehicle by this building I picked these specimens from the smartweed, ~Polygonum persicaria~. (Passing of specimens.) These insects also feed on the flowers and foliage of purple loosestrife, ~Lythrum salicifolia~, so plentiful and showy in our swampy fields. The most conspicuous damage is done to the foliage of wild grape vines. You will observe this when you visit Mr. Stephen Bernath's nut plantation. You will note the conspicuous defoliation of the vines on the fence rows. Willow is another host heavily attacked. I believe you have the beetles at your plantation at Wassaic, Mr. Smith?

MR. GILBERT SMITH: Plenty of them.

DR. ADAMS: You will also observe the damage at Mr. Smith's place. You will see that it is strictly a matter of skeletonization of the leaves.

A MEMBER: They eat the fruit, too.

DR. MacDANIELS: You have damage on fruit.

A MEMBER: They eat berries.

DR. ADAMS: Yes, but on nut plants the damage above ground is confined to leaf skeletonization. It varies widely, depending on the kind of nut plant. Before visiting Mr. Bernath's planting, I sought out the botanical names of the commoner nut plants in Dr. MacDaniels' Cornell Extension Bulletin No. 701, on "Nut Growing." Of the ~Juglans~ species, the black walnut, ~J. nigra~, is sometimes heavily attacked. There are large black walnut trees near one of our peach orchards. I have seen hordes of beetles gather in these trees in July and August, skeletonizing the leaves until the defoliation reached 40% or more. Late in August the beetles seemed to leave the walnut foliage and descend upon the ripening peaches. The heart nut, ~J. sieboldiana~ var. ~cordiformis~, was moderately fed upon at Mr. Bernath's nursery. The butternut, ~J. cinerea~, is only lightly attacked, as a rule.

The hickories and pecans are not attacked to any appreciable extent, but at least some of the chestnuts are very attractive to this pest. I have seen shoots of ~Castanea dentata~ with their foliage reduced to lace. Some of the small Chinese chestnuts, ~C. mollissima~, at Mr. Bernath's place, were about one-fourth defoliated in mid-August.

The hazels seem to be attractive to these beetles. When the Japanese beetle spreads to Prof. Slate's plantings of ~Corylus~ at Geneva, we may get more information on varietal preferences. I find that exposed foliage of ~C. americana~, the common wild hazel here, is sometimes fairly heavily fed upon. I am holding up to the window a portion of a hazel bush; you can see that the leaves along one side are skeletonized. It is probable that the species, hybrids, and varieties of ~Corylus~ will show the same marked variation in susceptibility that is shown in so many other genera of plants.

Among the oaks, the pin oak, ~Quercus Palustris~, and the English oak, ~Q. robur~, are commonly one-third defoliated while the common white and red oaks are almost immune. Among the maples—to go farther afield from nuts—the Norway, ~Acer platanoides~, and the Japanese, ~A. palmatum~, are often severely injured, where the sugar maple, ~A. saccharum~, is only lightly injured and the delicate-leaved red maple and silver maple, ~A. rubrum~ and ~A. saccharinum~, remain untouched.

Since the Japanese beetle is here to stay, and to spread, these differences are worth considering where plant materials are being selected for new ornamental plantings. In our bulletin on Japanese beetle (Cornell Extension Bulletin 770) we have to warn the reader that planting chestnuts may bring him trouble with the Japanese beetle, trouble which he would not have with flowering dogwood, ~Cornus florida~, or the common lilacs, ~Syringa vulgaris~, which are immune to this pest.

It may be, however, that some of the chestnuts carry immunity factors. In the U. S. Department of Agriculture Circular No. 547, published in 1940, "Feeding Habits of the Japanese Beetle," by I. M. Hawley and F. W. Metzger, ~Castanea crenata~, the Japanese chestnut, is listed with beech and chestnut oak as "generally lightly injured." I understand you consider the nut of this species poor, but if resistance factors are in the genus, there can be hope of finding or developing a chestnut resistant to Japanese beetle.

We might be able to do with chestnuts what has been done with poplars. The common poplars range from the Lombardy, ~Populus nigra italica~, which is heavily damaged by the beetle, to the white, ~P. alba~, which is immune. The forest geneticist, E. J. Schreiner, has written an article, "Poplars can be bred to order," which appears on pages 153 to 157 in "Trees," the Yearbook of Agriculture for 1949, published by the U. S. Department of Agriculture. Schreiner provides an interesting diagram of random planting of 102 poplar hybrids, in plots of 50 trees each, representing 30 parentages. He writes, "Japanese beetle infestation was heavy in ~1947~; as late as September 9 beetles were as numerous as 10 to 12 per leaf on the most susceptible plants. Although the insects were feeding everywhere on the sparsely scattered weeds growing under the hybrids, beetle feeding was found on only nine hybrids, representing four parentages. Three of these parentages include hybrids that were entirely free of beetle feeding during the entire infestation." Among five hybrids of ~P. charkowiensis~ and ~P. caudina~, three were highly susceptible, one moderately susceptible and one was non-susceptible.

Japanese beetles, when infesting rows of plants of the same variety, usually occur unevenly on the individual plants. Some of the factors have to do with the vigor or color of the tree. In my observation on peach, I have repeatedly seen a sickly, yellow and half-wilted tree with thousands of beetles in it, while other similar but healthy trees in the same row averaged only a few hundred beetles. You can make one branch of a tree more attractive to the insects than the rest of the branches by partly girdling it or permitting borers or cankers to damage the base of the branch. This observation suggests that the increased sugar content raises the attractiveness of the leaf. It coincides with what is already known that extracts of plants preferred by the Japanese beetle have, in general, a higher sugar content, or more of a fruit-like odor than those not attacked. (Metzger et al, Jour. Agric. Research, ~49~ (11): 1001-1008. 1934. Washington, D. C.)

There are other observations you can easily make yourselves. The Japanese beetle avoids shade, except on the hottest days, and its feeding in dense trees shows up most in the tops; its feeding on uniform plantings tends to show up most in the edge rows. Nursery-size trees are more extensively defoliated than larger ones. At this point we must consider that the insect usually has to fly into a planting from the outside, for it breeds chiefly in lawns and meadows. If the foliage mass of the nut planting is small and the grass areas nearby are large, the beetles are likely to do heavier damage than where the planting is very large and grass areas negligible. A small planting in a suburban area, beside a large golf course, cemetery or dairy farm, is going to be more heavily attacked than a large one set in a clearing in the woods.

~Control of the adult:~ The safest, most direct measure is to pick or knock the beetles off the plants, preferably in the early morning, when they are cool. They may be dropped in a pail with a little kerosene in it. Some plants can be shielded with thin nets which can be placed on them by day. We do not recommend Japanese beetle traps. These yellow traps, which are baited with geranoil and other essential oils, can draw beetles in from a considerable distance but we have found that, possibly because many beetles miss the trap, the population of beetles remains high near the trap, in spite of heavy daily catches. Although the use of one trap to the acre on a block 10 miles square would probably get results, the use of a few traps on a small nut planting is likely to be disappointing.

A MEMBER: Will birds or any kind of poultry eat them?

DR. ADAMS: Yes, poultry will eat them, as far as they can reach. Certain birds, of course, will feed on them to some extent, but birds, in summer, seem to have plenty of other things to eat, and they certainly leave plenty of beetles in plain sight uneaten. We can see that the birds are a fairly constant helpful factor, but are not to be relied upon to prevent injury occurring in a beetle outbreak.

Rotenone, which, I believe, is one of your main insecticides in nut culture, is fairly effective on Japanese beetles. It kills the beetles hit with the spray and gives protection for several days thereafter. If you apply it often enough, rotenone can take care of the plants so that they don't become disfigured by the beetles. Using cube powder, you may apply five ounces of 4% rotenone in 10 gallons of water. Of course, in many cases there is no objection to using DDT wetable powder or dusts, unless you are afraid of a mite problem arising after DDT is used. If DDT can be sprayed on the plants, it needs to be applied only about three times during a summer, or sometimes only twice. For plants that are growing very fast, the new growth, of course, has to be kept treated. You may prefer to spray once heavily over all the plants in July and then, after that, keep the beetles off by spraying or dusting the new growth, during August. For more directions see U.S.D.A. Farmers Bulletin No. 2004.

Now, there are new chemicals that will kill Japanese beetles very quickly. Parathion will kill them, but its toxicity necessitates great care in handling and, on peaches, we find it protects the plants for only a few days. Chlordane, which has a very important use in connection with these insects in the grub stage, is not recommended above ground; it is too brief in its action. Methoxychlor may be used instead of DDT. It is less effective, but much less poisonous, and should be applied more frequently.

Now, the other aspect of control is to try to reduce beetle production over the whole area so that you don't have so many beetles flying in to the plants during the summer and you don't have to spray so frequently, if at all. This is the phase to which I wish to give particular attention, after we consider the life history.

~Life history:~ The Japanese beetles in the adult stage are in evidence here from late June to late September, or, roughly, for the summer season. The adults lay their eggs in the soil, mostly in lawns, mowed grassy fields and pastures. The adults die but the eggs give rise to tiny, bluish-gray larvae which feed chiefly on grass roots. The larvae grow through the fall and spring, and, if more numerous than about 40 to the square foot in September, or about 25 in April and May, can cause severe lawn damage.

MR. CORSAN: That's the stage when the pheasants and starlings eat them.

DR. ADAMS: Yes, in the grub stage.

MR. CORSAN: I see thousands of starlings gorging themselves.

DR. ADAMS: Yes, scratching birds, crows and skunks can take them out; the starlings make a hole the size of a pencil point to do so. In our survey areas grub populations sometimes seem to drop rapidly in May, when the birds are feeding their nestlings. In June, the surviving larvae mostly change into pupae, and by July they are appearing as beetles. From the lawns and grassy fields they readily fly to weeds, shrubs, grapevines and trees. They fly at least a few hundred yards, if need be, to find their host plants. Well kept, sunny, lawns with good, moist soil, which carry 40 grubs to the square foot in the fall may still have plenty at transformation time in early summer. A lawn of 5,000 square feet could thus produce 100,000 beetles. Yards, roadways and pastures commonly produce as many as six beetles to the square foot, which means a quarter million to the acre.

~Chemical control in the grub stage:~ In New York we suggest that on a home property the more valuable sections of permanent lawn be grub-proofed with chemicals as soon as there are 5 to 10 grubs to the square foot. This grub-proofing has two effects: (a) it stops beetle production from that lawn, and (b) it prevents the lawn grass being damaged by the grubs of this and other annual grub species and by the birds and animals, including moles, which damage grubby turf. For grub-proofing I prefer to use chlordane. It may be applied in a spray, at 8 ounces of 50% wettable powder to 1,000 square feet, or it may be purchased in the more bulky 5% form and applied dry with a two-wheeled lawn fertilizer spreader. For each 1,000 square feet I take 5 pounds of 5% chlordane and, since it tends to clog the spreader, I mix it in a cardboard drum with 5 pounds of a dry, granular material such as the activated-sludge fertilizer known as "Milorganite." The ten pounds of mixture is then spread on the 1,000 square feet, half east and west, half north and south.

If applied in the fall or early spring there will be no beetles coming out in July and no grubs for several years. DDT at 6 pounds of 10% DDT to 1,000 square feet will give an even longer grub-proofing effect. Our plots so treated in 1944 are still grub-free. The possible trouble with DDT is that it is too nearly permanent, and if you should plow up a piece of lawn treated with it and try to raise tomatoes or strawberries, you might find the soil too toxic.

~Biological control in the grub stage:~ The chemical grub-proofing of the sunny parts of the front or main lawn on a property is desirable for the reasons stated, but it does not usually stop more than a fifth of the beetle production around the property, because there are usually plenty of neighbors' lawns, pastures, public grounds, and other beetle-producing turf areas nearby. How are you to reduce the beetle crop on these places, mostly on ground you don't control? Here is where biological control comes in, something which I feel will appeal to you in this group. The parasitic insects known as spring Tiphia, imported from the Orient and well established on hundreds of estates, golf courses, and cemeteries around Philadelphia and New York, may be introduced in your vicinity when grubs reach about 5 to the square foot. The parasites, which are like flying ants, appear above ground in spring and feed on honey-dew. The female burrows in the soil and attaches her eggs singly to Japanese beetle grubs. A maggot hatches and consumes the grub. I have charge of the distribution of these parasites in New York. I like to liberate at least one colony in each village or town division. Some of you may help me plan the liberation for your vicinity, possibly on a cemetery near your place. The colonies enlarge to about a square mile in 10 years, and may cut beetle production by 50%.

Another biological agent which can be added to grub-carrying turf is the bacterium causing Japanese beetle grubs to turn milky white and die. A powder is made from diseased grubs and talc and this milky disease spore inoculum is applied with a teaspoon in dots or spots over the turf. The important point is that the spore powder must be used on a plot where there are grubs to get the disease, and not on chemically grub-proofed soil. Milky disease spore powder is sold under three brand names, "Japidemic," "Japonex" and "Sawco-Japy." One-half pound, suitably applied, will cost you about $2.50 and be an act of good citizenship, for the disease slowly spreads to any grubby soil in surrounding properties. I can supply addresses of the producers and detailed reprints of my studies.

Discussion

MR. McDANIEL: Does this disease affect any other beetles we have in America, besides the Japanese?

MR. ADAMS: Yes, one other species; it causes some sickness in the grubs of the turf pest known as the Oriental beetle.

MR. McDANIEL: How about the green June beetle?

DR. ADAMS: No, unfortunately, it doesn't work on that beetle, which is a pest on Long Island and in the South.

A MEMBER: How much area would a (1/2-pound) can like that treat?

DR. ADAMS: It depends. You can apply a half-pound to a quarter acre, or any smaller space you want to put it on. If you want to put spots down closer together, say every three feet, it will treat about 1,000 square feet. It suggests on the label that you do. But if you treat a plot on a large field, I'd recommend you put it out at about a teaspoonful every ten feet. In other words, I wouldn't put less than a half-pound on the plot set aside for it on my place. The application is just a starter to introduce the disease in the area, and it doesn't matter too much whether you spot it at 10-foot intervals on a pasture or put it at fairly close intervals on an area about the size of this room. The point is that it mustn't be broadcast, because that spreads the spores too thin. Grubs don't get the disease if they eat only a few spores. We assume that where you put the spots down on the ground the grubs under those spots will get the disease and wander off and die. When a grub dies, it multiplies the number of spores up to many millions. That portion of soil becomes infective, and more grubs going through the infective portions carry the disease to intervening areas until the whole piece of turf is unhealthful to these grubs. Droppings of birds feeding on sick grubs spread the disease.

MR. FRYE: One application is all that's needed?

DR. ADAMS: One application is all that's needed. Control is slight at first, but increases with the passage of the years.

MR. CORSAN: Quail feed on them. Why can't we have quail around the farms instead of shooting them?

DR. ADAMS: I would be for that, but we have to find other methods for a lot of people. Besides, we need something that will intercept some of the grubs in the fall, before they get big. After all, by the time the quail are interested in them, they have already done some damage in the ground. In the ground the grubs can do two kinds of damage. They can make turf loose so it can be rolled back like a rug. Second, if you should plow up a piece of sod that has many grubs in it and try to plant row crops or nursery stock, they may eat the roots off the planting in the spring.

DR. McKAY: I'd like to ask what effect low temperature has on them and how far north you think will be their limit?

DR. ADAMS: The soil temperature at which the grubs begin to die in hibernation is 15 degrees, and I have never seen the soil temperature that low here under turf. (I operate a soil thermograph on my lawn.)

A MEMBER: How far down do they go?

DR. ADAMS: They hibernate at 4 to 8 inches in the ground. It's rare to have it drop below 27 degrees at these depths.

MR. STERLING SMITH: What do you mean, Fahrenheit?

DR. ADAMS: That is Fahrenheit.

A MEMBER: That's frozen solid. That's at 32 degrees.

DR. ADAMS: The deeper soil will drop only a few degrees below freezing. The soil here usually remains no lower than 32 degrees, except within an inch or two of the top.

A MEMBER: Do you think soil temperature is going to be a limiting factor?

DR. ADAMS: I think the limiting factor northward is the coolness of the summers. In Northern Japan their life history gets altered because of the shortness of the summer, and I think in the Adirondack area they won't be serious for that reason.

MR. WEBER: Will this spore powder kill other kinds of grubs that are in the sod?

DR. ADAMS: Not to any practical extent. It does not control the grubs of the "June bugs," or brown June beetles, or what are called "white grubs."

MR. LOWERRE: Would the DDT kill the parasitic wasps?

DR. ADAMS: Turf treated with chlordane or DDT is grub-proofed and is not of any use to the flying parasites as a place to lay eggs, or for bacteria to multiply. So we don't want to put chemicals on top of biological control plots. For instance, on an average home property I would treat the front lawn, the more valuable piece, with chemicals so that it would be 100% grub-proofed to protect the turf and to take that much turf out of beetle production. Then on the back lawns or grassy fields adjoining, I would apply at least a half-pound of this milky disease material, and in that way provide a complete treatment; the parasites can be added on some large public turf area nearby. And don't think you are going to stamp the Japanese beetle out just by spraying all the adult beetles you see each summer on the cultivated plants, because there are lots more on the shade trees, weeds and vines.

A new book, "The Insect Enemies of Eastern Forests," contains a great deal of information on the insects feeding on nut trees. Unfortunately, it isn't indexed to crops, so you can't look up "walnut" and find what insects bother you. You have to know what the insect is, and you will find it with its insect family. That is U. S. Department of Agriculture Miscellaneous Publication 657, by George E. Craighead. Price $2.50, from the Superintendent of Documents, Washington, D. C.

MR. CORSAN: What in the world has become of the black walnut caterpillar, that big, black fellow with the grey hairs?

DR. ADAMS: Maybe they are at a low point in a cycle. Mr. Bernath will show you a few of them.

MR. CORSAN: He might show me a few of them, but I have been pestered with them for years, and this year I haven't got any.

DR. ADAMS: I suppose natural conditions have taken care of them for a while, but they will come back again.

(Applause.)

DR. MacDANIELS: Thank you, very much, sir. We will take a few minutes recess now.

(Whereupon, a short recess was taken.)

Editor's Note: The following paper which was delayed, was originally scheduled for our 1949 Report.

Insecticides for Nut Insects

E. H. SIEGLER United States Department of Agriculture Agricultural Research Administration Bureau of Entomology and Plant Quarantine

Fortunately, the growers of nuts do not have to combat a large number of injurious insect species. However, some species do at times cause a heavy loss of nuts and may also damage the vegetation growth of the trees. Injury by insects will vary from year to year, due to various causes, and insects frequently show varietal host preferences. Timely use of insecticides is the most effective means of combating many harmful species.

Until the beginning of World War II a rather limited number of insecticides was available, such as lead arsenate, cryolite, nicotine, mineral oil emulsions, and rotenone. Some injurious insects were satisfactorily controlled through the timely application of one or the other of these materials, or combinations of them; others survived in damaging numbers in spite of all attempts to suppress them.

During and since World War II, both in the United States and abroad, work on insecticides has been stepped up markedly. As a result, many new insecticides have been developed and are available for general use.

The first of the new insecticides about which we heard was DDT. Actually, the compound itself was not a new one, since it was prepared by a German student chemist in 1874. However, no use was found for it until

1939, when a Swiss chemist found it promising as an insecticide against the Colorado potato beetle. It was first tested in the United States a few years later.

Since the successful introduction of DDT, promising new insecticides have become available more frequently and in greater numbers than ever before. Among these materials are certain chlorinated hydrocarbons related to DDT. These include methoxychlor and TDE, neither of which is, on the whole, as useful as DDT but both of which are of value and have an important advantage over DDT in that they are reported to be less toxic to warm-blooded animals. Other new chlorinated hydrocarbons include benzene hexachloride, synthesized in 1828 and first tested against insects in France in 1941 and discovered about the same time in England; chlordane, developed in the United States a few years ago; and toxaphene. Several organic phosphorus compounds, including hexaethyl tetraphosphate, tetraethyl pyrophosphate, and parathion, have also been developed.

Technical benzene hexachloride is a mixture of several isomers, the gamma isomer being the most toxic to insects. The practically pure isomer is known as lindane. A handicap to the general use of benzene hexachloride on fruit is its tendency to cause off-flavor condition when applied too close to harvest. Lindane is less likely to cause off-flavor in fruit than technical benzene hexachloride but may not overcome this fault altogether.

The organic phosphate insecticides, like DDT, were first found of value in Europe and were introduced into the United States after the close of World War II. Parathion in particular shows great promise for the control of many insect pests. Although these compounds are very poisonous and must be handled strictly in accordance with the manufacturers' recommendations, a recent announcement by Arnold J. Lehman, of the Food and Drug Administration, indicates that their residues are not likely to be harmful. He has stated that "parathion is not stored in the tissues to an appreciable extent —it is rapidly destroyed by the tissues of the body which in turn is an added mechanism for the prevention of tissue accumulation." Residues of hexaethyl tetraphosphate and tetraethyl pyrophosphate persist for only a short time and residues of parathion drop to a low level within 10 to 14 days after application. This information, however, does not make it unnecessary for the user to observe strictly all warnings and precautions issued by the manufacturers of parathion and of other organic phosphates. Serious effects and deaths have occurred though excessive exposures to parathion.

General Information Regarding the Use of the New Organic Insecticides

~Handling the insecticides.~ All the new organic insecticides, the organic phosphates in particular, are to some degree toxic not only to many insects but to man and animals as well. Even the most toxic ones can be used, however, without harmful effects on the operator, provided all the cautions issued by the manufacturer are properly followed. Special care must be taken in handling concentrated insecticides preparatory to making diluted spray or dust applications.

~Spray concentrations.~ DDT has been used more extensively than any of the other newer insecticides and for this reason there is considerable information relative to the spray concentrations known to be effective against insects susceptible to it. For spray purposes DDT is generally employed at the rate of 1-1/2 to 4 pounds of 50 percent wettable powder per 100 gallons of water.

Parathion is being used at 1/2 to 1-1/2 pounds of 15 percent wettable powder per 100 gallons of water for mites and up to 2 pounds to 100 gallons of water for insects more resistant to it. The occurrence of injury to the foliage and fruit of some varieties of apples when this insecticide is used is under investigation.

Benzene hexachloride (10 percent gamma isomer, wettable) is being used at 2 to 4 pounds, and sometimes less depending upon the insect, per 100 gallons of water. Wettable mixtures containing 25 percent of lindane (approximately pure gamma isomer) are used at dosages which would give an equivalent quantity of the gamma isomer in the diluted spray.

Chlordane is usually employed at the rate of 2 to 3 pounds of 50 percent wettable powder and toxaphene at 2 to 4 pounds of 40 percent wettable powder per 100 gallons.

These insecticides are also being sold for use as dusts, either ready to use or in a more concentrated form which can be reduced to dusting strength through the addition of inert material.

~Spray Residues.~ Spray residues are not important on nut crops, but on fruits it is important to time the insecticide applications so that harmful residues are avoided. Animals should not be allowed to graze vegetation beneath trees recently treated. Instructions on the packaged insecticide should be followed.

~Effect on beneficial insects.~ Since the more potent of the newer organic insecticides are toxic to many parasitic and predatory insects, all of which help to reduce the populations of injurious species, these insecticides, if used, must be largely relied upon to effect control by themselves. Often no immediate assistance is forthcoming from beneficial insects after these materials have been used.

Nut Insect Investigations

Except for studies on the chestnut weevils, nut insect investigations by the Bureau of Entomology and Plant Quarantine are being conducted primarily on the pecan at southern laboratories. Many of the remarks in this paper are therefore based on information obtained from these laboratories. In view of the short time the new organic insecticides have been available, work to determine their place in nut insect control programs is largely in the experimental stage. Much further work will be necessary before detailed instructions can be given for their general use.

Insects Attacking the Nuts

~The Pecan weevil.~ The adult of the pecan weevil[8] is a snout beetle that attacks not only pecan throughout the South but also hickory in the eastern half of the United States. During mid-season, previous to the formation of the kernel, nuts are frequently punctured for feeding purposes. This results in failure of the nuts to complete their development. The principal injury, however, is caused by grubs that develop from eggs laid in the nuts after the kernels have formed. This is usually during September on pecans in the South. The grubs feed on the kernels and may consume them completely (Fig. 1).

[8] ~Curculio caryae~ (Horn).

Applications of sprays containing 6 pounds of 50 percent DDT wettable powder per 100 gallons of water just previous to and during the oviposition period have proved effective against this pest.

[Illustration: Fig 1.—Nut infested with larvae of the pecan weevil.]

[Illustration: Fig. 2.—Larva of the butternut curculio in Japanese walnut shoot.]

[Illustration: Fig. 3—Adults of the walnut husk maggot on black walnut. Enlarged.]

[Illustration: Fig. 4.—Adult of a leaf-footed bug. Enlarged.]

[Illustration: Fig. 5.—Defoliation caused by the black pecan aphid.]

Nut curculios. Several species of curculios, such as the butternut curculio[9] (Fig. 2) and the hickorynut curculio,[10] infest the fruit of these and other nut trees. Their life histories and methods of attack are somewhat alike and for the purpose of this report the butternut curiculio is given as an example. This insect lays its eggs in both the young shoots and nuts, which usually drop as a result of the injury. The larvae then develop to maturity within the dying tissues after which they enter the soil and transform to adults. Subsequently they leave the soil to pass the winter above ground protected from low temperatures by weeds or other vegetation.

[9] ~Conotrachelus juglandis~ Lee.

[10] ~Conotrachelus affinis~ Boh.

Lead arsenate, 4 pounds per 100 gallons of water, has been relied upon in the past for control of various nut curculios. Among the newer insecticides, benzene hexachloride (6 percent gamma), 4 to 6 pounds per 100 gallons, has shown promise against a shoot curculio on pecans when applied soon after the trees start growth in the spring.

~Hickory shuckworm.~ The hickory shuckworm[11] is another serious pest of pecan and hickory nuts. Early in the year, previous to the hardening of the shells, the kernels are eaten. This injury causes many of the nuts to drop. In the fall, the later generations tunnel within and feed upon the shucks only. The affected nuts are usually smaller than normal; in addition the shells are often stained and are more difficult to separate from the husks.

[11] ~Laspeyresia caryana~ (Fitch).

Extensive experimentation in the control of this insect has been carried out without much success. No effective insecticide treatment can be recommended for its control.

~Walnut husk maggot.~ The adult of the walnut husk maggot[12] is a fly (Fig. 3); it is related to other injurious fruit flies such as the apple maggot, Mediterranean fruit fly, and the oriental fruit fly, which has recently been found in Hawaii. Adults emerge from the soil and fly to the trees in midsummer. Egg laying follows in 1 to 3 weeks, the eggs being deposited on the husks of several kinds of nuts. The maggots feed within the husks. Not only is the quality of infested nuts lowered, but, in addition, the husks are more difficult to remove. A closely related species is particularly damaging to the Persian or English walnut in California.

[12] ~Rhagoletis suavis~ Loew.

Lead arsenate, 2 to 4 pounds per 100 gallons of water, in combination with an equal quantity of hydrated lime is quite effective in destroying the adults of the walnut husk maggot when applied at the time they are present.

~Stinkbugs and leaf-footed bugs.~ There are a number of stinkbugs and leaf-footed bugs (Fig. 4), in addition to the species mentioned,[13][14] which are responsible for important injuries to pecans, filberts, and other nuts. These insects puncture the immature nuts with their beaks. The punctured areas become spongy, somewhat dark in color, and are bitter to the taste; on pecan the typical injury is referred to as black pit and kernel spot.

[13] ~Nezara vizidula~ (L.).

[14] ~Leptoglossus phyllopus~ (L.).

Crops of favorable host plants such as cowpeas and soybeans should not be planted in or adjacent to nut orchards subject to attack by these sucking bugs. In general, orchard sanitation should be practiced.

[Illustration: Fig. 6.—Galls produced by the pecan phylloxera.]

[Illustration: Fig. 7.—Injury to young pecan tree by the fall webworm.]

[Illustration: Fig. 8.—Larvae of the walnut caterpillar.]

[Illustration: Fig. 9.—Caterpillar of the hickory tussock moth.]

[Illustration: Fig. 10.—Rose chafer beetles on chestnut blossoms.]

Insects Attacking the Foliage

~Black pecan aphid.~ Pecan trees at times suffer sufficient damage from the black pecan aphid[15] to cause considerable defoliation (Fig. 5) during the latter part of the season. The injury to foliage in its earlier stages consists of irregularly shaped yellowish areas which turn brown when the tissues die.

[15] ~Melanocallis caryaefoliae~ (Davis).

This aphid is usually controlled with nicotine sulfate (40 percent nicotine), 3/8 pint plus summer oil emulsion, 2 quarts per 100 gallons of spray. Parathion and benzene hexachloride have given good results in experimental work but are not yet generally recommended.

~Pecan phylloxera.~ The pecan phylloxera[16] is related to aphids. It attacks principally the vegetative parts of the tree such as the leaves, petioles, and shoots on which galls (Fig. 6) are produced. Pecans, hickories, and other species of nuts are subject to infestation.

[16] ~Phylloxera devastatrix~ Perg.

In the past a spray of nicotine sulfate (40 percent nicotine) 13 ounces combined with either lime-sulfur solution, 2-1/2 gallons per 100 gallons of water, or lubricating-oil emulsion, 2 quarts per 100 gallons, applied in the late dormant period has been the standard recommendation. In recent experiments in the South with some of the new organic sprays, benzene hexachloride and some of the dinitro compounds have indicated good promise.

~Fall webworm,~[17] ~walnut caterpillar,~[18] ~and hickory tussock moth.~[19] The caterpillars of these species (Figs. 7, 8, 9) are frequent pests on the foliage of nut trees. They often defoliate entire branches.

[17] ~Hyphantria cunea~ (Drury).

[18] ~Datana integerrima~ (G. and R.)

[19] ~Halisidota caryae~ (Harr.)

The best time to apply control measures is as soon as possible after the caterpillars hatch. The insects can be readily destroyed with lead arsenate, 3 pounds, or DDT (2 pounds) of 50 percent wettable powder, per 100 gallons, applied when they appear. Other new organic insecticides may also be effective but have not been widely tested.

~The rose chafer and Japanese beetle.~ Adults of the rose chafer[20] (Fig. 10) and the Japanese beetle[21] are voracious feeders on the foliage of nut trees and must be destroyed if severe injury is to be avoided.

[20] ~Marcordactylus subspinosus~ (F.).

[21] ~Popillia japonica~ Newm.

Fortunately these insects may now be controlled by spraying with DDT, 2 pounds of 50-percent wettable powder per 100 gallons of water, when the beetles appear. In the case of the Japanese beetle a second application may be necessary if the infestation is heavy.

~Spider mites.~ Nut trees, especially those which have been sprayed with DDT, may become seriously injured by various species of mites.[22] DDT is very toxic to the natural insect enemies of plant-feeding mites and therefore the mites build up to injurious numbers.

[22] ~Tetranychus~ sp. and others.

Of the various miticides recently tested on pecan, a spray of parathion was the most promising. In some recent tests for the control of spider mites on chestnut trees, 1-1/2 pounds of 15 percent parathion wettable powder per 100 gallons of water was effective. Do not use parathion unless you observe all the precautions contained on the package label of the material.

[Illustration: Fig. 11.—Larva of the twig girdler. Enlarged.]

[Illustration: Fig. 12.—Adult of the flatheaded apple tree borer. Enlarged.]

[Illustration: Fig. 13.—Larvae of the flatheaded apple tree borer.]

[Illustration: Fig. 14.—Scars on trunk of pecan tree caused by cutting out flatheaded apple tree borers from their tunnels.]

[Illustration: Fig. 15.—Adult of the buffalo treehopper. Enlarged.]

[Illustration: Fig. 16.—Twig scarred as a result of egg laying by the buffalo treehopper.]

Insects Attacking the Trunk and Branches

A number of insects cause important damage to the trunk and branches of nut trees.

~Obscure scale and others.~ The obscure scale[23] infests a variety of nut trees. On pecan the chief injury results from attacks on branches under three inches in diameter.

[23] ~Chrysomphalus obscurus~ (Comst.).

The obscure scale and other scale insects can be controlled with lubricating-oil emulsion during the dormant period. However, nut trees are often susceptible to oil damage, especially at 3 percent concentration. Since healthy trees are more resistant to oil injury, it is therefore advisable to watch for scale infestations so as to spray them before the trees are weakened.

~Twig girdler.~ Nut trees are sometimes attacked by the twig girdler[24] (Fig. 11). This beetle lays eggs in the twigs, which are girdled so as to stop the flow of sap that would normally prevent hatching. The girdled twigs usually become detached from the trees and as a result the nut-bearing wood is reduced.

[24] ~Oncideres cingulata~ (Say).

The standard recommendation for control of this insect has been to gather and destroy the infested twigs in the orchard and from any infested trees nearby. Recent tests on pecan in northern Florida indicate that DDT and parathion may be effective against this insect. Three applications (the first on August 26 when the first girdled twigs were observed and the others on September 9 and 23) of DDT, 4 pounds of 50 percent wettable powder per 100 gallons of water, or parathion, 3 pounds of 15 percent wettable powder per 100 gallons, gave complete control. Further experiments will be required to determine the minimum effective concentration of spray and the number of applications needed for control. It is suggested that DDT be used for the control of this insect until more information is available on how to handle and to use parathion.

~Flatheaded apple tree borer.~ The adult beetle of the flathead apple tree borer[25] (Fig. 12) deposits its eggs throughout the summer season, preferably in the small grooves of bark on the unshaded portions of the trunk of pecan and other trees. The borers (Fig. 13) hatch and tunnel through the bark to the cambium layer. Young trees may readily be girdled (Fig. 14).

[25] ~Chrysobothris femorata~ (Oliv.).

To avoid this insect as far as possible, orchard sanitation should be practiced and the trees should be kept in a healthy condition. In some plantings wrapping the trunks with paper or burlap to protect against egg laying and maintaining low branches to shade the trunk have been helpful. Cutting out the borers with a knife has also been resorted to; trunk washes have likewise been used but have not been very effective.

~Buffalo treehopper and periodical cicada.~ Buffalo treehoppers[26] (Fig. 15) and the periodical cicada[27] weaken twigs by inserting their eggs in them. The injured bark becomes roughened as it heals (Fig. 16), and the growth of the limb is retarded.

[26] ~Ceresa bubalus~ (L.).

[27] ~Magicicada septendecim~ (L.).

Pruning of weakened twigs is recommended for wood injured by the cicada. If treehoppers are a pest, clean cultivation will help. Cover crops of cowpeas or clovers should not be planted. In preliminary tests two or three applications of tetraethyl pyrophosphate (20 percent), 3/4 pint per 100 gallons of water, have given promising results in controlling the periodical cicada. The first application should be made after the cicadas appear and the others as needed to prevent damage.

Observations on Effects of Low Temperatures in Winter 1949-1950 on Walnuts and Filberts in Oregon and Washington

JOHN H. PAINTER

Horticulturist, United States Department of Agriculture, Agricultural Research Administration, Bureau of Plant Industry, Soils, and Agricultural Engineering, Oregon State College, Corvallis, Oregon

In western Oregon and Washington, where the Japan Current is supposed to keep the winter temperatures moderate, something happens every now and then and we get really severe winters. We can't blame it on the "A" bomb because we had severe winter injury in 1919 and 1935 long before the "A" bomb.

The last two winters have been exceptionally cold, but this past winter of 1949-1950 was much more severe than the previous one.

In 1948-1949, the cold came rather suddenly in the latter part of December. In the past winter, 1949-1950, the real cold came on January 30, with temperatures ranging from 10 to 30 degrees below zero F. Most official temperatures were higher; but at Corvallis the official temperatures were taken at least 60 feet above the ground level, on the roof of the Agricultural Building, which is over a steam-heated building and is old enough to be not very well insulated. This cold continued in somewhat modified form for a week.

During the previous winter the lowest temperature recorded in the nut growing areas was about 10 to 11 degrees above zero F., and the severe cold lasted for only a couple of days.

In both winters the ground was fairly well covered with snow, but with considerably more snow this past winter than the previous winter.

No apparent damage to Persian walnuts was observed as a result of the cold in the 1948-1949 winter, but in certain low-lying areas catkins of Barcelona and Daviana filberts were killed, especially those of the latter. Considerable dieback of filberts occurred; but during the following growing season recovery was effected and at the end of the summer very little evidence of winter injury was visible.

The injury resulting from the cold weather of the past winter was much more severe than that of the previous winter. Whereas filberts were the only nut trees injured in 1948-1949, they escaped with relatively little damage in 1949-1950 in comparison with Persian walnuts.

On February 11, 1950, ten days after the really severe week, several walnut growers of long experience held grave fears for the entire industry. Peach and apple trees, which seem to exhibit winter injury more quickly than walnuts, showed so much damage then that walnut growers thought the injury to the Persian walnut would be even worse.

From February 11, 1950, to the present date (July 30) I have been making observations from time to time in different locations with special attention to walnuts and with some to filberts. It is thought that certain of these observations might be of interest to nut growers in other areas, even though there is nothing particularly new or startling about them. They do, however, tend to show how surprisingly well the Persian walnut trees can withstand severe cold if it occurs after they have once gone into dormancy.

Generally speaking, the winter injury to walnuts has been spotty. No areas of great size have been either free of injury or severely injured. Usually, where a difference in severity of damage is found between areas close together, some reason for the difference can be found, but it is not always evident on the surface.

Injury to Walnuts

With the possible exception of southern Oregon, it is safe to say that 100 percent of the walnut trees in Oregon and Washington suffered some twig injury as a result of last winter's cold. In many cases the subsequent dieback of the twigs may extend only a few inches, but sometimes the injury involves not only the past season's growth but that of three or even four years back.

As might well be expected, this twig injury of necessity has meant the loss of many terminal and lateral buds which bear the female flowers; so for that reason, if for no other, this twig injury has assumed serious aspects.

In many cases the catkins were severely injured even where there was little or no twig injury. The catkins of the Persian walnut seem to be extremely sensitive to cold. Many Persian walnut trees in Oregon this year failed to produce any catkins at all. Some produced very few normal catkins, but some half-developed and deformed catkins. An examination of these partially injured catkins, however, revealed the fact that they did produce some pollen. It will always remain a mystery to me how as many walnuts were pollinated and set as there were, with the scant crop of catkins.

In practically every orchard examined, where the temperature got as low as minus 10 degrees F., the pith cells were blackened. This is not uncommon in other tree crops following severe winter injury. Fairly good peach crops have been borne in Georgia on trees that had the pith cells completely blackened.

In the case of walnuts this year, many growers were considerably worried by the fact that even the wood tissue outside the pith region was black and watersoaked. However, to date (July 30, 1950) this condition has not proven serious; as long as the cambium cells were not injured no real trouble has developed. In some cases under observation, even where some injury to the cambium cells was known to have existed, enough live ones have been left to effect recovery. Compared to peach, holly, and even apple trees the Persian walnut has put up a marvelous fight to recover from the injury sustained.

Factors Accentuating Winter Injury in Walnuts

After the several months of observation, certain factors appear invariably to account for excessive damage to walnut orchards. Elevation seems to be a principal factor. The hillside orchards or those on upland sites (soils) were far less injured than the river-bottom or valley-floor orchards, even though the latter may be on a better soil as far as fertility is concerned. My early prediction of 50 percent of a crop in the hillside orchards seems now to have been about 10 percent short, unless other factors become involved. On the other hand, my early prediction of 25 percent of a crop in the valley-floor orchards has been close to correct. Of course, certain valley-floor orchards with a combination of adverse factors won't have even a 5 percent crop.

Older orchards were more severely hurt than younger orchards with otherwise similar conditions. This is possibly due to the lack of vigor and of reserve material, resulting from crowding and competition for elements of nutrition.

The size of the crop the preceeding year seems invariably to have had an effect upon amount of damage done. The matter of reserves is again involved. Two orchards that bore a reduced crop last year because of spring frost injury have come through much better than some other similar orchards, at practically the same elevation and age, that bore a crop last year.

Two adjacent hillside orchards show considerable difference in degree of winter injury and crop prospects for this year. It is believed that this difference was due to the fact that in one orchard 35 percent of the crop was destroyed by blight last year, in comparison with a 1 percent loss in the other. The owners and I estimate that there is at least 20 percent larger crop this year in the orchard which had the heavy loss from blight last year.

In several orchards where different levels of fertilization have been used by the grower, it appears that the more liberal the application of fertilizer, particularly nitrogen, the less severe was the winter damage sustained.

At the college orchard in Corvallis, the one tree that got no additional nitrogen last year and that bore the heaviest crop of nuts is outstandingly the most severely winter injured of the 17 trees involved.

Only two varieties of walnuts have been studied, Franquette and Mayette, and some Carpathian seedlings in one orchard. Here in Oregon the Mayette seems to have generally withstood the winter injury better than the Franquette. It is my belief that they are just naturally a little more vigorous than the Franquette. Yet they never seem to overproduce as the Franquette sometime does. Last year was the "on" year for Franquettes and that might easily account for the generally apparent better condition of the Mayettes this year.

Carpathians Resist Winter Best

Near Ontario, Oregon, I saw 7 seedling Carpathian walnut trees early this spring. They were leafed out and the catkins were elongated before any Franquettes, even in the Willamette Valley, had started breaking buds. No sign of winter injury was apparent on the Carpathian trees at that time, yet Franquettes at the Malheur Experiment Station, a mile away, were obviously killed to the groundline. The owner, Mr. Peter Countryman, says these trees are often damaged by spring frosts but they always produce some nuts.

A letter dated August 4, from Mr. Countryman, indicates that a hard frost on the morning of April 24 when the temperature dropped to 22 degrees, did considerable damage to the new growth and catkins on the lower half of the Carpathian walnut trees. He estimates not to exceed one-third of a crop on these Carpathian trees this year; but he says that since the freeze the trees have made good growth, the new terminals being about 18 inches in length and the nuts on them are very large.

To sum up the walnut situation, then, the encouraging thing is that no walnut orchards have been called to my attention that were completely killed. Several badly neglected orchards and two orchards where it is said that the temperature dropped lower than minus 25 degrees F. are so severely damaged that it is impractical to try to save them, but even these are not completely killed.

Injury to Filberts

From the less comprehensive observations made on filberts following the severe winter just past, it appears in general that when the filbert tree has gone into dormancy it is more tolerant of cold than the walnut. The difference of one month in time of occurrence of the cold in the two winters seems to have had more bearing on the damage to filberts than the difference in temperature. In the Forest Grove, Oregon, area, and in Clark County, Washington, filbert trees, however did suffer severely from the cold last winter, but these two areas were the "cold spots" of the Northwest.

It seems as if the same factors that accentuate winter damage in walnuts work in a similar way on filberts, except that the elevation factor does not seem to be of so great importance. Age of tree, level of nutrition, and size of preceeding year's crop seem to be more important than elevation. Young filbert orchards, on either hillside or valley-floor sites, seem to be much less severely hurt than older orchards on the same sites. It is the acreage of *young* filbert trees that will make good the agricultural statistician's estimate of 40 to 50 percent of a filbert crop this year.

I have seen one 32-acre orchard of 24-year-old filbert trees that was injured beyond repair, but they were crowded and unfertilized. At the very same location a 14-acre orchard of 15-year-old filberts with adequate spacing was not seriously injured, even though the trees were not fertilized.

One other orchard in a poor location and on waterlogged soil, which has had little or no care, has likewise been lost. Filberts definitely were hurt in the two "cold spots" previously mentioned, but official reports of minus 18 degrees F. were common in that area.

There was a noticeable difference in damage to catkins between Daviana and DuChilly. Very few Daviana catkins produced pollen; but DuChilly seemed to be fairly normal.

Injury in filberts was confined mostly to the catkins and twigs. Excessive sucker growth up and down the main trunk and branches has taken place in the filberts, as is the case in walnuts.

In neither walnuts nor filberts was there much splitting of the bark on the trunk. This was probably because there was no sudden fluctuation in temperatures and sunshine was not excessive during the critically cold days.

It has been previously stated that the filbert is possibly more tolerant of cold than the walnut. In spite of this there probably has been more extensive damage to filberts than to walnuts; but it must be remembered that filberts are the principal nut crop in those two "cold spots." Not many walnuts are grown there, but the ones that are were likewise injured.

Editor's Note: Mr. Gellatly's following papers were read by title.

Effects of the Winter of 1949-50 on Nut Trees in British Columbia

J. U. GELLATLY

Box 19, Westbank, B. C.

(Orchard at Gellatly, B. C.)

Our district is just recovering (in August) from the effects of the toughest winter we have experienced here in the past 50 years. This gave the weather test to the tune of -22° F., official. The unofficials were of 30 to 40 below—depending on distances and location from Okanagan Lake, a deep body of water three to four miles wide and eighty miles long. This lake rarely freezes over completely, especially near our section; so the open water acts as a thermostat during most winters. But the past one pulled a new stunt and it froze over completely giving zero winds a vast open sweep, so that to be near the lake was a disadvantage, for it was colder there than it was farther back, in more sheltered locations.

Heartnuts and Hybrids

The bright spot in the nut tree picture is our heartnut trees. They all came through in good shape, making rampant growths and carrying a heavy crop. These include: 2 Walters, 4 O.K. Heart, 1 Canoka, 1 Slioka, 1 Rover, 2 Calendar, 1 Westoka, 1 Nursoka, 1 Aloka, 1 Symoka, 15 select unnamed bearing seedlings, yet on trial. All are promising. Also we have three of the Elfin paper shell heartnut hybrids. I have failed to find a good pollinator for these Elfins, so they are shy croppers, although producing plenty of the female blooms. All of the above trees are 6 inches in diameter and up to 20 inches.

Then come the Buart nuts. I coined this name to designate the hybrids I had made having the butternut (~J. cineria~) as the pollen parent and Calendar heartnut (~J. sieboldiana cordiformis~) as the mother tree. Possibly the seven best of these are: Leslie, Dunoka, Fioka, Okanda, Kingsbury, Penoka, Flavo. These trees are all carrying crops and most of them are making good growth.

Filberts

Ackerman, Brag, Comet, Craig, Holder, Petoka, Carey, Baroka, Barcelona, Bawdin, Firstoka (Gellatly No. 1). These have made a good showing, as the majority of the trees or bushes under 4 to 6 inch crown diameter of these varieties, are doing well and carrying good crops, while many above these diameters suffered in varying degrees from slightly to severely, apparently regardless of variety, location, or soil on which they grew. It may be noted that all these varieties have been hardy in the past, but age was adding up and age evidently had somewhat to do with their inability to take the punishment they got this past winter. For all my large Bing and Lambert cherry trees were severely injured or entirely winter killed, as were nearby peaches, apricots, pear, and some apple trees, particularly in the larger sizes, while many of these younger trees were uninjured, except that they are fruitless this season.

Soft Shell Walnuts

(Juglans regia)

Broadview variety on Gellatly Farm, of 20 bearing trees, all suffered winter injury for first time in 20 years. This injury varied all the way from freezing back two to three feet of all higher branches and twigs, to an actual loss of one-third to two-thirds of entire tree and trunk. At date of writing all are staging a good comeback with no care but a "wait-and-see" policy as to final treatment. There was so much loss as to involve too much work if pruning and after care of sprouts were undertaken. It was decided to leave the dead limbs and branches as a protection to the fast growing new sprouts, which, without this protection, would probably have been badly damaged by wind and rain storms. Even large birds lighting on these new sprouts might break them down.

The dead limbs will be gradually removed later, as the new limbs harden up and take over. Many of these will be left as supports for at least two years, when I expect most of these trees will be back in production, if we get a return to normal (minus 10° F.) winters, many will produce in 1951, as the new wood is showing a good growth of catkins. Although all bearing trees on my place were injured, the younger trees in my nursery were not hurt to any noticeable extent. At Summerland Experimental Station, 25 miles south of Gellatly, grow two large Broadview walnut trees supplied by myself. I had grafted on these black walnut roots (~J. nigra~) at the ground line, in every respect like my own. These trees are carrying a good crop. One shows slight winter injury, the other none at all. The official low for their location was 22° below F. with nearby unofficials to 30° below.

Their present location is at least 200 feet above lake level, and on very well drained sandy loam. Mine are about 30 feet above the lake and on somewhat heavier loam. I note that trees on my more gravelly soil came through in the best shape at official-22° F., unofficial 24 to 28 below. My Broadview that made best survival had grown the previous year in a chicken yard. Ground was well scratched over and droppings incorporated in top 4 inches of soil. Tree was flood irrigated three or four times in dry season. On this tree only outer new branches were killed and tree gives every indication of being back in crop in 1951 season.

The crop record on this tree is from 1945 and reads '45—35 pounds; '46—75 pounds; '47—91 pounds; '48—36 pounds; '49—100 pounds. Weight is for clean, undried, and partly dried nuts at time of picking up. Some of the other Broadview trees have higher crop records, although of same age and size, with possibly a bit better soil, in same grove. One tree in six years, '44 to '49 inclusive, had an average of 74 pounds per year; another had an average for the same years of 104 pounds per year. Just recently I made a special trip to see how the parent Broadview tree had wintered. I found it had sustained severe damage to two-thirds of the upper part of the trunk and main branches. The lower third was staging a good comeback, despite unofficials of 35 to 40 below zero F. as reported by neighboring farmers.

The following varieties of soft shell bearing walnut trees were also winter injured: Munsoka, badly, top two-thirds of trunk; Linoka, badly, top two-thirds of trunk; Myoka (Jumbo type) one-third of top branches; Geloka (Jumbo type) frozen to ground line but sprouts two feet high now growing. On Sirdar (a Jumbo type long nut), only outer tips of branches were killed. This was a surprise to me, as it is a second generation seedling of Italian source. The parent tree grew and cropped well for many years on bench land at Sirdar, in southern interior of B. C. until the winter of 1935-36, when it was so badly damaged that the owner had it removed. I rather looked for a similar fate in this one. There is this difference: mine was not as old nor had it been cropping heavily as yet. The season here is barely long enough to develop fully the kernels of Sirdar.

Crath Carpathian Walnut No. 46

This walnut was grafted on black walnut (~J. nigra~) root in 1944 and planted here on low loam soil in 1945. It never has been hardy under our conditions, winter killing some every winter since it was planted. This past winter it was killed to below snow line 18 inches above union, whereas

Broadview trees alongside, which are the same in every respect, never were injured until this past winter. Then only minor damage to soft new growth was done. So it looks as though Broadview is still the best bet for our conditions.

I am of the opinion that extreme temperature is not the sole determining factor in causing winter injury to nut or other trees. This opinion is based on the behavior of trees that have winter killed continuously while in certain soil, but on being moved to another spot having enriched soil of similar make-up and drainage located only 200 yards away, have never winter killed since removal, and have taken much worse winters, including the one just past.

The fact that many of our introductions grow and thrive 150 to 200 miles north of here, where temperatures drop to minus 35° to 40°, with occasional drops to 54° below zero. Check this on your map of Interior of B. C. on 53° latitude at Quesnel, B. C. I see a geology map lists that district as sedimentary and volcanic rocks. My informant grows butternuts, chestnuts, and filberts. Another grower at Clinton, located on 50° latitude, central B. C. with temperatures to minus 40° F., grows Japanese and black walnuts, also Pioneer almond. We are sure that the same temperatures with our conditions would kill most of our trees.

Recipes

J. U. GELLATLY

Walnut Honey Sandwich

1 Teaspoon crystallized Honey (the coarser the crystal the better) 3 Broadview walnut half kernels or quarters.

Place honey on one-half kernel, then stick the other half on the honey, making a small sandwich, or kernel covered ball of honey. This is a delightful confection.

Potato Nut Soup

1. Grate 1 tablespoon onion. 2. Grate 1 good-sized potato.

Place in double boiler, stir while adding boiling water, to a thin paste. Stir until cooked clear like corn starch pudding. Add hot whole milk to bring to creamy soup. At this stage add one-fourth cup filbert kernels. First put nuts through one of the new nut planing gadgets. These are better than the old grinder shredders or choppers, as shavings are so thin and soft they just melt in hot liquid. (Also delightful on ice cream or fresh fruit.) Have potatoes well cooked before adding milk or nut flakes. Cooking nuts too long sets up some chemical change that thins the creamy texture of the soup.

Description of Filazel Varieties[28]

[28] Since the Peace River hazel is apparently ~Corylus rostrata~ these filbert hybrids of Mr. Gellatly belong to a different category from the "hazilberts" of Mr. Weschcke and the "Mildred filberts" which had ~C. americana~ parentage.—J. C. McD.

J. U. GELLATLY

The name (Filazel) I coined to designate those crosses I had made, having the Peace River hazel as the mother tree and Craig and others of our large filberts as the pollen parent.

Peoka

Has thin shell. Clean, well-filled kernel. Is heavy cropper and free husker. Nuts mature early. Are well filled by August fifth with shells starting to brown. Fully ripe by August tenth to fifteenth.

Manoka

One of the best of my first selections. Very attractive, heavy cropper, well-filled kernels by August first, shells coloring by August fifth. Ripe and falling August fifteenth.

Fernoka

Good cropper of roundish nuts, having short open husks and good clean kernels.

Myoka In clusters 1 to 6. Has short open husks. Leaves color well in the fall. Has ornamental value.

Fairoka

One to 7 nuts in cluster in fancy frilled and rolled back husk. Nuts roundish, of fair size and color. Flavor, good. Leaves color well in fall. Has ornamental value.

Maroka

Medium-sized nut exposed in clusters 4 to 6. Open husk, folded back.

Ureoka

Medium size for Filazel. Thin-shelled roundish nut, 4 to 6 in clusters. Very short, partially closed husk.

Orvoka

Two to 5 nuts in cluster. Clean kernels. Husk half-inch longer than nut. Has open side. Good cropper.

Brenoka

Long husk like parent hazel, but lacking prickles of the wild. Medium sized nut in clusters, 1 to 4.

Eloka

Two to 4 in cluster. Medium sized nut with clean kernels in open husks.

No. 500

Four to 10 nuts in cluster. Has short open husk. Good-sized nut of Barcelona type. Is a good cropper of clean kernels. Shell heavy.

No. 502

Largest Barcelona type Filazel that has fruited to date. Clusters contain 4 to 8 nuts enclosed in heavy medium-length closed husks.

No. 503

One to 9 nuts in cluster, having clean, full kernels in thick shells enclosed in short open husks.

No. 505

One to 6 nuts in cluster, having closed, medium length husks. A good cropper.

No. 509

Two to 6 roundish nuts in long closed husks free of prickles so common on wild hazels. A good cropper. The parent hazels used for these crosses mature the nuts by the first of August and were winter hardy at-60° F. in Peace River, Alberta.

Other Hazels

Manchurian short bush hazel, distinctive clipped off top on leaf with some colored (of reddish hue). This bush retains leaves all winter, and would make a good protective covering for wild life. Has well-flavored, clean kernels fully developed by August seventh, 1950. Kernel is enclosed in heavy, squat shells encircled with distinctive short closed husk, as if folded together just covering nut. The leaf shape and markings carry through and appear in the young seedlings.

Experiments with Tree Hazels and Chestnuts

J. U. GELLATLY

Corylus jacquemontii

(Smooth Bark) India Tree Hazel

Tree No. 1. Location—N.W. corner Lot 6, subdivision Lot 487, Scions from Kew Botanical Garden, England. Top grafted on Craig filbert 10 feet from ground line. This made good annual growth and compatibly well adjusted unions, which after many years are still in line and not readily detected except by difference in color and character of bark—the grafted top being smooth and lighter of color than Craig stock. Although stocks were bearing when cut for grafting, and scions were from bearing trees and had catkins on when received, grafts were trained to take over and become the main growth and leading tree from the Craig crown. This grafted tree did not produce catkins or nuts for four or five years, but branches on the stock went right on bearing, as did also other Craig sections on same root crown or filbert clump used for grafting above tree hazel. At date of writing, and following the severest winter of the past 45 years, when temperatures dropped to -24° F., followed by brief, bright sunshine and rapid rise of temperature, all ungrafted filberts of over three to four inches in diameter are dead or nearly so, while suckers 2-1/2 inches in diameter and smaller are quite sound and making good growth. So, also, are the stocks or sections top grafted to the tree hazel—even the larger 4 to 4-1/2 inches in diameter trunks. I ask why, as by all ordinary results the grafted trees should have been the easiest damaged. This tree, and the other sections of filberts on same crown, had cropped for three years past, so that from that angle they should have been on an equal footing. Only a few clusters of nuts grew on this ~Corylus jacquemontii~ this 1950 season.

Data on tree size: Height 32 feet—was grafted about 10 feet above ground line. Circumference of tree—12 inches above ground is 15 inches. At 4 inches below the graft, it is 10 inches, and the same four inches above graft union, which is very uniform, and if this combination could be reversed we would have an ideal non-suckering stock for commercial filbert orchards. ~Jacquemontii~ also buds well on cork bark ~C. colurna~ tree hazel.

Corylus jacquemontii
Smooth Bark India Tree Hazel on
Cork Bark Turkish Tree Hazel

Corylus colurna Stock

Tree No. 2 Location—S.W. corner of Lot 6, subdivision Lot 487. Budded August 15, 1941, at six feet from ground line, to one inch two year growth. Two years later top was removed and bud made to take over leadership. From then on it made good growth. Removal of top was not done at one operation, but first year leader was cut one-third way through, on long slope from bud downward on both sides, and allowed to callus over one year. Second year leader was cut further and when callused, top was then removed. This treatment gave good coverage of wound on trunk. Tree bore first crop 1949, eight years after budding. Nuts 1/2 inch in diameter, moderate shell of roundish form, well filled, with good flavor, clean kernels. August 4, 1950—Tree has a base circumference at ten inches above ground of 18-1/2 inches—at six feet above, 14 inches—below union circumference is 14 inches, while four inches above union it is 11 inches. No evidence of any winter injury after taking a-24° F. temperature. No crop this year, but has a good crop of catkins showing for 1951.

Corylus hetrophyllia Japanese Tree Hazel

Tree No. 3. Location—N. W. corner of Lot 6, subdivision Lot 487. Scions from Kew Botanical Gardens, England, top grafted on Craig Filbert stocks 10 feet from ground line. Made very good union. Present circumference four inches below union is 7-3/4 inches, and four inches above union is 8 inches.

The bark on this graft is similar to the Craig on which it is growing but lighter in color. There is no winter injury in evidence at this date except a very much lighter crop than usual. Has small, oval, light-colored nut of good flavor and color—clean kernels.

Corylus colurna (Thin Bark) Turkish Tree Hazel, also Cork Bark

Tree No. 4. Source of Scions—Oregon, U.S.A. Top graft on Craig stock six feet above ground. This Craig filbert clump has several divisions. Main one now six inches above ground. Has a circumference of 20 inches, and just above this branches into four main limbs of similar size, which at a height of six feet were grafted—two to the thin bark above, and two to the cork bark type. The thin bark type have made very compatible unions—well healed over. The circumference four inches below the graft is now 9-1/2 inches and at similar distance above is now 10 inches. July, 1950:—These are bearing a few nuts, following a winter temperature of-24° F. Although the two branches worked to the cork bark type have no crop this season, they have over-grown graft unions, and the tops are oversize for stocks. Circumference four inches below union is now 7 inches, and at same distance above is 9 inches. Both these types have thick shelled roundish nuts which are hard to get out of the husks, and so far have many blank nuts. India tree hazels also contain many blanks and are very difficult to separate from the husks. Trees are all hardy and vigorous.

Best of 25 seedling ~C. colurna~ (cork bark tree hazels). Circumference twelve inches above ground line is 31 inches, and at six feet above ground is 25 inches. Height about forty feet. On August 3, 1950, I climbed thirty feet into upper branches to see if there was any crop, but none was to be seen, but heavy crop of catkins was developing for 1951. I have many hybrids from all of these tree hazels and filberts, nearing the bearing age, and they give interesting promise of new strains, as all sorts of crossing are evident.

Tibet Hazel (C. tibicia)

Vigorous grower, upright, good cropper, fair size round nuts. Clean kernels, nut clusters, 4 to 6 nuts in open medium husks. Nuts fall free. These clusters differ from usual run of filberts or hazels in that each husk is separate on short neck from center of cluster.

Timber Type Tree Chinese Chestnut (Castanea mollissima)

Seed secured direct from China. All select large nuts. So far, only a very few produce trees that yield nuts of as large size as those planted. All that have are timber type trees. All the bush or dwarf spreading type trees yield small to medium-sized nuts, all of good quality and flavor. (Selection to 1950 date referred to.)

One Chinese Chestnut Selection Named

Skioka. Most promising timber type to date of this group of seedlings. Has one straight trunk 38 feet tall, base circumference 1 foot above ground, is 22 inches; and 6 feet above ground line circumference is 15 inches. To date, tree is sparse cropper. Started bearing in 1945, with three very large sized nuts in large fleshy burs. It has borne every year since, with gradual increase in number. In 1949 it matured 12 large nuts of 1-5/8 inch diameter. A good peeler and solid kernel. I have four other trees of similar size and all winter hardy this past winter, at 24° below. Skioka is the most promising to date of the four as to size of nut.

Bush or Peach Tree Type of C. mollissima

Of this type I have about 30 trees. Many seem 100% hardy and came through in good shape. However, for some years they, with the tree type, seemed to be having trouble with some soil deficiency or else some excess of soil salts which caused a lot of leaf fading, followed by browning and drying up. Some trees almost defoliate themselves, while others nearby and alongside are O.K., possibly due to individual tolerance of conditions.

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DR. MacDANIELS: The first paper after recess has to do with the varieties of hickory nuts. I know of no one who is in a better position to talk on this subject on their performance here in this part of New York State than Gilbert L. Smith of Millerton. He began a number of years ago topworking trees on a hillside and propagating trees as a nurseryman and probably is, as far as I know, one of the best men in nut shade trees and hickory varieties that there is anywhere in the country. Mr. Gilbert Smith.

MR. SMITH: I am no good at making a speech, so I am just going to read this. This is our experience with hickory varieties so far. That's just up to date, but not any further.

Our Experience with Hickory Nut Varieties

GILBERT L. SMITH, Route 2, Millerton, N. Y.

Because we are located so far north, 41° 45' North Latitude, we have paid particular attention to the earliness of ripening of the various varieties of hickory.

While we have living grafts of more than a hundred named varieties of hickory, only a comparative few have started to bear nuts. Of these, I will give a brief discussion, starting with the earliest and going through the list in order of their ripening.

ANTHONY, shagbark—We believe that this is Anthony No. 1 but as there are four or five varieties named Anthony with a number following the name, we are not absolutely sure. This variety has ripened very early with us. It is rather small but cracks very well and has borne well with us. We consider it to be an excellent variety.

WESCHCKE, shagbark—Is our second earliest variety so far. It is also rather small, with a distinctive shape, tapering from a rather broad blossom end to a sharp point at the stem end. Our graft has had one very good crop, but it is younger than many of our other grafts. We consider it a very good variety.

CROWN POINT, shagbark—Is our third variety in order of ripening. This is a rather small nut with some of them being very small; that is, there is quite a variation in the size of the nuts. It cracks quite well and is of very good quality. It has also borne as well or better than any other variety we have under test. We have never propagated it for sale as we have hardly thought it quite good enough.

In fourth place of ripening order, we have four ties, namely; Bauer, Cedar Rapids, Hines, and Independence.

BAUER, shagbark—Has borne well, is of good size, good quality and cracks well. It is also a very good shaped nut. We consider it to be one of the very good hickories.

CEDAR RAPIDS, shagbark—While our graft of this variety has borne but moderately, we consider it to be a very good variety. It is of good size, cracks well, is of good quality and attractive shape.

HINES, shagbark—While our graft of this variety has borne well, cracks well and is of good quality, it is so small that we have never propagated it for sale.

INDEPENDENCE, shagbark—The nuts of this variety are so small that we have paid little attention to it.

FOX, shagbark—This variety is in fifth place in order of ripening.

Fox won first prize in the 1934 N.N.G.A. contest. But there is a deep mystery connected with this variety as subsequent crops, grown on grafts, have not produced nuts of such top qualities. There have been many theories advanced but no one has solved the mystery yet. One theory is that there is bud variation in the parent tree and that Mr. Fox, quite naturally, cut scion wood from the lower parts of the tree, which were most readily accessible. During the war, I secured a special allotment of gasoline and made the trip to Fonda, N. Y., to cut scions from all parts of the tree. The scions from the various parts of the tree were labeled separately and were grafted on stocks in our test orchard. While not all of these grafts lived, we have living grafts from nearly all parts of the tree. I note that at least one of these grafts has nuts on it this year. If there is bud variation we hope that we will have at least some grafts of the superior Fox nuts.

In spite of all this, Fox is an excellent variety, being of good size, cracks well, and is of very good quality. While it is fifth in order of ripening, it is still an early hickory and will succeed considerably farther north than our location.

In sixth place we have two varieties, namely; Clark and Stocking.

CLARK, shagbark—Our graft of this variety has borne well, the nuts being of good size, crack well and are of good quality. We consider it to be a very good variety.

STOCKING, shagbark x bitternut—While our graft has grown very well, it has produced but very few nuts. We were not very greatly impressed with these.

In seventh place in order of ripening, we have two varieties, Camp No. 2 and Stratford.

CAMP NO. 2, shagbark—We did not find this variety good enough to interest us very much. Subsequent crops may show up better.

STRATFORD, not sure whether shagbark or hybrid[29]—Our Stratford graft has been poorly tended and has had little chance to show its merits. So while it has an excellent reputation, we know very little about it. However we have several good sized grafts of it, growing in nursery row, which have several nuts on this year, so we will find out more about it soon.

[29] It is a bitternut hybrid.—Ed.

In eighth place we have three varieties; Proper, Shaul, and Wilcox. While being in eighth place, these are still medium early varieties.

PROPER, shagbark—This is a little known variety, our graft is rather young and we have had too few nuts to form any opinion of this variety as yet.

SHAUL, shagbark—While this is a very good nut, being of good size, cracks well and of good quality, our graft on shagbark stock has grown slowly and it is the one variety so far that we have found will not do well on our bitternut stocks.

WILCOX, shagbark—So far this is our favorite variety. The graft has grown into a fine tree and has borne good crops of nuts which are of good size, crack almost perfectly and are of very good quality.

MINNIE, shagbark—While we have not had a crop of this variety since starting to keep a ripening record, it ripens about the same time as Wilcox and is a very good variety.

Ninth on our list we have two varieties; Davis and Peck Hybrid. It so happens that I discovered both of these varieties.

DAVIS, shagbark—First prize winner in the New York and New England Contest of 1934. Incidentally, a sample of Fox nuts was awarded tenth place in this same contest. You will note that this was the same year in which Fox won first place in the N.N.G.A.

Davis has pretty well lived up to expectations. Grafts of this variety are rapid growers. It is the only variety we have ever succeeded in making live on pignut stocks. While the grafts are slower growing on pignut stocks, they have lived for several years and have borne nuts. But as the squirrels have stolen all of the nuts, we do not know how they compare with the nuts grown on other stocks.

Our grafts of Davis have borne well, the nuts are of good size and crack well, although not as well as those of Wilcox. It is also of very good quality. We consider it to be a top rate nut.

PECK HYBRID, shagbark x bitternut—The nuts of this variety are large, thin shelled, crack well and are of good quality. It also bears well. The drawback is that only about one third to one half of the nuts are well filled. I can take freshly shucked nuts of this variety and by placing them in water can pick out a sample of nuts that are just about as good hickory nuts as you can find anywhere, but these will be only about one third of the nuts involved. For this reason we have never propagated it for sale.

In tenth place we have three varieties; Berger, Strever, and Triplett.

BERGER, shellbark—While this variety is quite small for a shellbark, it is quite large when compared with the shagbarks. Our graft of the Berger has borne fairly well, cracks well and is of very good quality. Incidentally our graft is the true Berger. There was some mix-up with the Berger wood, and some who thought they had Berger found that they had something else when their trees started to bear.

STREVER, shagbark—The original tree of this variety is growing near Pine Plains here in Dutchess County, on the Old Strever Homestead. This property was later sold to people named Owre, who tried to have the variety named after them. I believe that Strever is the more proper name.

While this variety is of good size and quality, it has not cracked quite well enough to rate it as a top flight hickory.

TRIPLETT, shagbark—This is a large shagbark which cracks well and is of good quality. Our graft bears well. I believe that it was discovered by Dr. Deming and the late Mr. Beeman. This is a variety which can well bear considerable attention in the future. We are propagating some of the trees for sale.

In eleventh place we have nine varieties, namely: Bridgewater, Griffin, Hagen, Harman, Kirtland, Lingenfelter, Manahan, Oliver, and Wampler.

BRIDGEWATER, shagbark—A large fine variety, cracks well, yields well and is of good quality. This is another discovery of Dr. Deming's and Mr. Beeman's. We have started to propagate it for sale.

GRIFFIN, shagbark—I have mislaid my comments on this variety and cannot remember much about it, except that it is of good size and bears well.

HAGEN, shagbark—We have not had enough nuts of this variety to enable us to form an opinion of it.

HARMAN, shagbark—A large nut. We did not think much of our first crop of this variety but the second crop was very good.

KIRTLAND, shagbark—This is a fine large nut, but with the one good crop, we have had, only about half of the nuts were well filled. The other half were floaters, only partly filled.

LINGENFELTER, shagbark—Here again we have had too few nuts to enable us to form an opinion. Mr. Reed thought very well of it.

MANAHAN, shagbark—This nut is of southern origin and I fear that we are too far north for it. However we have had one crop that was very good.

All other crops have not been matured. It is evidentally a very good nut where it can be grown.

OLIVER, shagbark—Too few nuts to form an opinion.

WAMPLER, shellbark—Too few nuts to form an opinion.

In twelfth place on our list, in order of ripening, we have Bowman and Redcay. These are both shellbarks and the nuts have not been well filled, as borne on our grafts.

In last place on our list, we have a southern shagbark, Booth, and two hicans, Bixby and Burlington. We have not been able to form an opinion of Booth. Bixby and Burlington have, so far, been very shy bearers and the nuts have not been well filled. They are of very large size and very excellent quality.

The time elapsed between the earliest and latest ripening of these different hickory varieties was 36 days. The time between the different steps were about three days. I do not give the dates because they will vary from year to year. In early years, Anthony has been ripe very early in September.

Summarizing this report shows that our tests so far indicate that the following varieties are good and well worthy of propagation: Anthony (probably No. 1), Weschcke, Bauer, Cedar Rapids, Fox, Clark, Wilcox, Minnie, Davis, Berger, Triplett, Bridgewater, Manahan (farther south). Instead of listing these 13 varieties alphabetically or in order of their merits, I have listed them in order of their ripening, earliest first, and so on. Those varieties in the first half of the list can be grown in locations considerably farther north than our location, which is 41° 45' North Latitude, while those in the last half of the list are not likely to be adapted to locations farther north than ours.

You will note that five of these varieties are not well known, but are good varieties. They are, namely; Bauer, Cedar Rapids, Clark, Triplett, and Bridgewater.[30]

[30] The Bridgewater pollenizes the male-sterile Weschcke variety in Wisconsin. See Mr. Weschcke's discussion, pp. 193-95 in NNGA Report for 1948.—Ed.

This is only a preliminary or progress report, and should not be taken as final in any respect. Neither does it cover all or near all, of the top-rate hickory varieties. For instance, you will note, the variety named Glover has not been mentioned. This is because our grafts of it have not started to bear yet, so we have no comparable basis for including it in this report. Yet there can be no question as to the merits of Glover, for it is one of the very best. There are, no doubt, many other very excellent varieties not mentioned here.

The hickory is the slowest growing, takes the longest to start to bear, is the nurseryman's headache (it taking about five years to grow stocks large enough to graft or bud, during which time they should have been transplanted at least twice to develop a better root system), they are about (the hardest of the nut species to transplant and their nuts are one of the smallest of the nut species only the filbert and the chestnut being as small). Yet because of their delicious flavor and other good qualities, hickories are probably the favorite nut of more people than any other of the nut species that can be grown in the northern part of this country.

(Applause.)

DR. MacDANIELS: I think we need more reports of that kind to get us oriented with our hickory varieties. I think when we get through with the walnut survey that the hickory nut survey would be next.

MR. CORSAN: Hickory was Dr. Charles S. Sargent's favorite tree, and he planted poison ivy under all of them, and it's there yet and they can't get rid of it. He wanted to keep the boys from gathering the nuts.

DR. MacDANIELS: I have poison ivy under some of mine, but not for that purpose.

MR. McDANIEL: It grows under all good trees.

DR. MacDANIELS: The next paper is one which George Slate kind of foisted off on me. He came around and said he thought something more should be said about the butternut and asked if I would get out a report and discuss the standards for evaluation. That is the reason for this paper, which I will read. It will take only about ten minutes.

How About the Butternut?

DR. L. H. MacDANIELS, Ithaca, New York

The purpose in presenting this paper is to summarize what is known about the butternut in the light of my own experience, and to find out from you in discussion what additional facts are available and what some of the problems in the culture of butternuts may be. A good summary by S. H. Graham is to be found in the 34th Annual report of the Northern Nut Growers Association, and short reports appear elsewhere. In general, however, judging from the proceedings of this Association, the butternut has not received much attention through the years. The lack of interest in the butternut indicates unsatisfactory experience with this nut on the part of those who have tried to grow and use it. An analysis of its good and bad characteristics is in order.

Of all the species of nuts with which the Association is concerned, the butternut is the most hardy and the most likely to succeed on poor soil. In general, the trees are easy to transplant, are early bearing, sometimes within two years from the graft, and are easy to grow. The flavor of the butternut is very distinctive and palatable, and usually much more flavorful than similar nuts derived from the Japanese butternut and the heartnut. Some people consider the butternut flavor the best of all nuts.

On the other hand, the butternut has a reputation for being short lived because of susceptibility to various diseases. The seedling trees which are usually sold are slow in bearing. The common wild nuts are hard to crack with a hammer, and the better named varieties are not well known or widely grown. The trees also have a reputation for being difficult to propagate. Of these faults, probably the difficulty of propagation and cracking are the most important in restricting its use.

Botanically the butternut (*Juglans cinerea*) belongs to a group of species within the genus Juglans that bears its fruit in long clusters or racemes, as contrasted with the walnut group which bears nuts singly or in clusters of two or three. The butternuts also have the fruit and leaves covered with sticky hairs instead of being smooth. The group is further characterized by having a cushion of hairs above the leaf scars and pointed terminal buds on the twigs. Other species within the group are the Japanese butternut *J. Sieboldiana*, its variety *cordiformis*, the heartnut, and several less well known species including *J. mandshurica* and *J. cathayensis*, both native to central Asia. These closely related species apparently hybridize with each other, but accurate information as to the nature and extent of such hybridization is not available.

The natural geographical range of the butternut covers a broad area of Northeastern North America, extending from New Brunswick southward to the mountains of Georgia and westward to Western Ontario, Dakota, and Arkansas. In this range it is most frequent in calcareous soils, reaching its best development in rich woodland, but persisting on poorer upland soils also. It thus has the most northern range of our native nut species, along with the Pignut, *Carya glabra*, and one species of hazelnut, *Corylus rostrata*. The other related species are of variable and uncertain hardiness and are not reliable in this northern range.

It is recognized that the butternut has little commercial value except as it is used in the New England states, particularly in Vermont, where it is combined with maple sugar in making maple-butternut candy. Anyone who has travelled through the New England states is familiar with the roadside advertising of this excellent product. On the general market, butternut kernels are not sold in quantity comparable to those of the black walnut, but are somewhat comparable to the kernels of the hickory which also do not have a commercial outlet except locally.

The greatest use of the butternut is, and will continue to be, for the home grounds and local consumption. I think it is highly probable that if the easy cracking varieties already named were better known, they would be much more widely planted. The common wild butternuts are really difficult to handle. They crack only after considerable hammering with a heavy hammer and then, when cracked, the kernels shatter to such an extent that recovery is very unsatisfactory for the labor expended. After butternuts have been gathered from the wild with some enthusiasm during the fall months, they often remain in the cellar or attic without ever being used. Even the squirrels and the rats will not go to the bother of extracting the kernels if other nuts are available.

For best results the nuts are usually cracked with a heavy hammer, the nut being held vertically against a solid vice or block, so it can be hit on the end. A glove to protect the fingers holding the nut is useful if many are to be cracked. Good results can be secured by holding the nut on its side and tapping it on the suture. This, however is difficult, as it necessitates shucking the nut and even then it is difficult to identify the suture.

Through the years many varieties of butternut have been named. Mr. R. L. Watts in the 35th annual report of the Association lists 26 names, and I am sure there are others. I personally have had experience with only three or four varieties. One of these, the Crax-ezy, has borne good crops and the nuts crack well. Another one, which I have named the Johnson, coming from Tonawanda, New York, cracks well but is a smaller nut. At one time I had Thill variety topworked on *Juglans Sieboldiana* stock, but the stock was killed by cold winter. Samples of Kinnyglen and Mandeville were furnished by Mr. Graham for testing. We do not, however, have any comparative rating of many varieties based on comparative tests, nor are there recognized standards of quality.

In order to set up standards of quality for butternuts, the following tentative schedule for judging has been worked out along the same lines as the schedule for judging black walnuts. Twenty-five nuts are used in a sample and the score is made up of the weight in grams of the kernels recovered on the first crack, plus total weight of kernels divided by 2, plus 1/2 point for each whole half kernel recovered. A nut should not be considered worthy of propagation unless practically all of the kernels come out in whole halves.

Proposed Schedule For Testing Butternuts

25 Nut Samples

Score = Wt. kernels first crack + total wt. kernels ÷ 2 + no. whole halves ÷ 2.

Variety	Weight Kernels 1st crack Grams	Total Weight Kernels Grams	No. Halves	Score	Remarks
Kinnyglen	52.0	57.5	36	98.8	
Crax-ezy	48.0	56.0	44	98.0	
Mandeville	53.6	66.0	10	91.6	
Johnson	38.5	45.5	40	81.3	
Seedling No. 1	36.5	45.0	7	62.5	
Seedling No. 2	26.0	43.0	22	58.5	
Seedling No. 3	20.0	44.5	10	47.3	

In this schedule the crackability of the sample is measured by the weight of first crack and the number of halves. The yield of kernels is measured by the total weight of kernels in the sample. The first crack includes only those kernels that either fall out or can be removed easily with the fingers. The remaining kernels are rescued with a pick or by recracking. In my judgment, the score accurately measures the merit of the samples. In the Mandeville, the large size is measured by the weight of kernels which in part offsets poor cracking quality. Poor cracking is usually caused by the edges of the halves being curved so as to be bound in the shell. Much more testing should be done to determine the value of the schedule.

Opinions regarding the ease of propagation of the butternut differ, but mostly it is considered difficult to propagate, with often complete failure. This merely means that the matter is not well understood. In my own experience I have had just about as many failures as successes, and must confess that I do not have much idea of what has been responsible for either success or failure. Best results have been secured by using inlay or bark slot grafts on stubs about 2 inches in diameter. This agrees with the experience of Mr. Burgart, of Michigan, and Mr. Weshcke, of Minnesota, who report that grafts must be made several feet from the ground and not at the crown.

Shield budding has apparently not been satisfactory. Mr. D. C. Snyder writes that chip budding is more successful. It is recommended by others and I agree that grafting should be done early, just as growth starts rather than later when trees are in leaf. Special care must be used in tying the new shoots of the graft to braces to prevent breakage by wind or birds. The butternut wood is very brittle and the grafts are often lost by breakage. The whole matter of butternut propagation merits further careful study.

Butternut varieties may be grafted on black walnut, butternut, or *J. Sieboldiana* stocks. Mr. Burgart, Mr. Weschcke, and Mr. D. C. Snyder consider black walnut to be better than the others, giving a more vigorous long lived tree. Varieties on butternut stocks are apparently relatively short lived and *J. Sieboldiana* stocks have a different growth rate and are not hardy. Mr. Burgart uses bark slot grafts on black walnut seedling stocks, 2-3 years old.

Butternut trees on their own roots transplant relatively easily because there is no taproot as with the black walnut and the hickory, and there are many fibrous surface roots that can be lifted when the tree is dug. Black walnut stocks are not difficult to manage, particularly if the taproots are cut on the seedlings. Culture is no special problem. Mulching and supplying nitrogenous fertilizer is good practice.

The butternut has the reputation of being susceptible to disease and hence being short lived as a tree. Whether or not this is actually the case is perhaps questionable. Many butternut trees, particularly those in favorable situations of soil and moisture, live to be of large size and old age. Trees on poorer, thinner soils apparently die off earlier than those under better conditions. In any case, it is well recognized that the butternut has a shorter life span on the average than the black walnut, which frequently lives to a large size and old age. There are two common diseases of the butternut. One is leaf spot caused by the fungus *Marsonia*, which defoliates the trees fairly early in the season and probably predisposes them to injury from other fungous attack. This is the same leaf spot that attacks the black walnut leaves. The other disease, which may cause trouble, is a fungous walnut blight known more specifically as Melanconis blight. It has not been established that this disease is an active parasite. The evidence indicates rather that it attacks trees that are already somewhat weakened by defoliation or other injury. It is a fact that many of the dead limbs on butternut trees are found to be affected with the disease. It is a matter of observation that trees growing under favorable conditions are less damaged by the disease than those growing under poor conditions of soil and water, therefore, keeping trees vigorous is good practice.

As with other nut tree species, there are troublesome insects. One of these, the butternut snout beetle or curculio, attacks both the butternut and the Japanese walnut. Control has apparently been secured by dusting foliage with DDT. Sometimes the leaves of butternuts are badly distorted with galls caused by mites. The bunchy top or witches'-broom caused by a virus, that is serious on the Japanese walnut, *Juglans Sieboldiana*, does not appear to be so virulent on butternut. This, however, is a matter of personal observation and is not based on a thorough study.

In conclusion, let me say that in my judgment, the butternut is worthy of more attention than it has had so far received, particularly by home owners in the northern states who would like to have trees in their yards that will bear nuts under conditions that are unfavorable for most other kinds. If it were publicized that varieties are available that will crack out in halves with relatively little effort, the chances are that with these facts in mind those interested in nut trees would give the butternut much more attention. The difficulty at the present time seems to be related to a lack of knowledge as to the relative merit of different varieties and a scarcity of trees because of difficulty of propagation. If we have time and the chairman will permit, I would welcome comments on the propagation problem and would also like to obtain any information on the merit of the named varieties. Let me also state that if any of you have a sample of 30 nuts of any named variety in this or last fall's crop that you can spare, I would be much pleased to have you send it to me for testing.

Discussion

MR. STOKE: It grows in New Brunswick, and I have had specimens from north of Lake of the Woods.

MR. CORSAN: They grow at Brooks, Alberta. I have the Helmick and it grows 14 to the cluster, has a thin shell and heavy meat, and the leaves are persistent. They don't drop off the first of September. That's the Helmick. It's grafted on black walnut stock, and the black walnut stock comes up like that (indicating) and the Helmick recedes.

DR. MacDANIELS: The black walnut overgrows it. There are about 40 varieties, and I would like very much to get hold of any of the samples I can get.

MR. CORSAN: Go up to Silver Bay, Lake George, and on the shore there the Indians have bred the butternut, and it's 10 to the cluster among those trees by Silver Bay, Lake George, New York. Ernest Thompson Seaton and I examined that grove years ago.

DR. MacDANIELS: Wish we had them where we could get at them. Any other comment on the butternut?

MR. McDANIEL: The Helmick is considered to be a "butter-jap" seedling of heartnut, possibly the other parent was a butternut.

DR. MacDANIELS: That is something we will have to decide in the Association, whether or not we are going to throw in these hybrids and the heartnut along with the butternuts in standards or try to keep them separate.

MR. CORSAN: Hybrid heartnut cross is very, very superior in every way to the butternut in my estimation, except for hardiness.

MR. STOKE: That is a hybrid. I have it. The Mitchell hybrid.

DR. MacDANIELS: The ordinary run of seedlings are not worth keeping, no question about that, and it's too much work to recover the kernels.

There are several announcements I'd like to make. One has to do with this hall. It is the American Legion hall, which they do not charge rent for. They do, however, and will expect some sort of a token of appreciation that will be fairly substantial. There is no provision for that in the budget, so any of you who are feeling a little mellow and flush, if you want to approach the treasurer with a contribution towards the use of this hall, that will be appreciated; otherwise, the matter will have to be settled out of the treasury as such.

MR. CORSAN: How about a dance in this hall?

DR. MacDANIELS: If we stay over, we might do something like that.

Then there is the other matter, and that is the prize for the proposed Carpathian walnut contest. There is no prize money available at the present time. If any of you wish to provide a first, second, or third prize, we might even tag it with your name, if that would be possible.

I think probably they will be able to get some publicity backing through farm papers and what not, but still if we have a backlog of prize money, why, that's much to your advantage.

Do you want to say anything further on that, Mr. Chase?

MR. McDANIEL: Mr. Sherman, I believe, has a word.

MR. SHERMAN: Not in this connection.

MR. PATAKY: Do any of the members here have shelled butternuts or hickory nuts that they would sell? If they do, I'd like to get their names and get in touch with them. I do have a demand for some shelled butternuts which I have trouble getting, and I do have trouble getting shelled hickory nuts. It is for the Wideman Company out of Cleveland. I got shelled butternuts before the war, but since the war they don't have the trade, but if they could get them, I think that would be the company that would take them. The Wideman Company of Cleveland, Ohio. They are a big wholesale house. Write to Christ Pataky, Mansfield, Ohio, R.D. 4.

MR. KINTZEL: Do you sell them in the shell?

MR. PATAKY: I do sell them in the shell, too, but there are a lot of people who won't buy them in the shell. We do have a demand for them, not too much on the butternut, but we do have for hickory nuts. I think we could sell a lot more hickory nut meats than hickory nuts even at the difference of the price. I know the price was quite high before the war. They paid somewhere around a dollar a pound before the war for shelled ones, and we even sold them at a profit for that, and we haven't been able to get any since the war. I don't know what happened, whether the kids are too busy playing basketball or football.

DR. MacDANIELS: They get too much for mowing lawns.

MR. WEBER: There is a nut crackery at Mitchell, Indiana. The man who cracks them cracks hickory nuts and puts them out in his name, John Eversol. Mr. Wilkinson can tell you exactly what his name is. He was down there last year. He is cracking walnuts, and in addition cracks hickory nuts and puts them in fine shape.

MR. CORSAN: Isn't it true that nuts have more Vitamin E than any other food in the world, and isn't Vitamin E the greatest antidote against anemia?

DR. MacDANIELS: I wouldn't know. You have a medical man here?

DR. WASHICK: I don't think you are right.

MR. CORSAN: In the West they say Vitamin E is a cure for anemia and they are having wonderful success, and they claim there is more vitamin E in nuts than any other food. I don't know, they are keeping me alive.

~Editor's Note~: Green walnuts are rich in Vitamin C. See 1942 Report, page 95.

DR. MacDANIELS: You are Exhibit 1.

I think Mr. Salzer has slides he wanted to show this afternoon.

MR. SALZER: I had a few. Perhaps we can use those blankets and just fix up, perhaps, a few of these windows in front, and I think we could probably show the slides.

DR. MacDANIELS: If you can leave the blankets here for a short time, we will get them later.

Any other questions?

I think our lunch is ready for us downstairs. We will come back up here at one o'clock.

(Whereupon, at 11:50 o'clock, a. m., the meeting was recessed, to reconvene at 1 o'clock of the same day.)

TUESDAY AFTERNOON SESSION

DR. MacDANIELS: Calling the afternoon session to order.

This afternoon I am going to turn the gavel over to our good friend, Spencer Chase, to carry on.

MR. CHASE: Thank you, thank you.

All of us are interested in the various experiment stations doing more work with nut trees, and we are very fortunate this afternoon in having two experiment stations represented, and we will first hear from Bill Clarke from Penn State, who will talk on, "Progress in nut culture at the Pennsylvania State College." Mr. Clark.

Progress in Nut Culture at the Pennsylvania State College

W. S. CLARKE, JR., State College, Pennsylvania

Work in nut growing at the Pennsylvania State College was formally begun in 1946, when a project on this subject was approved by the college authorities. A few acres of land were set aside for this work, and the following spring about half an acre was planted with a few nut trees of different species. At the present time an area of about twenty acres is set aside for nut plantings, although a few spots on this land are not plantable on account of rock outcrops.

We now have out in the field sixty black walnuts, all but three of them named varieties, which were received from Tennessee in 1949. Seventeen varieties are represented in this collection.

In the nursery are more than 200 seedling black walnuts. These were planted from nuts gathered from local trees in the fall of 1946. They were transplanted at the end of their first season and have remained in their present position for three years. They were planted largely for the sake of experience in handling the nuts and the young trees. Some of them have been grafted, and this year a few grafts of Thomas and Stabler were successful. On account of their size, all these trees will have to be taken out at the end of the present growing season.

About twenty Persian walnuts have been received from the United States Department of Agriculture. These are all budded trees, the buds having been taken from special selections with the best nuts from trees originally introduced from northern Europe and central Asia. Three out of four seedling Persian walnuts and one out of two Japanese walnuts planted in

1947 have survived and are included in our planting. One named variety of butternut is in our collection, and a number of seedlings in our nursery.

It has been our experience that walnut trees can be moved rather easily. The percentage of loss in transplanting has been negligible. On account of an emergency, this spring we had to move several walnuts which were already in full leaf. Some of the leaves were trimmed off, and the trees have survived and have even made some additional growth.

On our grounds is one Chinese chestnut left from a planting of eight in 1930. It was killed back to the ground in 1934 after winter temperatures of close to 30 degrees below zero, but it has since grown up to be a tree of moderate size. It suffered considerable injury to buds and twigs in 1948 from temperatures down to 23 degrees below zero, but has since recovered. In several years it has borne a crop of burs, but no other tree is available for cross-pollination, and the nuts have seldom filled.

Twelve seedling Chinese chestnut trees from different sources have been planted, and an area of several acres has been set aside to extend the work on chestnuts.

A start has been made toward a collection of filberts. Five named varieties of European filberts were planted in 1947. All have suffered from winter injury, but only one tree has been killed outright. Very few nuts have been produced. About 25 seedlings of European filberts and 25 of the American were received from Tennessee two years ago. About 90% have survived and are growing nicely.

Several other species of nuts have been tried without success. Two trees of the red hickory were set out several years ago, but they failed to leaf out. Four young trees of the golden chinkapin of the Pacific Coast were planted and grew well the first summer, but all four were killed by the first freeze in the fall. About a pound of nuts of the Turkish tree hazel were planted several years ago; these failed to come up the first year. The next winter the mice and rabbits discovered them and ate up most of the planting. A few germinated, but most of these were lost in transplanting, and today only two are left of the entire lot.

MR. CHASE: Thank you, Mr. Clarke.

(Applause.)

Discussion

MR. SHERMAN: I'd like to say, just before you leave this subject, that the speaker barely mentioned the fertilization experiment that was started in Pennsylvania on black walnuts. I think the members of the nut survey stuck their necks out and got their heads hit a little bit when we said that the black walnut as an orchard industry in Pennsylvania was sick. We hadn't been able to find crops of black walnuts. We found individual trees, but we couldn't find orchards of black walnuts, and as a result of that, this fertilization experiment was started, in a 55-acre black walnut orchard with Ohio, Stabler and Thomas varieties.

The owner, Truman Jones, said, "I don't care what you do with the Stablers, you can't hurt them, anyway; they are no good to begin with." But this orchard, evidently from all outward appearances, has been growing very slowly for quite a number of years. It isn't the size it should be, and we think the main trouble there is lack of fertility, and that's the reason why this fertilization experiment was started.

It's quite an ambitious experiment. It takes in about 93 trees in the center of a 55-acre planting of black walnuts. They haven't had a crop, I think, for five or six or seven years. They don't have a crop this year, but we are hoping that some of them next year will have a crop, but if not then the year following.

They are asking about the cultivation. There has been no cultivation there in the orchard for a number of years. It's down in a pretty heavy bluegrass sod. In a portion of that we put the disc in on the tractor and disced and redisced until we got what we thought was a pretty fair seedbed. They found that vertical profile a mixture, and we are hoping to have clover sod instead of bluegrass sod. That's combined with fertility work. I won't take time to go into that, but I think this group is interested in knowing that there is quite an extensive fertility experiment on black walnuts to see why the large plantings are not producing.

I might say in this connection, Mr. Hostetter isn't here this afternoon, hasn't been here, but he has a dandy bang-up nice crop of nuts this year, and Ohio and Thomas are his main varieties.

MR. CRAIG: Did he use any fertilizers?

MR. SHERMAN: Yes, the fertilizer was disced in, and he tried to disc under that bluegrass sod and get that rotting under there. There are quite a few ramifications to that program.

MR. CORSAN: Did you mention Turkish tree hazel?

MR. CLARKE: Yes, we have two trees of it left.

MR. CORSAN: It takes two years to sprout from the time you plant the seed. Have you tried the European beechnuts in your locality?

MR. CLARKE: No, we haven't.

MR. CORSAN: It will produce far more than the American beechnut and is more successful in every way. They can be gotten from Holland quite cheaply. They sell the European beech, and they are beautiful and loaded with nuts and the Europeans think far more of them than the Americans do. The cut-leaf beech is an European beech, and I have seen the tree in Southern Michigan and at the Old Soldiers' Home at Dayton, Ohio, loaded with nuts. And frequently, not just once in every 13 years, like our beechnut. And they are a bigger nut.

Nut Tree Culture in Missouri

T. J. TALBERT, University of Missouri, Columbia, Mo.

The wide interest now being shown in the planting of nut trees throughout the State emphasizes the need of information on nut culture. Although nut trees may be grown with less care and attention than fruit trees, yet to be successful in starting plantings a knowledge of successful practices developed by the Missouri Agricultural Experiment Station at Columbia should prove of great value.

The information which follows applies particularly to the native black walnuts, butternuts, hardy northern pecans, hickories, chinkapins, and hazelnuts. All these nut plants are native to Missouri and may do well if given proper attention in the various districts of the state to which they are adapted.

NUTRITIVE VALUE OF NUTS

Nuts are now given in the diet a higher rating than ever before. This is true because recent studies in nutrition show that they supply not only the elements needed for health and growth, proteins, oils, and carbohydrates but also an abundance of vitamins A, B1, and G. In fact, the nuts compare very favorable with meats in rankings for the above vitamins. Most of the nuts are especially noteworthy in high vitamin A and B1 content. It is also believed generally that nuts contain nearly all of the mineral essentials demanded for the promotion of healthy nutrition.

Moreover, nuts are usually palatable in the raw stage and are prized most highly for dessert purposes. The black walnut is particularly outstanding because it retains its flavor after cooking. Nuts now have a very extensive use in the preparation of confectioneries, cakes, breads, and salads. They enhance the flavor of many other foods.

The value of nuts as food accessories has long been recognized. They also supply so much body fuel in so compact a form that they are particularly well suited for the use of mountain climbers, "hikers," and even soldiers engaged in long marches and maneuvers.

USES FOR NUT TREES

~As Shade Trees~—If during the past 40 or 50 years, a large portion of the shade trees planted had been nut trees like the native walnut, pecan, hickory, chestnut, and chinkapin of the better varieties, it is easy for anyone to see that great benefits would have resulted.

~For Highway Planting~—No other native trees lend themselves so admirably to highway use as the so-called northern or native pecan, the black walnut, and the hickories. These nut trees are all generally well-shaped, reach considerable heights particularly on fertile soils, are stately in appearance, and add beauty and attractiveness to the landscape wherever they are grown.

SOILS AND FERTILIZERS

~Soils Needed for Good Growth~—The nut trees adapt themselves to a very wide range of soil conditions. In fact, few other trees are capable of such a wide range of adaptability to soil types. The uplands usually planted to corn and wheat and the flood plains of the river basins may both be well suited to nut growing.

For good growth and production deep well-drained soils are required. Under proper conditions the trees develop rapidly, have an extensive root system, and eventually may reach a great age. Furthermore, nut trees cannot grow successfully on wet poorly-drained land where water stands on or just beneath the surface a considerable portion of the year. Lowlands which may be found well adapted to the growth of willow and gum trees, may be too wet and sour for the growth of nut trees. It would also be well to avoid dry, very thin, and very sandy soils.

In their native range the pecan, hickory, and walnut thrive on the alluvial soils of the Missouri and Mississippi River Valleys. They grow well also on the upland sandy loam soils adapted to the growth of corn, oats, and wheat. All of these nut trees are usually influenced more by the fertility, humus, and moisture content of the soil, than by any particular soil type.

~Fertilizers for Nut Trees~—The deep rich alluvial soils of river and creek valleys do not present the same fertilizer problems as light and heavy upland soils. Manure supplemented with superphosphate at the rate of about 20 to 30 pounds to a ton should prove to be a satisfactory fertilizer on depleted soils. It is spread in a circle around the trees extending out about twice the spread of the branches and plowed or harrowed into the soil. A moderate application would range from 8 to 12 tons to the acre.

Leguminous cover crops are particularly valuable for building up the nitrogen and humus content of the soil when plowed under. Their judicious use with non-leguminous cover crops and supplemented with commercial fertilizers to increase the tonnage for plowing under, will usually bring good returns in growth and production.

CARE OF THE PERMANENT PLANTINGS

Since but few diseases and insects attack nut trees in Missouri, very little if any spraying work will be required while the trees are young. As the trees grow older, however, it may be necessary to give pest control more attention. Caterpillars that infest the foliage of the trees in late summer and early fall can usually be destroyed by cutting off the comparatively few branches on which the worms have clustered and burning them. The pest may also be destroyed on high branches by means of torches. If the trees can be sprayed thoroughly, arsenicals and other insecticides used in spraying apple orchards will be found very effective while the worms are small.

As in the care of a young apple or peach orchard, it is important that the young trees for at least the first two or three years be given cultivation and some fertilization on lands of lower fertility if a good growth is not being made. A heavy mulch of straw or litter around the trees may prove very satisfactory.

Moreover, livestock should be kept away from the trees until they are established and the branches of sufficient height to be out of danger of injury. It is a serious mistake to plant or grow from seed small nut trees and leave them unprotected from farm animals. If the land is to be grazed, each tree may be guarded with strong posts and barbed or woven wire spaced about 8 to 10 feet from the trees.

Once the young nut orchard is thoroughly established and growing thriftily, grass may be grown beneath the trees and furnish nearly as much hay or pasture as though the trees were not present. If livestock is allowed to graze in the orchard, which is a questionable practice while the trees are young, the trees should be pruned and trained to fairly high heads.

~Spacing for Nut Trees~—The growing of nut trees for timber alone requires a spacing of about 25 to 35 feet apart with other species of trees common to the area growing up later between the nut trees to facilitate the development of tall clean trunks. Under such conditions nut production is inhibited and harvests may be comparatively small. Nut trees grown mainly for nut production rather than for timber may be planted 60 to 80 feet apart on the square plan.

The Thomas black walnut may bear a few nuts the second year following transplanting. Different varieties and species of grafted walnuts, pecans, and hickories often begin bearing from two to four years after setting. Chestnut seedlings may also bear in the second or third year. Black walnuts from seed sometimes bear a few nuts at 8 to 10 years of age. Profitable bearing, however, may not be expected in the average nut orchard until the trees are at least 10 to 12 years old.

PRUNING WALNUT, PECAN AND HICKORY NUT TREES

For the most part these nut trees do not require heavy pruning. Superfluous branches, dead limbs, and unsymmetrical ones, should be removed from time to time while the trees are young and becoming established. A uniform top is desirable. The pruning is begun when the trees are 2 or three years old by removing the lowest branches. The rule is to cut away only one branch a year. But trees making a very strong growth may stand more pruning and those making a poor growth may need none.

Cultivation and other orchard practices may be greatly simplified in commercial plantings by pruning and training the tree heads to heights of six or eight feet. Even then the lower branches will ultimately be pressed downward by the weight of nuts and foliage when bearing begins.

Regular annual pruning is required generally to prevent the limbs from interfering with orchard practices. Furthermore, branches lower than six or eight feet high, should be subdued by cutting back while the trees are young. These limbs should be removed ~only~ when the trees have become anchored strongly enough in the soil to prevent the directions of the trunk being influenced by the prevailing winds.

THE BLACK WALNUT

There is something about the distinctive flavor of our native black walnut kernels that appeals to the American people. And there is much about the black walnut tree itself that makes it much admired and respected.

It grows rapidly, and yet it is one of our most valuable timber trees. It is an excellent tree for the grounds about the home. Not only does it yield an annual crop, but it is a lovely shade tree—beautiful to look at—and has the further advantage that the lawn grasses grow well beneath it.

~Has Wide Distribution~—It is a very cosmopolitan tree in that it will thrive almost anywhere if given half a chance. From lower Canada to the Gulf of Mexico and from the Atlantic to the Pacific Coast, it may be found in various states of production. On the fertile lands, however, of the Mississippi and Ohio River basins it reaches perhaps its highest development. The 10 high ranking states in walnut lumber production are as follows, in order of their importance: Missouri, Illinois, Kentucky, Ohio, West Virginia, Iowa, Tennessee, Arkansas, Indiana, and Texas.

~Valuable Timber Tree~—Some of the main or principal uses of the wood may be enumerated as follows: For the making of gun stocks, it stands supreme. Since walnut does not warp or swell when wet it does not interfere with the action of the gunlock in gun stocks. The wood also may be made into a sharp edge and fit snugly against the metal parts, while the dark color and beautiful grain produces an attractive implement. It is a standard and a favorite for musical instruments notably pianos and organs; sewing machine tables, cases, small airplane propellers, picture frames, caskets, cabinet work, moldings and many forms of ornaments. The shells of the nuts were, during World War I, manufactured into carbon and used for gas masks.

The wood possesses unusual and rare combinations of qualities which make it superior in the manufacturing of the articles mentioned above. Its freedom from warping, checking, or splitting when subjected to alternate wetting and drying is an unusual quality. It works easily with all kinds of tools, has remarkable durability in the presence of wood-decaying fungi and insects. Moreover, it is hard, durable, heavy, stiff and strong. The dark color of the wood does not allow soiling stains to show and the grain of the wood and its texture make it easy to grip.

~Produces a Nutritious Food~—The kernels of the black walnut are now used not only in candy making but to a large extent in breads, cakes, salads, waffles, and other forms of food. In the cities the kernels are sold yearly in increasing amounts not only from wholesale and retail grocers but by street venders as well. One may often find the kernels for sale at food stands and in other places where fruits and vegetables are sold.

~Changing Seedling Trees to Named Varieties~—On nearly every farm, walnut trees are growing along ravines, fence rows, and on rough land which is more or less out of the way and inaccessible. Most of these may be top-worked by one or more methods to the named and more desirable kinds of black walnuts without imparing the value of the timber. In 5 to 7 years seedling trees ranging in age from 15 to 40, if topworked, may produce crops equal to untreated trees. Still younger and smaller trees from one to

10 or 12 years old, may generally be top-worked with less difficulty than older trees.

~Results from Top-working Experiments~—Cleft grafting work performed at the Missouri Agricultural Experiment Station has been very successful. In fact, walnut top-working has been but little if any more difficult than apple or pear top-working. With reasonable care and fairly good technique the grafting operation is not difficult to perform. It is believed, however, that the common practice in top-working pecan, hickory, and walnut has been to dehorn too severely. This may induce insect and disease injury which often results in a very poor tree after 10 or 12 years. For good results, six inches in diameter should be the maximum size of the limb for top-working.

~Encourages New Industry~—A wider interest in black walnut kernels has caused a new industry to spring up. This consists of nut cracking or shelling establishments which have been located in the walnut growing districts. The plants in many instances buy walnuts in large quantities. The nut meats are removed and sold at wholesale, usually in barrel lots containing 180 pounds of nut meats. In most districts the new industry is in operation for most of the year.

Power driven machines feeding from large hoppers are used for cracking the nuts. Nearly all the workers pick the meats from the cracked nuts. Women are generally employed and are paid on a piece-work basis or by the pound. Moreover, employees are often given a premium for nut meats removed from the shells with the "halves" unbroken.

This new black walnut industry has increased and heightened the interest in planting the trees for both nut and timber production. Consequently, in the districts where these nut cracking mills have been established, many producers are planting either small or large blocks of black walnut trees. In some cases the plantings are made up of grafted or budded trees of named varieties, while in others the nuts are planted and the seedlings later top-worked to the kinds desired.

The named varieties and better seedling sorts bring the highest price in the form of nuts and as kernels. In fact, the nuts of the named varieties usually sell for twice the price paid for the average seedling nuts. Some of the chief varieties most highly prized for their thin shells, weight of kernels, cracking quality, and flavor are Thomas, Stabler, Tucker, Ohio, and Miller.

To obtain a marketable and paying product, care in the gathering, husking and extracting of kernels, is necessary. Culling the nuts and cracking none but the good ones are also important. Through such methods, many producers are able to supply city markets and roadside stands with kernels which sell readily and at good prices.

~Returns from Trees~—Walnut trees will give returns in general in proportion to the care given. They are fairly rapid growers under good culture. At an age of 20 years the trees may reach a height of 35 feet with 50 feet at 30 years and about 70 feet at 50 years. In other words, a growth of about 2 feet a year for 20 years is not unusual. After this age the trees slow down gradually to about a foot of growth a year.

It is estimated that walnut trees from 60 to 70 years of age will produce on the average from 100 to 150 board feet of lumber. Trees of such an age may also produce an average of all the way from four or five bushels of nuts per tree each year up to as many as ten to fourteen or more bushels per year.

THE BUTTERNUT

Among our native walnuts the butternut is valued highly especially for home use. On the markets, however, the rough shell and comparatively small size of the kernel have in general tended to keep prices low and the demand limited. There are now prospects for the introduction and growing of superior hybrid varieties. Grafted varieties which bear particularly good nuts are becoming more available through nut nurseries.

The trees may become very large in height, spread and trunk diameter. They are attractive and stately in appearance and it is the hardiest member of the walnut genus as its native range extends well into Canada. The bark is gray in color and the wood is soft. Heartwood decay is common in old trees, although they may reach great age. The species has a rather restricted range within the Eastern states, but it occurs naturally as far west as eastern Kansas and Nebraska. In Missouri, its growth is confined largely to the central and northern areas where black walnuts are plentiful.

The nuts are oblong, sharp-pointed at the apex, cylindrical, bluntly rounded at the base, rough and jagged over the surface, and as a rule thick-shelled. In spite of this, some varieties have good shelling quality, and the kernels possess usually a rich, agreeable flavor. In confections the butternut kernel may compete successfully with the popular flavor of the black walnut kernels. The butternut may be propagated and grown successfully by adopting the practices suggested for the culture of the black walnut. As is true with the black walnut it may be inter-grafted upon other walnuts or used as a stock for them, but its propagation, particularly as an understock, is more difficult.

THE PECAN

The pecan is a member of the hickory group and its range in this continent extends from Iowa to Mexico. Other hickories extend into Canada. The hickories are valuable for both nuts and timber. Fifteen different species of the hickory group have been recorded. Of these only three or four produce nuts of outstanding value. In nut production, the pecan hickory is the most important of all the hickories. For crop value of nuts it rivals the Persian (English) walnut and the tree is one of the largest east of the Rocky Mountains. The pecan tree is native to the south and south central parts of the United States and it is found in the forests as a native tree throughout Missouri.

Commercial production within the state may reach 800,000 pounds or more in good crop years, and according to the State-Federal Crop Reporting Service there are now about 88,000 pecan trees in the State of bearing age. All of these consist of seedling groves except the comparatively recent orchard plantings of the southeastern area. Commercial culture of standard varieties in the United States is confined largely to Georgia, Texas, Oklahoma, Louisiana, Alabama, Mississippi and Florida.

The natural habitat is along streams and on river bottom lands. At the present time the commercial varieties consist mainly of the large so-called "paper-shell" sorts of southern origin. These require a comparatively long growing season for their development. Consequently the southern types may not be productive in the more northern regions.

The cultural range of the pecan may be divided into two rather large belts, known as southern and northern. In fact, pecan culture is sometimes designated as "southern" and "northern" due to differences in size of nut, thickness of shell, and time required for maturity of nuts. The approximate northern limit of the southern area is near the extreme southeastern boundary line between Missouri and Arkansas. The northern belt extends into Nebraska and Iowa and includes approximately the entire state of Missouri.

The chief difference between these areas is the length of the growing season. In general, the southern or "paper-shell" varieties require from 240 to 250 days to mature their nuts, while the northern varieties which produce usually nuts of smaller size with somewhat thicker shells need from 180 to 200 days.

VARIETIES

There is no factor in pecan growing of greater importance than the proper selection of varieties for planting. Fertile soils and good culture will not make poor varieties profitable or low yielding kinds fruitful.

Only in southeast Missouri are the southern varieties such as Stuart, Pabst, Moneymaker, Success, Schley, and others a success. This is true because the fruit buds of these varieties in other sections of Missouri are generally killed by winter cold. Furthermore even if they escape the winter cold, the growing periods for all sections except southeast Missouri may not be long enough for the full maturity of the nuts.

Since none of the sorts adapted to the southern belt are sufficiently hardy to justify their planting in Missouri except in the southeastern section, growers in other parts of the state should confine their interests and selections to the so-called northern varieties. Some of the best of these are the Major, Niblack, Giles, Indiana, Busseron, Greenriver, and Posey.

Chance seedlings which have not been named are now and then found that may be equally as worthy or better for planting locally than any of the named varieties listed above. In fact, these suggested sorts were derived from chance seedling trees. Producers generally, therefore, should be on the lookout for seedling trees of merit. When so discovered, the Missouri Agricultural Experiment Station at Columbia will be glad to make tests free of charge and report upon the cracking percent, amount of kernel, appearance, flavor, texture, quality, oil content, etc.

The nuts produced by the hardy varieties adapted generally to Missouri conditions are usually smaller in size and have somewhat thicker shells but may possess equally as high or even higher oil content and kernel quality than the southern sorts. The better varieties of this group, however, rank high enough to compete favorably on the markets of the country in both shelled and unshelled state with the southern varieties.

A full crop of pecans would run from 30 to 35 carloads, the majority of which are produced along the Mississippi river in the bottom lands from Ste. Genevieve southward. Heavy shipments are made in a good year especially from Ste. Genevieve, St. Mary's, Menfro, Caruthersville and Hornersville, and in these sections are some of the largest and best nuts.

Pecans are found along the Mississippi river from St. Charles north to Hannibal, but too generally in that area the trees are scarce and the production smaller, with nuts of thicker shells.

Pecan trees are also found growing wild along the Missouri river bottom as far west as Lexington, and up the Grand river bottoms to Chillicothe, and the nuts in this area are about the size of those in the north Mississippi valley section, but are sweet with high oil content.

There is a pecan production district along the Osage river and the Kansas border, with heavy shipping section at Rockville and Schell City.

Missouri pecans are classed as Westerns in the commercial market. They are favored by the confectionery trade. A great many native trees are found in the south Mississippi section, but there is a growing interest in budded pecan trees, especially around Caruthersville.

The total of the budded varieties of pecan trees in Missouri does not constitute more than approximately one per cent of the total of growing trees.

Many years ago a large acreage of the bottom lands along the Mississippi river were thick with immense, heavy-producing pecan trees—but most of this pecan timber was cut down either for fuel wood or saw timber. Short-sighted people have been known to chop down trees simply to secure the nuts.

THE HICKORIES

The native hickories of Missouri have been held in high esteem since early settlements were established. They are notorious on account of their slow rate of growth yet they offer greater possibilities to nut growers than is usually believed. As shade trees they have a high ranking.

Promising varieties may now be had by obtaining scions from superior bearing seedling trees and from young named and grafted trees in the nurseries of commercial concerns. Grafted trees may come into bearing in three or four years after the operation.

Perhaps as many as five species are native of Missouri. The big shellbark or kingnut is common to the south and southwest regions, but its range is not as wide as others. The shagbark which is the most valuable nut producer of all the hickories, is rather widely distributed particularly in northern and central Missouri. Numerous varieties have been described and named because of their particular merits. Shellbark nuts may be large and attractive, but are often poorly filled.

The pignut, mockernut, and bitternut have a rather general distribution especially in the central and northern parts of the state. These nuts are not considered of great value except for their hybrids with other species. Perhaps the most natural type of hybrid occurring among the hickories is crosses between the shagbark and shellbark, one of the best varieties of which is Weiker.

The pecan and shellbark hybrids include McAllister, Nussbaumer, and Rockville, while the Burton is believed to be a pecan-shagbark cross. The natural crosses of the pecan and hickory found in the wild have not been entirely satisfactory. The trees vary greatly in fruitfulness and the nuts in thickness of shells, size, shape, and kernel quality. A strong tendency to produce nuts with imperfect kernels is common among the pecan-shellbark crosses.

Local varieties selected from the wild may have merit for use in top-working hickories or pecans. The pecan is suggested because it makes a good stock for the hickories and as it grows more rapidly. Some of the best of the older named sorts for planting or for use in top-working appear to be the following: Barnes, Fairbanks, Stanley, Weiker, Kentucky, Swain, Laney, Kirtland, and Rieke.

THE CHINKAPINS

The chinkapin is related closely to the chestnut and resembles it strikingly in most of the important characteristics. It occurs in two well known forms. West of the Mississippi River, the Ozark chinkapin tree may reach a height of sixty feet in good soil, while the other form (Allegany chinkapin) in the eastern range grows to a height of about 15 feet. Each may be grafted or budded upon the other without difficulty. Named varieties of the chinkapin are not available at this time. The Japanese, Chinese, and European chestnuts are introduced species.

The blight disease has almost wiped out the great American chestnut forests of the East. As yet, however, the malady has not been introduced into Missouri. (The oak wilt, however, has been found there.—Ed.) The chinkapin of this area is highly resistant to the blight and some of the hybrids carry the resistant quality and bear nuts of good size and high quality. The native chinkapin forests especially of southwest Missouri are valued highly not only for their nuts but particularly for post timber.

The native chinkapin tree in Missouri grows to large size in good soil and it may be found as one of the largest forest trees on the stony ridge lands of southwestern sections of the Ozark Mountains. The nuts are very much like those produced by chestnut trees except they are smaller. In flavor and quality the nuts may be found equal or superior to the chestnuts.

Both the chinkapin and chestnut may be grafted or budded one upon the other. In fact, the western chinkapin may be used successfully as a stock for the chestnut.

The European chestnut is very susceptible to the blight. A very large coarse nut is produced by the Japanese chestnut and it does not blight quite as readily as the American sorts. The Chinese chestnut is the most resistant to blight and it is admired for its beauty as a lawn tree. Promising varieties include Abundance, Nanking and Meiling.

Some desirable varieties of the American and hybrid chestnuts for growing in Missouri are as follows: Boone, Fuller, Paragon, Progress, Rochester, and Champion.

FILBERTS AND HAZELNUTS

The European filbert which is grown so successfully in Oregon and Washington has not been generally successful in Missouri. This has been due mainly to winter injury, resulting either in the killing of the staminate catkins by cold, or of the developing catkins by late spring freezes and frosts. For good fruiting they need cross pollination. Some of the well-known and popular filbert varieties are Barcelona, Du Chilly, Medium Long and Italian Red. Rush, Winkler, and others, are partly or purely American hazelnuts.

The native hazelnut which may be found throughout the State is hardy and generally a fairly regular cropper. Seedling nuts, while not as large usually as the Northwestern filbert, are found now and then that approach them closely in size and cracking quality. Furthermore, the native seedling nut kernels may excel occasionally in flavor and quality.

Interested nut growers are, therefore, urged to perpetuate the most promising hazelnuts of the wild by simple layerage. Until hardier varieties of the filbert are found, the chief attention may be well spent on the propagation and culture of the native seedling sorts of merit. As yet none of the Missouri native seedlings have been described, named and propagated for sale and distribution.

Tip or simple layering seems to be the most satisfactory method of propagating the hazelnut and filbert. Shoots or suckers, one-year old and arising from the base of the plant are used. They are left attached to the mother plant and are bent over until the ends of tips rest upon the soil.

To encourage root growth, the underside of the branch to be covered with soil is frequently notched or ringed. The part of the branch in contact with the moist soil is then covered leaving a small portion of the end protruding. The branches are sometimes pegged down with forked sticks or weighted with stones. After one season's growth, the branch should be established with roots and top. It is then cut from the parent and removed for transplanting to its permanent location.

Well, now, my good friends, I have talked about five or ten minutes longer than I intended to, but you just listened so attentively you encouraged me, so it's your fault. I am happy to be here. Show me an organization like the Northern Nut Growers Association, as full of vim and vigor and vinegar and going ahead, and I will show you a successful organization.

Thank you.

MR. CHASE: Thank you, Professor Talbert, for a very nice message.

I am still a little angry at Professor Talbert because I realize now that if he had accepted my invitation to come to another good southern state two years ago our meeting would have been a much better one at Norris.

Now, we have several papers here which deal with chestnuts, and there seems to be a good deal of interest among the membership concerning chestnuts this year, and perhaps before we get into chestnuts for nut production we might hear a short resume of Dr. Graves' breeding work for timber type chestnut. This problem of chestnut for timber purposes, of course, accounts for the presence of Chinese and Japanese chestnuts in the country today, and yet most of our efforts to establish chestnut plantings for timber purposes have been unsuccessful. You heard from Dr. Diller last year concerning these efforts.

This paper will deal with the breeding work which is now under way by Dr. Graves in Connecticut, and I have asked Dr. McKay to give us a brief digest of this paper.

Chestnut Breeding Work: Report for 1950

ARTHUR HARMOUNT GRAVES
Connecticut Agricultural Experiment Station, New Haven, Conn.
and
Division of Forest Pathology, U.S.D.A. Plant Industry Station, Beltsville, Maryland

In southern Connecticut the 1950 season for vegetative growth and development was excellent except for the dry period in September. The chief fault lay in much more cloudy weather than usual,[31] and the deficiency in sunlight coupled with a slightly lower average temperature in the spring, and cool nights, combined to delay the chestnut flowering season for as much as ten days. The main body of our cross pollination experiments did not begin until July 4, whereas last year it began on June 23 and 24, and was nearly completed by July 4.

[31] For example, the report of the U. S. Weather Bureau at New Haven, Conn., for May, 1950, says, "The feature of the month was the lack of sunshine which retarded the growth of crops in this area." See also report of the New York City Station for April, 1950.

This year 103 crosses were made, not all different combinations, but each one with either different or reciprocal parents. The principal combination was a cross of Japanese chestnut with Chinese-American or American-Chinese, a mixture that in recent years has given excellent results. This year also, as in the past, our CJA's were crossed with American chestnut.

[Illustration: Fig. 1. Cross pollinating Chinese chestnuts. Sleeping Giant Plantation, Hamden, Conn. Trees near left of center and at left, with drooping catkins, are Japanese-American hybrids. Photo July 13, 1950, by B. W. McFarland.]

~Cooperation with Italy.~ A considerable part of the cross pollination work this year has been undertaken for the benefit of the Italian authorities, namely experiment stations at Florence and Rome. This has been done at the suggestion of the Division of Forest Pathology, Beltsville, Md., which has been working along the same line.

As is now generally known, the chestnut blight was discovered in Italy in 1938, and has been making rapid headway in a country 15 percent of whose forests are in chestnut. To the Italians the chestnut means much as an article of food. They use the timber also, and the various ages of coppice growth in many ways[32]. Particular effort this year has been directed toward the breeding of promising nut-bearing types for them and especially resistant strains that bear large nuts like the cultivated European chestnut.

[32] Graves, Arthur Harmount. Breeding Chestnut Trees: Report for 1946 and 1947. 38th Ann. Rept. Northern Nut Growers Assn. p. 85. 1947.

Now, we have found that many of our Chinese chestnuts are practically immune to the blight. Even if the disease does appear, in most cases it is in the outer bark only, and is soon healed over. Moreover, the Chinese chestnut has a large nut, comparable in size to the cultivated Europeans with pollen from our best Chinese trees, and at the same successful crosses of the European and Chinese are made.

Last fall, as a result of an article in the *New Haven Register* by Mr. A. V. Sizer, I learned of two European chestnut trees of bearing age in New Haven back yards. So, this summer we have crossed these Europeans with pollen from our best Chinese trees, and at the same time have taken the pollen from one of them (in the other the pollen was sterile) and applied it to the female flowers of our Chinese trees. Most of the resulting nuts have been sent to the Italian scientists in the hope that some of them will develop into desirable nut-producing, disease-resistant hybrids. Some will be retained for testing here. If the resulting trees are not sufficiently blight-resistant, they will be crossed again with the Chinese.

In the summer we received by air mail from Dr. Aldo Pavari, of the *Stazione Sperimentale di Selvicoltura* in Florence, Italy, two tubes of pollen of the European chestnut, *Castanea sativa*, of the varieties *pistolese* and *selvatico*. These pollens were also applied to our best Chinese trees. They resulted in 12 good nuts which have been shipped to Dr. Pavari.

Further, we have on our Sleeping Giant Plantation, Hamden, Conn., several hybrids, now 16 years old, of the Seguin and the Chinese chestnuts, the former species being also a native of China, but dwarf and everblooming and remarkably prolific. These hybrids are excellent as nut producers, since they inherit the large-sized nut of the mollissima parent, combined with the increased productivity of the Seguin parent. Furthermore they are extremely blight-resistant.[33] These hybrids have therefore been intercrossed among themselves this year, chiefly for the benefit of the Italian people. One hundred and eight nuts from reciprocal crosses of these hybrids were shipped to Italy. Also, in response to a request, we sent nuts of our best Chinese and Japanese trees and of the *mollissima-seguini* hybrids to M. C. Schad of the *Station d'Amelioration du Chataignier*, Clermont-Ferrand, France.

[33] These hybrids will shortly be put on the market, under the sponsorship of the Conn. Agr. Expt. Sta. and the Division of Forest Pathology, U.S.D.A. As regards the everblooming habit of the Seguin parent, that character seems to be lost or at least partly suppressed. A second flowering of one of the hybrids usually occurs in August.

~Other crosses.~ Two Chinese-American trees in our plantation at the White Memorial Foundation near Litchfield, Conn., bore a considerable number of female flowers this year for the first time. They have been crossed with the fine Japanese tree of Mr. A. N. Sheriff at Cheshire, Conn., figured in my report for 1948-49. (P. 92, fig. 3, of 40th Rept. of N.N.G.A.) From them, 75 nuts were harvested of the combination CAxJ. Four crosses were made on the trees at Redding Ridge, Conn., in the cooperative plantation of Mr. Archer M. Huntington, resulting in 73 nuts. Also, the resistant Americans on Painter Hill, Roxbury, Conn., were again crossed with CJA's and Chinese from our Sleeping Giant Plantation and from these were obtained 247 nuts. Finally, we have this year succeeded in making a cross between *Castanea henryi*, the Henry Timber Chinkapin from southern and central China, which is said to attain a height of 90 feet, and *C. mollissima*, the Chinese chestnut. Since *henryi* blooms very early, much before our *mollissima*, the Division of Forest Pathology mailed us pollen of *C. mollissima*, which reached us just in time to be applied to *henryi*. Seven good nuts of this cross were gathered.

Altogether, as the overall result of our cross pollination work, we harvested 1259 nuts, more than twice as many as obtained in any other year since we began this work in 1930.

TABLE 1

Heights of Some of Largest Trees, as of Oct. 1, 1950.
All at Sleeping Giant Plantation, Hamden, Conn.

Species or Hybrid	Location	Age in yrs.	Height in ft.	Remarks

J × A	Row 4	Tree 10	19	30	Repeatedly inarched
J × A	" 4	" 4	14	33	Grafted on Jap. stock, Apr. 1937
J × A	" 4	" 12	19	29	Repeatedly inarched
J	" 7	" 5	20	23	
C	" 1	" 4	24	30-3/4	
CJA	" 60	" 39	13	29	
CJA	" 61	" 48	13	24	
CJA	" 8	" 8	4	14	Grafted on Chinese stock, spring, 1947. Fruited this yr. 1st time.

J=*Castanea crenata*
A=*Castanea dentata*
C=*Castanea mollissima*

~Nuts, Scions and Pollen Received.~ During the fall of 1949 we received nuts from New Hampshire, Mass., Conn., N. Y., N. J., W. Va., N. C., Ohio, and Ill. Scions were received in March and April from Mr. R. M. Viggars of the Bartlett Tree Expert Co. station at Wilmington, Del. (*C. dentata*); and from Messieurs Schad and G. A. Solignat, *Centre de Recherches Agronomiques*, Clermont-Ferrand, France, (*C. crenata* and *sativa*.) During June and July, pollen of *C. dentata* came from Mr. E. J. Grassmann, Elizabeth, N. J., Mr. Paul Maxey, Montcoal, W. Va., Mr. Malcolm G. Edwards, Asheville, N. C.; *C. mollissima* and *dentata* from the Division of Pathology, U.S.D.A.; *C. sativa*, vars. *pistolese* and *selvatico* from Dr. Aldo Pavari, *Stazione Sperimentale di Selvicolture,* Florence, Italy; and *C. pumila* and *dentata* from Mr. Alfred Szego, Flushing, N. Y. This list is presented as evidence of the widespread interest in our work. It is a pleasure to acknowledge this cooperation and to thank the many donors. We are especially glad to report that several "catches" have been made with the C. sativa scions from France and those of the tall *mollissimas* at Mt. Cuba, Del., from Mr. Viggars.

May I again caution those who send us nuts not to allow them to become dried out. The embryos, when dried, are killed. The nuts should be wrapped in moist cotton, peat moss, or something similar, and mailed to me not later than a few days after harvesting, at 255 South Main Street, Wallingford, Conn.

~Insects, bad and good.~ The cankerworms were rather destructive in May at our Sleeping Giant Plantation (not at the others) but fortunately later than usual. The mite, *Paratetranychus bicolor*, attacked the leaves of some of the trees on the Sleeping Giant Plantation rather late in the season, so that on September 8 we sprayed with the Station's power sprayer, using Aramite effectively. Shade and humidity seem to favor the spread of this pest. Japanese beetles appeared but have never been very destructive with us. As happened last year, we sprayed twice for the weevils, August 14 and September 8, with excellent results.

This spring in early June, four hives of bees were placed in one of our Sleeping Giant Plantations by bee experts of the staff of the Conn. Expt. Station. Improved results in pollination and the resulting nut harvest cannot be affirmed with only one season's trial.

A Method of Controlling the Chestnut Blight on Partially Resistant Species and Hybrids of Castanea

ARTHUR HARMOUNT GRAVES
Connecticut Agricultural Experiment Station, New Haven
and
Division of Forest Pathology, U.S.D.A. Plant Industry Station, Beltsville, Maryland

This method has been in use since 1937 on our chestnut plantations, and has been so remarkably successful that we believe all chestnut growers should be thoroughly acquainted with it.

[Illustration: Fig. 1.]

Whenever chestnut trees are attacked by the blight fungus, suckers arise below the lesion, and if the lesion is at or near the base of the tree, as often happens, these suckers grow from the base of the tree, i.e. at the root collar. It is then a simple matter to cut out the diseased bark of the lesion with a sharp knife, paint over the wound, and graft the tip of one or more of these suckers *above* the lesion, into the healthy bark. Of course the sucker must be long enough to reach the healthy part of the bark above the lesion. It is measured roughly by the eye and then cut off at a proper length, usually a little longer than seems necessary. The tip is then sharpened into two beveled surfaces coming up to a thin sharp transverse edge like a long wedge. (Fig. 1a.) The tip edge must be very sharp in order to push up easily between the bark and wood. Now, or rather, before trimming the sucker, in the healthy bark above the blight lesion cut an inverted T, making the cut into the bark as far as the wood and then cut a gradual slope from the the surface of the bark down to the horizontal part of the inverted T. Next, lift the bark gently from the wood above the horizontal cut and insert the end of the sucker. If the sucker, or scion, is slightly longer than the upper end of the cut, it can be bent outward at the same time that the scion is being inserted and thus a spring is secured making it easier to force the scion up between bark and wood. I should add that if the lesion is not at the base of the tree, suckers usually arise just below it in any case, and these can be inarched in the same way as the basal shoots.

[Illustration: Fig. 2

Fig. 2 Showing inarching method of controlling the chestnut blight a Chinese-Japanese hybrid chestnut, 13 yrs. old, infected toward base with Chinese type of blight, i.e. in outer bark only. Right: sucker inarched in spring of 1946; left, inarched spring of 1950. (The black figure resembling an arrow, about half way up, is accidental, being a cluster of labels.) b. Grafted tree (the large tree of Japanese-American chestnut on Japanese stock); graft made in 1937 where finger is pointing; left: inarch of 1947, itself inarched near base in 1950; right, inarch of 1949. c. Japanese-American hybrid chestnut with principal inarch made in 1943; other later inarchings showing in part. All photos by Louis Buhle, Brooklyn Botanic Garden, and loaned courtesy of the Garden.]

The next step is to bind together the parts being grafted, winding strong, cotton string firmly around the cut with its scion enclosed, covering practically all of the vertical cut of the inverted T. Finally, melted paraffin—not too hot—is applied to the union, every part being carefully covered in order to exclude air and thus prevent drying out. We use Clarke's melter which, with adjustment of the flame, will keep the paraffin at a temperature slightly above the melting point and thus will not get too hot. Grafting wax may also be used instead of paraffin. The best time to perform the operation in Connecticut is during April or early May.

Our first scions or inarches, grafted in 1937, are now 6 inches in diameter at ground level and constitute the main tree. If they become blighted, other suckers are inarched into them, and so on. The purpose of the inarching is to restore the communication between leaves and roots, which is so essential to the life and health of the tree, and which the diseased bark of the blight lesion interrupts, eventually causing girdling and death of the trunk or branch attacked. A series of these inarchings of different ages is shown herewith. (Fig. 2.) On our plantations we no longer dread the chestnut blight, since we can usually circumvent it by this method. However, with the American chestnut, because the fungus advances rapidly in this species, the girdling is often completed before the scions can take hold. Therefore, with that species or with the least resistant hybrids the method is often though not always ineffectual.

This method of grafting is not new. It is similar to bridge grafting and has been known and practiced for centuries. The only credit we can claim is for its application to the chestnut blight as a method of control.

MR. CHASE: We will now hear from Mr. George Salzer, Rochester, New York,
"Experiences with Chestnuts in Nursery and Orchard in Western New York."
Mr. Salzer.

Experiences with Chestnuts in Nursery and Orchard in Western New York

GEORGE SALZER, Rochester, New York

My work with Chinese chestnut trees during the past ten years has been most interesting. The first trees were grown in our back-yard garden; then, when more seed was available locally, a building lot was purchased for use as a nursery. Seed is planted in the spring because when fall planting was tried, the rodents took most of the nuts.

Up until last year, chestnut seed was stratified in perforated cans in the open ground with fairly good results. Last fall, we tried the method used and described by Dr. Crane and Dr. McKay in the 1946 report of this Association. Crimp top cans were used with nail holes in the top and bottom. Instead of using regular storage facilities, the cans were stored in a concrete block storage pit built below the floor of the garage. This proved very successful. Not only were the nuts in excellent condition for eating in the spring, sweet and of good flavor, but a much larger percentage of the seed germinated. This storage pit also serves to hold trees dormant and in good planting condition from digging time in March until early June.

Last year, many young seedlings were lost during the dry weather and hand weeding between the trees was next to impossible. This spring, we tried the method of planting used and described by Mr. Sam Hemming in the 1947 report of this Association. We planted the seed in a narrow trench two inches deep; then filled the trench with saw dust; level with the surface. The saw dust serves as a mulch to hold moisture for the young seedlings and hand weeding between trees is reduced to a minimum. It is also possible to use the wheel cultivator between the saw dust marked rows before the shoots appear. This was a great help in controlling early weed growth.

We were troubled with cutworms cutting off the new seedlings close to the ground, the same as they cut off young tomato plants. We controlled them by using a poison-bran bait as described in Leaflet Number Two issued by the Department of Agriculture.

All trees are grown from seed of trees growing in the Rochester area. These had their origin from north of Pekin, China. Most of the trees are three years old when sold and have been transplanted at least once. This gives us a good sized tree that transplants well and should bear some chestnuts in three or four years. Sales are to people in our locality, although a few mail orders have been filled. So far, we have had no complaints. These are all seedling trees and until grafting or budding of named varieties becomes stabilized, I believe we should concentrate on growing large numbers of seedlings at a price within the reach of all who want chestnut trees.

This spring some large chestnut seed received from a southern grower was planted for experimental purposes. I will bring them into bearing to learn whether they will bear as large a nut in our climate as they do in the southern states, and whether the kernel will be as sweet and have as good flavor as those grown in upstate New York. I have yet to see a tree growing in the Rochester area bearing as large a nut as those grown in the southeast, and all the large nuts I have tasted did not seem to be as sweet as ours. Probably the old saying "the smaller the nut, the sweeter the nut" is true. Of course these are all seedling trees, but by this time we should know whether size of nut and sweetness of kernel are determined by climate or individual trees.

Our largest trees are eleven-year old seedlings of unknown origin. One is, I believe, outstanding. It started bearing when four years old and has consistently been a good producer. The nut is real chestnut in color and good size, running about seventy to the pound. I have not found a tree in this area bearing a larger nut. The kernel is sweet and the flavor excellent. The tree has good shape and limb structure, always sending up a central leader. This is the tree I would like to propagate.

Small Nuts Sell Better

Last fall, I tried a selling experiment with chestnuts for eating, and sold small quantities of small and medium sized nuts at the rate of $1.50 a pound. However, no one seemed interested in the larger ones. They thought they were European chestnuts that sold here for $.25 a pound. I did not have many for sale, but I am convinced there is a market for good sweet chestnuts. It seems useless to compete with those imported from Italy. Ours are far superior, and many who remember the American chestnut, will, I believe pay a luxury price for good quality chestnuts.

In 1946, we purchased a 10-1/2 acre piece of land, 16 miles southwest of Rochester for the purpose of planting a chestnut orchard. This land had not been worked for about twelve years. The soil is heavy and fertile, typed as Poygan clay loam. Bed rock is sixty feet below the surface. The following spring, we planted about 300 trees and each year more are set out. There are now about 700 trees from two to five years old, and most of them are growing well.

The rows are twenty feet apart and the trees stand fifteen to twenty feet apart, in the row. I know this will be too close when the trees are full grown, but we have the trees and I want to bring as many into bearing as possible, searching for the ideal tree. We also expect to lose some trees through wild life and other causes.

Many of the first trees planted were lost the following year due to excessive rainfall, poor surface drainage, rabbit and meadow mouse damage. In 1948 two 400 foot drainage ditches were dug across the property. This made it possible to plant trees successfully on most of the land. However, another ditch is needed to eliminate a low spot, then all of the land can be used.

The meadow mouse that girded so many trees could not be controlled by the use of poison bait and the rabbit also did considerable damage. Through the wild life service of the Department of the Interior, we obtained a repellant that was effective. It is distributed in the eastern states by the Rodent Control Fund of the University of Massachusetts. We have used it now for two years and have no more mouse or rabbit damage.

The woodchuck does considerable damage even though we have eliminated all their dens on our land. They come in to feed from the neighboring areas and will have to be controlled by shooting. Deer are also present but have as yet caused no damage. Probably, they are waiting for the trees to grow larger.

Last spring, new growth on the trees was killed by a late freeze—a most unusual occurance for this area. This was caused by an excessively warm April, followed by below-freezing temperature in the middle of May. It was the first time in the memory of the oldest residents that black locust and native black walnut trees were damaged by a spring freeze. However, most of the trees recovered, but their growth was retarded.

This spring several of the trees blossomed, but set no burs. In a few years, I hope to have more to report on this orchard project.

(Here was shown a chestnut tree picture.)

MR. SALZER: If anyone has any comments, if they think it has good limb structure, that's what we are looking for.

MR. SHERMAN: We could tell you better if we could see it when it's dormant.

MR. WEBER: What sort of a cultivator do you use?

MR. SALZER: Wheel cultivator.

MR. WEBER: Why don't you get a Wheelmaster? You may not want to cultivate as often as if you had a power one.

MR. CHASE: We shall now have another chestnut paper by Alfred Szego of
Long Island.

Chestnuts in Upper Dutchess County, New York

ALFRED SZEGO 77-15A 37th Ave., Jackson Heights, New York City

Pulvers' Corners, a collection of farmhouses, a gas station and ice cream parlor is located about 8 miles from the northern Connecticut border not too far from the southwestern tip of Massachusetts.

The Berkshire hills roll through here and at this point we find ourselves at approximately the northern limits of the deciduous hardwood forest belt.

Here the American chestnut is native formerly growing in great abundance until stricken a mortal blow by the invincible chestnut blight.

Just a few hundred feet north of here on a hilltop, I started in 1945, a different kind of nut tree plantation.

Placing main emphasis on the chestnut, a start was made on the cultivation of the thousands of sprouts and seedlings on my 43 acre coppice forest.

A cluster of ~Castanea dentata~ seedlings that appeared promising was selected. The following practices proved fairly successful in keeping a few trees healthy, and bringing one into bearing in 1950. For the interest of fellow members working along a similar line, I enumerate the following practices.

1. Clean and thorough tree surgery, cutting out blight cankers immediately upon discovery.

2. Removal of all very blight susceptible nearby sprouts and the burning of all infected branches and material.

3. Artificial watering during drought periods.

4. Application of superphosphate, muriate of potash and trace elements. Es-Min-El was used in our case. Our soil tests high in nitrogen.

5. Removal of all overstory trees and other interfering growth.

It may be noted that the importance of hygiene and sanitation cannot be stressed too strongly.

Our own native chinkapin, ~Castanea pumila~ when brought up north proves itself a delightful subject. Outside of the weevil-infested area, it becomes a hardy producer of superb little chestnuts. This species offers great promise to the plant breeder because of its very early bearing (3-4 years from seed). Perhaps hybridization with ~Castanea mollissima~ varieties may bring something very fine and valuable. This species is tender during its first year but perfectly hardy afterwards. Northern growers require special techniques to grow chinkapins from seed.

The strains of Chinese chestnut, ~Castanea mollissima~ in most cases do not seem extremely happy here. The trees appear to sustain varying degrees of winter injury. The tips of the branches often freeze. Usually the branch comes into leaf on the lower part first and then upwards. However, a few individuals appear perfectly hardy. The outlook is excellent for the discovery of exceptional individuals suitable for the northern zones.

The Japanese chestnut, ~Castanea crenata~ shows very good adaptation to this region. Although my trees of this species are young, very vigorous growth indicates some value here. Unfortunately, the nuts have a bad after-taste when eaten raw thus limiting its commercial possibilities. I have noticed this undesirable characteristic in tasting hybrid nuts derived from trees possessing ~Castanea crenata~ parentage. I was informed at Beltsville that the hybrid known as S8, a cross between ~Castanea pumila~ and ~C. crenata~, was rejected for its poor quality nuts.

I have established many other species of chestnuts and their hybrids. Some of these are from seed obtained from the Bell experimental plot of the U.S.D.A. at Glenn Dale, Maryland. Seed from this source has produced a much better grade of seedlings than those from anywhere else.

A somewhat different version of the tin can planting method is now being used here. Number two size and larger tin cans have a few punctures made with a hammer and nail in the bottom. These have their tops removed, of course, and after being filled with loose soil, are used as pots in which to start chestnuts.

In the early spring germinating chestnuts are removed from jars, kept in my refrigerator. One is planted in each can flat side down, barely beneath the soil level.

After the season has warmed up these "canned plants" are set out in a trench, buried to the rim. Rock wool is placed around the stems of the seedlings covering the soil and the nut. This has acted as a rodent deterrent.

The "canned plants" are then, at leisure, set out in their permanent places. Just before doing this an ordinary beer can opener is used to enlarge the punctures in the bottom of the can to permit the roots to penetrate better. In a few years the can should disintegrate entirely and at no time will interfere with root growth.

By holding the chestnuts under refrigeration and not planting in the fall I have kept my plantings free of the chestnut weevils.

I found that by planting the flat side down, the stem seems to go down very easily, and the sprout coming up from it seems to go up more easily, also.

Discussion

MR. RICK: Are they planted permanently in the can?

MR. SZEGO: Yes, they are planted in the can. The can will disintegrate in two or three years.

MR. RICK: Don't you have those in rows?

MR. SZEGO: No, I sometimes place them on the grass. The morning dew seems to provide enough moisture to carry through the dry spells. But, again, I live in a mountainous area. This may not apply out in Oklahoma.

MR. CHASE: Next on the program is a demonstration of his method of propagating nut trees in pots in the greenhouse by Mr. Bernath, who has been very successful with this method. Mr. Bernath.

Demonstration of Method of Propagating Nut Trees in Greenhouse

STEPHEN BERNATH, Poughkeepsie, N. Y.

Here is the way I handle the nut trees when we propagate under glass in the greenhouse. These are two-year seedlings potted up. That root is cut away and any large lateral roots that are too large to bend well we cut them off, and we take all the fibrous roots we can and put them in this pot. Put your soil around it first, and when you have it nearly full, just the same as if you take your son and lay him on your knee and spank the butt good and put the soil around the roots. Then pack it with your thumb and your potting is done.

(Taking scion) I use only one bud. One bud is good as a dozen.
(Cutting-with pruning knife.)

MR. WEBER: How do you cut above the bud that you use above the graft?

MR. BERNATH: If the nodes are far enough apart I put it farther, but I like to put it as short as I can but allow not less than half inch or an inch or more on top, and you cut it away after the union has taken and the growth started. Sometimes some of them may have a growth of two inches before you take them out of the case. They are not uniform. Some of them are way in advance of some of the others. Some of them are tardy, slow.

This is my budding knife, here, which is about 40 years old.

MR. CHASE: The question is asked, this isn't the time of year that you would do this, is it?

MR. BERNATH: No, sir. I start in January. You can continue into April. You can take a batch out and put another batch in.

MR. RICK: How many weeks, usually, before you graft, after these are put in the case?

MR. BERNATH: I would say that with most of your varieties it's from four to six weeks, with the exception of ornamentals. That will take six to eight, sometimes longer, but nut trees generally come on quickly. I have known them to have two inches of growth, I think, in three weeks. (Sharpening knife.)

A MEMBER: You are like the violinist. You have to tune up first.

MR. BERNATH: Yes, and never forget to wipe your knife. And remember not to put your finger on the fresh cut. (Cutting). Here is the cut before I insert the scion. In cutting your scion wood, now here is the butt. Cut on the inside. When you cut on this side it throws the bud a little bit far out because it's on an angle. You know about the depth of the cut here, and you go like this: (Cutting).

A MEMBER: Do you come down to a pretty good point?

MR. BERNATH: (Holding up scion.)

A MEMBER: Is that a side graft you are making there?

MR. BERNATH: Yes. (Inserting scion in cut.) Now, on this one I am going to use a rubber strip.

DR. MacDANIELS: Hold it up so we can see the whole thing as you have it stuck in there. That is a side graft with the bud next to the stock.

MR. BERNATH: That's right.

MR. RICK: The scion was cut on both sides, was it, or one side?

MR, BERNATH: Yes, on both sides.

MR. WEBER: Wedge shape.

MR. KINTZEL: An inch below the bud.

MR. BERNATH: (Wrapping graft) Here is where your thumb comes into play. As you put this on, start right here (stretching rubber). See how far that can stretch? You cross it and you can take your finger off. Now release it. Have your finger on it. Put this finger right here. All right, you see you get under, pull right up there. There it is, the graft is done.

MR. EMERSON: You don't use any wax?

MR. BERNATH: No wax whatsoever. Never use any.

MR. CORSAN: Or any latex?

MR. BERNATH: No, nothing at all.

MR. RICK: How do you slope this?

MR. BERNATH: I have a little, miniature box here, and that would represent a bench in the greenhouse. (Demonstrating).

Here is another one (taking another scion).

MR. CORSAN: That's used by dentists and plastic surgeons.

MR. BERNATH: Now watch the difference. If the scion wood happened to be smaller than your stock, you cut accordingly. In other words, you are not going in as far. See (showing). Or else you can cross it. Now, just a minute, we will get that (making cut in stock; slicing scion off diagonally). You don't go up as high on this side. Now, then, you take it, if you are a pretty good hand with a knife. That's all right, even if it's not shaped at all. There it is (inserting in cut). But one thing—I want to warn you, if you want to follow this, be careful not to rub the bud off in handling it. If you do, you might as well throw it away, because you are licked.

MR. WEBER: That is one reason for having the bud face the stock?

MR. BERNATH: No, but makes a better growth.

Persian walnut, I find, unless it's way far down on the trunk of a tree, will not form adventitious buds. Now, you can do it with a chestnut. You can rub the main bud off and you will find two or three of them coming, or more, right around that place. But one of these walnuts will not form an adventitious bud, so you might as well throw it away, or if you knock off even the new growth on it, you might as well dump it, because it will not form a tree.

Now here is a tape that I use.

MR. KINTZEL: Rubber tape?

MR. BERNATH: No, no, cloth.

MR. STOKE: That's about the same as surgical tape?

MR. BERNATH: Made especially for grafting, Mr. Stoke.

Now, you have to watch it closely because this is a tricky thing.

MR. CORSAN: This is not called Scotch Tape?

MR. BERNATH: No, this is made especially for grafting. You can get this from some of the boys.

MR. WEBER: A. M. Leonard and Son, Piqua, Ohio.

MR. RICK: That will require more attention than the rubber. The rubber takes care of itself, where this one you have to take off.

MR. WEBER: No, this decays.

MR. BERNATH: You start right here on the stock. Now you make sure that the scion—

MR. WEBER: You start at the top?

MR. BERNATH: The top, always on the top.

MR. WEBER: And that has a tendency to keep the scion worked down, whereas if you started at the bottom you might push it up.

MR. BERNATH: You have quite a pressure right around there—watch it, because it will tear, and if it tears with you, why, it's so hard to get straightened out—and then press together.

MR. WEBER: And you don't wax either the top, or anything?

MR. BERNATH: No. Now, the reason for leaving this under stock that long: if you are not careful, fungus growth will set in. If you cut right here, then the whole thing is affected with it, see. Wrap it firmly and that is there on both sides, and when the union forms and the growth begins here, when you take them out of the case, for instance, now, you take a sharp pair of shears and cut as close as you can. (Removes top of understock.) Never mind if you cut the cloth, it doesn't make any difference. Just cut it right there. Snip it right off. But that is when you take them out of the grafting case.

A MEMBER: Wouldn't it also be all right to leave that stub on to tie your sprout to so it won't want to break?

MR. BERNATH: No, you might be better off if you had a stake. Put a stake on the side of it. When everything is right that surface will callus over right quickly. It may not seem so. It does make a perfect union unlike a graft of some other types.

MR. WEBER: When you make that cut of the excess understock, you don't even wax?

MR. BERNATH: No. You can if you want to, but I don't wax. Just leave it like that.

Now the next operation. Here is this miniature greenhouse. It's moist peat. That's just about the right substance. Would anybody like to look at this? Don't get it too wet. Just walk right up here.

MR. WEBER: It feels as if it's ground up.

MR. BERNATH: It is.

MR. CORSAN: Mr. Bernath, would that be the right stuff to put sweet chestnuts in in the fall?

MR. BERNATH: You mean for sprouting?

MR. CORSAN: Yes.

MR. BERNATH: That would be all right.

MR. CORSAN: That's not too damp?

MR. BERNATH: No.

MR. CORSAN: I have put it in that and had the greatest success.

MR. CHASE: Now, folks, let's everybody sit down, and please keep quiet and try to absorb what's going on here. We can't have 10 or 15 individual conversations going on.

MR. BERNATH: Now here we have two pots grafted. Now, of course, the bench in the greenhouse is wider and longer. Here is what you do. You start the first row, just move the peat back like that, and you lay them in like that, one after the other, the pots on the side.

MR. WEBER: With the bud side up?

MR. BERNATH: That's right. Now, you go right along. When you come to the next row, here is what you do (piling up peat) like that. If you want to cover the scion, all right; if you don't, perfectly all right. You can put electric heating coils under it.

MR. RICK: Is there any advantage in sloping the top? Would it matter if it was flat?

MR. BERNATH: No, no, doesn't matter. This just happened to be an old melon box. I had started melons early in the spring.

Now, while the grafts are in the process of forming the unions, that is, when the cambium begins to form, you do not water until you take these out of the case. Add no more water, but make sure your pots are moist enough. For instance, in this one, there is plenty of moisture for the period of incubation.

MR. KINTZEL: How long? Couple of weeks?

MR. BERNATH: No. Sometimes they start to grow in three weeks, but generally four weeks, maybe a little over. Sometimes less; depends on everything.

MR. SHERMAN: What temperature in the greenhouse?

MR. BERNATH: Well, if you note in the springtime when the trees are beginning to grow, you know the night temperature goes down, while daytime may go up to 75, 80 in the spring. All right, you follow nature, and you'll never go wrong.

Now, the temperature, at night, if it does go down around the fifties, or even less, doesn't do any harm. That's the house temperature. But under the benches where you have your heat coils, that's of course, at least 60, maybe a little better, and, of course, in daytime it may—well, it's all right if it goes up to 70, 75. Then, of course, you have to ventilate through the house, and as a matter of fact, under the benches. Take a lot of bags and nail them along the walk to keep the heat under the benches. That gives you the bottom heat.

Now, as I understand, some of our members have tried this method, but they applied too much heat. They burned them. If they didn't burn them, fungus growth set in, because there's high humidity in that box. You will see the moisture condensation on the glass. Drops of water accumulate, and that's a thing you will have to guard against. So every morning give it at first about a 5-minute period when you take a dry cloth and wipe the surface moisture off the glass, the under side, to prevent the water from dripping on the unions here, to keep it dry. Then as you go along you can increase that period, but not over 15 minutes, until around the fourth week, you can generally put a stick under the glass to give more ventilation. When you see that the union is formed and everything is all right, take the glass off, take your grafts out and stand them up straight, and from there on you can water them, but not before.

And then you cut these stocks off right there as close as you can get it, sort of an upward movement, like that (demonstrating with knife).

MR. WEBER: It doesn't make any difference if you cut the rubber band that's on it or not?

MR. BERNATH: No, not too much, if it's callused up good, if the union is hard enough. And then, of course, you put the glass on, and then you keep these grafts in the greenhouse. But don't forget now, something that is important, when you graft these. Here we have a greenhouse over us. This little box represents the batch of grafts. Don't forget you have to shade them. If you didn't shade these, they would burn to a crisp. I have lost several hundred blue spruce grafts by going away on a day when it was cloudy and I forgot to tell Mrs. Bernath, "If the sun comes out, raise the sash." When I came home, this part of the greenhouse was shaded; now, in this corner here I think it was around 250 beautiful grafts but the next day I was going to take them out. They were burnt to a crisp. I saved a few trees right where it was shady.

MR. CALDWELL: The blue spruce are grafted by the same method?

MR. BERNATH: Yes, I use this method for inside grafting for everything.

MR. CALDWELL: Use this method for shagbarks the same way?

MR. BERNATH: Yes, same way with hickories and oaks.

MR. WEBER: What sort of shading element do you use? Anything real tight, or how?

MR. BERNATH: Yes, air tight. The grafting case has got to be air tight.

MR. WEBER: The shade?

MR. BERNATH: Oh, any kind of cloth, cheesecloth, muslin. I know that will do it.

MR. CHASE: Whitewash?

MR. BERNATH: That's all right, too. If you use whitewash, I would recommend using white lead with gasoline and just spray it on. That will help a lot, but I generally use a cloth for shade.

MR. O'ROURKE: Why do you place the scions so that the bud is on the inside?

MR. BERNATH: It makes a straighter tree. The other way it's inclined to grow out this way (indicating). It grows toward the stock, makes a straighter tree.

MR. STOKE: I think there is one more advantage there. On the edge next to the stock you get a better contact than you do on that lip on the outside, and it leads more directly into the bud.

DR. CRANE: Less danger, too, that that bud will rub off.

MR. BERNATH: Keep them shaded, but only 50 per cent shade. And then in about two weeks you take the shade off, let the sun shine on it. It doesn't hurt—over the glass. And then you take these pots when danger of frost is over, plant them out, in nursery rows, or, if you want to put them in permanent places, it's perfectly all right. Take this, put your finger under like that (demonstrating), give her a tap, and the ball comes out of the pot in your hand. And if it's permanent, plant it down to here; cover the union.

MR. WEBER: And the scion eventually forms its own root?

MR. BERNATH: It will. You will find that pot will be filled up with fibrous root.

MR. SZEGO: When do you take the tape off?

MR. BERNATH: Don't take it off at all. It will decay.

MR. MILLER: But the same graft can't be used outside without grafting wax, can it?

MR. BERNATH: Yes, you have to wax outside. That's right, you have to use wax. Otherwise the grafting method is the same for top-working.

MR. MILLER: Because in there you have it air tight. Outside you have to wax.

MR. BERNATH: You can't do it without wax, not outside. But budding you can do without wax outside.

This is a whole plant right here. That's a whole plant root, and this is right in this four-inch pot. That tap root is cut away and all the lateral roots, finer roots, put right in there and put in soil like any transplanted plant.

DR. ROHBACHER: When do you put that stock in the house?

MR. BERNATH: If you want to start work in January, towards the end of December after the understock has had the rest period. You can store them, unless you are in a place where you don't get much frost in your ground.

DR. ROHBACHER: You have to dig those up in the fall?

MR. BERNATH: You have to dig these up about three weeks before you want to graft. There is another point I should have been wide awake enough to tell you in the beginning: when you put these in the bench put them in peat moss like that, because otherwise it would be next to impossible to keep those plants moist enough.

MR. WEBER: That's standing upright.

MR. BERNATH: Upright until you graft. That's only the understock. Watch them closely, say about two weeks, and you may test it. In other words, knock these out and examine the root system. When you see those little white rootlets beginning to grow like thin macaroni, white, most of them, that's a sign that you had better get busy grafting.

MR. WEBER: But not until you see the edges of those roots poking through.

MR. RICK: And the stock isn't in the case until you are ready to graft?

MR. BERNATH: They are in the benches, but not in the case. No outside cover except just the glass of the house.

That's about all there is to it. It isn't much.

MR. RICK: It's been a wonderful demonstration.

MR. SZEGO: When do you cut your scion wood?

MR. BERNATH: Oh, I get scion wood from December on, late December,
January and February.

MR. RICK: It would be all right just to go out to the tree and cut your scions and bring them in and the next day graft?

MR. BERNATH: Yes. Well, no. I like to store them a little bit, for the reason that the starches will form. It's amazing how wood will act after you cut it, provided it doesn't dry out. All those cells, you know, in that they form what we call a certain type of starch. You can do it all right with apple trees and pear trees. You can put it right on the tree right from the tree, but I wouldn't advise it on the nut trees.

MR. RICK: Do you keep your scions cool until you are ready to use them?

A MEMBER: My way of keeping it is in fresh sawdust. That's the best means.

MR. WEBER: Do you dampen it any?

MR. BERNATH: Yes. And I have nothing but an earth cellar where I store my scion wood, and they keep well until June.

MR. RICK: To prevent fungus would it be a good idea to dip them in a weak solution of Bordeaux?

MR. BERNATH: I never tried it. I couldn't say. That's one reason why sometimes some of our members here wonder why I write and say, "Please do not wax." I do not want a waxed scion. As far as I am concerned, I would throw them right out. I wouldn't bother to graft them.

MR. CORSAN: You just put them in damp sawdust?

MR. BERNATH: Yes, put them in damp peat or even damp newspaper, wrap it and ship it.

(Newspaper is very good for this purpose.—J. C. McD.)

MR. CORSAN: And no waxing.

MR. BERNATH: No.

MR. STOKE: I agree with you. I got some scions that were waxed, and the scion was beautifully green and every bud was dead.

MR. BERNATH: That's it again. The reason for that is that you have to heat the wax to make it thin enough, and the reaction of the heat is bad for the scion wood.

MR. STOKE: I don't believe it's that alone. I believe a bud can't go without air for a great length of time. It is a living organism and needs the air. Those scions had come from Europe, and every one was dead.

MR. BERNATH: Mr. Silvis will tell you how he keeps his scions good.

MR. SILVIS: Through Goodrich Chemical Company I was interested in what Dr. Shelton, another Ohio member who is a chemist, had available, an emulsion called "Goodrite Latex VL-600." That's the agricultural and horticultural designation for its use. Otherwise, industrially it's known as Geon 31 XX, and some other names.

MR. CORSAN: That is the latex that congeals quickly?

MR. SILVIS: Yes. It's water soluble and makes a very stiff; impervious water barrier on everything it becomes attached to. Therefore, if you dipped the entire scion—usually I go out and cut scion wood and maybe even as late as the next day dip it in the latex. Then after it's dried for five minutes, I can take and throw it in the garage and leave it there until June, July and

August, and I can take it to the refrigerator, the same thing. I think the refrigerator is the best place.

MR. SHERMAN: You know last March, at the Ohio meeting there was some wood dipped there, and the latter part of May I came through and picked up a piece and brought it in to Harrisburg in the back of my car in the window where it was cooked in transportation, and it made two inches of growth in the Harrisburg office just lying on my desk.

MR. SILVIS: I have seen it happen, and it doesn't restrict the growth. I have had it on filberts, Persian walnut, and hickory. Then when I cut my stock by using a simple splice graft, in grafting it I use a rubber band, same rubber band they used here, tie it and just forget about it. You don't need the additional shading, and you don't need additional waxing.

DR. MacDANIELS: Can you use that material as a wax? Do you put on additional wax?

MR. SILVIS: It isn't necessary in a splice graft, because you have got a good union.

DR. MacDANIELS: Suppose you haven't got a good union?

MR. SILVIS: I wouldn't use it anyway, because you are covering the cut portion pretty well anyhow.

MR. RICK: Is this outside or inside?

MR. SILVIS: I would say outside. You dip the wax at 70 degrees temperature. Any colder than that would allow it to congeal. It's thick. I am not sure about this, but I think you can dilute it with about eight parts water, if you wish, six or eight parts water to one part latex. It still will make a complete coverage.

That's for scion storage, and it does eliminate making boxes in some places where they have storage problems. It eliminates the storage problem and eliminates waxing immediately after grafting.

MR. WEBER: Your method completely shuts off the air from the bud the same as waxing would do.

MR. SILVIS: And any water going in.

MR. STOKE: I was wondering how long you kept it. You said it was soluble in water. You mean before it sets up?

MR. SILVIS: Before it sets up.

MR. LOWERRE: That's if it's a suspension. It is some time before the water sets up.

MR. STOKE: Retaining moisture and yet being soluble, and that's the thing I wanted to clarify.

MR. SILVIS: If you leave it out, it is a dispersal, let's call it, but it appears like shellac after it is dry.

(Editor's Note: See fuller discussion in 1949 Report, pp. 30-37.)

MR. CHASE: I think we all owe Mr. Bernath a vote of thanks for showing us this. (Applause.) We will visit his place tomorrow, and if you have additional questions, I am sure he will be glad to answer them for you. He has left the grafting case over here for anyone to see.

MR. SHERMAN: In case of heavy rain tomorrow, what are the plans?

MR. SALZER: Wear rubbers.

MR. CHASE: We are not going to have any rain tomorrow.

(He was right.—Ed.)

We have a short paper here that I have asked Dr. Anthony to summarize for us, "Experiences in Nut Growing Near Lake Erie," by Ross P. Wright, Erie, Pennsylvania.

DR. ANTHONY: Mr. Wright is a very interesting man and has a very interesting plantation. He is a manufacturer and fortunately has a son who is mature and married and as interested in the work as he, so there is a continuity that we are pretty sure of.

Experiences in Nut Growing Near Lake Erie

ROSS PIER WRIGHT, Erie, Pennsylvania

This report should be made by my son Richard Wright. He is in charge of the farm but is on a trip to Europe with his family and will not return in time for your meeting.

The farm is located in the Chautauqua Grape Belt; due to the proximity of Lake Erie, which acts as a heat reservoir, it is not as a rule bothered by the late frosts in the Spring or early frosts in the Fall, this making it a very satisfactory climate for Concord grapes. Peaches are also grown commercially.

The village of Westfield is located on the main road between Erie and Buffalo, and the Wright family has lived there for the past 136 years. We have several hundred acres and really started the endeavor more with the idea of seeing if nuts might be profitably grown, without any idea of going into the nut business.

In 1915, 35 years ago, we planted a three acre plot with several varieties of nut trees obtained from nurseries. They were black walnuts, hickories, hazel nuts, pecans, English walnuts, and Japanese heartnuts.

The black walnuts are native of Westfield and the trees we planted have done well. The only hickories that survived were two Siers hickories. We did not think much of them until recently as they did not fill out any too well, but the last three or four years they have for some reason decided to fill better. Due to the extremely thin shell they are very easily cracked and at the moment we think quite highly of these Siers hickories.

We have a nut cracker made by the Dazey Corporation of St. Louis, Missouri, which costs $5.00 or $6.00. It is very effective with the Siers but does not crack thick shelled hickories very well. On the other hand it is ideal for pecans and English walnuts.

The filberts in this field are not very satisfactory, with the exception of the Winkler hazel. These usually bear very well. The trouble with the filberts is that the catkins are quite prone to winter kill but the Winkler hazel seems to be more hardy. There again we think more of them since we have used the Dazey Nut Cracker. The Winkler nuts are rather small and have quite a hard shell and if a hammer is used it is quite likely to crush the kernel.

The English walnuts we planted at that time were not of a hardy type and were prone to winterkill. There are really only two stunted trees left.

The pecans do not winterkill but the nuts do not fill.

The Japanese heartnuts we planted were successful. One of them we consider very satisfactory and is worthy of propagation. We call it the Lobular heartnut.

In the Spring of 1923, 27 years ago, we obtained a half bushel of heartnuts from our representative in Japan and planted them. Three years later we interplanted some of the trees in a four acre field in which we were planting as permanent trees some Snyder and Thomas black walnuts. Reporting on that field as it is today we will say that these walnuts and heartnuts, up to five years ago, bore very well indeed and the nuts filled properly, but the last few years the nuts have not filled properly although they have received nitrate of soda. We are somewhat in a quandry as to the reason for it.

Adjoining the field is a black walnut tree, probably 150 years old, which always bore nuts and they have always filled up to the last few years. In this field where the majority of the seedling heartnuts have been planted there was the usual interesting difference in the nuts. Some were of the true heartnut variety, some had the rough shaggy shell and shape of a butternut and others were round and looked like English walnuts. Some of the heartnut trees have developed a disease called witches'-broom or bunch disease. There does not, to date, seem to be any cure for it. We used some heavy applications of zinc sulphate and thought the trouble had improved but the improvement seems to have been only temporary.

In this field also are the trees which Clarence Reed designated as the Wright heartnut and the Westfield heartnut.

In 1933 to 1935, 15 to 17 years ago, we grafted about 35 hickories with various varieties. They were grafted in a grove of hickories which were on our farm and which were perhaps eight inches in diameter. This endeavor did not prove to be much of a success. Some of the grafts died after a year or two and the others which have continued to live do not appear to bear to any extent. We would have to mark that particular endeavor down as very close to a failure.

Perhaps if we had given the grafting endeavor more attention we might have had different results but we are in the manufacturing business in Erie, Pennsylvania, and really look upon the Westfield, New York, farm as a type of relaxation. In those years 1933 to 1935 industry was experiencing a major distress and I am afraid most of our attention was given to our factory rather than our farm. In fact, that situation applies very largely to all of our nut endeavors. There is an old Scotch saying "The eye of the master fattens the kine," and during the last 15 or 20 years when we in industry have experienced a tremendous depression followed by a war it has meant that those interested have had to watch their manufacturing plants to the detriment of their other interests regardless of how much they regretted it.

In 1934, 16 years ago, we became interested in chestnuts as a possible commercial crop. We purchased a quantity from J. Russell Smith, interplanting them in a vineyard we expected to pull out as it was getting too old. Two years later, through the cooperation of Clarence Reed, Dr. Gravatt, also others at Beltsville, Maryland, we got some 2,000 seedlings of various types, some being hybrids. As some of these bore we planted what we thought were the best nuts in a nursery and at present have about 3000 chestnut trees ranging from three years old up to 16 years. There is some blight occasionally showing which appears to be on the hybrids. About 35 acres of the chestnuts were interplanted in vineyards which we were planning to pull out. During the war, however, the price of grapes was quite high and we left the grapes, pulling the last of them out this Spring. Due to cultivation of the grapes an appreciable number of the nut trees were cut out accidentally, and have later been filled in with seedlings, with the result that the orchard has a rather peculiar appearance. The mature trees, this year, have been doing, we think, very well, and a great majority of them are bearing from a light crop to a rather heavy crop.

Up to date we have had no trouble with worm in our chestnuts. In fact we have not found a single wormy chestnut. This interests us appreciably, as when the old American chestnuts were common on our farm it would seem as if hardly a chestnut escaped a worm hole if you kept them long enough. If you ate the chestnuts immediately it wasn't so bad—the worms were probably too small to be observed.

We understand that in some sections Chinese chestnuts are attacked by worms but I repeat we haven't had one to date.

Our chestnuts are planted largely in Volusia clay loam on fields where chestnuts formerly flourished. This soil is not fertile, as soils go, and the trees will probably not grow as large nor will they grow as fast as if planted in a more fertile soil. At first we used a spacing of 36 feet but we now use 24 feet, which we think will be satisfactory for our farm.

Since the chestnuts have come into bearing and the project has become to some extent a commercial one, we are more interested in doing what we can for the trees. We are convinced that the mulching process is to be recommended. There is some sawdust to be obtained in this section and as far as it goes we have covered the ground under the branches of the trees with a mulch of sawdust about five or six inches deep. We will not know how successful that is for a few years.

We have planted the fields with a cover crop of rye grass and orchard grass, and this month are cutting it and throwing it under the trees. We have some adjoining fields which were in hay but which had rather run out. We are cutting these likewise and throwing the hay under the trees. We believe if we keep this practice up for a few years we will have a reasonable mulch under the trees. We have become interested in Reed canary grass. We have had a few sample patches of it and are going to plant a couple of outside fields with it to be used for mulch. It grows stronger than any other northern grass with which we are conversant, and therefore would produce more mulch. We are also giving the land two rather heavy applications of mixed fertilizer each year.

We think the chief thing we have learned about chestnuts is that the first few years the trees should be cultivated, fertilized, watered, and mulched. You cannot handle them the way you could, for instance, Christmas trees by simply sticking them in a field of grass. The first year they should be watered every ten days if they require it, and watered the second year if there is a real drought.

In closing we would say that as far as our immediate section is concerned, it is our guess that chestnuts are the only nuts which might appear to have commercial possibilities. Of course, at present, the nuts sell at quite a high price and I fear beyond their value. What will happen when the numerous orchards which have been planted in the last few years come into bearing is any man's guess.

We do not believe that the black walnuts would ever prove a commercial success here, although they normally do well. Of course the trouble is the competition of the wild nuts from other sections. On the other hand, if some one had the time to give to working up a market for the improved black walnuts, he might get some profit out of it.

If I were younger, I might want to try growing a number of Winkler hazel nuts. I think hazel nuts covered with chocolate make a very attractive candy, and here, in this section, the Winkler seems to be immune to blight and other troubles. This year, for the first time in our recollection, the frost got them and the crop is very light.

I do not know just what to say about the heartnuts. They might not have enough flavor to suit some people, but when eaten with salt I think they are delicious. They are very free cracking. We have one, the Lobular, which as soon as they are cracked can be shaken out of the shell. I am disturbed however over the bunch disease to which some of them are subject.

Please note that our remarks in regard to the commercial possibilities of these various nuts has reference to our farm at Westfield and to no other place.

I regret I am not going to be at your meeting to endeavor to answer any question which might be asked.

Discussion of Mulches

DR. ANTHONY: Mr. Sherman and I were there a few years ago, and he has very definitely given up the heartnut and black walnut. Many trees in this area are affected with this bunch disease, which caused failure to set, and he has very definitely decided that he is out of those two nuts.

MR. FRYE: That sawdust, how old must it be, and how green have you used?

DR. ANTHONY: We have used sawdust in our fruit tree work. There is a period when I don't like it. When it's raw and going down, it uses a good deal of nitrogen. Also, if it gets dry, it will blow. Also when it gets dry it will run off with the water, and I would like to use it pretty well rotted down when I get it, and usually you can find old rotted piles. If you do use it on trees where nitrogen is a factor, you probably will have to use additional nitrogen.

Now, with the chestnut where you want to mature them fairly early in the fall, it might work all right, because it will withhold the nitrogen in the breakdown of your sawdust. But apparently, it works pretty well. I think it was Mr. Sam Hemming who suggested using it in the rows. Most of our State Forests and Waters nurseries in their seedling beds, plant their seedlings, including chestnuts, make a mixture of sawdust and sand, about one of sawdust and two of sand, and then broadcast that right over their seeds. The seeds are broadcast on the firm soil, then this mixture of sawdust and sand is broadcast over the seeds. That gives a uniform planting of your seeds and gives a very nice protection. There is one place that I think sawdust works very nicely.

Straw mulch, any material of that kind, in breaking down takes nitrogen from the soil. They are all good if you balance that loss of nitrogen that is lost during the period of breakdown. Now, there comes a time, if you put a mulch on the soil and let it stay there for six or eight years and keep building it up, when you pass imperceptibly from straw into soil, and when you reach that time, your breakdown of your straw is usually done without taking nitrogen from your soil itself, and from that time on you may release nitrogen. But until you get that imperceptible transformation from straw to soil, there is a time when the breakdown of the straw uses your nitrogen, which is all right, if it's late in the season, but not early. I'd want to watch my trees and get my nitrogen on early, then let the straw use it later on.

A MEMBER: The migration of nitrogen—is there some such migration, and is it just in the case of the sawdust?

DR. ANTHONY: You put it right on top, it's much worse. You can put it right on top and it will take a year or two to pass through that period where the utilization in the breaking down of the straw is greater than the release of nitrogen. If it's mixed in the soil, the tree gets more of it.

MR. STOKE: How deep is that effect on the soil?

DR. ANTHONY: We have used straw, hay, weeds, sawdust, chips, anything of the kind, putting on a 5 to 6-inch layer. As I say, it takes from one to three years to get through that period.

Now, Massachusetts has the longest continuous use—all of New England has—of mulch, and they are reaching a point now where some of the mulches are ten years old where the release of nitrogen is too much and they get poor color on McIntosh. I think with the Chinese chestnut this is one thing we have got to watch to get good maturity. Going farther and farther south, you have more trouble. As you go to the north, our trees color more easily, and there you wouldn't want to force them, as our New England people find. They are releasing too much nitrogen late in the season. So I would not want to use long, continued mulch in the chestnut, I'd watch my maturity, and the minute they get a little slow in maturing, I'd quit.

MR. BERST: How about corn cobs?

MR. JAY SMITH: How about anything in the street, leaves?

DR. ANTHONY: Anything like that, whether it's oak or maple. One goes down as quickly as the other.

MR. CORSAN: On the way down here I called in to see Rodale, and we found him in a mass of brewer's hops and ground up corn cobs. He had them in the chicken house, and you know how a chicken house smells. He had no smell in the chicken house. We looked all through his place, and we saw another big pile of furs, mink, and such trimming off of them, a big pile about that high (indicating), and that will go down. He had everything under the sun in the way of mulch, but corn cobs ground up fine was the chief one in sight.

Personally, I like to grow the mulch on the land right there. We can grow it—up to 10 ton of green mulch to the acre. I have done it many, many times. You have something there that goes down quickly. The very growing of that through the latter part of the summer also uses the nitrogen and hardens up your trees. Then we turn it down and within two to three weeks we have it reseeded, and so we are growing a constant supply in the soil-itself. You get the same effect as hauling in your mulch. It's cheaper, usually, and you get, I think, a little bit better control. Your mulches are not dry, they are turned under when—well, it's crimson clover in the red, right in the blossom. They go down very quickly. We leave as much as possible on the surface. I think it's a little cheaper and a little more satisfactory control. I put them on quite green. I find they rot much quicker.

MR. CHASE: I will now turn the gavel back to Dr. MacDaniels, who will take over.

DR. MacDANIELS: Thank you, very much, Mr. Chase.

Perhaps we had better take a 10-minute recess.

(Whereupon, a short recess was taken.)

Nominating Committee Elected

DR. MacDANIELS: We will proceed with the election of a nominating committee. That committee is elected. It is a committee of three, and the nominations come from the floor. The present nominating committee is Mr. Stoke, Mr. Sylvester Shessler, and Mr. Sterling Smith. Now, I guess it is a good plan to change the nominating committee, and I think we ought to have regional representation. I think that is important. Does anybody have a nomination? Say we start in the Middle West.

A MEMBER: Mr. Silvis.

DR. MacDANIELS: He will take it. That's middle. Another nomination from the farther west.

MR. CHASE: Mr. Chairman, I nominate Dr. Crane.

DR. MacDANIELS: That would be South Atlantic.

MR. WEBER: I nominate Mr. Chase.

DR. MacDANIELS: Do you wish to nominate more than three and have a ballot?

MR. FRYE: I move nominations be closed.

DR. MacDANIELS: Nominations closed. Do you move to have the secretary cast a unanimous ballot?

DR. McKAY: So move, Mr. Chairman.

MR. WEBER: Proceed with the election.

DR. MacDANIELS: The motion is that nominations be closed and the secretary be instructed to cast a ballot for the slate as nominated. Any further discussion? If not, all in favor say "aye."

(A vote was taken on the motion, and it was carried unanimously.)

DR. MacDANIELS: Carried.

Resolutions

DR. MacDANIELS: Is the Resolutions Committee here? Mr. Allaman, I believe you are president of the Pennsylvania group, are you not?

MR. ALLAMAN: Yes.

"In the passing of Clarence A. Reed, who was a nut culturist of the United States Department of Agriculture, we not only lost a friend in the experimental field, but also a dear personal friend. Mr. Reed was keenly interested in all phases of nut culture, devoting practically his entire life to this work. We are more deeply indebted to him than can be expressed. Paraphrasing what Lincoln said of the dead soldiers at Gettysburg, it remains for us to continue the effort and build upon the foundation to which he so largely contributed.

"Therefore, be it resolved that the secretary of this Association spread upon the record this resolution and send a copy to Mrs. Reed."

DR. MacDANIELS: You have heard this resolution. I think it would be appropriate we move to accept and adopt this by a rising vote.

(Whereupon, a rising vote was taken.)

DR. MacDANIELS: There are two other resolutions Mr. Allaman will read.

MR. ALLAMAN: "The Northern Nut Growers Association in its forty-first meeting expresses its appreciation for the fine accomodations for its meeting place supplied by Post No. 739 of the American Legion. The Association also desires to compliment the Post on its foresight in providing this community with such a satisfactory meeting place.

"May it therefore be resolved that the secretary spread this upon the minutes and send a copy to the Legion."

Another resolution: "We, the members of the Northern Nut Growers Association, express our keen appreciation of the very efficient services of Mrs. Stephen Bernath and Gilbert L. Smith and others for their splendid accommodations at this convention."

DR. MacDANIELS: These two resolutions, do you wish to accept them or adopt them together?

DR. CRANE: Move that they be adopted as a whole.

DR. MacDANIELS: Moved that they be adopted together. Any discussion? If not, all in favor say "aye."

(Whereupon, a vote was taken on the motion, and it was carried unanimously.)

DR. MacDANIELS: Passed without dissent.

Are there other resolutions anyone has from the floor?

(No response.)

Report of Auditing Committee

DR. MacDANIELS: The auditing committee's report.

MR. WEBER: I have it. "We have found from our examination of the treasurer's records that his accounts are in proper balance and that the statement of his bank account, issued by his bank as of August 11, 1950, shows he had on deposit in the Erie County United Bank of Vermilion, Ohio, the sum of $2280.37. We feel our treasurer, Mr. Sterling A. Smith, has faithfully discharged his duties during the current year and recommend his continuance in that office, nomination for which has already, of course, taken place. Royal Oakes, Chairman, Auditing Committee." (Applause.)

DR. MacDANIELS: It all sounds very legal. I think it's all right. I take it that applause indicates the acceptance of the report. Unless I hear dissent, we will take that to be so.

DR. CRANE: Move the report of the Auditing Committee be accepted.

DR. MacDANIELS: O.K., we will make it legal. Who will second the motion?

MR. STOKE: Second.

DR. MacDANIELS: Moved and seconded that the Auditing Committee report be accepted.

(A vote was taken on the motion, and it was carried unanimously.)

Election of 1950-51 Officers

DR. MacDANIELS: Next will be the election of officers, and we will ask the chairman of the Nominating Committee to give his report. Inasmuch as I am apparently concerned, I will hand the gavel to Mr. Chase for the election.

MR. CHASE: We'd like to hear the report from the chairman of the Nominating Committee, Mr. Stoke.

MR. STOKE: Most of you no doubt heard the report of the Nominating Committee at our first session, but we nominated Dr. William Rohrbacher of Iowa City, Iowa, for president, and for vice-president our perennial candidate here, who has disappeared from the scene, renominating Dr. L. H. MacDaniels. We hope to make him president next time. If he doesn't make it next time, I think we will have to throw him out. And for the secretary, our friend, Joe McDaniel. They are not relatives. And the treasurer, repeating officer, Sterling Smith. The secretaryship and treasurership shouldn't change any more often than necessary.

MR. STERLING SMITH: I object.

Before you move on that, I'd like to say that it isn't really legal, I think, that I should have been on the nominating committee, and being one of the officers, it would be very well taken on my part if there were any nominations from the floor.

MR. CHASE: We are coming to that.

Any objections that we have nominations from the floor? Are there any nominations for president?

MR. WELLMAN: Move nominations be closed.

MR. CHASE: Are there any other nominations for vice-president? (No response.) I am sure we must have one for the treasurer. (No response.) Do we have any for secretary?

MR. CORSAN: Why not have the former Miss Jones president again?

MR. STOKE: She becomes a member of the Board of Directors, and I think it would be out of order to elect her to another office.

MR. CORSAN: I withdraw it.

MR. CHASE: Now I will entertain your motion, Mr. Wellman.

MR. WELLMAN: I move it.

MR. CHASE: It has been moved that the slate by the nominating committee be accepted.

DR. CRANE: Second.

(Whereupon, a vote was taken on the motion, and it was carried unanimously.)

MR. CHASE: Dr. MacDaniels, you may come in now.

DR. CRANE: We moved that nominations be closed. We haven't accepted them.

MR. STOKE: When you are through, I have a resolution to offer.

DR. CRANE: Move that the report of the nominating committee be accepted and we proceed with the election by voice vote. All in favor of having the secretary cast a ballot for the slate nominated by our nominating committee please signify by saying "aye."

(A vote was taken on the motion, and it was carried unanimously.)

MR. STOKE: I would like to make a motion that we elect a parliamentarian, and I wish to nominate Dr. Crane.

MR. STERLING SMITH: Second the motion.

(A vote was taken on the motion, and it was carried unanimously.)

MR. FRYE: We elected a parliamentarian last year. I wonder how it's coming on.

DR. CRANE: I have a report on it.

MR. WEBER: Mr. John Davidson, Xenia, Ohio.

MR. McDANIEL: He was parliamentarian before we made him our president.

MR. WEBER: That's passed on to Dr. Crane.

MR. CHASE: Now, Dr. MacDaniels, you may come in.

DR. MacDANIELS: Hope it's legal.

Is there any further business? Do you think of any, Mr. Weber?

MR. WEBER: Hold it open until after the banquet. Then if we think of something that we have left out, we haven't adjourned.

DR. MacDANIELS: I will adjourn this particular session and give the gavel to our new president.

MR. WEBER: We adjourn until this evening at the banquet.

DR. ANTHONY: Before you bang it down, may I make one announcement? I thought you would be interested in an action that the Pennsylvania Nut Growers have taken. Mr. Allaman, it is O.K. to report that committee appointment?

DR. MacDANIELS: The question is raised as to the time of the next meeting. The place has been decided. The time, I think, has to be left to be worked out with the authorities at Illinois, is that right? Do you want to say a word, Dr. Colby?

DR. COLBY: It is difficult, if not impossible, to give an exact date right now, because we don't know at this time what our facilities for meeting rooms and lodging will be on any particular date in the latter part of the month of August. We will have to check and find out the best days, if that is agreeable to the group.

DR. MacDANIELS: Does this group wish to express a preference as to the last week in August or the first week in September? In other words, it would be the week before Labor Day, or the week after. That wouldn't necessarily fix it, but it would give the committee, if there were no other restrictions as to available facilities, would be a guide for a choice.

MR. WELLMAN: Call for a show of hands.

DR. MacDANIELS: I will do that. Those who would prefer a meeting date comparable to this year? (Showing of hands.)[34] Those who prefer the week after Labor Day? (No hands raised.)

[34] The 1951 meeting will be at the University of Illinois in Urbana, August 28 and 29, to be followed with a tour in western Illinois for those who can stay through the morning of August 31.

MR. STERLING SMITH: Maybe those who prefer the after Labor Day date aren't here now.

DR. ROHRBACHER: I just want to say I appreciate very much the honor that has been bestowed upon me. I appreciate the fact that the president is purely an emblem, a figurehead, but with the staff that's under him, it's the same as in the Post Office Department of the United States, the head receives all the salary and his understudies do all the work. So it's a very appropriate setting, and we should go forward under a very good staff of men that have been elected to the positions under that of the president.

One thing I want to say in regard to the problem that came up last night that was discussed: that as the president, I can assure you that the vice-presidents are certainly not going to be emblems if they expect to continue on in their positions in the various states that are in the group, because the working out of this problem, the success of it, is going to depend on how well these vice-presidents carry out their work.

I thank you.

DR. MacDANIELS: We will close this session until tonight. I will give Dr. Rohrbacher the gavel.

(Whereupon, at 4:50 o'clock, p.m., the Tuesday afternoon session of the Northern Nut Growers Association was closed.)

Note on the Annual Tour, August 30, 1950

The third day of the Annual meeting, as is customary with the Association, was spent touring interesting nut plantings in the vicinity. The first stop was Bernath's Nursery, southwest of Pleasant Valley, where he has his greenhouse, young nut plants, and a number of fruiting trees. The second stop was on the grounds of the State School at Wassaic, where many grafted nut trees, particularly walnuts, are thriving, due to the interest and activity of Gilbert L. Smith, when he was on the staff there. A picnic lunch was served in the recreational area of the school grounds. Here Dr. W. C.

Deming of Hartford, Conn., Dean of the Association, was on hand to greet many of his old friends. After lunch we visited Mr. Stephen Bernath's farm nut planting, then the topworked hickory woods on Mr. Wm. A. Benton's farm out of Millerton. At the Benton and Smith Nut Nursery, also on the farm, the tour was concluded.

OBITUARIES

Harry R. Weber

Members were saddened to hear of the death, on his way home, of Harry R.
Weber, who had taken an active part in the meeting at Pleasant Valley,
as he did in most of the meetings since the very earliest years of the
Association. We shall have a more complete obituary in the next volume.

George B. Rhodes

COVINGTON, Tenn., Dec. 16, 1950—Services for George B. Rhodes of Mt. Carmel who died Saturday at 5:15 p.m. at his home will be held Sunday afternoon at 3 at the Clopton Methodist Church. The Rev. David Olhansen, pastor of the church, assisted by the Rev. E. D. Farris of Henning will officiate. Burial will be in the Clopton Cemetery.

Mr. Rhodes, who was 82, was born at Clopton, Tenn., and spent his entire lifetime in Tipton County. He was the first county agent of Tipton County. He was interested in the budding of pecans and had operated a nursery for the past 20 years. He was a member of the Clopton Methodist Church.

He leaves his wife, Mrs. Ivie Drake Rhodes of Covington; two sons, Sol Rhodes of Tampa, Fla., and Marion Rhodes of Beverly Hills, Calif.; two daughters, Mrs. R. B. Davie of Covington and Mrs. Lillian Bringley of Memphis; two sisters, Mrs. Pauline Meacham of Senatobia, Miss., and Mrs. Mattie Nelson of Forrest City, Ark., and two brothers, Sam Rhodes of Bolivar, and Duke Rhodes of San Francisco, Calif.; seven grandchildren and five great grandchildren.—Reprinted from a Memphis paper.

Mr. Rhodes' greatest contribution to nut growing was the discovery and first propagation of a heartnut variety mow called Rhodes. It is the most successful heartnut yet tried in western Tennessee, a reliable and heavy cropper, and one of the best cracking varieties of all known heartnuts. It deserves testing in other areas.

Note: The following members of the N. N. G. A. have died recently, and we hope to have fuller obituaries on them in the next volume:

Charles C. Dean, of Anniston, Ala. (Died September 21, 1950.)

Henry Gressel, of Mohawk, N. Y. (Died in June, 1951.)

W. N. Achenbach, of Petoskey, Mich.

L. B. Hoyer, of Omaha, Nebr.

Life Member Wang Is in Hong Kong

In our 1942 Report there was a note that our only Chinese member, P. W. Wang, had probably died, since he had not been heard from since 1930. Mr. Wang, we are happy to report, has recently written to us from Hong Kong. Many of the nut trees he planted while secretary of the Kinsan Arboretum at Chuking (not Chungking) in Kiangsu Province had survived the Japanese invasions and were fruiting in 1945, but are now in Communist hands. Mr. Wang hopes some day to be able to send to America scions of a fine pecan (seedling of Teche variety) which he fruited at Chuking. Meanwhile, he wishes to have nut literature and catalogues sent to him at his present address: P. W. Wang, c/o China Products Trading Corporation, 6 Des Voeux Road, Central, Hong Kong.

Letters

Nuts in Quebec

July 16, 1950

Dr. George L. Slate,
Associate Professor,
New York State Agricultural Experiment Station,
Geneva, New York

Dear Dr. Slate:

I am very much flattered by your invitation to prepare a paper on nut culture in Quebec. My only regret is that for two reasons I am unable to comply with your request.

The first is that I am quite ignorant on the subject. It is only lately that I have developed an interest in this matter when I suddenly found myself responsible for a so-called "arboretum" which is now mainly empty space that I am endeavoring to fill. The fact that shagbark hickory and butternut were common in our woods and that some of our neighbors have apparently flourishing individual trees of black walnut served to arouse my interest in the question. One neighbour has a tree of what he calls "French walnut" because they came from near Lyons, France, which are evidently the ordinary English or Persian walnut. Furthermore, I have been advised that there is quite a grove of black walnut near Lotbiniere, Quebec, which is on the south shore of the St. Lawrence not far from the city of Quebec. I understand that it was planted some seventy-five years ago and trees are now timber size. Indeed, I was told that the owner was offered a considerable sum during the war—the wood was wanted for gun stocks. I have not been there to verify this. However it occurred to me that it would be a good idea to get several specimens of various nut species that might grow here to place in the arboretum—this might incidentally give some information on what species would survive our winters.

The second reason that I am unable to write any article on nut culture in Quebec is because as far as I know there is no nut culture here. Most of the trees I refer to were simply planted as ornamentals. I have never been able to locate anyone who has taken any particular interest in growing them for the nuts.

I would like very much to extend my knowledge on the subject by attending your meeting at Poughkeepsie, New York, on August 28th to 30th, but unfortunately I will be absent in Nova Scotia on those dates.

Following your information I secured some literature on northern nut culture and will look forward to receiving any further information along this line that may be forthcoming.

Again thanking you for your courtesy and assuring you of my continued interest, I am,

Yours very truly,

W. H. BRITTAIN
Vice-Principal,

Macdonald College of McGill University

Macdonald College, Quebec, Canada

Note: I believe that perhaps the things mentioned in his second paragraph should be followed up.—H.L.S.

Pecans Produce Poorly in Middle Atlantic States

November 13, 1950

Dr. Lewis E. Theiss
Lewisburg, Pennsylvania

Dear Dr. Theiss:

Speaking of pecans, we have harvested the first crop this year here on the station, from trees planted in 1932, of the varieties Indiana, Greenriver, Busseron and Major. Even though these nuts were not harvested until November 9 they are poorly filled. It seems that we just cannot mature them here in an average season. Our trees have not grown satisfactorily and although they may bloom, the nuts normally fail to mature.

Our summers are not long enough and the day and night temperatures are not high enough uniformly to satisfactorily produce pecans even in this area.

Very truly yours,

H. L. CRANE
Principal Horticulturist,
Division of Fruit and Vegetable
Crops and Diseases

U. S. Plant Industry Station.
Beltsville, Maryland

~Editor's Note:~ Dr Crane's experience is exactly similar to my own. The pecans in the grounds at my country home were well loaded with nuts this year, 1950. I doubt if a single nut was half filled.—L. E. T.

Nut Tree Diseases in Europe and Turkey

November 17, 1950

Dr. Lewis E. Theiss
Lewisburg, Pennsylvania

Dear Dr. Theiss:

I have only recently returned from three and one-half months spent in Europe, primarily on chestnut problems, as a consultant for the Economic Cooperation Administration. The trip was made at the request and expense of European interests, except while I was up in the Scandinavian countries and at the 7th International Botanical Congress. I gave a paper at the Congress, entitled "The world-wide spread of forest diseases," in which chestnut blight received limited attention.

In Italy, chestnut blight, ~Endothia parasitica~, was first reported at Genoa in 1938, although it started there much earlier. It is now widely distributed here and there as far south as the Naples area. No confirmed infections have been reported from Sicily, Sardinia, or French Corsica, though inspection work has been very, very limited. In all the places where I saw it, the disease was increasing rapidly, with numerous recently-blighted trees. It is expected that the disease will ultimately kill the 988,000 acres of coppice growth, which produces few nuts, and the 1,111,500 acres of grafted orchards. The time of death of isolated stands like the two islands and many other areas can be materially decreased by careful inspection and removal of the earliest infections, just as we have held the disease under control in the European chestnut orchards in California. It is doubtful if this will be done however, in spite of their large unemployment problem.

As the blight continues its rapid spread over Italy, the production of nuts will steadily decrease. The Italian exports to this country will decrease, and the market for the rapidly expanding production of Chinese chestnuts in the eastern United States will improve. The Italian foresters are growing large quantities of Chinese chestnuts which they purchased in this country, but the difficulties of quickly reestablishing a large nut industry are very great. This Bureau, including Dr. Graves, has been sending pollen, scions, and plants of our selections to help with this work. It is of vital importance to have a sound economy in Italy to help prevent the Communists from taking over, and loss of their forest and nut orchards and part of their oaks from the blight will be a sad blow to their economy.

The chestnut blight fungus in Italy is attacking three important European oaks, ~Quercus ilex~, ~Q. Pubescens~, and ~Q. sessiliflora~. These are more important in some countries than chestnuts. For instance, Spain has 3,705,000 acres of ~Q. ilex~ orchards, grown largely for acorn hog feed. This will interest Dr. Smith. Possibly the disease may be less destructive to oaks in other countries than I fear, my opinion being based on the examination of only a limited number of diseased oaks in Italy.

I assume you have heard that Mr. Bretz of our Division has found that the oak wilt fungus has attacked some of our Chinese chestnuts in Missouri. What it will amount to, no one knows. The oak wilt continues to spread southward and eastward, and this year one infection was reported by the State authorities on oaks in your own Pennsylvania.

In Switzerland, in Tessin province, which is along the Italian border, the blight is spreading rapidly. The disease undoubtedly is in Yugoslavia, as there is so much infection in nearby Italy, but I was not in Yugoslavia. In Spain, there are several infections of blight that came in on the original importations of chestnuts directly from Japan. I made two trips into Spain and the authorities there have promised to do everything possible to eradicate these small spot infections.

In Denmark, England, France, Germany, Portugal, and Turkey no blight had been reported by the authorities with whom I conferred, but in most of these countries very little inspection work has been conducted. Any inspection for blight in southern Europe is complicated by the presence of the ink root rot disease, which from a distance looks like the blight. I remember one grafted orchard planting, in the Asia Minor part of Turkey, where a large proportion of the trees were dead or dying, with yellow leaves hanging, resembling the blight. Incidentally, here, as at a number of other places in different countries, orchards, forest, and nearby agricultural land was owned by the village itself.

In southern France I was impressed by a most serious and widely distributed disease of Persian walnuts. Vigorously growing trees start to decline and within a year or two they are dead. The French authorities had no satisfactory explanation of the trouble. I informed them that it looked a lot like trees killed by ~Phytophthora cinnamomi~, the cause of the chestnut root and ink disease in America and Europe. This fungus also attacks both Persian and black walnuts and other trees (including apples) under certain conditions.

Sincerely,
G. F. GRAVATT

Senior Pathologist,
Division of Forest Pathology

U. S. Plant Industry Station, Beltsville, Md.

Nut Work of the Minnesota Experiment Station

March 27, 1950

Mr. Gilbert Becker,
Climax, Michigan

Dear Mr Becker:

I have heard that not long ago you sent out a questionnaire relative to nut growing and grafting. Perhaps you would like to include the work which has been going on at the Minnesota Agricultural Experiment Station since 1918.

When this study was started, we had no information to give to many who came to us with questions on nut growing possibilities in this state. At no time have we attempted to promote commercial development as the interest here seems to be almost wholly amateur.

Our first efforts, begun in 1918, were designed to test kinds and varieties which could be grown in Minnesota. Black walnut varieties such as Thomas, Ohio, Ten Eyck, Stabler and Miller were planted at University Farm. Also sweet chestnuts Boone, Rochester, Cooper, Paragon, Fuller and Progress were set out. Hickory varieties and hybrids planted in 1918 and 1919 were Kirtland, Weiker, Stanley, Siers, Hales and McCallister. We planted a few trees of the Franquette Persian walnut, the Indiana, Niblack and Posey pecans and a few filberts such as Minnas Zellernuss, Daviana, and Large Globe. Some seedling trees of the shagbark hickory also were set out in 1918 and 1919.

To supplement this test somewhat similar collections were sent to cooperators in what seemed to be favorable locations.

We had the usual difficulty in establishing these trees and winter temperature eliminated all the pecans, sweet chestnuts, Persian, walnuts and filberts. Some of the seedling hickories survived and have grown vigorously but after thirty-two years have borne no nuts.

Since 1939 cooperative work has been under way with Professor R. E. Hodgson at the Southeast Experiment Station, Waseca. Efforts there mainly have been to establish varieties of black walnut and hickory by grafting. Black walnut and hickory varieties have been grafted also at the Fruit Breeding Farm, Excelsior.

The accompanying record is taken from a report for the Experiment Station in 1949. It should tell you in brief the status of our investigations at present.

Very truly yours,
W. G. BRIERLEY

University of Minnesota
Department of Agriculture
Division of Horticulture

Nature and Extent of Work Done this Year

All black walnut and hickory trees made fairly satisfactory growth in 1948 in spite of deficient rainfall. The "Gideon Seedling Hickories"

(~Carya laciniosa~) planted in 1945 have become established at Waseca, Rochester, Lakeville, Mound and at the Fruit Breeding Farm.

Attempts to establish nut varieties by top-working on seedling trees again met with poor success. At Waseca 5 of 14 hickory grafts and 4 of 25 black walnut grafts grew. At the Fruit Breeding Farm only 6 of 33 hickory grafts grew. In this case, the poor results were due in large part to use of an asphalt grafting compound which injured the callus tissue at the union. Better than usual success was obtained with black walnuts as 19 of 37 grafts grew.

As in previous seasons, the best temperature for storage of scion wood was 34 to 36 degrees F.

Major Results

The best black walnut varieties for Minnesota are Thomas, Ohio, Stambaugh, Smith and Schwartz. Of these Thomas produces the best nuts, but the tree is somewhat straggly in growth. The Ohio produces large nuts of good quality and is by far the best tree in ornamental value. It also is the hardiest of all varieties tested as it has shown no injury during 16 winters. Of lesser value are Ten Eyck which apparently is not fully hardy, and Mintle in which quality is poor here. Varieties which have not shown sufficient merit to warrant recommendation here are Stabler, Monterey, and Clark. Varieties which have not fruited are Allen, Cochrane, Huber, Kraus and Myers.

Practical Application of Results or Public Benefits

Results obtained have been used frequently as basis for recommendations relative to kinds and varieties for planting, and for grafting methods. Scionwood of the better varieties has been distributed to interested growers.

Progress of Work

Success with walnut grafts under all conditions during 16 years at the Fruit Breeding Farm has averaged only 32 per cent. In individual seasons success has varied from zero to 54 per cent.

Hickories not only are grafted with difficulty but also are very slow to reach bearing age. No nuts have been produced as yet from the following varieties grafted on the dates shown: Anthony (1939) Lingenfelter (1942) Burlington (1944) Gerardi hican (1944) Miller (1947) Barnes (1948) Last (1948) Marquette (1948) and Schinnerling (1948). Some seedling trees planted in 1948-1949 have produced no nuts in 32 years.

Hickory varieties established at Waseca by grafting are Beaver (1939), Fairbanks (1939), Burlington (1939), Anthony (1947), Billeau (1947), Hagen (1947), Wilcox (1947), Last (1948). Marquette (1948) and Stratford (1948). A tree of Hales planted in 1921, which grew very slowly for several years has borne no nuts in 27 years. One tree of Fairbanks grafted in 1939 bore a few nuts in 1944 but has not borne since then.

There has been a long-standing belief among horticulturists that grafts of ~Carya ovata~, the shagbark hickory are incompatible on bitter hickory ~C. cordiformis~. At Waseca, grafts of Beaver, Burlington and Fairbanks make in 1939 have healed completely and made excellent unions with the bitter hickory stock. That the varieties named are of hybrid origin may account for the compatibility apparent in this case.

Vegetarian, 93, and Bride, 60, Honeymoon Among Bananas, Nuts

MIAMI, Fla., Jan. 4—(UP)—A 93-year-old vegetarian and his 60-year old bride settled down today for a honeymoon among the nuts and bananas they say keep them young.

George Hebden Corsan and Lillian Armstrong, whose pert looks belie her years, were married here Tuesday. Wedding guests were served orange juice and coconut cream milk.

The bridegroom has been wintering here for the past 13 years. His home is Echo Valley, Islington, Toronto. His wife retired last month after 30 years of teaching in Toronto public schools.

"I'm sure we'll be happy," Mrs. Corsan said. "We have mutual interests"

Both credit their youthfulness and agility to vegetarianism, drinking gallons of fruit juices and staying outdoors as much as possible.

Corsan, whose sturdy 155 pounds are stretched on a six-foot frame, can husk a coconut with his bare hands in less than two minutes, no mean feat.

He operates a large experimental nut farm in Toronto, and has a 16-acre tract just south of here where he grows seven varieties of bananas and experiments with macadamia nuts, furnished him by the University of Hawaii. He works the farm singlehanded.

Corsan says he taught another physical culturist, Bernarr MacFadden, to swim in 1909 when he was an instructor at a Brooklyn YMCA. He says swimming helps keep him in shape and takes a daily dip in the ocean.

The Corsans will spend their honeymoon right on the nut farm.

"We might have a few fights," he said. "But they won't last long. She's too young to fight. And besides, she can outrun an English hare."

Broken Neck Fails to Halt Plans of "Youngster", 94

TORONTO, June 12—Physical Culturist George Hebden Corsan—just turned 94—says he is going to throw a birthday party Saturday, Right now he's in the hospital recovering from a broken neck suffered when he fell 20 feet from a tree May 27.

Mr. Corsan—a vegetarian who once labeled medicine "a jumbled heap of ignorance"—didn't want to go to the hospital at all. But doctors thought he'd better, since the fracture was about like that suffered by a man hanged on the gallows. He agreed to go after being assured the visit would only be for X-rays.

Since he's been in the hospital Mr. Corsan has fared—over the protest of dietitians—on nothing but orange juice. Yesterday he observed his birthday by eating a banana and a little black bread.

Doctors said Mr. Corsan missed severing his spinal cord by a quarter inch and had two skull fractures. To almost any other person, they said, the injury would be fatal.

Mr. Corsan was married for the third time last January in Florida.—Washington Evening Star, June 13, 1951.

Membership List

As of July 3, 1951

*Life member
**Honorary member
§Contributing member
+Sustaining member

ALABAMA

Deagon, Arthur, 128 Broadway, Birmingham. ~Farm in Penna.~
Hiles, Edward L., ~Hiles Auto Repair Shop~, Loxley

BELGIUM

R. Vanderwaeren, Bierbeekstraat, 310, ~Korbeek-Lo.~

CALIFORNIA

Armstrong Nurseries, 408 N. Euclid Avenue, Ontario
~General nurserymen, plant breeders~
Brand, George, U.S.N.G.B.C, Mob. 5, Port Hueneme
Buck, Ernest Homer, Three Arch Bay, 16 N. Portola, South Laguna
Deckard, L. A., 741 La Verne Avenue, Los Angeles 22
Flagg, Dr. Don P., 10365 Fairgrove Ave., Tujunga
Haig, Dr. Thomas R., 3021 Highland Avenue, Carlsbad, California
Linwood Nursery, Route No. 2, Box 476, Turlock
Parsons, Charles E., Felix Gillet Nursery, P. O. Box 1025, Nevada City.
~Nurseryman~
Pentler, Dr. C. F., 806 Arguello Blvd., San Francisco 18.
~American Friends Service Committee~
Pozzi, P. H., 2875 S. Dutton Ave., Santa Rosa. ~Brewery worker~, ~farmer~

Serr, E. F., Agr. Experiment Station, Davis. ~Associate Pomologist~
Welby, Harry S., 500 Buchanan Street, Taft. ~Private and Corp. Hort.~

CANADA

Brown, Alger, Route 1, Harley, Ontario. ~Farmer~
Collins, Adam H., 42 Seaton St., Toronto 2, Ont.
Cornell, R. S., R.R. No. 1, Byron, Ontario
Corsan, George H., Echo Valley, Toronto 18, Ontario. ~Nonagenarian.~
**Crath, Rev. Paul C., 299 Rosewell Ave., Toronto 12, Ontario
Crisp, Dr. Allan G., Suite 204, 160 Bloor St. W., Toronto, Ontario
English, H. A., Box 153, Duncan, B. C. ~Farmer~, ~fruit and nut grower~
Filman, O., Aldershot, Ontario. ~Fruit and veg. grower~
Gellatly, J. U., Box 19, Westbank, B. C. ~Plant breeder~, ~fruit grower~, ~nurseryman~
Goodwin, Geoffrey, Route No. 3, St. Catherines, Ontario. ~Fruit grower~
Harrhy, Ivor H., Route 1, Burgessville, Ont. ~Fruitgrower and poultry~
Housser, Levi, Route 1, Beamsville, Ontario. ~Fruit farmer~
*Neilson, Mrs. Ellen, 5 Macdonald Avenue, Guelph, Ont.
Papple, Elton E., Route 3, Cainsville, Ont.
Porter, Gordon, 258 McKay, Windsor, Ont. ~Chemist~
Smith, E. A., Sparta, Ont. Farmer
Snazelle, Robert, Forest Nursery, Route No. 5, Charlottetown, P. E. I. ~Nursery Supt.~
Short, J. R., 70 Wickstead Ave., Leaside, Ont.
Trayling, E. J., 509 Richards St., Vancouver, B. C. ~Jeweller~
Wagner, A. S., Delhi, Ont.
Walker, J. W., c/o McCarthy & McCarthy, 330 University Ave., Toronto, Ont.
Wharton, H. W., Route No. 2, Guelph, Ont. ~Farmer~

White, Peter, 30 Pear Ave., Toronto 5, Ont.
Willis, A. R., Route No. 1, Royal Oak, Vancouver Island, B. C. ~Accountant~
Woods, David M., 48 South Front St., West, Toronto, Ont. ~Vice President, Gordon McKay, Ltd.~
Young, A. L., Brooks, Alta.

CONNECTICUT

Daniel, Paul C., Lakeville
**Deming, Dr. W. C., 141 Fern St., Hartford. (Summer address: Litchfield) ~Dean of the Association~
Frueh, Alfred J., Route 2, West Cornwall
Graves, Dr. Arthur H., 255 S. Main St., Wallingford. ~Consulting Pathologist, Conn. Agr. Expt. Station, New Haven, Conn.~
Henry, David, Blue Hills Farm, Route 2, Wallingford.
*Huntington, A. M., Stanerigg Farms, Bethel. ~Patron~
Lehr, Frederick L., 45 Elihu St., Hamden
*Newmaker, Adolph, Route No. 1. Rockville
Pratt, George D., Jr., Bridgewater
Risko, Charles, City Tobacco & Candy Co., 25 Crescent Ave., Bridgeport
White, George E., Route No. 2, Andover. ~Farmer~

DELAWARE

Brugmann, Elmer W., 1904 Washington St., Wilmington. ~Chemical Engineer~
Logue, R. F., Gen. Mgr., Andelot, Inc., 2098 du Pont Bldg., Wilmington
Wilkins, Lewis, Route 1, Newark. ~Fruit grower~

DENMARK

Granjean, Julio, Hillerod. (See New York.)
Knuth, Count F. M., Knuthenborg. Bandholm

DISTRICT OF COLUMBIA

American Potash Inst., Inc., 1155-16th St., N.W., Washington
Ford, Edwin L., 3634 Austin St., S.E., Washington
Kaan, Dr. Helen W., National Research Council, 2101 Constitution Avenue, Washington. ~Research Associate~
Reed, Mrs. Clarence A., 7309 Piney Branch Rd., N.W., Washington 12

ECUADOR, SOUTH AMERICA

Acosta Solis, Prof. M., Director del Departmento Forestal, Ministerio de Economia, Quito. (~Exchange.~)

ENGLAND

Baker, Richard St. Barbe, The Gate, Abbotsbury, Weymouth, Dorset. (~Founder, Men of The Trees.~)
The Gardeners Chronicle, London. (~Exchange.~)

FLORIDA

Avant, C. A., 940 N.W. 10th Ave., Miami ~Real Estate, Loans.~ (~Pecan orchard in Ga.~)
Estill, Gertrude, 153 Navarre Dr., Miami Springs. (Summer address under Mich.)

GEORGIA

Edison, G. Clyde, 1700 Westwood Ave., S.W., Atlanta.
Hardy, Max, P. O. Box 128, Leeland Farms, Leesburg. ~Nurseryman~, ~farmer~
Hunter, Dr. H. Reid, 561 Lake Shore Dr. N.E., Atlanta. ~Teacher~, ~nut farmer~
Noland, S. C., Box 1747, Atlanta 1. ~Owner, Skyland Farms~
Wilson, William J., North Anderson Ave., Fort Valley. ~Peach and pecan grower~

HOLLAND

Institute for Horticultural Plant Breeding. Herenstraat 25. Wageningen. (~Exchange~)

HONG KONG

*Wang, P. W., c/o China Products Trading Corp., 6 Des Voeux Rd., Central

IDAHO

Baisch, Fred, 627 E. Main St., Emmett
Dryden, Lynn, Peck. ~Farmer~
Hazelbaker, Calvin, Route No. 1, Box 382, Lewiston

ILLINOIS

Albrecht, H. W., Delavan
Allen, Theodore R., Delavan. ~Farmer~

Andrew, Col. James W. (See under Washington)
Anthony, A. B., Route No. 3, Sterling. ~Apiarist~
Baber, Adin, Kansas
Best, R. B., Eldred. ~Farmer~
Blodgett, Thomas, 3610 Pine Grove Ave., Chicago 13
Blough, R. O., Route No. 3, Polo
Blyth, Colin R., Math. Dept., U. of I., Urbana. (Farm in northern Ontario)
*Boll, Herschel L., 2 Hort. Field Lab., Univ. of Ill., Urbana.
~Asst. in Pomology~
Brock, A. S., 1733 North McVicker Ave., Chicago 39
Churchill, Woodford M., 4323 Oakenwald Ave., Chicago 5
Colby, Dr. Arthur S., U. of Illinois, Urbana
Daum, Philip A., North Sixth St., Carrollton
Dietrich, Ernest, Route No. 2, Dundas. ~Farmer~
Dintelman, L. F., State Street Road, Belleville
Douglass, T. J., 309-1/2 North St., Normal
Fordtran, E. H., Route No. 2, Box 197-A, Palatine
Frey, Frank H., 2315 W. 108th Place, Chicago 43.
~Asst. to V. P., CRI & P RR.~
Frey, Mrs. Frank H., 2315 W. 108th Place, Chicago 43. ~Housewife~
Gerardi, Louis, Route No. 1., Caseyville. ~Nut and fruit nurseryman~
Grefe, Ben, Route No. 4, Box 22, Nashville. ~Farmer~
Heberlein, Edward W., Route No. 1, Box 72A, Roscoe
Helmle, Herman C., 526 S. Grand Ave., W., Springfield. ~Div. Eng.,
Asphalt Inst.~
Hockenyos, G. L., 213 E. Jefferson St., Springfield. ~Business man~
Jungk, Adolph E., Route No. 1, Jerseyville, Illinois
Kammarmeyer, Glenn, 1711 E. 67th St., Chicago 49
Kreider, Ralph, Jr., Route No. 1, Hammond. ~Farmer~
Langdoc, Mildred Jones (Mrs. Wesley W.) P. O. Box 136, Erie. ~Nursery~,

~farm~, ~housewife~
McDaniel, J. C., c/o Hort. Field Lab., U. of I., Urbana. ~Horticulturist. (Sec'y of Ass'n.)~
McDaniel, J. C., Jr., Urbana
Oakes, Royal, Bluffs (Scott County)
Pray, A. Lee, 502 N. Main St., LeRoy
Robbins, W. J., 885 N. LaSalle St., Chicago 10. ~Insurance~
Sonnemann, W. F., Experimental Gardens, Vandalia. ~Lawyer~, ~farm operator~
Spencer, H. Dwight, 275 W. Decatur St., Decatur. ~Attorney~
Warnecke, Martin H., 714 First Avenue, Maywood
Whitford, A. M., Farina. ~Nurseryman~
Zethmayr, Gordon, Route No. 1, Box 130, West Chicago

INDIANA

Aster Nut Products, Inc., George Oberman, Mgr., 1004 Main St., Evansville
Bauer, Paul J., 123 S. 29th St., Lafayette
Bolten, Ferd, Route 3, Linton. ~Farmer, fruit grower. (Carpathian walnut seeds.)~
Boyer, Clyde C., Nabb
Buckner, Dr. Doster, 421 W. Wayne St., Ft. Wayne 2. ~Physician and Surgeon~
Clark, C. M., C. M. Clark & Sons Nurseries, Route 2, Middletown ~Nurseryman~, ~fruit farmer~
Dooley, Kenneth R., Route No. 2, Marion. ~Gardener~
Eagles, A. E., Eagles' Orchards, Wolcottville. ~Walnut grower~, ~apple orchardist~
Eisterhold, Dr. John. A., 220 Southwest Riverside Drive, Evansville 8.

~Medical Doctor~

Fateley, Nolan W., 26 Central Avenue. Franklin. ~Auditor and cashier. (Carpathian walnut seeds.)~

Glaser, Peter, Route No. 9, Box 328, Koening Road, Evansville

Grater, A. E., Route 2, Shipshewana.

§Johnson, Hjalmar W., Rt. 4, Valparaiso. ~V. P. Inland Steel Co.~

Pape, Edw. W., Route 2, Marion

Prell, Carl F., 1414 E. Colfax Avenue, South Bend 17

Richards, E. E., 2712 South Twyckenham Drive, South Bend. ~Studebaker Corp.~

Russell, A. M., Jr., 2721 Marine St., South Bend 14

Skinner, Dr. Chas. H., Rt. 1, Thorntown

Sly, Miss Barbara, Route No. 3, Rockport

Sly, Donald R., Route 3, Rockport. ~Nurseryman,~, ~nut tree propagator~

Wallick, Ford, Rt. 4, Peru

Ward, W. B., Horticulture Bldg., Purdue University, Lafayette. ~Ext. Horticulturist~, ~Vegetables~

Whitsel, Gilbert L., Jr., 515 S. 15th Street, Lafayette

Wichman, Robert P., Route No. 3, Washington. ~General farming~

Wilkinson, J. F., Indiana Nut Nursery, Rockport. ~Nurseryman~

IOWA

Berhow, Seward, Berhow Nurseries, Huxley

Boice, R. H., Route No. 1, Nashua. ~Farmer~

Cole, Edward P., 419 Chestnut Street, Atlantic

Ferguson, Albert B., Center Point. ~Nurseryman~

Ferris, Wayne, Hampton. ~President of Earl Ferris Nursery~

Huen, E. F., Eldora. ~Farmer~

Inter-State Nurseries, Hamburg. ~General nurserymen~
Iowa Fruit Growers Assn., W. H. Collins, Sec'y, State House,
Des Moines 19. ~Cooperative buying organization~
Kaser, J. D., Winterset. ~Farmer~
Knowles, W. B., Box 476, Manly
Kyhl, Ira M., Box 236, Sabula. ~Nut nurseryman~, ~farmer~, ~salesman~
Martazahn, Frank A., Route No. 3, Davenport. ~Farmer~
McLeran, Harold F., Mt. Pleasant. ~Lawyer~
Orr, J. Allen, 535 Frances Bldg., Sioux City 17
Rohrbacher, Dr. William, 811 East College Street, Iowa City.
~Practice of Medicine~ (~President of the NNGA.~)
Schlagenbusch Brothers, Route No. 2, Fort Madison. ~Farmers~
Snyder, D. C., Center Point. ~Nurseryman, nuts and general.~
Tolstead, W. L., Central College, Pella
Wade, Miss Ida May, Route No. 3, LaPorte City. ~Bookkeeper~
Watson, Vinton C., 106 E. Salem St., Indianola
Welch, H. S., Mt. Arbor Nurseries, Shenandoah
White, Herbert, Box 264, Woodbine. ~Rural Mail Carrier~
Williams, Wendell V., Route No. 1, Danville. ~Farmer~

KANSAS

Baker, Fred C., Troy. ~Entomologist~
Borst, Frank E., 1704 Shawnee Street, Leavenworth
Breidenthal, Willard J., Riverview State Bank, 7th and Central,
Kansas City. ~Bank President~
Funk, M. D., 612 W. Paramore Street, Topeka. ~Pharmacist~
Gray, Dr. Clyde, 1045 Central Avenue, Horton. ~Osteopathic Physician~
Harris, Ernest, Box 20, Wellsville. ~Farmer~
Leavenworth Nurseries, Carl Holman, Proprietor, Route No. 3,

Leavenworth.
~Nut nurseryman~
Mondero, John, Lansing
Thielenhaus, W. F., Route No. 1, Buffalo. ~Retired postal worker~
Underwood, Jay, Riverside Nursery, Uniontown

KENTUCKY

Alves, Robert H., Nehi Bottling Company, Henderson
Armstrong, W. D., West Ky., Exp. Sta., Princeton. ~Horticulturist~
Magill, W. W., Horticulture Dept., U. of Ky., Lexington
Miller, Julian C., 220 Sycamore Drive, Paducah
Moss, Dr. C. A., Willlamsburg. ~Bank President~
Rouse, Sterling, Route No. 1, Box 70, Florence. ~Fruit grower~,
~nurseryman~
Tatum, W. G., Route 4, Lebanon. ~Commercial orchardist~
Tallaferro, Philip, Box 85, Erlanger
Usrey, Robert, Star Route, Mayfield
Walker, William W., Route No. 1, Dixie Highway, Florence

LOUISIANA

Hammar, Dr. Harald E., USDA Chemical Lab., 606 Court House, Shreveport
~Chemist~
Perrault, Mrs. Henry D., Route No. 1, Box 13, Natchitoches. ~Pecan grower~

MARYLAND

Case, Lynn B., Route 2, Box 208, Federalsburg
Crane, Dr. H. L., Bureau of Plant Industry Station, Beltsville.
~Principal Horticulturist, USDA.~
Eastern Shore Nurseries, Inc., P. O. Box 743, Easton. ~Chestnut growers~
Graff, George U., Harding Lane, Rt. 3. Rockville
Gravatt, Dr. G. F., Plant Industry Station, Beltsville.
~Research Forest Pathologist~
Hodgson, William C., Route No. 1, White Hall. ~Farmer~
Kemp, Homer S., (Proprietor) Bountiful Ridge Nurseries, Princess Anne
McCollum, Blaine, White Hall. ~Retired from Federal Government~
McKay, Dr. J. W., Plant Industry Station, Beltsville.
~Government Scientist~
+Negus, Mrs. Herbert, 4514 32nd Street, Mt. Rainier
Porter, John J., 1199 The Terrace, Hagerstown. ~Farm Owner~
Shamer, Dr. Maurice E., 3300 W. North Avenue, Baltimore 16. ~Physician~

MASSACHUSETTS

Babbit, Howard S., 221 Dawes Avenue, Pittsfield. ~Service station owner and part time farmer~
Bradbury, H. G., Hospital Point, Beverly
Brown, Daniel L., Esq., 60 State Street, Boston
Bump, Albert H., P. O. Box 275, Brewster
Davenport, S. Lathrop, 24 Creeper Hill Road, North Grafton. ~Farmer~, ~fruit grower~
Fitts, Walter H., 39 Baker St., Foxboro. ~General foreman, instrument company~
Kendall, Henry P., Moose Hill Farm, Sharon
Kerr, Andrew, Lock Box 242, Barnstable

La Beau, Henry A., North Hoosic Road, Williamstown. ~Stat. engineer~
O'Brien, Howard C., 25 Irvington Street, Boston 16
Rice, Horace J., 515 Main Street, Wilbraham. ~Attorney~
*Russell, Mrs. Newton H., 12 Burnett Avenue, South Hadley
Wellman, Sargent H., Esq., Windridge, Topsfield. ~Lawyer~
Weston Nurseries, Inc., Weston
Wood, Miss Louise B., Pocassett, Cape Cod

MICHIGAN

Ainsworth, Donald W., 5851 Mt. Elliott, Detroit 11
Andersen, Charles, Route No. 2, Box 326, Scottsville, ~Nurseryman~
Barlow, Alfred L., 13079 Flanders Avenue, Detroit 5
Becker, Gilbert, Climax
Boylan, P. B., Route No. 1, Cloverdale. ~Homesteader~
Bumler, Malcolm R., 2500 Dickerson, Detroit 15. ~Insurance trustee~
Burgart, Harry, Michigan Nut Nursery, Box 33, Union City. ~Nurseryman~
Burgess, E. H., Burgess Seed & Plant Company, Galesburg
Burr, Redmond M., 320 S. 5th Avenue, Ann Arbor. ~General Chairman, The
Order of Railroad Telegraphers, Pere Marquette District, C&O Ry. Co.~
Cook, Ernest A., M.D., c/o County Health Dept., Centerville
Corsan, H. H., Route No. 1, Hillsdale. ~Nurseryman~
Dennison, Clare, 4224 Avery, Detroit 8
Emerson, Ralph, 161 Cortland Avenue, Detroit 3
Estill, Miss Gertrude. (See under Florida, Summer Address: Route 4,
Box 762, Battle Creek)
Hackett, John C., 3321 Butterworth Rd., S.W., R. R. 5, Grand Rapids 6
Haseler, L. M., Route No. 4, Box 130 South Haven
Hagelshaw, W. J., Route No. 1, Box 394, Galesburg. ~Grain farmer~,

~contractor~
Hay, Francis H., Ivanhoe Place, Lawrence. ~Farmer~
**Kellogg, W. K., Battle Creek
Korn, G. J., 309 N. Church Street, Kalamazoo 11. ~Shop worker~
Lee, Michael, P. O. Box 16, Milford
Lemke, Edwin W., 2432 Townsend Ave., Detroit 14. ~Engineer~, ~nut orchardist~
McCarthy, Francis W., Box 392, Algonac
Miller, O. Louis, 417 N. Broadway, Cassopolis. ~Forester~
O'Rourke, Prof. F. L., Hidden Lake Gardens, Tipton. ~Professor of ornamental horticulture, Mich. State College~
Pickles, Arthur W., 760 Elmwood Avenue, Jackson
Prushek, E., Route No. 3, Niles. ~Plant breeding~
Sherman, L. Walter, 3308 Mackinaw St., Saginaw
Simons, Rev. R. E., Flat Rock
Somers, Lee, Route No. 1, Perrinton
Tate, D. L., 959 Westchester St., Birmingham
Ullrey, L. E., 1209 Cambridge Drive, Kalamazoo 27

MINNESOTA

Hodgson, R. E., Dept. of Agriculture, S.E. Experiment Station, Waseca
Tulare, Willis E., 300 3rd Avenue, S.E., Rochester
Weschcke, Carl, 96 S. Wabasha St., St. Paul. ~Proprietor Hazel Hills Nursery Co.~

MISSISSIPPI

Gossard, A. C., U. S. Hort. Field Station, Route No. 6, Meridian. ~Associate Horticulturist, USDA~

Meyer, James R., Delta Branch Experiment Station, Stoneville.
~Cytogeneticist (cotton)~

MISSOURI

Bauch, G. D., Box 66, Farmington. ~Farm Forester~
Hay, Leander, Gilliam
Howe, John, Route No. 1, Box 4, Pacific
Huber, Frank J., Weingarten. ~Farmer~
James, George, James Pecan Farms, Brunswick
Logan, George F., Oregon
The M-F-D Co., 1305 Moreland Ave., Jefferson City
Nicholson, John W., Ash Grove. ~Farmer~
Ochs, C. Thurston, Box 291, Salem. ~Foreman in garment factory~
Richterkessing, Ralph, Route No. 1. St. Charles. ~Farmer~
Rose, Dr. D. K., 230 Linden, Clayton 5
Stark Bros. Nursery & Orchard Co., Attn. Mr. H. W. Guengerich, Louisiana
Wuertz, H. J., Route No. 1, Pevely

NEBRASKA

Brand, George. (See under California.)
Caha, William, 350 W. 12th, Wahoo
Hess, Harvey W., The Arrowhead Gardens, Box 209, Hebron
Sherwood, Jack, Nebraska City

NEW HAMPSHIRE

Demarest, Charles S., Lyme Center
Lahti, Matthew, Locust Lane Farm, Wolfeboro. ~Investment banker~

NEW JERSEY

Anderegg, F. O., Pierce Foundation, Raritan
Blake, Harold, Box 93, Saddle River
Bottoni, R. J., 41 Robertson Road, West Orange. ~President of Harbot Die Casting Corp.~
Brewer. J. L., 10 Allen Place, Fair Lawn
Buckwalter, Mrs. Alan R., Route. No. 1, Flemington
Cox, Philip H., Jr., 30 Hyde Rd., Bloomfield
Cumberland Nursery, William Wells, Proprietor, Route No. 1, Millville. ~Nurserymen~
Donnelly, John, Mountain Ice Company, 51 Newark St., Hoboken
Dougherty, William M., Broadacres-on-Bedens, Box 425, Princeton ~Secretary, U. S. Rubber Co.~
Ellis, Mrs. Edward P., Strawberry Hill, Route No. 1, Box 137, Keyport
Kass, Leonard P., 82 E. Cliff St., Somerville
Lamatonk Nurseries, A. H. Yorks, Proprietor, Neshanic Station
Lippencott, J. C., 15 Mundy Ave., Spotswood
McDowell, Fred, 905 Ocean Avenue, Belmar
Parkinson, Philip P., 567 Broadway, Newark 4. ~Engineer and appraiser~
Ritchie, Walter M., Route No. 2, Box 122-R, Rahway
Rocker, Louis P., The Rocker Farm, Box 196; Andover. ~Farmer~
Sheffield, O. A., 283 Hamilton Place, Hackensack. ~Dunn & Bradstreet~
Sorg, Henry, Chicago Avenue, Egg Harbor City. ~Manufacturer~
Van Doren, Durand H., 310 Redmond Road., South Orange. ~Lawyer~
Williams, Herbert H., 106 Plymouth Ave., Maplewood

NEW MEXICO

Gehring, Rev. Titus, Box 177, Lumberton

NEW YORK

Barton, Irving Titus, Montour Falls. ~Engineer~
Bassett, Charles K., 2917 Main St., Buffalo. ~Manufacturer~
Beck, Paul E., Beck's Guernsey Dairy, Transit Road, East Amherst. ~Dairy Executive~
Benton, William A., Wassaic. ~Farmer, and Sec'y, Mutual Insurance Co.~
Bernath, Stephen, Bernath's Nursery, Route No. 3, Poughkeepsie. ~Nurseryman~
Bernath, Mrs. Stephen, Route 3, Poughkeepsie
Bixby, Henry D., East Drive, Halesite, L. I. ~Executive V.P., American Kennel Club, N. Y. City~
Brook, Victor, 171 Rockingham Street, Rochester 7. ~Sales Engineer~
Brooks, William G., Monroe. ~Nut tree nurseryman~
Bundick, Clarkson U., 35 Anderson Ave., Scarsdale. ~Mechanical engineer~
Caldwell, David H., N. Y. State College of Forestry, Syracuse. ~Instr. in wood technology~
Carter, George, 428 Avenue A, Rochester 5
Cassina, Augustus, Valatie, Columbia County
Feil, Harry, 1270 Hilton-Spencerport Road, Hilton. ~Building contractor~
Ferguson, Donald V., L. I. Agr. Tech. Institute, Farmingdale
Flanigen, Charles F., 16 Greenfield St., Buffalo 14. ~Executive manager~
Freer, H. J., 20 Midvale Rd., Fairport. ~Typewriter sales and service~
Fribance, A. E., 139 Elmsdorf Ave., Rochester 11
Glazier, Henry S., Jr., 1 South William St., New York 4
Graham, S. H., Bostwick Road, Route No. 5, Ithaca. ~Nurseryman~
Granjean, Julio, c/o K. E. Granjean, 9406 6th Ave., Forest Hills
Gressel, Henry, Route 2, Mohawk. ~Retired chief lock operator, N. Y. S. Barge Canal~
Hasbrouck, Walter, Jr., 19 Grove St., New Paltz. ~Post office clerk~

Hill, Francis S., Sterling. ~Letter carrier on rural route~
Iddings, William A., 1931 Park Place. Brooklyn 33
Irish, G. Whitney, Fruitlands, Route No. 1, Valatie. ~Farmer~
Kettaneh, F. A., 745 Fifth Ave., New York 22
Knipper, George M., 333 Chestnut Ridge Rd., Churchville
Knorr, Mrs. Arthur, 15 Central Park, West, Apt. 1406, New York
Kraai, Dr. John, Fairport. ~Physician~
Larkin, Harry H., 189 Van Rennsselaer Street, Buffalo 10
*Lewis, Clarence (Retired.)
Lowerre, James, Route 3, Middletown
*MacDaniels, Dr. L. H. Cornell University. Ithaca. ~Head, Dept. of Floriculture and Ornamental Hort.~
Miller, J. E., Canandaigua. ~Nurseryman.~
Mitchell, Rudolph, 125 Riverside Drive, New York 24. ~Mechanical engineer~
*Montgomery, Robert H., 1 E., 44th Street, New York
Mossman, Dr. James K., Black Oaks, Ramapo
Newell, Palmer F., Lake Road, Route No. 1. Westfield
Owen, Charles H., Sennett. ~Superintendent of Schools~
Pura, John J., Green Haven, Stormville
Salzer, George, 169 Garford Road, Rochester 9. ~Milkman~, ~chestnut tree grower~
Schlegel, Charles P., 990 South Ave., Rochester 7
Schlick, Frank, Munnsville
Schmidt, Carl W., 180 Linwood Avenue, Buffalo
Shannon, J. W., Box 90, Ithaca
Sheffield, Lewis J., c/o Mrs. Edna C. Jones, Townline Road, Orangeburg
Slate, Prof. George L., Experiment Station, Geneva. ~Fruit Breeder~
Smith, Gilbert L., Benton & Smith Nut Tree Nursery, Route 2, Millerton. ~Nurseryman~, ~retired teacher~

Smith, Jay L., Chester. ~Nut tree nurseryman~
Spahr, Dr. Mary B., 116 N. Geneva St., Ithaca
Steiger, Harwood, Red Hook. ~Artist-designer~
+Szego, Alfred, 77-15 A 37th Avenue, Jackson Heights, New York
Timmerman, Karl G., 123 Chapel St., Fayetteville
Wadsworth, Willard E., Route No. 5, Oswego
Wheeler, Robert C., 36 State Street, Albany
Windisch, Richard P., c/o W. E. Burnet Company, 11 Wall St., New York 5
*Wissman, Mrs. F. De R. ~(Retired.)~

NORTH CAROLINA

Brooks, J. R., Box 116, Enka
Dunstan, Dr. R. T., Greensboro College, Greensboro
Finch, Jack R., Bailey. ~Farmer~
Parks, C. H., Route No. 2, Asheville. ~Mechanic~

NORTH DAKOTA

Bradley, Homer L., Long Lake Refuge, Moffit. ~Refuge Manager~

OHIO

Ackerman, Lester, Route No. 3, Ada
Glen Helen Department, Antioch College, Yellow Springs
Barden, C. A., 215 Morgan Street, Oberlin. ~Real Estate~
Beede, D. V., Route No. 3, Lisbon
Bitler, W. A., R. F. D. I, Shawnee Road, Lima. ~General contractor~
Borchers, Perry E., 412 W. Hillcrest Ave., Dayton 6
Brewster, Lewis, Route No. 1, Swanton. ~Vegetable grower~

Bridgewater, Boyd E., 68 Cherry St., Akron. ~V. P. Bridgewater Machine Co.~
Bungart, A. A., Avon
Cinadr, Mrs. Katherine, 13514 Coath Ave., Cleveland 20. ~Housewife~
Clark, Richard L., 1517 Westdale Rd., South Euclid 21. ~Sales manager~
Cook, H. C., Route No. 1, Box 125, Leetonia
Cornett, Charles. L., R. R. Perishable Inspection Agency, 27 W. Front St., Cincinnati. ~Inspector~
Craig, George E., Dundas (Vinton County). ~Fruit and nut grower~
Cranz, Eugene F., Mount Tom Farm, Ira
Cunningham, Harvey E., 420 Front Street, Marietta
Daley, Jame R., Route No. 3, Foster Park Road, Amherst. ~Electrician~
Davidson, John, 234 East Second Street, Xenia. ~Writer~
Davidson, Mrs. John, 234 East Second Street, Xenia
Diller, Dr. Oliver D., Dept. of Forestry, Ohio Exp. Sta., Wooster
Distelhorst, P. E., 3532 Douglas Road, Toledo 6
Dowell, Dr. Lloyd L., 529 North Ave., N. E., Massillon. ~Physician~
Farr, Mrs. Walter, Route No. 1, Kingsville
Garden Center of Greater Cleveland, 11190 East Blvd., Cleveland
Gerber, E. P., Kidron
Gerstenmaier, John A., 13 Pond S. W., Massillon. ~Letter carrier~
Goss, C. E., 922 Dover Avenue, Akron 20
Grad, Dr, Edward A., 1506 Chase Street, Cincinnati 23
Hansley, C. F., Box 614, Sugar Grove. ~Contractor~
Hawk & Son Nursery, Route No. 2, Beach City. ~Chestnut trees~
Hill, Dr. Albert A., 4187 Pearl Road, Cleveland
Hornyak, Louis, Route No. 1, Wakeman
Howard, James R., 2908 Fleming Road, Middletown
Irish, Charles F., 418 E. 105th St., Cleveland 8. ~Arborist~
Jacobs, Homer L., Davey Tree Expert Company, Kent

Kappel, Owen, Bolivar
Kerr, S. E., M. D., Route No. 1, North Lawrence
Kintzel, Frank W., 2506 Briarcliff Ave., Cincinnati 13 ~Principal~, ~Cincinnati public schools~
Laditka, Nicholas G., 5322 Stickney Ave., Cleveland 9. ~Electrician~
Leaman, Paul Y., Route No. 1, Creston
Lorenz, R. C., 121 North Arch Street, Fremont
Machovina, Paul E., 1228 Northwest Blvd., Columbus 12. ~College professor~
McKinster, Ray, 1632 South 4th Street, Columbus 7
Meister, Richard T., ~Editor, American Fruit Grower~, Willoughby
Metzger, A. J., 724 Euclid Avenue, Toledo 5
Oches, Norman M., R. D. 1, Brunswick. ~Mechanical Engineer~
Osborn, Frank C., 4040 W. 160th St., Cleveland 11. ~Tool and die maker~
Page, John H., Box 34, Dundas (Vinton County)
Pataky, Christ, Jr., 492 Hickory Lane, Route No. 4, Mansfield. ~Produce market~, ~grocer~
Pattison, Aletheia, 5 Dexter Place, E. W. N., Cincinnati 6
Pomerene, Walter H., Route No. 3, Coshocton. ~Agricultural Engineer, Hydrological Research Station~
Purdy, Clyde W., 19 Public Square, Mt. Vernon
Ranke, William, Route No. 1, Amelia
Roberts, J. Pearl, Rt. 3, Freeport
Rummel, E. T., 16613 Laverne Avenue, Cleveland 11. ~Sales manager~
Schoenberger, L. Roy, Green Pines Farm, Route No. 2, Nevada
Seas, D. Edward, 721 South Main Street, Orrville
Sebring, R. G., 1227 Lincoln Road, Columbus
Shelton, Dr. Elbert M., 1468 W. Clifton Blvd., Lakewood 7
Shessler, Sylvester M., Genoa. ~Farmer~
Silvis, Raymond E., 1725 Lindbergh Avenue, N. E., Massillon. ~Realty~

Smith, Sterling A., 630 W. South Street, Vermilion. ~Telegrapher, NYC RR (Treasurer of the Assn.)~
Spears, Ernest G., 4326 Forest Ave., Norwood 6
Spring Hill Nurseries Company, Tipp City. ~General nurserymen~
Steinbeck, A. P., East Nimisilla Rd., North Canton. ~Rubber worker, Firestone Tire & Rubber Co.~
Stocker, C. P., Lorain Products Corp., 1122 F. Street, Lorain
Stolz, Thomas O., 334 Claranna Ave., Dayton 9
Thiesing, J. R., 113 S. Washington, New Bremen
Thomas, Fred, Route No. 1, Bedford Road, Masury
Toops, Herbert A., 1430 Cambridge Blvd., Columbus 12. ~College Professor~
Underwood, John, Route No. 4, Urbana
Urban, George, 4518 Ardendale Road, South Euclid 21. ~Mayor~
Van Voorhis, J. F., 215 Hudson Avenue, Apt. B-1, Newark
Von Gundy, Clifford R., R. F. D. No. 8, Cincinnati 30
Walker, Carl F., 2851 E. Overlook Rd., Cleveland 18. ~Consulting engineer~
Weaver, Arthur W., R.F.D., Box 196B, Cass Rd., Maumee
*Weber, Harry R., Esq. (Deceased.)
Weber, Mrs. Martha R., Route No. 1, Mahawe Farm, Cleves
Willett, Dr. G. P., Elmore
Williams, Harry M., 221 Grandon Road, Dayton 9. ~Engineer~
Wischhusen, J. F., 15031 Shore Acres Drive, N. E., Cleveland 10
Yates, Edward W., 3108 Parkview Avenue, Cincinnati 13. ~Mechanical engineer~
Yoder, Emmet, Smithville

OKLAHOMA

Butler, Roy, Route No. 2, Hydro. ~Farmer~, ~cattleman~

Cross, Prof. Frank B., Dept. of Horticulture, Oklahoma A&M College, Stillwater.
~Teaching and Experiment Station Work~

Gray, Geoffrey A., 1628 Elm Ave., Bartlesville

Hartman, Peter E., 3002 S. Boston Pl., Tulsa 5. ~Nurseryman~

Hirschi's Nursery (A. G. Hirschi), 414 North Robinson, Oklahoma City
~Dry cleaning business, nurseryman~

Hughes, C. V., Route No. 3, Box 614, 5600 N. W. 16th Street, Oklahoma City

Keathly, Jack, Marland. ~Farmer~

Kissick, E. A., State Board of Agr., 122 State Capitol Bldg., Oklahoma City. ~Marketing Specialist~

Meek, E. B., Route 2, Wynnewood

Pulliam, Gordon, 1605 Osage Ave., Bartlesville

Scales, Charles D., 3200 N. W. 26th St., Oklahoma City 7

OREGON

Miller, John E., Treasuredale, Route No. 1, Box 312-A, Oswego

Pearcy, Harry L., Route 2. Box 190, Salem. ~H. L. Pearcy Nursery Co. (Nut trees.)~

PENNSYLVANIA

Allaman, R. P., Route 86, Harrisburg. ~Farm superintendent~

Amsler, E. W., 707 Main St., Clarion

Anthony, Roy D., 215 Hillcrest Ave., State College. ~Tree Crops Advisor, Pa. Dept. of Agr.~

Arensberg, Charles F. C., First Nat'l Bank Bldg., Pittsburgh 22

~(Chinese chestnut seed grower.)~

Banks, H. C., Route No. 1, Hellertown

Beard, H. K., Route No. 1, Sheridan. ~Insurance agent~

Beck, Dr. William M., 200 Race St., Sunbury

Berst, Charles B., 11 W. 8th Street, Erie. ~Inspector, Lord Mfg. Co., Erie, Pa.~

Bowen, John C., Route No. 1, Macungie

Brown, Morrison, Ickesburg

Buckwalter, Geoffrey R., c/o F. H. Levey Co., Inc., 1223 Washington Ave., Philadelphia 47

Clarke, William S., Jr., P. O. Box 167, State College

Colwell, Dr. Frederick A., R.F.D. No. 1, Collegeville

Damask, Henry, 1632 Doyle Street, Wilkinsburg 21. ~Telephone man~

Ebling, Aaron L., Route No. 2, Reading

Etter, Fayette, P. O. Box 57, Lemasters. ~General foreman for an electric company~

Gage, Charles K., 1429 Newman Road, Havertown

Gardner, Ralph D., 4428 Plymouth St., Harrisburg. ~Assistant State Fire Marshal~

Good, Orren S., 316 N. Fairview Street, Lock Haven. ~Retired~

Gorton, F. B., Route No. 1, East Lake Road, Harborcreek. ~Electrical contractor~

Hammond, Harold, 903 South Poplar Street, Allentown

Hershey, John W., Route No. 1, Downingtown. ~Nurseryman~

Hostetter, L. K., Route No. 3, Lancaster. ~Farmer~, ~black walnut grower~

Hughes, Douglas, 1230 East 21st Street, Erie

Johnson, Robert F., 1630 Greentree Road, Pittsburgh 20

Jones, Mildred M. (See Mrs. Langdoc—under Illinois)

Jones, Dr. Truman W., Walnut Grove Farm, Parksburg

Kaufman, Mrs. M. M., Box 69, Clarion

Knouse, Charles W., Colonial Park, Harrisburg. ~Coal dealer~
Laboski, George T., Route No. 1, Harborcreek. ~Fruit grower and nurseryman~
Leach, Will, 406-410 Scranton Life Bldg., Scranton 3. ~Lawyer~
Mattoon, H. Gleason, Box 304, Narberth. ~Consultant in Arborculture~
McKenna, Philip M., P. O. Box 186, Latrobe
Mecartney, J. Lupton, 918 W. Beaver Ave., State College. ~Pomologist~
Miller, Elwood B., Mill and Chapel Sts., Hazleton
Miller, Robert O., 3rd and Ridge Streets, Emmaus
Moyer, Philip S., 80-82 U. S. F. & G. Bldg., Harrisburg. ~Attorney~
Niederriter, Leonard, 1726 State Street, Erie. ~Merchant~
Nonnemacher, H. M., Box 204, Alburtis. ~Line foreman, Bell Tel. Co. of Pa.~
Ranson, Flavel, 728 Monroe Avenue, Scranton. ~Farmer~
Reidler, Paul G., Ashland. ~Manufacturer of textiles~
Rick, John, 438 Penna. Sq., Reading. ~Fruit grower and merchant~
Schaible, Percy, Upper Black Eddy. ~Laborer~
Scott, J. Lewis. 5-A Camberwell Drive, R.F.D. No. 2, Pittsburgh 15
Shade, Earl L., 1027 E. 26th St., Erie
Sherman, L. Walter. (See under Michigan.)
Smith, Dr. J. Russell, 550 Elm Ave., Swarthmore. ~Retired teacher, writer and nurseryman~
Stewart, E. L., Pine Hill Farms Nursery, Route No. 2, Homer City
Theiss, Dr. Lewis E., 110 University Ave., Lewisburg. ~Retired professor~
Thompson, Howard A., 311 West Swissvale Ave., Pittsburgh 18
Twist, Frank S., Box 127, Northumberland. ~Salesman~
Waite, Knighton V., M. D., Renton
Washick, Dr. Frank A., S. W., Welsh & Veree Roads, Philadelphia 11. ~Surgeon~
Weaver, William S., Weaver Orchards, Macungie

Weinrich, Whitney, P. O. Box 225, Wallingford. ~Chemical engineer~
Wister, John C., Scott Foundation, Swarthmore College, Swarthmore. ~Horticulturist~
Wright, Ross Pier, 235 W. 6th Street, Erie. Manufacturer
Zimmerman, Mrs. G. A., R. D., Linglestown

RHODE ISLAND

Allen, Philip, 178 Dorance Street, Providence

SOUTH CAROLINA

Bregger, John T., Clemson. ~Research Supervisor (Soil Conservation), Orchard Erosion Investigations~
Gordon, G. Henry, 13-1/2 Main St., Union. ~Returned Mariner~

SOUTH DAKOTA

Richter, Herman, Madison. ~Farmer~

TENNESSEE

Alpine Forest Reserve, c/o J. Edwin Carothers, Alpine
Boyd, Harold B., M. D., 3418 Waynoka St., Memphis 11. ~Physician~
Chase, Spencer, T. V. A., Norris. ~Horticulturist~
Garrett, Dr. Sam Young, 1902 Hayes St., Nashville. ~Surgeon~
Holdeman, J. E., 855 N. McNeil St., Memphis 7
Howell Nurseries, Sweetwater. ~Ornamental and chestnut nurserymen~
McDaniel, J. C. (See under Illinois)
Meeks, Hamp, c/o Jackson Elec. Dept., Jackson. ~Electrical Engineer~

Murphy, H. O., 12 Sweetbriar Avenue, Chattanooga. ~Fruit grower~
Richards, Dr. Aubrey, Whiteville. ~Physician~
Roark, W. F., Malesus. ~Farmer~, ~chestnut grower~
Robinson, W. Jobe, Route No. 7, Jackson. ~Farmer~
Sammons, Julius, Jr., Pecan Row Farm, Whiteville. ~Farmer~, ~orchardist~
Saville, Chris, 118 Church St., Greeneville
Shipley, Mrs. E. D., 3 Century Court, Knoxville 16. ~Housewife~
Smathers, Rev. Eugene, Calvary Church, Big Lick. Minister, farmer
Southern Nursery & Landscape Co., Attn. Hubert Nicholson, Winchester. ~General nurserymen~

TEXAS

Arford, Charles A., Box 1230, Dalhart. ~R. R. engineer~, ~amateur horticulturist~
Brison, Prof. F. R., Dept, of Horticulture, A. & M. College, College Station
Florida, Kaufman, Box 154, Rotan
Kidd, Clark, Arp Nursery Co., P. O. Box 867, Tyler. ~Nut nurseryman~
Winkler, Andrew, Route 1, Moody. ~Farmer and pecan grower~

UTAH

Petterson, Harlan D., 2076 Jefferson Avenue, Ogden. ~Highway engineer~

VERMONT

Aldrich, A. W., R.F.D. No. 3, Springfield
Collins, Joseph N., Route No. 3, Putney. ~Civil engineer~, ~farmer~

~Ellis, Zenas H., Fair Haven. Perpetual member, "In Memoriam."~
Holbrook, F. C., Scott Farm, Brattleboro

VIRGINIA

Acker Black Walnut Corp., Box 263, Broadway. ~Walnut processors~
Burton, George L., 722 College Street, Bedford
Curthoys, George A., P. O. Box 34, Bristol
Dickerson, T. C., 316-56th Street, Newport News. ~Statistician~, ~farmer~
Dudley, Charles L., Glen Wilton
Gibbs, H. R. Linden. ~Carpenter~, ~wood worker~
Gunther, Eric F., Route No. 1, Box 31, Onancock. ~Retired business man~
Lee, Dr. Henry, 806 Medical Arts Building, Roanoke 11
Narten, Perry F., 6110 N. Washington Blvd., Arlington 5
Pinner, Henry, P. O. Box 155, Suffolk
Stoke, H. F., 1436 Watts Avenue N. W., Roanoke
Stoke, Mrs. H. F., 1436 Watts Avenue, N. W., Roanoke
Stoke, Dr. John H., 21 Highland Avenue, S. E., Roanoke 13. ~Chiropractor~
Thompson, B. H., Harrisonburg. ~Manufacturer of nut crackers~

WASHINGTON

Andrew, Col. James W., Hqts. 39 Wing, A.P.O. 942 c/o P. M., Seattle. (Farm in Illinois.)
Bartleson, C. J., Box 25, Chattaroy. ~Office worker~
Brown, H. B., Greenacres
Bush, Carroll D., Grapeview. ~Chestnut grower and shipper~, ~nurseryman~
Denman, George L., East 1319 Nina Avenue, Spokane 10. ~Dairyman~

Eliot, Craig P., P. O. Box 158, Shelton. ~Electrical engineer~, ~part time farmer~

Erkman, John O., Apt. 85, 1219 Washington Way, Richland. ~Physicist~

Kling, William L., Route No. 2, Box 230, Clarkson

Latterell, Miss Ethel, 408 N. Flora Rd., Greenacres. ~Greenhouse worker~

Linkletter, Frank D., 115 4th Ave. North, Seattle 9. ~Retired~

Naderman, G. W., Route 1, Box 381, Olympia. ~Caretaker of summer resort~

Ross, Vevel C., 4025 Rucker Ave., Everett

Shane Brothers, Vashon

Shepard, Will, Chelan Falls

Tuttle, Lynn, Nursery, The Heights, Clarkston. ~Nut nurseryman~

WEST VIRGINIA

Cannaday, Dr. John E., Charleston General Hospital, Charleston 25. ~Physician~

Engle, Blaine W., Mutual Fire Ins. Co. of W. Va., Goff Bldg., Clarksburg

Frye, Wilbert M., Pleasant Dale. ~Retired~

Gold Chestnut Nursery, c/o Mr. Arthur A. Gold, Cowen. Chestnut nurseryman

Haines, Earl C., Shanks

Long, J. L., Box 491, Princeton. ~Civil engineer~

Mish, Arnold F., Inwood. ~Associational farmer~

Reed, Arthur M., Moundsville. ~Proprietor, Glenmount Nurseries~

WISCONSIN

Ladwig, C. F., 2221 St. Laurence, Beloit. ~Grocer and (hobby) farmer~

Mortensen, M. C., 2117 Slauson Avenue, Racine

Raether, Robert, Route No. 1, Augusta (Eau Claire County)

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